ENT ATLAS

CLINICAL

Nose & sinuses

21 articles · imaging key facts · November 2026

For personal study. Follow local protocols and senior advice for patients.

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CLINICAL · NOSE & SINUSES

Epistaxis

Bleeding from the nose is a sign, not a diagnosis. Most bleeds are small, anterior and self-limiting; a few are posterior, brisk and dangerous. The skill lies in telling the two apart, and in climbing the ladder of treatment one rung at a time.

AT A GLANCE

- Resuscitate while you stop the bleeding.
- Pressure, vasoconstrictor, cautery, then packing.
- Then ligation or embolisation.
- Check anticoagulants and blood pressure.
- Unilateral bleeding with blockage needs a scope.
- SPA dominant in 85%; landmark the crista ethmoidalis; electrocautery beats silver nitrate.

1 CHAPTER ONE Overview

A common bleed with an uncommon tail.

1.1 Definition

Epistaxis is bleeding from inside the nose. The blood may run forward out of the nostril, backward into the nasopharynx and throat, or both. It is always a sign of something else, never a final diagnosis, and Dhingra's rule holds on every shift: an attempt should always be made to find its local or systemic cause.

1.2 Epidemiology

Most people have at least one nosebleed in their lives, and Lee estimates that around one in six of the population bleeds in any given year. Only a small fraction ever seek medical help. The distribution is bimodal: children and young adults, who bleed from the front of the septum, and adults past middle age, who make up most hospital admissions and most posterior bleeds.

1.3 Why it matters

The patients who reach the emergency department are older, often hypertensive, frequently on an anticoagulant or antiplatelet drug, and bleeding from further back in the nose, where blood runs silently down the pharynx and is easy to underestimate. In them, epistaxis is as much a resuscitation problem as a nasal one.

CHAPTER TWO

2 Applied anatomy

Two carotid systems meeting in the nose.

2.1 Arterial supply

The nasal cavity draws blood from both the internal and the external carotid arteries, and the two systems anastomose freely on the septum and the lateral wall. That double supply is why nosebleeds can be persistent, and why ligating one vessel sometimes isn't enough.

2.1.1 Internal carotid system

- **Anterior ethmoidal artery,** from the ophthalmic artery, entering the nose through the roof.
- **Posterior ethmoidal artery,** also from the ophthalmic, smaller and further back.

These supply the upper septum and lateral wall above the middle turbinate.

2.1.2 External carotid system

- **Sphenopalatine artery,** terminal branch of the maxillary artery: the main supply of the nose.
- **Greater palatine artery,** from the maxillary, reaching the septum through the incisive canal.
- **Superior labial artery,** from the facial, supplying the front of the septum and vestibule.

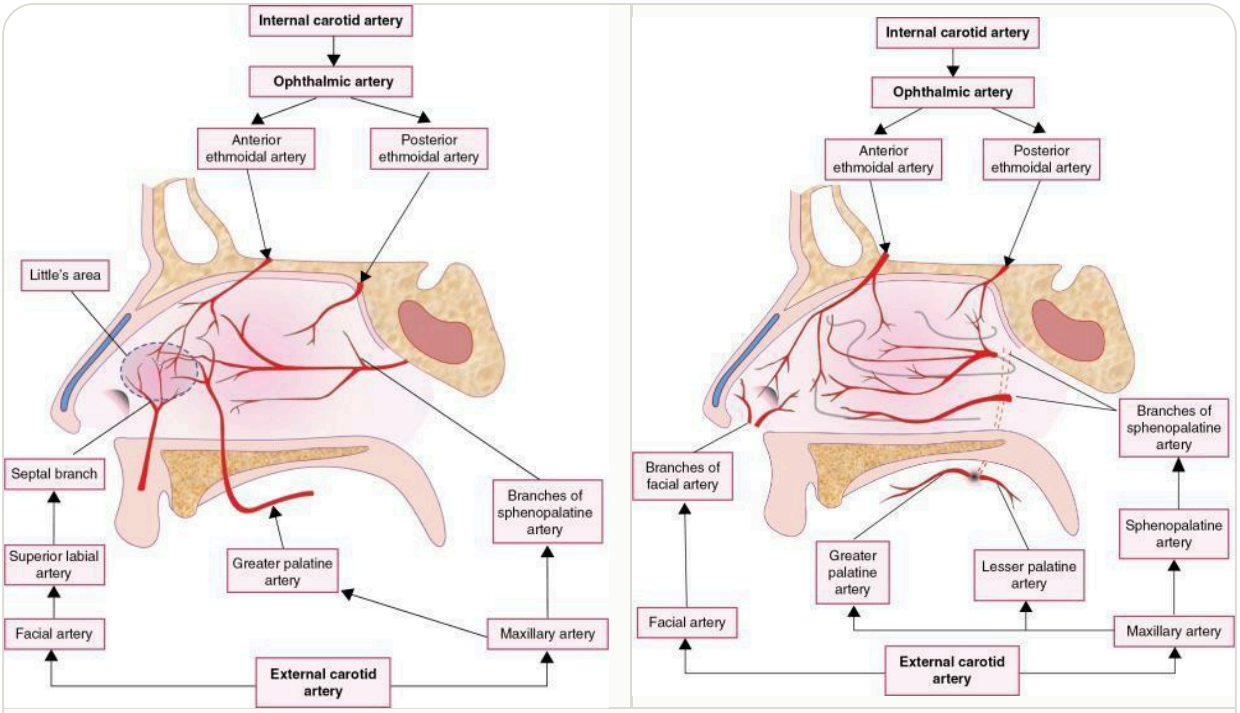


FIGURE 1 Arterial supply of the nasal septum (first) and the lateral nasal wall (second). The anterior ethmoidal, septal branch of the sphenopalatine, greater palatine and superior labial arteries converge on Little's area.

Dhingra, figs 33.1 and 33.2

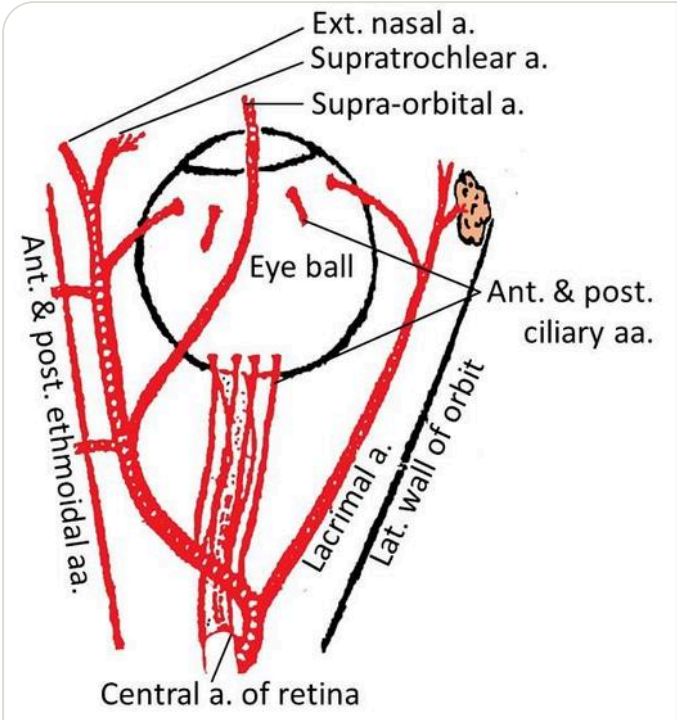


FIGURE 2 The ophthalmic artery giving off the anterior and posterior ethmoidal arteries in the orbit, on their way to the roof of the nose.

Anatomy atlas · Doss p. 100

2.2 Little's area and Kiesselbach's plexus

On the anteroinferior septum, just above the vestibule, four arteries meet to form *Kiesselbach's plexus*; the region it occupies is *Little's area*.

2.2.1 The four contributors

- Anterior ethmoidal (internal carotid)
- Septal branch of the sphenopalatine (external carotid)
- Greater palatine (external carotid)
- Septal branch of the superior labial (external carotid)

2.2.2 Why it bleeds

It sits exactly where inspired air dries and cracks the mucosa, where the mucosa is thin and closely bound to cartilage, and where a fingernail reaches. About nine in ten bleeds start here.

2.3 The posterior territory

2.3.1 Sphenopalatine artery

It enters the nose through the sphenopalatine foramen, just behind the *crista ethmoidalis* of the palatine bone, and divides into the posterior lateral nasal and posterior septal arteries. Bleeding from these branches is the classic posterior epistaxis.

2.3.2 Woodruff's plexus

A plexus on the lateral wall below the posterior end of the inferior turbinate, and the other classic posterior bleeding site in adults.

2.3.3 Ethmoidal territory

Bleeding above the level of the middle turbinate points to the ethmoidal arteries and therefore to the internal carotid system, which matters because embolisation cannot safely reach it.

WHERE THE BOOKS DIFFER

Dhingra	Lee
Woodruff's plexus is a plexus of veins below the posterior end of the inferior turbinate.	Woodruff's is the naso-nasopharyngeal plexus over the posterior centimetre of the

nasal floor, inferior meatus and middle meatus.

For the exam, give Dhingra's definition, and add that posterior arterial bleeds come from sphenopalatine branches.

2.4 Venous drainage

Veins follow the arteries. Dhingra highlights the *retrocolumellar vein*, running vertically just behind the columella across the floor of the nose, as a common source of venous bleeding in young people.

3 CHAPTER THREE Aetiology

Local, systemic, or neither.

CAUSES OF EPISTAXIS

- LOCAL **Trauma** inflammation and drying, structural, neoplastic, iatrogenic
- SYSTEMIC **Drugs** coagulation and platelet disorders, vascular disorders, cardiovascular, organ failure
- IDIOPATHIC **No cause found** despite assessment

3.1 Local causes

3.1.1 Trauma

The commonest cause overall. Digital trauma (nose picking) dominates in children; in adults add forceful blowing, blows to the face, fractures of the nasal bones, midface and skull base, and foreign bodies.

3.1.2 Inflammation and drying

Dry, heated air in winter, acute viral rhinitis and sinusitis, rhinitis sicca, atrophic rhinitis, and the crust-forming granulomatous infections all leave a fragile, cracked mucosa.

3.1.3 Structural

A septal spur or deviation creates turbulent, drying airflow over its edge; a septal perforation bleeds from its crusted rim.

3.1.4 Neoplastic

- **Benign:** juvenile nasopharyngeal angiofibroma, haemangioma, inverted papilloma.
- **Malignant:** carcinoma of the nose, sinuses and nasopharynx.

Any unilateral, recurrent bleed, especially with obstruction, must be examined endoscopically.

3.1.5 Iatrogenic and topical

Nasal and sinus surgery, nasogastric and nasotracheal tubes, nasal oxygen cannulae, topical steroid and decongestant sprays, and intranasal cocaine.

3.2 Systemic causes

The logic is simple: anything that impairs clotting, platelets or vessel walls will show up in the nose.

3.2.1 Drugs and supplements

Warfarin and the direct oral anticoagulants, aspirin, clopidogrel and other antiplatelets, NSAIDs, and, as Lee reminds us, over-the-counter vitamin E, fish oil, garlic, ginkgo and ginseng.

3.2.2 Coagulation and platelet disorders

Haemophilia, von Willebrand disease (the commonest inherited bleeding disorder), thrombocytopenia, leukaemia and aplastic anaemia.

3.2.3 Vascular disorders

Hereditary haemorrhagic telangiectasia (see
8.3
), scurvy, and the vessel fragility of old age.

3.2.4 Cardiovascular

Hypertension and atherosclerosis are associated with posterior bleeds in older adults. Whether raised pressure causes the bleed or simply keeps it going is still debated; either way, it should be measured and treated.

3.2.5 Organ failure

> Liver failure (reduced clotting factors) and renal failure (uraemic platelet dysfunction).

3.3 Idiopathic

In a large share of patients no specific cause is found despite a careful history, examination and blood tests. The label is earned only after the search, not instead of it.

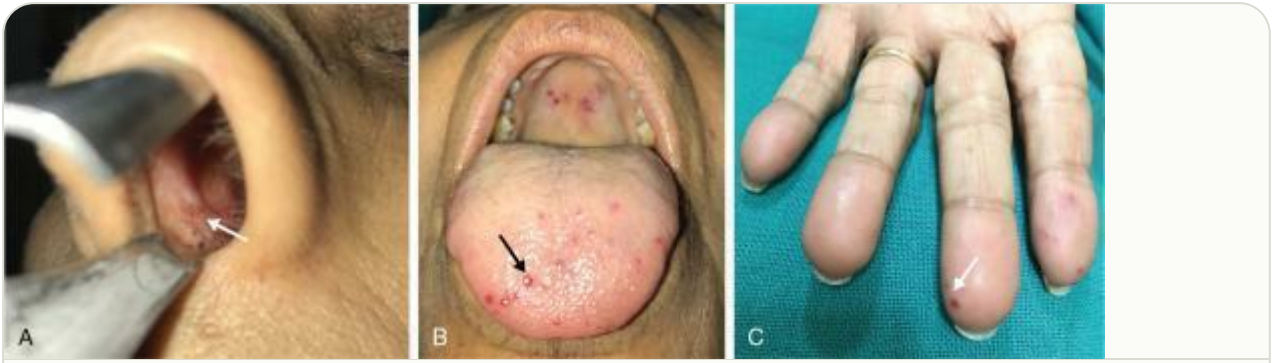


FIGURE 3 Hereditary haemorrhagic telangiectasia: lesions on the nasal septum, tongue, palate and fingertips (arrows). Always look at the lips, tongue and fingers of a recurrent bleeder.
Dhingra, fig. 33.3

RED FLAGS

- **Adolescent boy, unilateral bleeding and obstruction:** juvenile angiofibroma until proven otherwise. No biopsy in clinic.
- **Adult with unilateral bleeding, obstruction, crusting or facial pain:** endoscopy for a sinonasal tumour.
- **A child under two:** rare; think of non-accidental injury or a bleeding disorder.
- **Massive or recurrent bleeding after trauma or skull base surgery:** carotid pseudoaneurysm; needs angiography.

4 CHAPTER FOUR Classification

The distinction that drives everything that follows.

4.1 By site: anterior or posterior

With the patient sitting up, an anterior bleed runs mostly out of the nostril. A posterior bleed runs back into the throat, often from both sides, and no source is seen on anterior rhinoscopy.

TABLE

	ANTERIOR	POSTERIOR
Frequency	Common, about 90%	Less common
Site	Little's area or the front of the lateral wall	Posterosuperior cavity; often hard to localise
Age	Children and young adults	Usually over 40
Cause	Mostly trauma, dryness	Often spontaneous; hypertension, atherosclerosis, anticoagulation
Severity	Usually mild; pressure, cautery or an anterior pack	Often severe; admission, posterior pack or surgery

4.2 By severity

There is no universal scale; in practice the useful split is whether first aid is enough.

4.2.1 Minor

Stops with pressure or a single cautery, haemodynamically stable, can go home.

4.2.2 Major

Continues despite first aid, needs packing, shows haemodynamic change or a falling haemoglobin, or occurs in a frail or anticoagulated patient. These patients are admitted.

4.3 By pattern: single or recurrent

Recurrence changes the question from how to stop it to why it keeps happening. Recurrent unilateral bleeding needs endoscopy; recurrent bilateral bleeding, especially with a family history, needs HHT to be considered.

5 CHAPTER FIVE Clinical evaluation

Resuscitate first, then find the bleeding point, then ask why.

5.1 Immediate assessment

The first question is not where the blood is coming from, but how much has been lost and whether the airway is safe. A posterior bleed can fill the stomach before it fills a basin, and swallowed blood that returns as dark vomit is easily mistaken for haematemesis.

ON THE WARD: THE FIRST FIVE MINUTES

1. Sit the patient up and leaning forward. Check pulse, blood pressure and saturation; secure IV access and send a full blood count, group and save, and clotting if anticoagulated or bleeding heavily.
2. Firm pressure on the soft lower third of the nose for at least five minutes without peeking; many units ask for 10 to 15. Ice in the mouth adds a little reflex vasoconstriction.
3. Suction the clots, spray topical vasoconstrictor and anaesthetic, and look with a headlight and speculum.
4. Ask about anticoagulants, antiplatelets, previous bleeds, family history and alcohol, and which side bled first.
5. Don't stop or reverse anticoagulants for a bleed that local measures can control. Discuss with the prescribing team if it is life-threatening.

5.2 History

5.2.1 The bleed

Side of onset, duration, estimated volume, whether it runs forward or into the throat, and previous episodes and their treatment.

5.2.2 Local symptoms

Obstruction, discharge, crusting, facial pain or numbness, and trauma.

5.2.3 Systemic background

Medications including supplements, hypertension, liver or renal disease, easy bruising, and a family history of bleeding or nosebleeds.

5.3 Examination

5.3.1 General

Pallor, pulse and blood pressure, signs of liver disease, bruising, and telangiectasias on the lips, tongue and fingertips.

5.3.2 Anterior rhinoscopy

After clot removal and vasoconstriction: inspect Little's area first, then the front of the lateral wall. The oropharynx is checked for blood running down from behind.

5.3.3 Nasal endoscopy

For recurrent bleeds, for bleeding not localised on rhinoscopy, and for any unilateral bleed with suspicious symptoms. It finds points hidden lateral to the turbinates or high in the nose and allows directed cautery instead of a blind pack.

5.4 Investigations

5.4.1 Blood tests

Not needed for a minor anterior bleed in a healthy patient. For significant bleeds: full blood count, group and save or crossmatch, clotting with INR in warfarinised patients, and liver and renal function where relevant.

5.4.2 Imaging

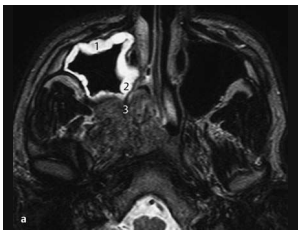
CT or MRI when a tumour is suspected; for juvenile angiofibroma, contrast imaging replaces biopsy.

5.4.3 Angiography

Diagnostic and therapeutic: for suspected pseudoaneurysm, vascular tumours, and as the first step of embolisation.

IMAGING IN THE APP 1 case

Juvenile angiofibroma, the tumour to think of in a teenage boy with one-sided bleeding.



Angiofibroma, Juvenile

Tumours · 12 images

6 CHAPTER SIX Local management

The ladder of control.

6.1 The treatment ladder

1. **First aid and topical vasoconstriction**
   Pressure on the soft nose, leaning forward; pledgets of vasoconstrictor and local anaesthetic.
2. **Cautery of a visible point**
   Silver nitrate or bipolar, to the bleeding point only.
3. **Anterior packing**
   Ribbon gauze, expanding sponge, or resorbable material.
4. **Posterior packing**
   Postnasal pack, Foley or dual balloon; admit and monitor.
5. **Endoscopic cautery**
   Directed at the source under local or general anaesthesia.

6 **Ligation or embolisation**
For bleeding that persists despite packing.

6.2 First aid

Trotter's method: the patient sits up, leaning slightly forward over a basin, breathing through the mouth, while the soft part of the nose is pinched firmly against the septum. Leaning forward stops blood being swallowed and lets you see how much is lost. Cold compresses add reflex vasoconstriction.

6.3 Cautery

Cautery works when you can see what you are treating. Anaesthetise first, and treat only the bleeding point.

6.3.1 Chemical

Roll a silver nitrate stick over a small ring around the vessel, edge first and then inwards, before touching the vessel itself. Best for small, visible anterior points in a field that is not actively pouring.

6.3.2 Electrocautery

Bipolar diathermy, usually under endoscopic view, is more effective for brisker or deeper points and for posterior sites.

PEARL

Never cauterise both sides of the septum at the same sitting: the mucoperichondrium on both sides may necrose and leave a perforation.

6.4 Anterior packing

6.4.1 Ribbon gauze

About a metre of ribbon gauze, 2.5 cm wide in adults and 12 mm in children, soaked in liquid paraffin or bismuth iodoform paraffin paste and laid in layers so that it presses evenly on the whole cavity. Usually removed after 24 hours; antibiotics are given if it is left longer.

6.4.2 Expanding sponges and balloons

Compressed sponges that swell when wetted, and inflatable balloon packs: quicker to insert, more comfortable, and now the standard in most emergency departments.

6.4.3 Resorbable packs

Preferred when the patient is anticoagulated or has a bleeding disorder, because removing a non-resorbable pack can restart the bleed.

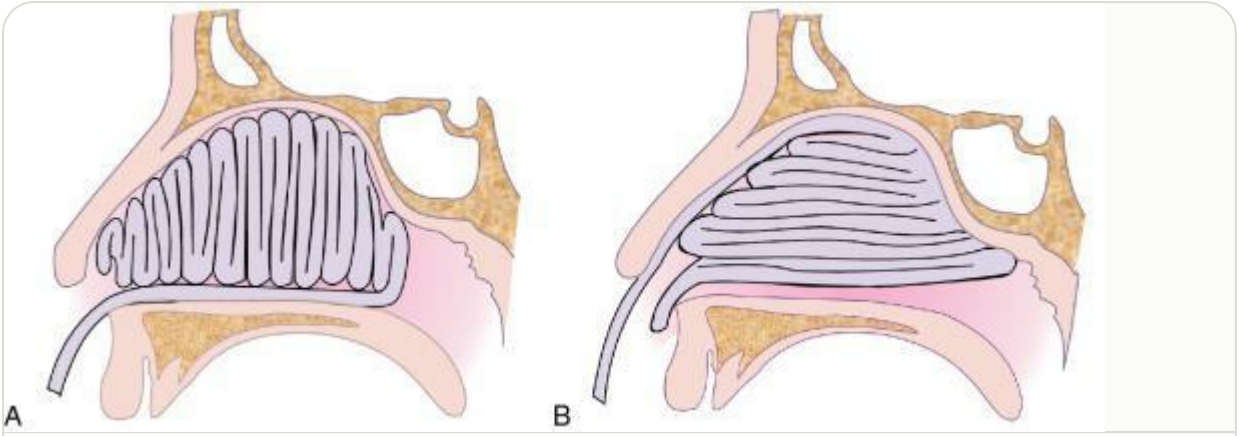


FIGURE 4 Anterior nasal packing. (A) Vertical layers from the back forwards. (B) Horizontal layers from the floor upwards.
Dhingra, fig. 33.4

6.5 Posterior packing

When blood runs back into the pharynx and an anterior pack fails, the choana must be blocked.

6.5.1 Postnasal gauze pack

A rolled gauze cone with three silk ties is drawn into the nasopharynx on a catheter passed through the nose and out of the mouth, then pushed into place with a finger. An anterior pack goes in front, and two ties are secured over a dental roll at the columella. The third tie is left in the oropharynx, taped to the cheek, to pull the pack out later.

6.5.2 Foley catheter

A 12 to 14 F Foley is passed until the balloon lies in the nasopharynx, inflated with saline (not air), drawn forward against the choana and backed by an anterior pack. Protect the ala and columella from pressure.

6.5.3 Dual-balloon devices

Commercial catheters with a posterior and an anterior balloon, and a central tube that preserves a nasal airway.

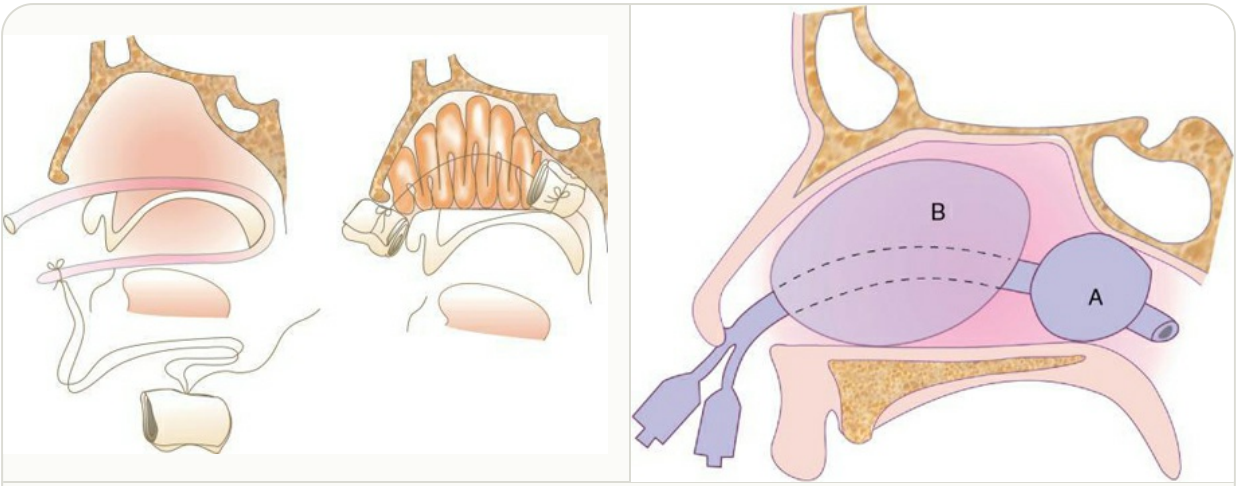


FIGURE 5 First, inserting a postnasal pack on a catheter. Second, a dual-balloon catheter with the posterior balloon at about 10 mL and the anterior at about 30 mL.
Dhingra, figs 33.5 and 33.6

6.5.4 Care of the packed patient

- **Admit and monitor:** posterior packs fail in about a third.
- **Oxygen:** bilateral packs can cause hypoxia through the nasopulmonary reflex, especially in the elderly.
- **Watch for:** alar and columellar necrosis, sinusitis, otitis media, aspiration and toxic shock syndrome.
- **Antibiotics** are traditionally given while a non-resorbable pack is in, though the evidence for routine use is weak.
- **Explain the pack:** what type it is, when it comes out, and what should bring them back.

6.6 Medical adjuncts

6.6.1 Blood pressure

Treat pain and anxiety first; control persistent hypertension, but avoid dropping the pressure sharply.

6.6.2 Tranexamic acid

Topical tranexamic acid did not reduce the need for packing in the NoPAC trial; it should not delay standard treatment.

GUIDELINE UPDATE

The **AAO-HNS Nosebleed guideline (2020)**: compression for at least five minutes; packing if the site cannot be seen; resorbable packs on anticoagulants or antiplatelets; endoscopy for recurrent or difficult bleeds; cautery confined to the bleeding site; ligation or embolisation when these fail; and first-line local treatment before transfusion, reversal or stopping anticoagulants unless the bleed is life-threatening.

CHAPTER SEVEN

7 Surgery and intervention

Taking out the supply: the vessel follows the site.

7.1 Choosing the vessel

PERSISTENT BLEEDING

- SOURCE SEEN **Endoscopic cautery** under local or general anaesthesia
- POSTERIOR, BELOW THE MIDDLE TURBINATE **Sphenopalatine ligation** or embolisation
- ABOVE THE MIDDLE TURBINATE **Ethmoidal ligation** not embolisation
- SEPTAL DEVIATION **Septal surgery** to reach the bleed

7.2 Endoscopic sphenopalatine artery ligation

7.2.1 Indications

Posterior bleeding that persists or recurs despite adequate packing. TESPAL is now the operation of choice and has replaced the older open procedures.

7.2.2 Technique

Under general anaesthesia, a mucosal incision on the lateral wall in front of the crista ethmoidalis; a mucoperiosteal flap is raised back to the crista, behind which the artery and its branches emerge from the foramen. They are clipped or coagulated, and the flap is replaced.

7.2.3 Outcomes and complications

Lee quotes control in 95 to 98%, usually with discharge the next day. Complications are minor: crusting, adhesions and numbness of the palate or nose.

7.3 Ethmoidal artery ligation

7.3.1 Approach

An external (Lynch) incision at the medial canthus, raising the periosteum of the medial orbital wall, usually for high bleeds after trauma.

7.3.2 The 24-12-6 rule

The anterior ethmoidal foramen lies about 24 mm behind the anterior lacrimal crest, the posterior foramen about 12 mm further back, and the optic canal about 6 mm beyond that. The posterior artery is approached with great care, or left alone.

7.4 Embolisation

7.4.1 Method

The internal maxillary artery is catheterised by the femoral route and blocked with particles, gelfoam or coils, often bilaterally because of cross-flow.

7.4.2 Who it suits

Patients unfit for anaesthesia, vascular tumours and traumatic pseudoaneurysms. It cannot treat the ethmoidal arteries, which arise from the internal carotid system.

7.4.3 Outcomes and risks

Success around 88%. Complications include stroke, blindness, facial pain and skin necrosis.

7.5 Septal surgery

Raising a mucoperichondrial flap divides the vessels feeding a recurrent septal bleed; septoplasty also straightens a deviation that hides a bleeding point or

prevents adequate packing.

7.6 Historical operations

7.6.1 Maxillary artery ligation

Through a Caldwell-Luc approach and the posterior antral wall. Failure in 10 to 15%, with infraorbital numbness, oroantral fistula and dental injury.

7.6.2 External carotid ligation

Above the origin of the superior thyroid artery. Too proximal to be reliable, given the collateral flow; of examination interest now.

PEARLS

- Bleeding above the middle turbinate is ethmoidal: embolisation won't help, think ligation.
- The crista ethmoidalis is the landmark for the sphenopalatine foramen: the artery lies just behind it.
- If endoscopic ligation fails, look for an ethmoidal source or a missed accessory branch before repacking.

OPERATIVE GUIDES IN THEATRE 4 operations

Anterior and Posterior
Anterior and posterior ethmoid artery ligation
Sphenopalatine Artery Ligation for Epistaxis
Sphenopalatine artery ligation
Surgical Management of Hereditary Hemorrhagic Telangiectasia
HHT
Juvenile Angiofibroma
Angiofibroma

8 CHAPTER EIGHT Special situations

Children, the anticoagulated, HHT, and the bleed after trauma.

8.1 Children

8.1.1 Typical picture

Bleeding from Little's area, from crusting and picking, that stops on its own.

8.1.2 Treatment

Antiseptic cream such as chlorhexidine with neomycin to the vestibule twice daily for four weeks settles most cases; silver nitrate for a visible vessel that keeps bleeding. Avoid creams containing arachis oil in peanut allergy.

8.1.3 When to investigate

Recurrent bleeding with easy bruising or a family history needs a clotting screen including von Willebrand studies. Under two years of age, think of a bleeding disorder or non-accidental injury.

8.2 The anticoagulated patient

Patients on warfarin, a direct oral anticoagulant or dual antiplatelets bleed longer and re-bleed more often, but stopping their drugs risks stroke or stent thrombosis. Control the bleed locally first with resorbable packing and check the INR in warfarinised patients. Reversal and withholding are for life-threatening bleeds, decided with the prescribing team.

8.3 Hereditary haemorrhagic telangiectasia

Recurrent bleeding from multiple telangiectasias, usually from childhood, with a family history. Repeated packing and aggressive cautery make things worse; laser, bevacizumab, septodermoplasty and Young's procedure are the tools. See the HHT topic for the full account.

8.4 Trauma and pseudoaneurysm

Delayed, massive or recurrent bleeding days to weeks after head injury or skull base surgery suggests a pseudoaneurysm of the internal carotid artery. It is a neurovascular emergency: pack, resuscitate and arrange urgent angiography.

9 CHAPTER NINE Complications and outcome

Of the bleed and of its treatment.

9.1 Of the bleed

Hypovolaemic shock, anaemia, aspiration, and myocardial ischaemia in patients with coronary disease.

9.2 Of treatment

- **Cautery:** septal perforation.
- **Packing:** pressure necrosis, sinusitis, otitis media, hypoxia, toxic shock syndrome.
- **Surgery and embolisation:** see 7.2 to 7.4 .

9.3 Prognosis

Most bleeds settle with simple measures. Recurrence is common in the elderly, the anticoagulated and those with HHT, which is why the cause matters as much as the cure.

PRACTICAL POINTS (OXFORD TEXTBOOK)

About 95% of bleeds arise from Little's area, but the **sphenopalatine artery** is the dominant supply to the nose in about 85% of people (the rest are anterior ethmoid-dominant), which is why endoscopic sphenopalatine ligation works so well for posterior bleeds. At surgery the foramen lies just behind and below the **crista ethmoidalis**, a bony ridge on the lateral wall near the posterior end of the middle turbinate, and the artery often divides into several branches before emerging, all of which must be controlled. Risks to explain include rebleeding, crusting, palatal numbness, adhesions, epiphora and, very rarely, visual loss.

Where available, **electrocautery** is preferred to silver nitrate because it fails less often and reduces packing, admission and pain; after silver nitrate, irrigate with saline to

prevent skin burns. Ice on the nasal bridge has no proven benefit. A patient whose anterior bleed is stopped by cautery can go home with antiseptic or antibiotic cream for two weeks; after a significant posterior bleed, observe for 6 to 12 hours; anyone with a pack is admitted. Many specialists give antibiotics when a pack stays more than 48 hours. For warfarin with major bleeding, give intravenous vitamin K and prothrombin complex concentrate (fresh frozen plasma only if that is unavailable); for minor bleeding with a high INR, withhold warfarin and give oral vitamin K. Ask about "natural blood thinners" such as garlic, ginger, ginkgo, turmeric and vitamin E.

10 CHAPTER TEN For the examination

What the examiner wants to hear.

10.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Little's area:** the four arteries, their parent vessels, and why it bleeds.
- **Trotter's method:** position, mouth breathing, and pinching the soft nose.
- **Anterior versus posterior:** the five-row comparison.
- **Postnasal pack:** technique, the third tie, and why the patient is admitted.
- **Which artery for which site,** and why ethmoidal bleeds can't be embolised.
- **Teenage boy, unilateral recurrent bleed:** angiofibroma, and no biopsy in clinic.

10.2 Common pitfalls

COMMON PITFALLS

- Underestimating blood loss in an elderly patient who is swallowing blood.
- Packing blindly before suctioning and looking, when a cauterisable point was visible.
- Cauterising both sides of the septum at once.
- Stopping anticoagulation reflexively for a bleed a pack would have controlled.

- Repeated packing in recurrent bilateral bleeds that are really HHT.

Written for this app from Oxford Textbook of Otolaryngology ch. 27, 77, Oxford Specialist Handbook ch. 20, Oxford Handbook of ENT ch. 9, 25, Dhingra ch. 33 and K. J. Lee ch. 31. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Hereditary haemorrhagic telangiectasia

Hereditary haemorrhagic telangiectasia is a genetic disease of blood vessels that presents to the rhinologist as nosebleeds that never stop coming back. The nose is the visible part of a systemic disorder: every patient also needs screening for silent arteriovenous malformations in the lungs and brain.

AT A GLANCE

- Recurrent bleeds, telangiectasia and family history.
- Screen for lung and brain AVMs.
- Use gentle care; avoid repeated cautery.
- Three genes (ENG, ACVRL1, SMAD4); step up from KTP laser to septodermoplasty to nasal closure; resorbable packs only.

1 CHAPTER ONE Overview

A vascular disease that bleeds through the nose.

1.1 Definition

HHT, also called *Osler-Weber-Rendu syndrome*, is an autosomal dominant disorder of angiogenesis. It produces fragile *telangiectasias* in the skin and mucous membranes and larger *arteriovenous malformations* (AVMs) in internal organs.

1.2 Epidemiology

About 1 in 5,000 to 10,000 people, reported mainly in white populations, affecting men and women equally. Penetrance increases with age.

1.3 Why it matters

Recurrent epistaxis causes chronic iron-deficiency anaemia and repeated transfusions, and treating it badly (repeated packing and aggressive cautery)

makes it worse. Pulmonary AVMs cause paradoxical embolism, stroke and brain abscess; cerebral AVMs can bleed.

2 CHAPTER TWO Pathogenesis

A defect in vessel signalling.

2.1 Genetics

Mutations disturb transforming growth factor-beta signalling in endothelial cells, so the connective tissue and matrix that should support new vessels during angiogenesis fail to form.

- **HHT type 1:** mutation in *endoglin* (chromosome 9q34); pulmonary and cerebral AVMs are commoner.
- **HHT type 2:** mutation in *ACVRL1*, also called ALK1 (chromosome 12q); hepatic AVMs are commoner.
- **Juvenile polyposis–HHT overlap:** mutation in *SMAD4*; these patients need colonoscopy because of colonic polyps and cancer risk. Together, the three genes explain about 90% of clinically diagnosed cases, and genetic testing is mainly used to screen relatives.

2.2 The lesion

Postcapillary venules dilate and connect directly to dilated arterioles, losing the capillary bed between them. The walls lack elastic tissue and smooth muscle, so they cannot contract after injury: minor trauma from dry air or a sneeze causes brisk bleeding that does not stop easily.

3 CHAPTER THREE Clinical features

Nose, skin and organs.

3.1 Epistaxis

The commonest and earliest feature: recurrent, spontaneous bleeding from childhood, from both sides, becoming more frequent with age. About 80 to 90% have nosebleeds by age 21.

3.2 Mucocutaneous telangiectasias

Small red spots that blanch with pressure, on the lips, tongue, palate, fingertips, face and nasal mucosa. About 80 to 90% develop them by their thirties or forties.

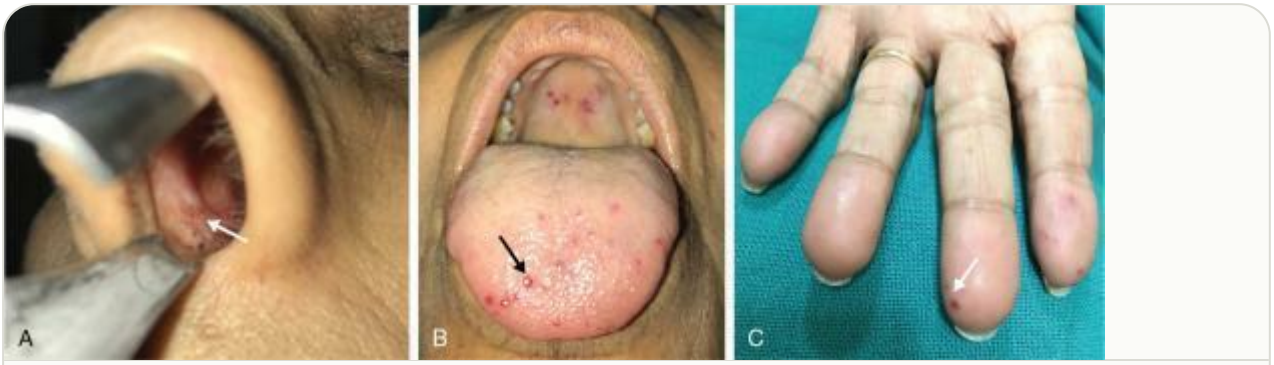


FIGURE 1 A patient with recurrent epistaxis from HHT, with telangiectasias on the nasal septum, tongue, palate and fingertips (arrows).
Dhingra, fig. 33.3

3.3 Visceral arteriovenous malformations

ORGAN INVOLVEMENT IN HHT (LEE)

ORGAN	FREQUENCY	WHAT IT CAUSES
Lungs	About 30%	Hypoxia, paradoxical emboli, stroke, brain abscess, haemoptysis
Brain	5–15%	Haemorrhage, seizures
Liver	40–70%	Usually silent; high-output heart failure, portal hypertension

ORGAN	FREQUENCY	WHAT IT CAUSES
Gut	20–25%	Bleeding and anaemia
Spine	Rare	Spinal haemorrhage

3.4 Anaemia

Chronic iron-deficiency anaemia from nasal and gut bleeding.

4 CHAPTER FOUR Diagnosis

The Curaçao criteria.

4.1 Criteria

- **Epistaxis:** spontaneous and recurrent.
- **Telangiectasias:** multiple, at characteristic sites (lips, mouth, fingers, nose).
- **Visceral lesions:** gastrointestinal telangiectasia, or pulmonary, hepatic, cerebral or spinal AVMs.
- **Family history:** a first-degree relative with HHT.

4.2 Interpretation

Three or more criteria: *definite*. Two: *possible* or suspected. Fewer than two: unlikely. Genetic testing confirms the diagnosis in uncertain cases and in relatives.

4.3 Work-up

- Full head and neck examination with nasal endoscopy.
- Full blood count and iron studies.
- **Contrast echocardiography** (bubble study) to screen for pulmonary AVMs, in every patient.
- **MRI brain with contrast** for cerebral AVMs.

- Endoscopy for symptomatic gastrointestinal bleeding; Doppler ultrasound or CT for the liver.
- Genetic counselling.

SCREEN THE LUNGS

A pulmonary AVM can cause a stroke or brain abscess in a patient who has only ever complained about nosebleeds. Screen every patient with contrast echocardiography and treat significant AVMs by embolisation.

4.4 Differential diagnosis

Bleeding disorders (von Willebrand disease, platelet disorders), anticoagulant use, and a bleeding nasal tumour in one-sided bleeding.

5 CHAPTER FIVE Management of the nose

Reduce bleeding without damaging the mucosa.

5.1 Principles

The goal is fewer and smaller bleeds and a better haemoglobin, not cure. Avoid trauma to the mucosa: repeated packing and aggressive cautery destroy healthy mucosa, encourage new telangiectasias and can cause septal perforation.

5.2 The acute bleed

Gentle pressure; resorbable packing if needed, because removing a conventional pack tears lesions and restarts bleeding; avoid cautery of large areas.

5.3 Conservative and medical treatment

1. **Humidification and emollients**
   Saline sprays, ointments and humidified air keep the mucosa moist.

2 **Iron and transfusion**
Correct anaemia; intravenous iron when oral iron fails.

3 **Tranexamic acid**
Oral antifibrinolytic reduces the frequency and severity of bleeds.

4 **Hormonal treatment**
Oestrogen or tamoxifen convert the mucosa to a thicker squamous type.

5 **Bevacizumab**
An anti-VEGF antibody, given topically, by submucosal injection or intravenously, reduces bleeding in severe disease.

5.4 Procedures

5.4.1 Photocoagulation

KTP, argon, Nd:YAG or diode laser, bipolar diathermy or coblation, aimed at individual lesions and repeated every few months as new ones appear.

5.4.2 Septodermoplasty

The bleeding anterior septal mucosa is replaced with a split- or full-thickness skin graft, which is tougher. It halves the number of later laser treatments.

5.4.3 Young's procedure

Closure of the nostrils, one or both sides, by suturing flaps inside the vestibule. Bleeding stops in about 83%, at the price of nasal breathing and smell. Rebleeding after reopening is troublesome.

5.4.4 Embolisation

Gives only temporary benefit because new collaterals form; reserved for emergencies.

OPERATIVE GUIDES IN THEATRE 1 operation

Surgical Management of Hereditary Hemorrhagic Telangiectasia
Laser, septodermoplasty, Young's closure

PEARLS

- Repeated cautery makes HHT worse.
- Resorbable packs, never conventional ribbon gauze, if packing is unavoidable.

- Treat the nose and the haemoglobin together.

A STAGED APPROACH (OXFORD TEXTBOOK)

Treatment is stepped up by severity, judged by bleeding frequency, the haemoglobin and transfusion needs; the Epistaxis Severity Score helps track it. The first intervention is usually an examination under anaesthesia with **KTP or argon laser** to stage and treat the disease: these green wavelengths are absorbed by haemoglobin, each lesion is circled and then ablated, tranexamic acid and antibiotic cover are given at induction, and no pack is needed. Bipolar diathermy works better than laser in a bloody field. If laser controls bleeding for less than about three months, move to **septodermoplasty** or systemic treatment; benefit lasts months after laser, years after septodermoplasty, and permanently after **nasal closure** (the Lund-modified Young's procedure with three vestibular flaps), which is offered to transfusion-dependent patients who accept the loss of nasal breathing and smell.

In an acute bleed, bleeding is almost always from the anterior third, so posterior packs are rarely needed. Use resorbable packs such as Nasopore or Surgicel, or a haemostatic matrix, rather than expanding sponges that tear other lesions; give antibiotic cover for any pack; and if packing fails, embolise the facial and maxillary feeders without coils. Recent trials of systemic anti-angiogenic drugs, including intravenous bevacizumab and oral pomalidomide, have reduced bleeding in severe disease, and these are now used by specialist HHT centres.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Inheritance and genes:** autosomal dominant; endoglin (type 1) and ACVRL1 or ALK1 (type 2).
- **Curaçao criteria** and how many make the diagnosis definite.
- **Screening** for pulmonary and cerebral AVMs.

- **The treatment ladder,** septodermoplasty and Young's procedure.

6.2 Common pitfalls

COMMON PITFALLS

- Treating every bleed as simple epistaxis without asking about family history and looking at the lips and fingers.
- Repeated packing and wide cautery.
- Forgetting to screen the lungs.

Written for this app from Oxford Textbook of Otolaryngology ch. 43, Oxford Specialist Handbook ch. 20, Oxford Handbook of ENT ch. 9, Dhingra ch. 33 and K. J. Lee ch. 31. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Deviated septum, septal haematoma and abscess

The septum is the midline wall that holds up the lower nose and divides the airway. A deviated septum only matters when it blocks the patient. A swollen septum after a blow is the emergency: a haematoma drained the same day is a minor procedure, while one left alone becomes an abscess, a dead cartilage and a saddle nose.

AT A GLANCE

- Treat deviation only if it causes symptoms.
- Always examine the septum after trauma.
- Drain a haematoma the same day.
- Keep an L-strut in septoplasty.
- Large perforations can now be closed with an anterior ethmoidal artery flap; buttons are no longer a routine first step.

1 CHAPTER ONE Overview

The partition of the nose.

1.1 Applied anatomy

The septum has three parts: the *columellar septum* (the columella, with the medial crura of the alar cartilages), the *membranous septum* (skin between the columella and the septal cartilage), and the *septum proper*, made of cartilage and bone covered by mucoperichondrium and mucoperiosteum.

1.1.1 The framework

- **Quadrilateral (septal) cartilage:** the front of the septum, supporting the nasal tip and dorsum.
- **Perpendicular plate of the ethmoid:** above and behind.
- **Vomer:** below and behind.
- **Small contributions:** the crest of the maxilla and palatine bones, the nasal spine of the frontal bone, and the rostrum of the sphenoid.

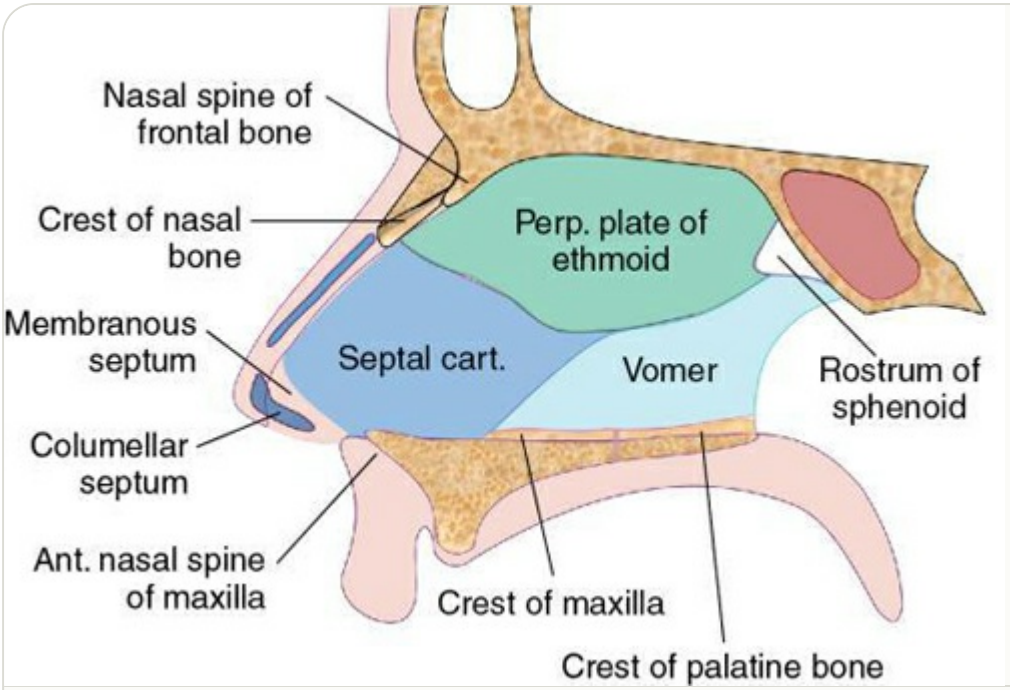


FIGURE 1 Anatomy of the nasal septum: septal cartilage, perpendicular plate of the ethmoid and vomer.

Dhingra, fig. 26.1

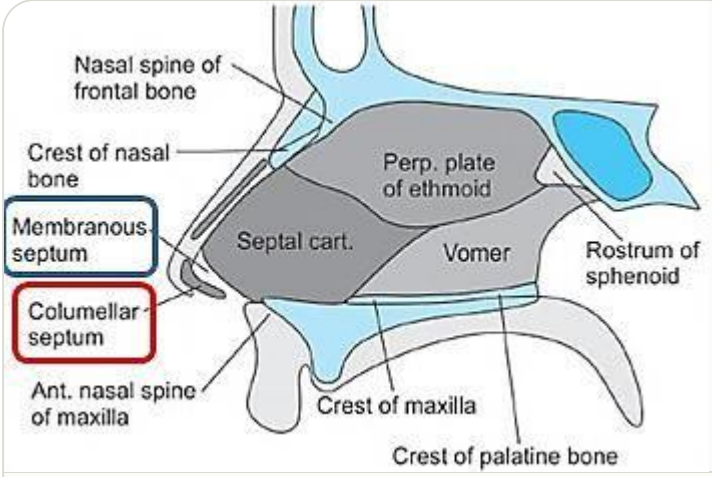


FIGURE 2 The septal cartilage and bones of the septum.

Anatomy atlas · ENT Steps p. 104

1.1.2 Blood supply

Little's area on the anteroinferior septum, where four arteries meet, is the commonest site of epistaxis (see the epistaxis topic).

1.2 Why it matters

Septal problems cause nasal obstruction, sinusitis, epistaxis and external

deformity, and septal surgery is one of the commonest operations in ENT.

2 CHAPTER TWO Septal trauma

Fractures of the septum.

2.1 How it happens

Blows from the front, side or below buckle, fracture, dislocate or crush the septum. The septal cartilage can dislocate from the vomerine groove without any fracture of the nasal bones. Mucosal tears cause profuse epistaxis; an intact mucosa traps blood as a haematoma.

2.2 Named fractures

- **Jarjaway fracture:** from a frontal blow, running horizontally backwards just above the junction of the septal cartilage with the vomer.
- **Chevallet fracture:** from a blow from below, running vertically from the anterior nasal spine up to the junction of the bony and cartilaginous dorsum.

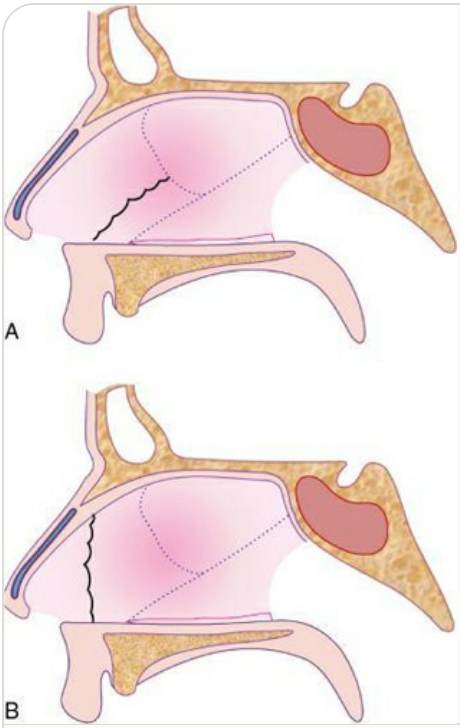


FIGURE 3 Septal fractures: (A) Jarjaway type, (B) Chevallet type.
Dhingra, fig. 26.2

2.3 Treatment

Drain any haematoma; reposition dislocated or fractured fragments, support them between the flaps with mattress sutures and packing, and treat any nasal bone fracture at the same time. Ignored septal injuries leave a deviated cartilaginous nose and an asymmetric tip, columella or nostril.

CHAPTER THREE

3 Deviated nasal septum

When the partition is crooked.

3.1 Causes

3.1.1 Trauma

A lateral blow dislocates the cartilage from the vomerine groove and maxillary crest; a frontal blow buckles and telescopes it. Childhood injuries, and birth trauma from a difficult delivery, are often forgotten.

3.1.2 Developmental

Unequal growth of the palate and the skull base buckles the septum. Mouth breathers with adenoid hypertrophy often have a high-arched palate and a deviated septum; it is also seen with cleft lip and palate and dental anomalies.

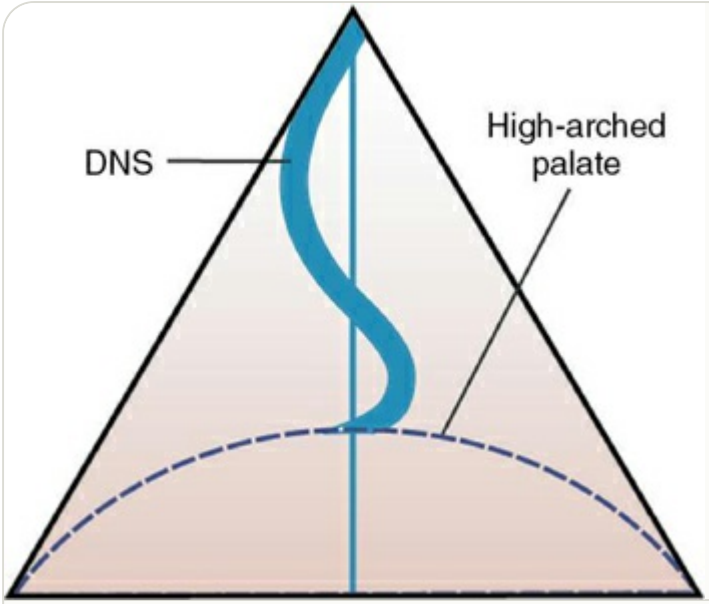


FIGURE 4 Deviated septum with a high-arched palate.
Dhingra, fig. 26.3

3.1.3 Racial and hereditary

Commoner in white populations, and often in several members of a family.

3.2 Types

- **Anterior (caudal) dislocation:** the lower border of the cartilage projects into one nostril, best seen with the head tilted back.
- **C-shaped deviation:** a single curve; the wider side often shows compensatory turbinate hypertrophy.
- **S-shaped deviation:** two curves, which can block both sides.
- **Spur:** a shelf-like projection, usually at the junction of bone and cartilage, which may press on the lateral wall.
- **Thickening:** from an organised haematoma or overlapping fragments.

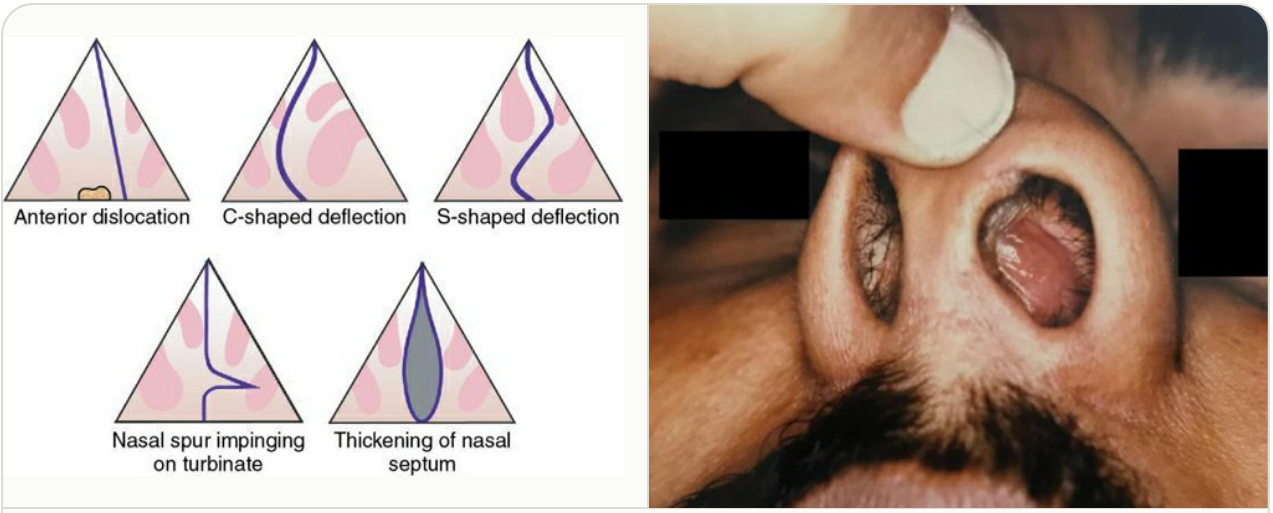


FIGURE 5 First, the types of septal deviation. Second, a C-shaped deviation nearly blocking the left side.

Dhingra, figs 26.4 and 26.6

3.3 Clinical features

3.3.1 Nasal obstruction

One-sided or both; high deviations block more than low ones, because the main airflow passes through the upper nose. Paradoxical obstruction on the wider side comes from compensatory turbinate hypertrophy.

3.3.2 Other symptoms

- **Headache,** from a spur pressing on the lateral wall.
- **Sinusitis,** from obstructed ostia and poor ventilation.
- **Epistaxis,** from crusting over the convexity or vessels stretched over a spur.
- **Reduced smell,** when air cannot reach the olfactory cleft.
- **External deformity** of the dorsum, tip or columella.
- **Middle ear infections.**

3.4 Where is the obstruction?

LEVELS OF NASAL OBSTRUCTION

LEVEL	CAUSES
Vestibule	Caudal septal dislocation, stenosis, synechiae

LEVEL	CAUSES
Nasal valve	Narrow valve angle, collapse after rhinoplasty, synechiae
Attic (upper septum)	High septal deviation
Turbinates	Hypertrophy, concha bullosa
Choana	Choanal atresia, choanal polyp, adenoids

3.5 The nasal valve

The narrowest part of the airway, bounded by the septum, the lower edge of the upper lateral cartilage (normally at an angle of about 10 to 15 degrees) and the head of the inferior turbinate. *Cottle's test*: pulling the cheek laterally opens the valve; if breathing improves, the valve is part of the problem.

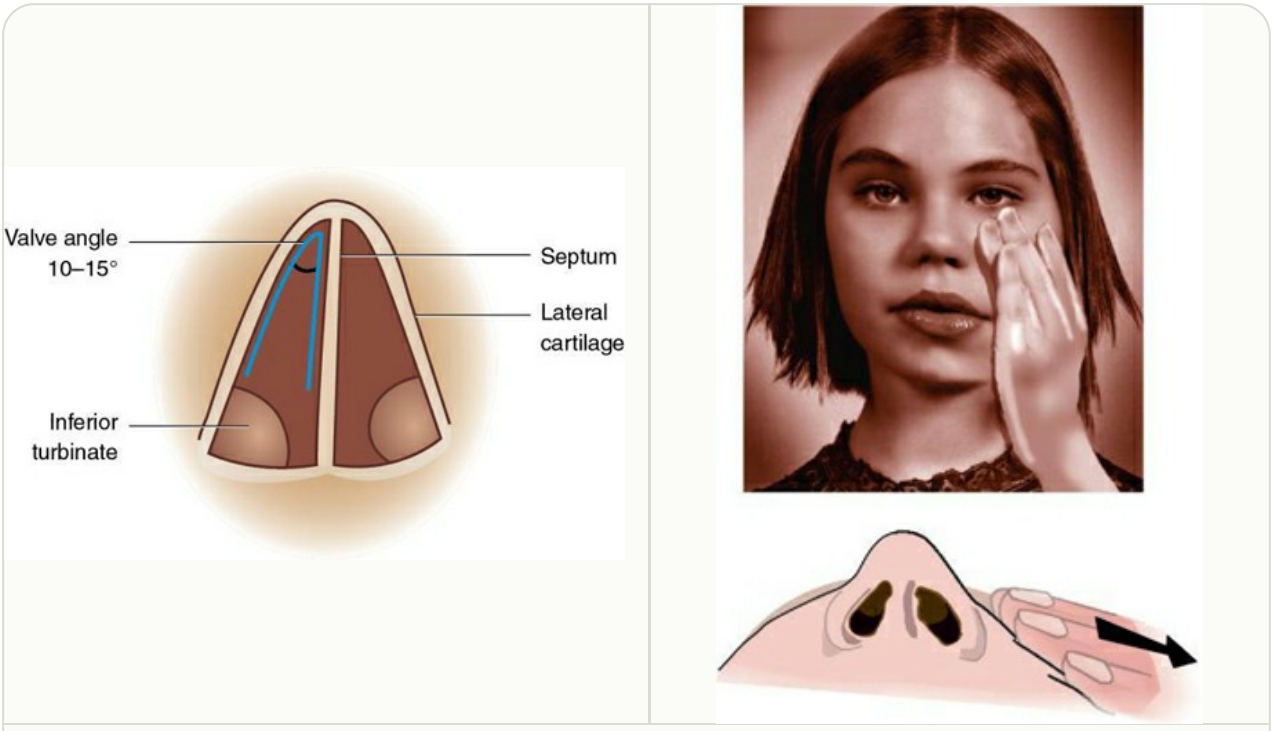


FIGURE 6 First, the nasal valve area. Second, Cottle's test: pulling the cheek laterally opens the valve.

Dhingra, figs 23.3 and 26.7

3.6 Management

Minor, symptomless deviations need nothing. Treat coexisting rhinitis first. Operate when the deviation causes obstruction or the problems listed above.

3.6.1 Septoplasty

The conservative operation: only the most deviated parts are removed; the rest is straightened (by scoring, morselising or repositioning) and put back. A flap is usually raised on one side only. It has largely replaced submucous resection, and can be done in children with severe obstruction because it preserves growth centres.

3.6.2 Submucous resection

Flaps are raised on both sides and the deviated cartilage and bone are removed, leaving dorsal and caudal struts; now mostly historical (see the nose and sinus operations topic).

3.6.3 Associated procedures

Turbinate reduction for hypertrophy, and septorhinoplasty when the external nose is deviated.

OPERATIVE GUIDES IN THEATRE 4 operations

Septoplasty—Classic and Endoscopic
Septoplasty
Nasal Septal Perforation
Perforation
Surgery of the Turbinate
Turbinate surgery
Rhinoplasty and Surgery of the Nasal Valve
Nasal valve work

4 CHAPTER FOUR Septal haematoma and abscess

The emergency.

4.1 Septal haematoma

4.1.1 Cause

Blood under the perichondrium or periosteum, after nasal trauma or septal surgery, or spontaneously in bleeding disorders.

4.1.2 Clinical picture

Bilateral nasal obstruction, frontal headache and pressure over the bridge, with a smooth, rounded, soft swelling of the septum in both nostrils.

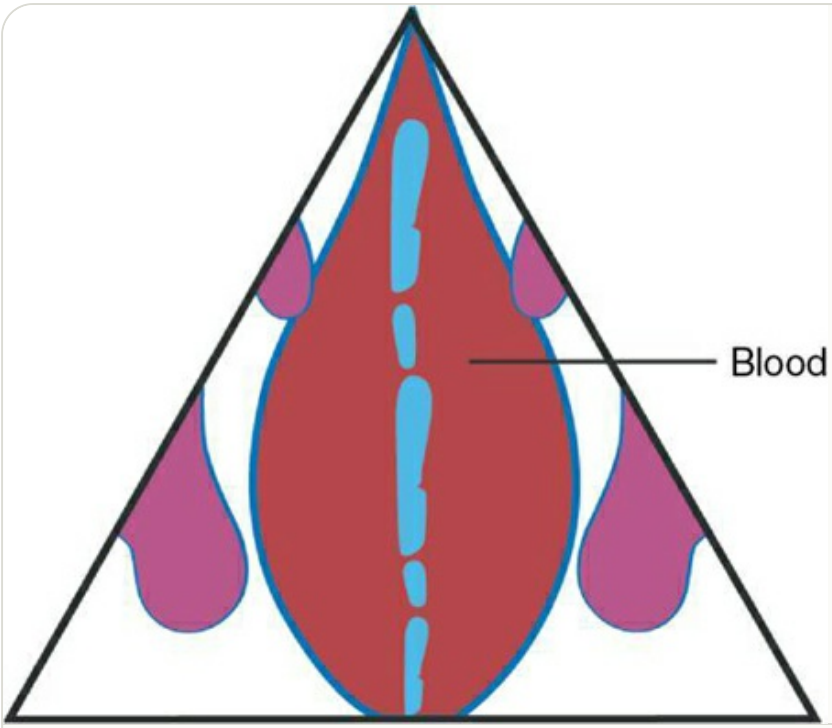


FIGURE 7 Septal haematoma: smooth swellings of the septum in both nostrils.
Dhingra, fig. 26.8

4.1.3 Treatment

Aspirate small collections with a wide-bore needle; incise larger ones parallel to the floor, excise a small strip of mucosa to keep it draining, pack both sides to prevent reaccumulation, and give antibiotics to prevent abscess.

4.2 Septal abscess

4.2.1 Cause

Usually an infected haematoma; occasionally after a furuncle of the nose or lip, or systemic infections such as typhoid or measles.

4.2.2 Clinical picture

Severe obstruction, pain and tenderness over the bridge, fever and chills, a red, swollen nose, fluctuant septal swelling and tender submandibular nodes.



FIGURE 8 Septal abscess.
Dhingra, fig. 26.9

4.2.3 Treatment

Drain urgently at the most dependent point, excise a piece of mucosa, remove pus and dead cartilage, reopen daily for two or three days if needed, and give antibiotics guided by culture for at least ten days.

4.3 Complications

- Cartilage necrosis and *saddle nose* (supratip depression), corrected by augmentation rhinoplasty after two or three months.
- Septal perforation.
- Meningitis and cavernous sinus thrombosis, rare but serious.

NEVER SEND HOME A SWOLLEN SEPTUM

A child or adult with nasal obstruction after a blow needs the septum examined. A missed haematoma in a child may only show itself years later as a saddle nose.

5 CHAPTER FIVE Septal perforation

A hole in the partition.

5.1 Causes

CAUSES OF SEPTAL PERFORATION

- TRAUMA **Septal surgery** repeated cautery, nose picking, piercing
- INFECTION AND INFLAMMATION **Septal abscess** myiasis, rhinolith or foreign body, tuberculosis, lupus, leprosy, syphilis
- DESTRUCTIVE DISEASE **Granulomatosis with polyangiitis** NK/T-cell lymphoma, carcinoma
- DRUGS AND CHEMICALS **Cocaine** long-term steroid sprays, chromium, arsenic, soda ash
- UNKNOWN **Idiopathic**

Lupus, tuberculosis and leprosy perforate the cartilaginous septum; syphilis perforates the bony septum.

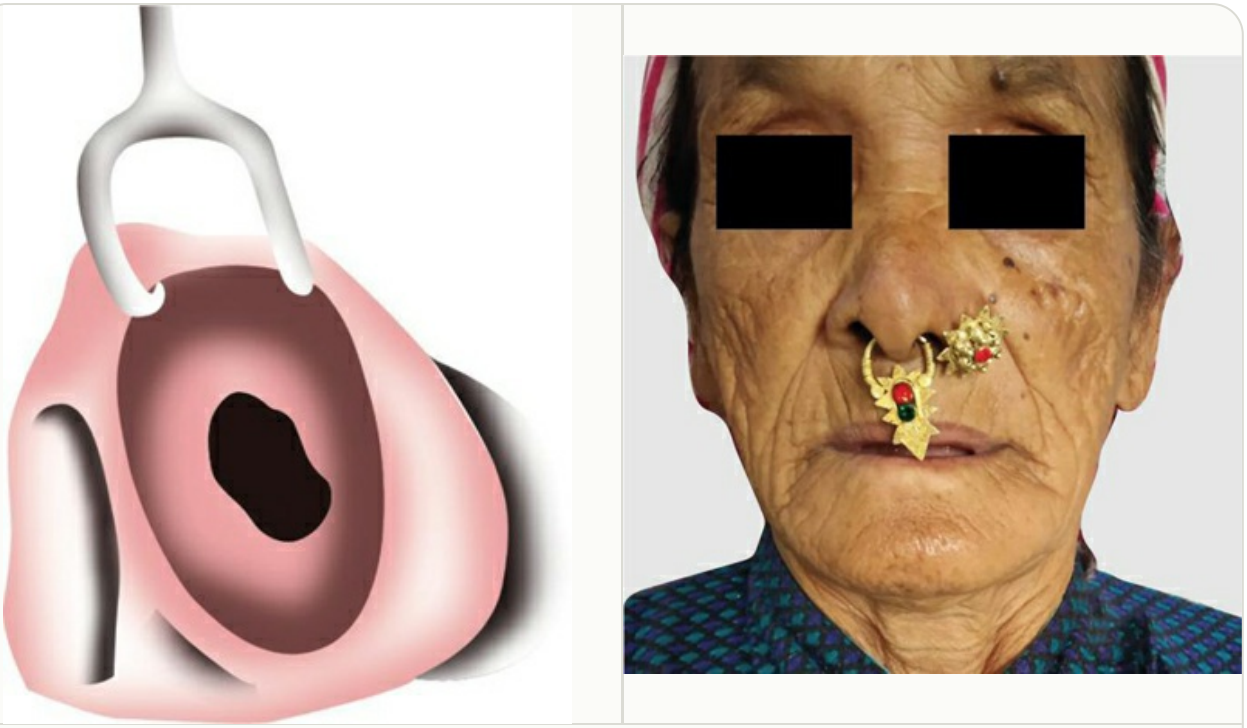


FIGURE 9 First, a septal perforation. Second, a septum deliberately perforated for an ornament.
Dhingra, figs 26.10 and 26.11

5.2 Clinical picture

Small anterior perforations whistle with breathing. Larger ones crust, obstruct and bleed when crusts are removed; very large ones can cause a saddle nose.

5.3 Investigations

Look for the cause: blood tests (ANCA, ACE, syphilis serology), urine drug screen where relevant, and biopsy of granulation or of the edge of the perforation (taken from the posterosuperior edge, away from the healed anterior rim).

5.4 Treatment

- **Conservative:** saline or alkaline douches and bland ointment to keep the nose free of crusts.
- **Septal button:** a silicone obturator.
- **Surgical closure:** for small, inactive perforations, with local rotation or advancement flaps and an interposition graft.

CURRENT PRACTICE

The Oxford Textbook reflects a move away from the routine "trial" of a silastic button: some patients find it irritating and obstructive, and a button can enlarge the perforation. Custom-made obturators are more comfortable but lack strong evidence. Perforations larger than about 2 cm are hard to close with local advancement flaps, but techniques borrowed from endoscopic skull base surgery, such as a pedicled flap based on the **anterior ethmoidal artery**, now allow closure of larger defects. Cocaine use should be asked about at any age, and posterior perforations at the bony–cartilaginous junction are the ones that bleed heavily.

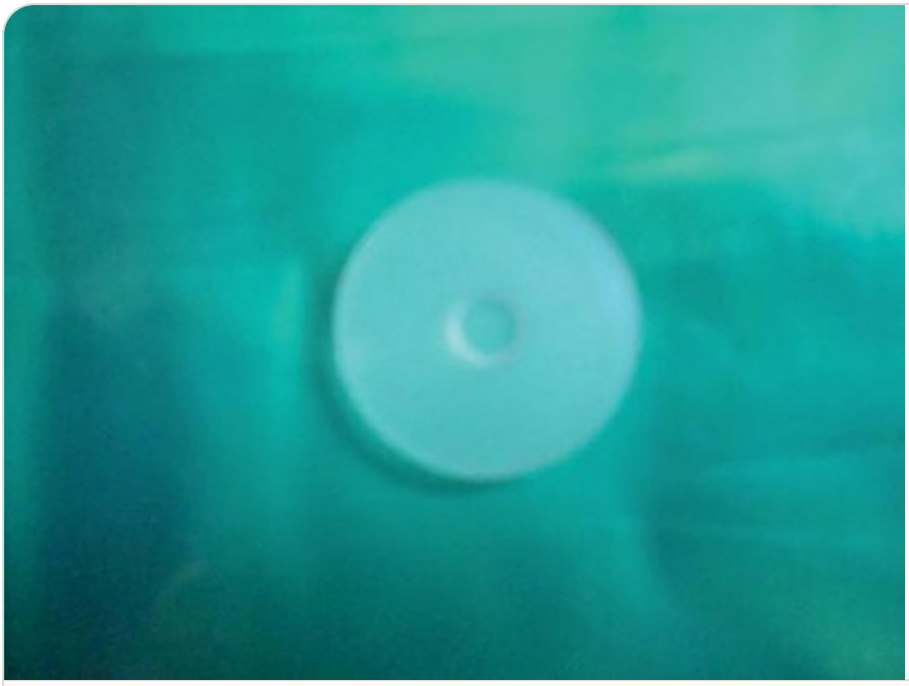


FIGURE 10 A silicone septal button closing a perforation.
Dhingra, fig. 26.12

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 On call

A BLOCKED NOSE AFTER A BLOW

1 Look at the septum in every patient with nasal trauma.

2. A boggy, swollen septum is a haematoma: drain it the same day.
3. Pack both sides and give antibiotics.
4. Review for reaccumulation and abscess.

6.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Parts of the septum** and its framework.
- **Types of deviation,** and the symptoms each causes.
- **Cottle's test** and the nasal valve.
- **Septoplasty versus submucous resection.**
- **Septal haematoma and abscess:** treatment and complications.
- **Causes of septal perforation,** and which diseases perforate the bony or cartilaginous part.

6.3 Common pitfalls

COMMON PITFALLS

- Missing a septal haematoma after trauma.
- Operating on a septum when rhinitis is the real cause of obstruction.
- Biopsying the healed anterior edge of a perforation.

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CLINICAL · NOSE & SINUSES

Allergic and non-allergic rhinitis

Rhinitis is the commonest complaint in the nose clinic: sneezing, a running nose, itch and a blocked nose. The history sorts allergic from non-allergic. Steroid sprays used properly control most patients, and turbinate surgery helps only when medicines have genuinely been tried. One-sided symptoms are never rhinitis until the nose has been examined.

AT A GLANCE

- Allergic or non-allergic by history.
- Topical steroid first, taught properly.
- Avoid decongestants beyond a week.
- One-sided symptoms aren't allergy.
- Immunotherapy is the only disease-modifying treatment; treat the nose to help asthma.

1 CHAPTER ONE Overview

Inflammation of the nasal lining.

1.1 Definition

Rhinitis is inflammation of the nasal mucosa, defined clinically by two or more of nasal obstruction, rhinorrhoea, sneezing and itching. *Rhinosinusitis* adds sinus involvement (see the sinusitis topics).

1.2 Classification

RHINITIS

- ALLERGIC **Seasonal or perennial** intermittent or persistent (ARIA)
- NON-ALLERGIC, NON-INFECTIVE **Vasomotor (idiopathic)** drug-induced, rhinitis medicamentosa, hormonal, NARES, gustatory, emotional, occupational

INFECTIVE **Acute and chronic rhinitis** atrophic rhinitis and specific infections (separate topics)

1.3 Why it matters

Allergic rhinitis affects up to a quarter of many populations, disturbs sleep, school and work, and travels with asthma: rhinitis patients are about four times more likely to develop asthma, and 20 to 30% of patients with rhinitis have asthma.

2 CHAPTER TWO Applied physiology

Why the nose runs and blocks.

2.1 The erectile mucosa

The turbinate mucosa contains venous sinusoids surrounded by smooth muscle that fill and empty them. Sympathetic stimulation empties them and opens the airway; parasympathetic stimulation fills them and drives the glands to secrete. Their alternating congestion from side to side is the *nasal cycle*.

2.2 Hyperreactivity

In both allergic and non-allergic rhinitis the mucosa over-reacts to ordinary stimuli: temperature change, dust, smoke, perfume, cold air.

3 CHAPTER THREE Allergic rhinitis

IgE in the nose.

3.1 Types

- **Seasonal:** symptoms in the season of the pollen the patient reacts to (tree, grass or weed pollens).

- **Perennial:** symptoms throughout the year, from house dust mite, moulds, cockroaches and animal dander.

3.2 Genetic predisposition

The chance of a child developing allergy is about 20% with one allergic parent and 47% with two.

3.3 Pathogenesis

3.3.1 Sensitisation

In predisposed people, inhaled allergens drive production of specific IgE, which binds by its Fc end to mast cells and basophils.

3.3.2 Re-exposure

Allergen bridges two IgE molecules on the cell surface; the mast cell degranulates, releasing preformed mediators (histamine) and newly made ones (leukotrienes, prostaglandins).

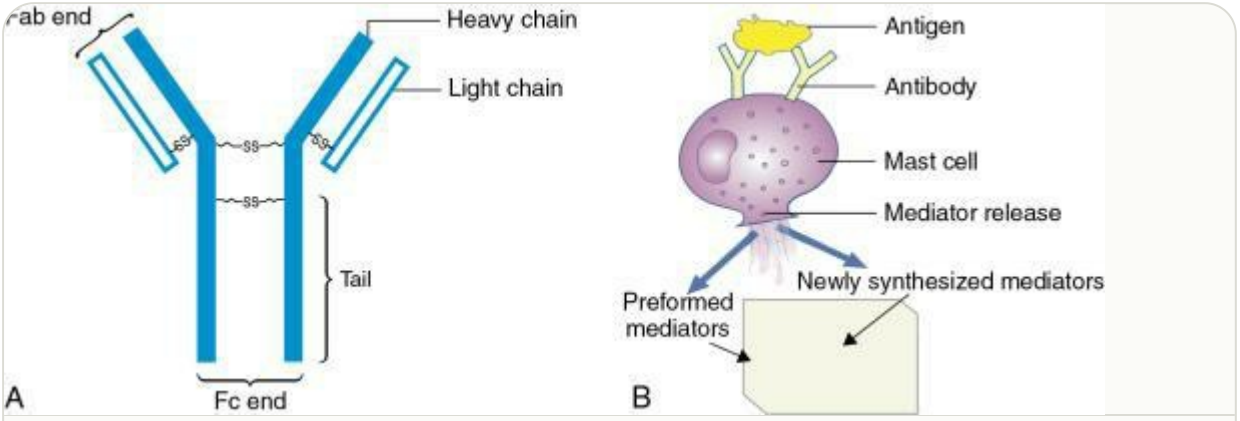


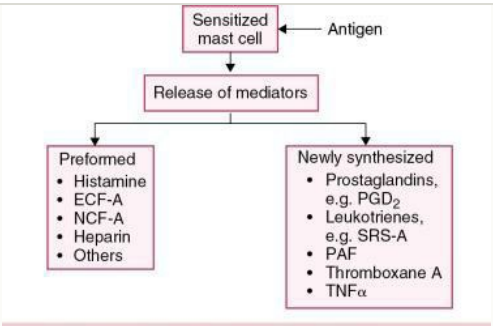
FIGURE 1 (A) The IgE antibody: the Fc end fixes to the mast cell, the Fab end binds allergen. (B) Release of mediators when allergen bridges two IgE molecules.
Dhingra, fig. 30.1

3.3.3 Early and late phases

- **Early phase:** within 5 to 30 minutes: sneezing, rhinorrhoea, blockage, from histamine.
- **Late phase:** 2 to 8 hours later, without further exposure: eosinophils, basophils and T cells infiltrate, causing congestion and thick secretions.

3.3.4 Priming

Mucosa sensitised by one allergen reacts to smaller doses later, and to non-specific stimuli as well.



Histamine	Vasodilatation, bronchospasm
ECF-A	Eosinophil chemotactic factor of anaphylaxis—attracts eosinophils to the site of reaction
NCF-A	Neutrophil chemotactic factor—attracts neutrophils
Heparin	Enhances phagocytosis
Prostaglandins	Vasoactive and bronchospastic
Leukotriene	Vasoactive and bronchospastic
PAF	Platelet aggregating factor. Histamine and serotonin are released from platelets. Causes chemotaxis of neutrophils and eosinophils
Thromboxane A	Spasmogenic
TNFα	Tumour necrosis factor. Helps transmigration of neutrophils and eosinophils and attracts them to the site of reaction

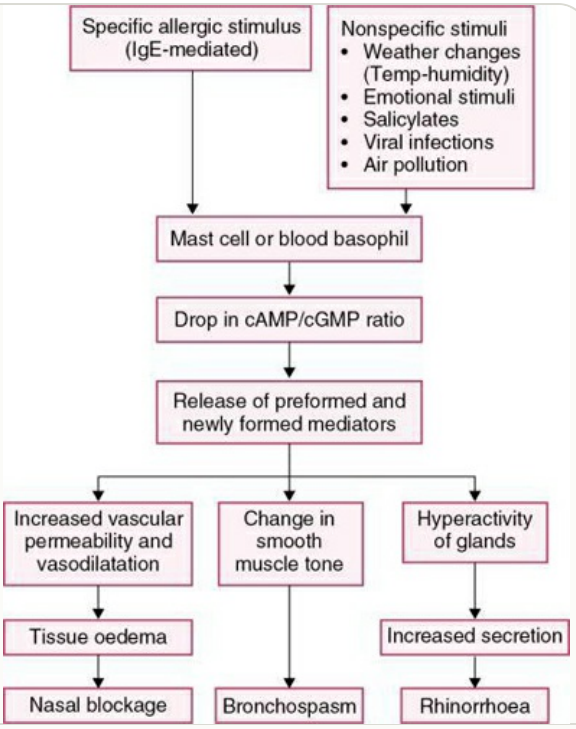


FIGURE 2 First, mediators released from a mast cell. Second, allergic and non-specific stimuli acting through the same mast cells.

Dhingra, figs 30.2 and 30.3

3.4 Clinical features

3.4.1 Symptoms

Usually starting at 12 to 16 years, though any age is possible. Seasonal disease: paroxysmal sneezing (10 to 20 at a time), watery discharge, blockage and itching of the nose, eyes, palate or throat. Perennial disease is milder but persistent: frequent colds, a stuffy nose, reduced smell, postnasal drip, cough and ear symptoms.

3.4.2 Signs

- **Nose:** pale, oedematous, sometimes bluish mucosa; swollen turbinates; watery discharge; a transverse nasal crease from the *allergic salute*.
- **Eyes:** lid oedema, cobblestone conjunctiva, dark circles (*allergic shiners*).
- **Ears:** retracted drum or effusion.
- **Pharynx and larynx:** granular pharyngitis, hoarseness.

- **Children:** the long face and open mouth of chronic mouth breathing.

3.5 ARIA classification

ARIA CLASSIFICATION

CATEGORY	DEFINITION
Intermittent	Symptoms less than 4 days a week, or for less than 4 weeks
Persistent	Symptoms more than 4 days a week and for more than 4 weeks
Mild	None of: disturbed sleep, impaired daily activities or sport, impaired school or work, troublesome symptoms
Moderate to severe	One or more of these

3.6 Investigations

- **Nasal smear:** eosinophils, best taken during active symptoms (also seen in NARES).
- **Blood count:** peripheral eosinophilia, inconsistent.
- **Skin prick test:** allergen pricked into the skin of the forearm or back; a wheal and flare at 10 to 15 minutes is positive, with histamine and diluent as controls.
- **Specific IgE in serum** (formerly RAST): correlates well with skin tests; useful when skin testing is impossible.
- **Nasal provocation test:** reproducing symptoms with allergen.
- **Nasal endoscopy:** to exclude polyps and structural causes.

3.7 Complications

Recurrent sinusitis, nasal polyps (in about 2%), otitis media with effusion, dental and facial growth problems from mouth breathing, and asthma.

3.8 Treatment

3.8.1 Avoidance

Most effective with a single allergen: remove the pet, encase pillows and mattresses, reduce dust mites, change work if necessary.

3.8.2 Drugs

- **Intranasal corticosteroids:** the most effective single treatment for persistent or moderate to severe disease. Teach technique: aim away from the septum.
- **Antihistamines:** oral non-sedating or intranasal, for sneezing, itch and rhinorrhoea; less effect on blockage.
- **Decongestants:** a short course only; oral ones are combined with antihistamines.
- **Sodium cromoglycate:** stabilises mast cells; useful before exposure.
- **Ipratropium bromide:** for watery rhinorrhoea.
- **Leukotriene receptor antagonists:** montelukast, especially with asthma.
- **Oral steroids:** a short course for severe, uncontrolled episodes only.
- **Anti-IgE (omalizumab):** for severe allergic asthma; not a routine treatment for rhinitis.

3.8.3 Step care (ARIA)

1. **Mild intermittent**
   Oral antihistamine or intranasal cromoglycate.
2. **Moderate or persistent**
   Intranasal corticosteroid.
3. **Severe**
   Intranasal steroid with a non-sedating antihistamine.
4. **Uncontrolled**
   Short oral steroid course; consider immunotherapy.

3.8.4 Immunotherapy

Gradually increasing doses of the allergen, by subcutaneous injection or sublingually, until a maintenance dose is reached; it raises specific IgG (blocking antibody) and reduces IgE. Continued for about three years; stopped if no benefit after uninterrupted treatment. The only treatment that alters the course of the disease, for patients whose symptoms are not controlled by drugs or who are sensitised to a single dominant allergen.

PEARLS

- Rhinitis and asthma are one airway: treat both.

- Most steroid-spray failures are technique failures.
- Allergic children with blocked noses may also have big adenoids.

4 CHAPTER FOUR Vasomotor rhinitis

Non-allergic, hyperreactive.

4.1 What it is

A non-allergic rhinitis that mimics allergy, with negative allergy tests, from autonomic imbalance with parasympathetic overactivity and a hyperreactive mucosa. Emotions influence it through the hypothalamus.

4.2 Clinical features

- Paroxysmal sneezing on getting out of bed.
- Profuse, watery rhinorrhoea, sometimes the main complaint; dripping on leaning forward must be distinguished from CSF rhinorrhoea.
- Blockage alternating from side to side, worse at night and on the dependent side.
- Postnasal drip.

The turbinate mucosa is congested and hypertrophic, or normal.

4.3 Complications

Hypertrophic rhinitis, polyps and sinusitis.

4.4 Treatment

4.4.1 Medical

Avoid triggers; antihistamines and oral decongestants; topical steroids; ipratropium for rhinorrhoea; a short systemic steroid course in severe cases; address psychological factors.

4.4.2 Surgical

Reduce the turbinates when obstruction persists; correct a deviated septum or polyps. *Vidian neurectomy*, cutting the parasympathetic secretomotor fibres, for intractable rhinorrhoea; now largely replaced by endoscopic posterior nasal nerve procedures.

5 CHAPTER FIVE Other non-allergic rhinitis

Named causes.

5.1 Drug-induced

Antihypertensives that block sympathetic activity (reserpine, methyldopa, beta-blockers), anticholinesterases, oral contraceptives, and aspirin or NSAIDs in sensitive patients.

5.2 Rhinitis medicamentosa

Rebound congestion from overuse of topical decongestant drops beyond about five to seven days. Treated by stopping the drops, a topical or short systemic steroid course, and turbinate reduction if hypertrophy persists.

5.3 Hormonal

- **Rhinitis of pregnancy:** oedematous mucosa from oestrogen; treated with saline, topical steroids and limited decongestant use; avoid newer antihistamines.
- **Hypothyroidism:** parasympathetic predominance; relieved by thyroxine.
- **Honeymoon rhinitis:** congestion with sexual excitement.

5.4 NARES

Non-allergic rhinitis with eosinophilia syndrome: eosinophils on smear but negative allergy tests; responds to topical steroids; may precede aspirin sensitivity and

polyps.

5.5 Gustatory rhinitis

Watery rhinorrhoea with spicy food, a cholinergic reflex; ipratropium before meals.

5.6 Emotional rhinitis

Anxiety, tension or grief causing rhinitis; counselling.

5.7 Non-airflow rhinitis

After laryngectomy or tracheostomy, when air no longer passes through the nose.

5.8 Local allergic rhinitis

Allergy-like symptoms with negative skin and blood tests but specific IgE produced in the nasal mucosa, shown by nasal provocation; managed like allergic rhinitis.

6 CHAPTER SIX Turbinate surgery

When medicines are not enough.

6.1 Indications

Persistent obstruction from inferior turbinate hypertrophy despite a proper trial of topical steroids, or rhinitis medicamentosa with fixed hypertrophy.

6.2 Methods

Submucosal reduction (microdebrider or radiofrequency), outfracture, partial turbinectomy, laser or cryosurgery. Preserve the mucosa: excessive removal leaves an *empty nose*, with crusting and paradoxical blockage.

OPERATIVE GUIDES IN THEATRE 1 operation

Surgery of the Turbinate

Turbinate reduction after medical care

ALLERGY TESTING, TREATMENT AND THE UNITED AIRWAY (OXFORD TEXTBOOK)

Allergic rhinitis affects about one person in five in Western countries and usually starts around three years of age, after atopic dermatitis and before allergic asthma, the sequence known as the **allergic march**; it is rare before two because sensitisation needs prolonged exposure. Non-allergic rhinitis typically begins in early adulthood and is linked to adult-onset asthma, whereas vasomotor rhinitis is not.

Testing. A skin-prick wheal more than 3 mm larger than the negative control is positive, but only about half of people with positive tests have allergic disease, so tests are interpreted with the history and requested mainly when immunotherapy or targeted avoidance is planned. A small panel (house dust mite, grass, tree and weed pollen, cat, dog and moulds) covers most allergy. Serum specific IgE is a little less sensitive than skin testing because tissue IgE levels are much higher than serum levels.

Treatment. Intranasal steroids take days to start working and about two weeks to reach full effect; modern sprays with low bioavailability are preferred in children. Steroid sprays combine well with oral antihistamines, and a combined steroid–azelastine spray helps patients not controlled by either. Depot steroid injections are not recommended. **Immunotherapy**, subcutaneous or sublingual, is the only disease-modifying treatment and is given for about three years; the sublingual route avoids injections and has an almost negligible anaphylaxis risk. Allergen-impermeable mattress and pillow covers are the best-studied avoidance measure for house dust mite. In turbinate surgery, reducing the **conchal bone** gives the longest-lasting effect.

The united airway. Most patients with allergic asthma have allergic rhinitis, and treating the nose often improves asthma control, so immunotherapy and turbinate reduction can be offered earlier in patients with both conditions.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Pathogenesis** of allergic rhinitis, and early versus late phase.
- **Signs:** allergic salute, transverse crease, allergic shiners, pale boggy mucosa.
- **ARIA classification** and step care.
- **Skin prick test** technique and controls.
- **Immunotherapy:** mechanism and duration.
- **Vasomotor rhinitis** and the forms of non-allergic rhinitis.
- **Rhinitis medicamentosa.**

7.2 Common pitfalls

COMMON PITFALLS

- Calling one-sided symptoms allergy without examining the nose.
- Prescribing decongestant drops without a time limit.
- Sending a patient for turbinate surgery before a proper trial of steroid spray.
- Missing CSF rhinorrhoea in a patient with clear, one-sided dripping.

Written for this app from Oxford Textbook of Otolaryngology ch. 29, 42, Oxford Specialist Handbook ch. 20, Oxford Handbook of ENT ch. 9, Dhingra ch. 27, 30, 31 and K. J. Lee ch. 27, 31, 59. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Acute rhinosinusitis and orbital complications

Most acute rhinosinusitis is a cold that has reached the sinuses, and most of it settles without antibiotics. The rhinologist's real job is to recognise the minority that turn bacterial, and the few that break out of the sinuses into the orbit, the bone or the brain. A swollen eye in a child with a cold is an emergency until the orbit has been assessed.

AT A GLANCE

- Viral first; bacterial if over 10 days or worsening.
- Any eye swelling is urgent: check vision.
- Use Chandler stages, CT and IV treatment.
- Drain abscesses.
- Post-viral ARS: worse after 5 days or beyond 10; bacterial needs 3 of 5 EPOS features; antibiotics are overused.

1 CHAPTER ONE Overview

A cold that reaches the sinuses.

1.1 Definition

Inflammation of the nose and paranasal sinuses. The term *rhinosinusitis* is preferred, because the nasal mucosa is always involved too.

1.2 Classification by time

CLASSIFICATION OF RHINOSINUSITIS

TYPE	DURATION
Acute	Under 4 weeks, with complete resolution
Subacute	4 to 12 weeks
Chronic	12 weeks or more

TYPE	DURATION
Recurrent acute	4 or more episodes a year, each lasting 7 to 10 days or more, with full recovery between
Acute on chronic	Sudden worsening of chronic disease, returning to baseline

1.3 Symptoms

Nasal blockage, anterior or posterior discharge, facial pain or pressure, and reduced smell, with cough, fever, fatigue, dental pain, halitosis, headache or ear fullness.

2 CHAPTER TWO Applied anatomy

The sinuses and their drainage.

2.1 Drainage

The frontal, anterior ethmoid and maxillary sinuses drain into the middle meatus through the *ostiomeatal complex*; the posterior ethmoids drain into the superior meatus and the sphenoid into the sphenoethmoidal recess. Mucociliary clearance in each sinus follows a fixed path towards its natural ostium.

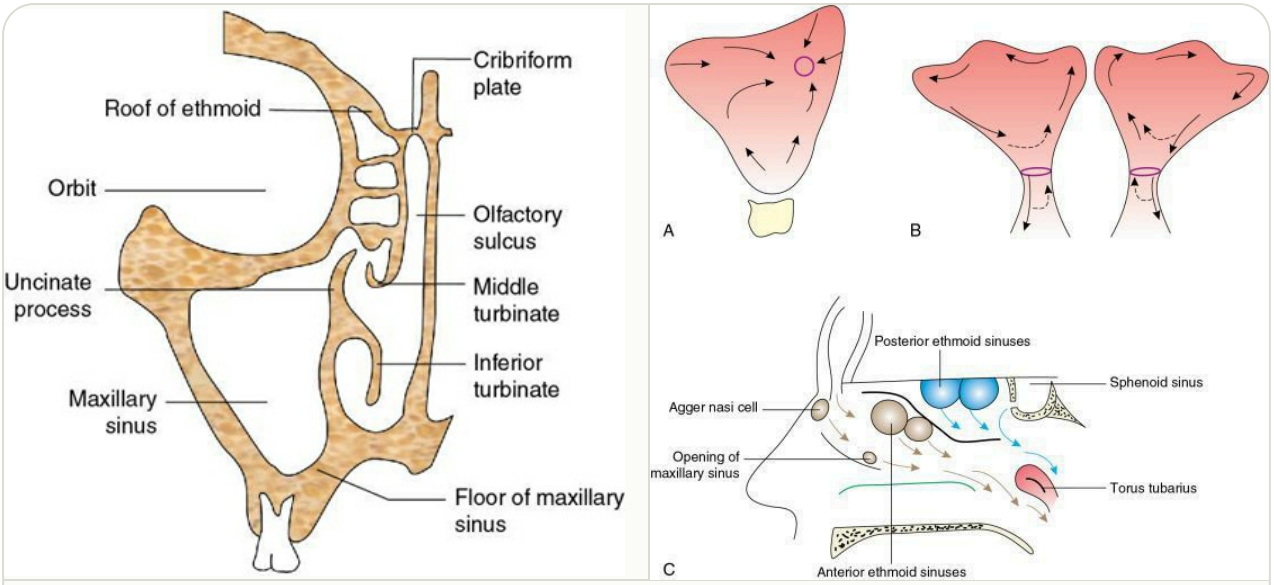


FIGURE 1 First, a coronal section showing the maxillary and ethmoid sinuses in relation to the orbit and nasal cavity. Second, mucociliary clearance patterns of the sinuses.

Dhingra, figs 35.1 and 35.4

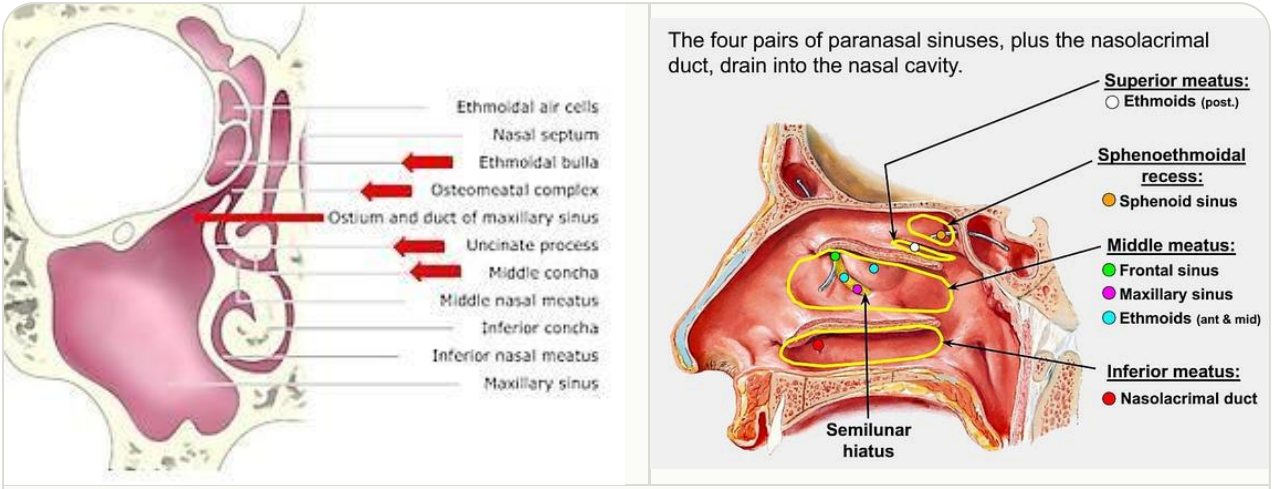


FIGURE 2 The ostiomeatal complex in coronal section, and the drainage of each sinus into its meatus.

Anatomy atlas · ENT Steps p. 108, ENT Steps p. 101

2.2 Development

The maxillary and ethmoid sinuses are present at birth; the frontal sinus appears around the sixth year and the sphenoid later in childhood. So ethmoiditis is the sinusitis of infants, and frontal sinusitis an adult disease.

2.3 The thin walls

The ethmoids are separated from the orbit only by the paper-thin *lamina papyracea*, and the frontal, ethmoid and sphenoid sinuses all border the anterior cranial fossa. This is why complications go to the eye and the brain.

CHAPTER THREE

3 Pathogenesis

From virus to bacteria.

3.1 Acute viral rhinosinusitis

Rhinoviruses, influenza and parainfluenza viruses, spread by droplets, with an incubation of one to four days. Congestion, rhinorrhoea, sneezing and low fever improve within seven to ten days.

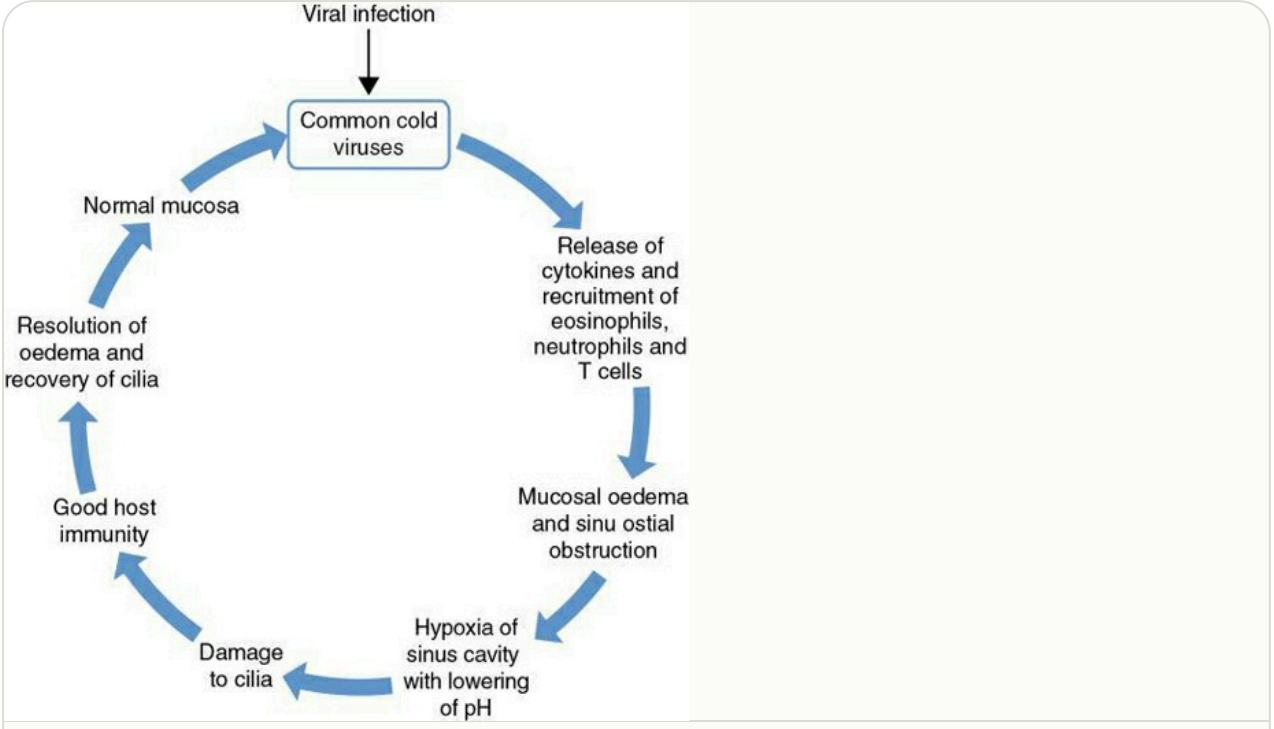


FIGURE 3 Pathophysiology of acute viral rhinosinusitis.
Dhingra, fig. 36.1

3.2 Acute bacterial rhinosinusitis

The virus damages cilia and epithelium; oedema blocks the ostia; secretions stagnate and bacteria multiply: *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Moraxella catarrhalis* and *Staphylococcus aureus*. Anaerobes accompany dental infection.

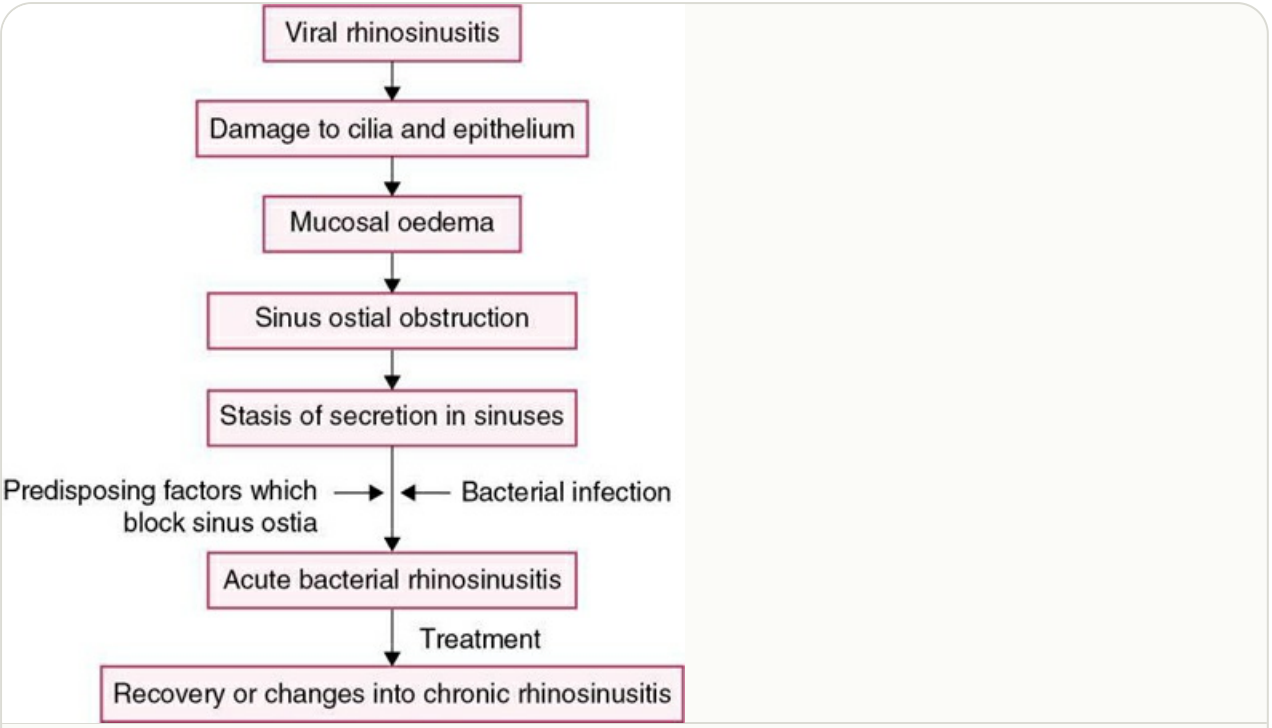


FIGURE 4 The sequence from acute viral to acute bacterial rhinosinusitis.
Dhingra, fig. 36.2

3.3 Diagnosing bacterial infection

Suspect it when symptoms persist beyond ten days without improvement, or worsen after initial improvement (*double worsening*), or with severe symptoms (high fever and purulent discharge) from the start.

3.4 Predisposing factors

Viral infections, swimming and diving, dental infection, trauma, a deviated septum or concha bullosa, polyps, allergy, adenoids, immune deficiency and cystic fibrosis.

4 CHAPTER FOUR Clinical picture by sinus

Where it hurts tells you where it is.

4.1 Acute maxillary sinusitis

4.1.1 Causes

Spread from a cold; swimming in contaminated water; dental infection from the premolar and molar roots, which lie close to the sinus floor, or a root pushed into the sinus, or an oroantral fistula after extraction; trauma.

4.1.2 Clinical features

Fever and malaise; pain over the cheek, often referred to the teeth or gums (so the patient may see a dentist first), worse on stooping, coughing or chewing; tenderness over the anterior wall; redness and swelling of the cheek in children; pus in the middle meatus and postnasal discharge.

4.1.3 Complications

Chronic sinusitis, frontal sinusitis, osteomyelitis of the maxilla, and orbital cellulitis.

4.2 Acute frontal sinusitis

4.2.1 Clinical features

Severe frontal headache with typical periodicity: it starts on waking, peaks around midday and fades by evening (the *office headache*). Exquisite tenderness on pressing upwards on the sinus floor above the medial canthus; upper lid oedema; a streak of pus high in the middle meatus.

4.2.2 Complications

Orbital cellulitis, osteomyelitis of the frontal bone, meningitis, extradural and frontal lobe abscess, and chronic sinusitis.

4.3 Acute ethmoiditis

4.3.1 Clinical features

The sinusitis of infants and young children. Pain over the bridge of the nose and deep to the eye, worse on eye movement; swelling of both lids; a swollen middle turbinate. Orbital cellulitis comes early.

4.4 Acute sphenoiditis

4.4.1 Clinical features

Rarely isolated. Headache at the occiput or vertex, sometimes referred to the mastoid, with postnasal discharge seen on the roof of the nasopharynx. Isolated sphenoid opacity needs a mucocele or tumour excluded.

5 CHAPTER FIVE Diagnosis

Clinical first, imaging for complications.

5.1 Examination

Nasal endoscopy shows pus in the middle meatus; a swab from the middle meatus can be cultured.

5.2 Imaging

Not needed for uncomplicated acute disease. CT of the sinuses (with contrast and of the orbits and brain) is requested when a complication is suspected or treatment fails. Plain radiographs (Waters' view) and transillumination are historical.

IMAGING IN THE APP 7 cases

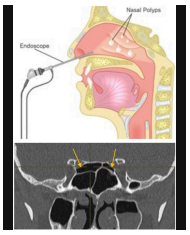
Maxillary, frontal and odontogenic sinusitis, and the orbital and intracranial complications on CT and MRI.



Maxillary Sinusitis
Infection & inflammation · 10 images



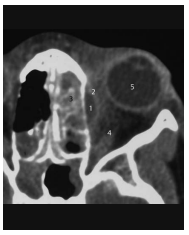
Frontal Sinusitis
Infection & inflammation · 3 images



Sinusitis and its orbital and intracranial complications
Infection & inflammation · 12 images



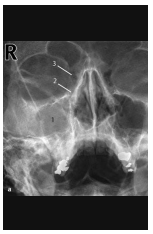
Orbital Cellulitis
Infection & inflammation · 6 images



Subperiosteal Orbital Abscess
Infection & inflammation · 9 images



Intracranial Abscess from a Sinonasal Focus
Infection & inflammation · 2 images



Odontogen Sinusitis
Infection & inflammation 3 images

CHAPTER SIX

6 Management

Most need no antibiotic.

6.1 Viral rhinosinusitis

Symptomatic: saline, analgesics (non-aspirin), short-term decongestants and fluids. Antibiotics are not needed.

6.2 Bacterial rhinosinusitis

- **Analgesics:** NSAIDs or paracetamol.
- **Antibiotics:** amoxicillin, with or without clavulanate, first line; doxycycline or a respiratory quinolone in penicillin allergy. Watchful waiting is reasonable in mild disease with a safety net.
- **Saline irrigation.**
- **Intranasal steroids:** reduce oedema and shorten the illness.
- **Decongestants:** topical for a few days only; oral ones where safe.
- **Antihistamines:** only with coexisting allergy, as they thicken mucus.
- **Steam inhalation:** symptomatic relief.

6.3 Surgery for acute disease

6.3.1 Maxillary

Antral washout is rarely needed now; endoscopic middle meatal antrostomy if drainage is required.

6.3.2 Frontal

If pain, fever or lid swelling worsen despite 48 hours of treatment, *trephine* the frontal sinus through a short incision below the medial end of the eyebrow, take pus for culture, and irrigate through a tube until methylene blue reaches the nose. Endoscopic frontal drainage is the modern alternative.

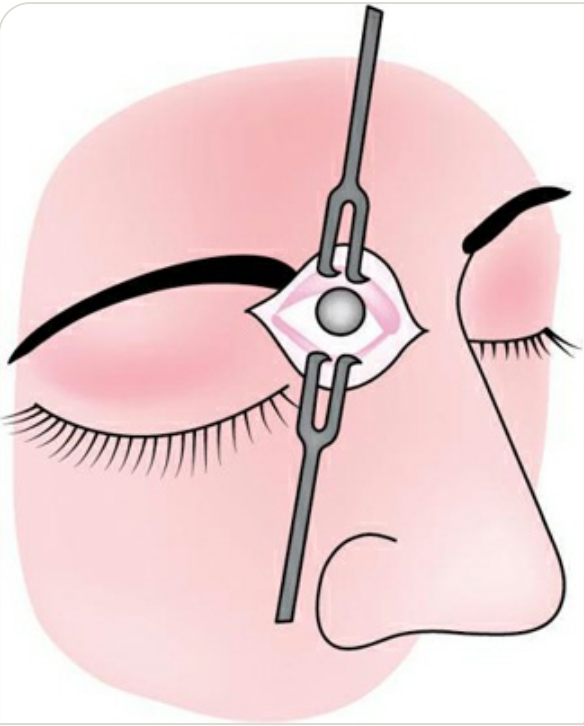


FIGURE 5 Trephination of the right frontal sinus through its floor.
Dhingra, fig. 36.3

6.3.3 Ethmoid

Drain an orbital abscess and the ethmoids, endoscopically or through an external ethmoidectomy.

OPERATIVE GUIDES IN THEATRE 4 operations

Endoscopic Ethmoidectomy
Endoscopic ethmoidectomy

Orbital Abscess
Orbital abscess drainage

Trephination of the Frontal Sinus
Frontal trephination
Frontal Sinus Cranialization
Pott's puffy tumour and cranialisation

7 CHAPTER SEVEN Complications

Local, orbital and intracranial.

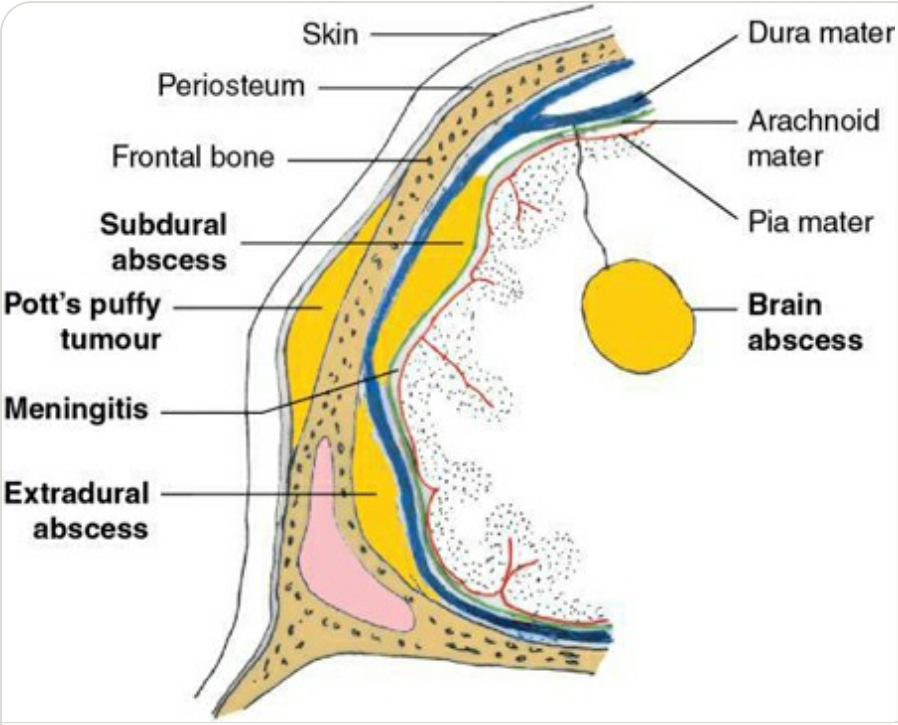


FIGURE 6 Complications of sinusitis.
Dhingra, fig. 38.1

7.1 Local complications

7.1.1 Mucocele and pyocele

An expanding, mucus-filled sinus, from long-standing ostial obstruction, that slowly thins and erodes its bony walls; infected, it is a *pyocele* (mucopyocele).
Frequency: frontal, ethmoid, maxillary, sphenoid.

- **Frontal:** a non-tender cystic swelling above the medial canthus (90%), pushing the eye forward, down and out; sometimes a forehead swelling; egg-shell crackling.
- **Ethmoid:** expands the medial orbital wall and bulges into the middle meatus.

- **Sphenoid:** presents with superior orbital fissure or orbital apex syndrome and proptosis.

Treatment is endoscopic marsupialisation into the nose, which has replaced external operations.

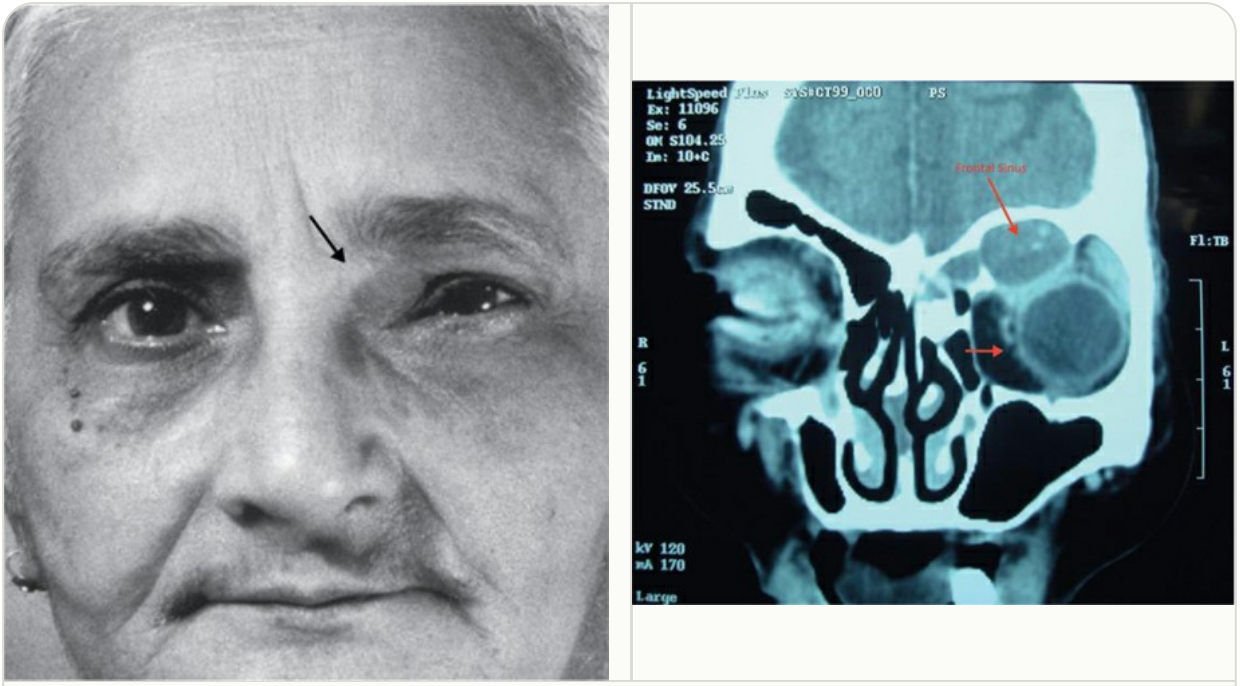


FIGURE 7 First, a frontal mucocele presenting above the left medial canthus. Second, CT of a frontoethmoid mucocele displacing the eye down and out.

Dhingra, figs 38.2 and 38.3

7.1.2 Retention cyst

A cyst of an obstructed mucous gland, usually in the maxillary sinus, found incidentally; it does not erode bone and needs no treatment.

7.1.3 Osteomyelitis

- **Maxilla:** in infants, because the anterior wall contains spongy bone; cheek swelling, lid oedema and fistulae in the infraorbital region, alveolus or palate, with risk to tooth buds.
- **Frontal bone:** in adults; pus under the pericranium forms a soft, doughy forehead swelling, *Pott's puffy tumour*, and often an extradural abscess beneath. Antibiotics, drainage, trephination and removal of sequestra.



FIGURE 8 First, osteomyelitis of the maxilla with an infraorbital fistula. Second, chronic frontal sinusitis with a fistula in the floor of the sinus.

Dhingra, figs 38.4 and 38.5

7.2 Orbital complications

Most come from the ethmoids, through the lamina papyracea or by thrombophlebitis of the ethmoidal veins.

CHANDLER CLASSIFICATION

STAGE	NAME	FEATURES
I	Preseptal cellulitis	Lid oedema; vision and eye movements normal
II	Orbital cellulitis	Diffuse orbital oedema, proptosis, chemosis, restricted movement
III	Subperiosteal abscess	Pus between bone and periosteum; eye displaced away from the abscess
IV	Orbital abscess	Pus within the orbit; severe proptosis, ophthalmoplegia, falling vision
V	Cavernous sinus thrombosis	Spread to the cavernous sinus; signs in both eyes

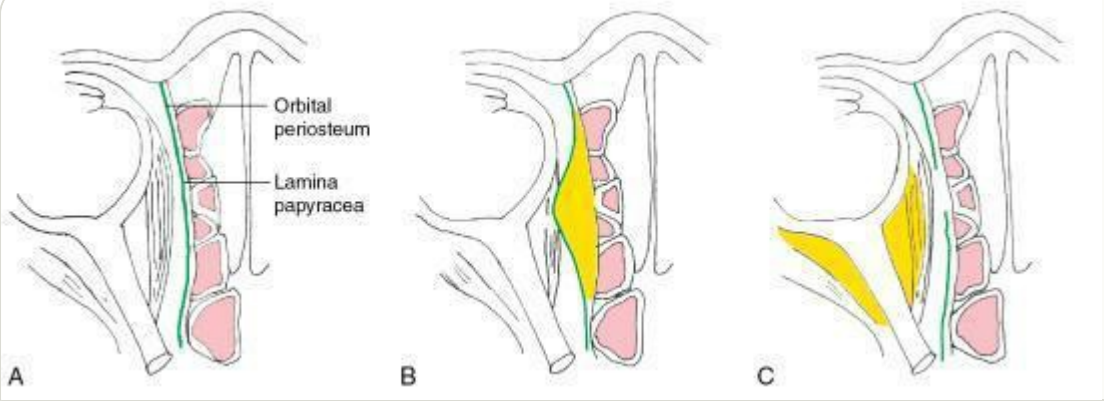


FIGURE 9 Orbital complications: (A) normal, (B) subperiosteal abscess, (C) orbital abscess. Dhingra, fig. 38.7

7.2.1 Displacement by the source

An ethmoid subperiosteal abscess pushes the eye forward, down and out; a frontal one, down and out; a maxillary one, up and forward.

7.2.2 Orbital apex and superior orbital fissure syndromes

Posterior ethmoid or sphenoid infection: deep orbital pain and paralysis of the sixth, third and fourth nerves (superior orbital fissure syndrome), with loss of vision and of V2 sensation in the orbital apex syndrome.

THE SWOLLEN EYE

- Check visual acuity, colour vision, pupils, eye movements and proptosis.
- Any loss of vision, colour vision or movement, or a relative afferent pupil defect: urgent contrast CT and surgical drainage.
- Involve ophthalmology early; examine both eyes for cavernous sinus spread.

7.2.3 Treatment

Admission, intravenous antibiotics and nasal decongestion; contrast CT. Drain a subperiosteal or orbital abscess (endoscopically or externally) when vision is threatened, there is no improvement in 24 to 48 hours, or the abscess is large.

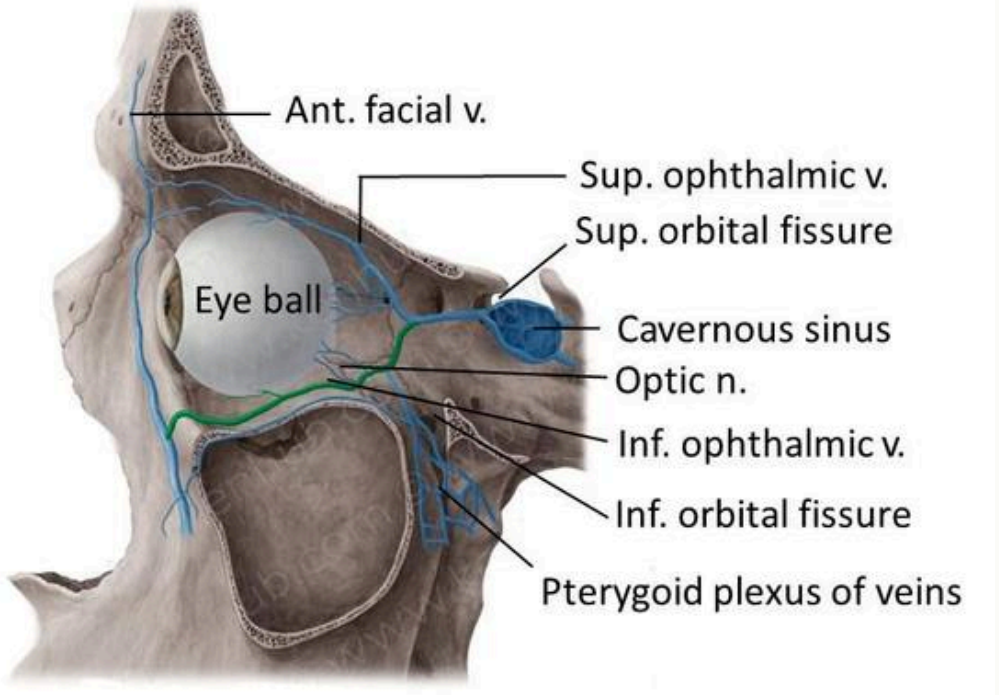


FIGURE 10 Orbital veins draining to the cavernous sinus and pterygoid plexus: the valveless route for spread.

Anatomy atlas · Doss p. 101

7.3 Intracranial complications

Meningitis, extradural, subdural and brain (frontal lobe) abscess, and cavernous sinus thrombosis, from the frontal, ethmoid and sphenoid sinuses.

7.3.1 Cavernous sinus thrombosis

Spread through the valveless ophthalmic veins from the ethmoids, sphenoid, orbit, the danger area of the face, or the pharynx. Abrupt onset with rigors; both eyes become involved; chemosis and proptosis; sequential palsies of the third, fourth and sixth nerves up to total ophthalmoplegia; a fixed, dilated pupil; reduced V1 sensation; papilloedema. Diagnosed with contrast CT or MRI.

ORBITAL CELLULITIS OR CAVERNOUS SINUS THROMBOSIS?

	ORBITAL CELLULITIS	CAVERNOUS SINUS THROMBOSIS
Source	Usually ethmoid sinus	Nose, sinuses, orbit, face, pharynx, ear
Onset	Gradual	Abrupt, with rigors
Eyes	One	Spreads to both

	ORBITAL CELLULITIS	CAVERNOUS SINUS THROMBOSIS
Chemosis and proptosis	Moderate	Marked
Cranial nerves III, IV, VI	Movement limited by swelling	Progressive palsies
Pupil	Usually normal	Dilated, fixed
General state	Unwell	Toxic, very ill

7.4 Descending and focal infections

Infected discharge running down the pharynx causes pharyngitis, laryngitis and bronchitis; sinus infection can also act as a focus for distant disease.

EPOS 2020 AND ANTIBIOTIC STEWARDSHIP (OXFORD TEXTBOOK)

EPOS 2020 defines three phases of acute rhinosinusitis (symptoms for less than 12 weeks). The **common cold** (acute viral rhinosinusitis) lasts less than 10 days. **Acute post-viral rhinosinusitis** is when symptoms worsen after five days or persist beyond ten. **Acute bacterial rhinosinusitis** is suggested by at least three of: discoloured discharge (often one-sided), severe local pain (often one-sided), fever above 38 °C, raised ESR or CRP, and "double sickening" (worsening after an initial improvement). Only 0.5 to 2% of viral episodes become bacterial, and discoloured mucus alone does not mean bacteria. EPOS 2020 dropped the term "subacute". **Recurrent acute rhinosinusitis** means four or more episodes a year with symptom-free intervals.

Antibiotics are greatly overused, given on the first day in many episodes and to over 90% of patients in UK primary care, although early antibiotics do not prevent complications. Most patients need only analgesia, saline and, for post-viral symptoms, a nasal steroid spray; antibiotics are kept for severe or bacterial disease. The usual organisms remain *Streptococcus pneumoniae*, *Haemophilus influenzae* and *Moraxella catarrhalis*, but resistance to penicillin and macrolides has risen sharply. *Streptococcus anginosus* (milleri group) is the commonest organism in complicated sinusitis in children and adolescents, with a tendency to form intracranial abscesses.

8 CHAPTER EIGHT
For the examination

What the examiner wants to hear.

8.1 On call

SWOLLEN EYE WITH SINUSITIS

1. Check vision, colour vision, pupils, movements and proptosis, and examine the other eye.
2. Admit; start intravenous antibiotics and nasal decongestion.
3. Contrast CT of the sinuses, orbits and brain.
4. Involve ophthalmology; drain an abscess if vision is threatened or there is no improvement.
5. Headache, drowsiness or meningism: image the brain and involve neurosurgery.

8.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Definitions** by duration, and when to diagnose bacterial infection.
- **Organisms** and first-line antibiotic.
- **Features of each sinus,** including the office headache.
- **Frontal trephination.**
- **Chandler's classification** and the direction of eye displacement.
- **Mucocele and Pott's puffy tumour.**
- **Orbital cellulitis versus cavernous sinus thrombosis.**

8.3 Common pitfalls

COMMON PITFALLS

- Antibiotics for every cold.
- Calling an orbital abscess preseptal cellulitis without checking vision and movement.
- Missing the extradural abscess beneath a Pott's puffy tumour.

Written for this app from Oxford Textbook of Otolaryngology ch. 36, Oxford Specialist Handbook ch. 21, Oxford Handbook of ENT ch. 9, Dhingra ch. 36, 38 and K. J. Lee ch. 28. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Chronic rhinosinusitis and nasal polyps

Chronic rhinosinusitis is a long-standing inflammation of the nose and sinuses that lasts beyond twelve weeks, with or without polyps. It is not simply an infection that refuses to clear: anatomy, cilia, allergy, asthma, biofilms and the patient's own immune pattern all play their part. Most patients are controlled with saline and steroid sprays used properly; surgery opens the sinuses so that the medicine can reach them.

AT A GLANCE

- 12 weeks of symptoms with objective findings.
- Saline and topical steroids first.
- Surgery helps medical treatment reach the mucosa.
- Children with polyps need a CF test.
- EPOS 2020: primary localised or diffuse, type 2 or not; biologics for recurrent type 2 polyps meeting 3 of 5 criteria.

1 CHAPTER ONE Overview

Inflammation beyond twelve weeks.

1.1 Definition

Inflammation of the nose and sinuses lasting 12 weeks or more, with at least two cardinal symptoms, one of which must be blockage or discharge, plus objective evidence on endoscopy or CT.

1.1.1 Cardinal symptoms

- Nasal blockage or congestion.
- Anterior or posterior (postnasal) discharge, often purulent.
- Facial pain or pressure.
- Reduced or lost sense of smell.

1.1.2 Objective signs

> Endoscopy showing oedema or polyps in the middle meatus, or mucopus there; or CT showing mucosal changes in the ostiomeatal complex or sinuses.

1.2 Two phenotypes

- **CRS without nasal polyps:** more often infective and neutrophilic.
- **CRS with nasal polyps:** usually type 2 (eosinophilic) inflammation driven by IL-4, IL-5 and IL-13, often with asthma.

1.3 Acute exacerbation

A sudden worsening that returns to the chronic baseline after treatment.

2 CHAPTER TWO Causes and pathogenesis

A vicious cycle.

2.1 Predisposing factors

2.1.1 Anatomical

Structural variants that narrow the ostiomeatal complex: septal deviation, *concha bullosa* (a pneumatised middle turbinate), a paradoxical middle turbinate, Haller (infraorbital ethmoid) cells, prominent agger nasi cells, and a pneumatised or medially bent uncinate process.

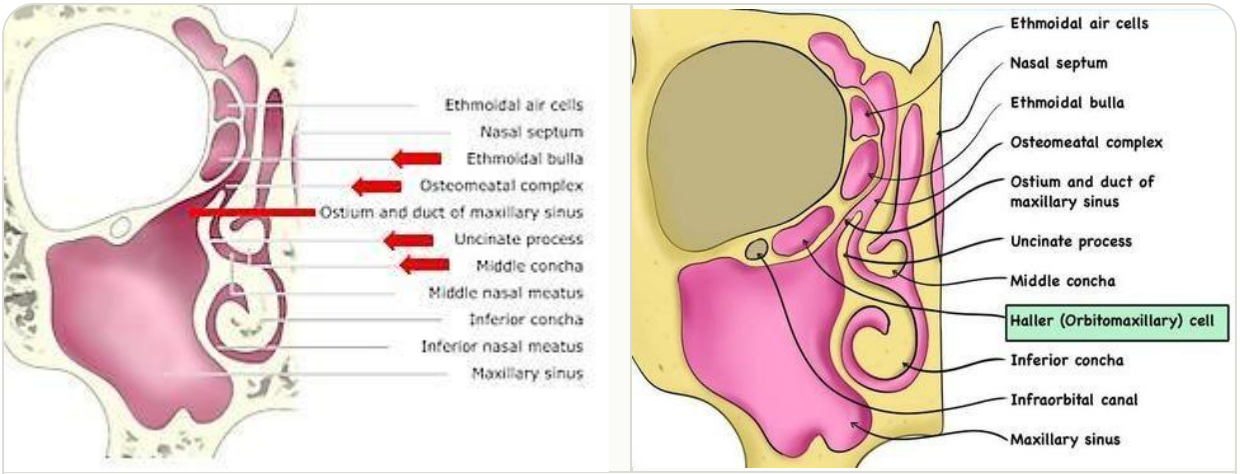


FIGURE 1 The ostiomeatal complex in coronal section, and a Haller cell narrowing the maxillary outflow.

Anatomy atlas · ENT Steps p. 108, ENT Steps p. 146

2.1.2 Mucociliary failure

Infection, smoking and pollution impair cilia; in *primary ciliary dyskinesia* (Kartagener syndrome with situs inversus and bronchiectasis) cilia do not beat; in *cystic fibrosis* the mucus is too thick to move; *Young syndrome* combines sinopulmonary disease with azoospermia.

2.1.3 Host and environment

Allergy, asthma (about half of CRS patients), aspirin sensitivity, immune deficiency (IgG deficiency, HIV, chemotherapy, transplant drugs), smoking, occupational dusts and fumes, dental infection, and osteitis of the sinus bone after surgery.

2.1.4 Biofilms

Colonies of bacteria in a protective polysaccharide matrix that resists antibiotics and keeps the disease going.

2.2 Organisms

Polymicrobial: *Staphylococcus aureus* (sometimes methicillin-resistant), *Pseudomonas*, *Klebsiella*, *E. coli* and anaerobes, unlike the organisms of acute disease.

2.3 The cycle

Ostial obstruction leads to stasis, low oxygen and infection; infection and inflammation damage cilia and swell the mucosa further, which blocks the ostium more.

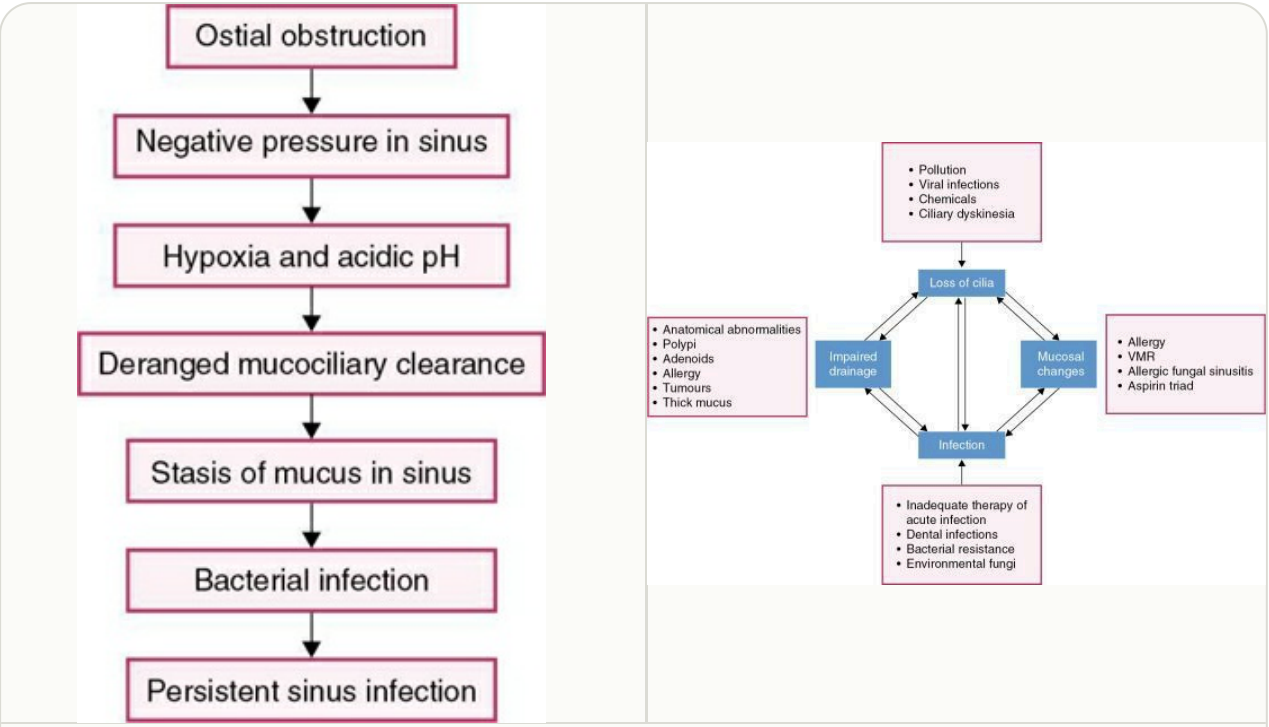


FIGURE 2 First, the sequence of events after sinus ostial obstruction. Second, the factors in the vicious cycle of chronic sinusitis.

Dhingra, figs 37.1 and 37.2

3 CHAPTER THREE Nasal polyps

Grapes in the nose.

3.1 Two kinds

ANTROCHOANAL OR ETHMOIDAL POLYPS?

	ANTROCHOANAL POLYP	ETHMOIDAL POLYPS
Age	Children and young adults	Adults
Cause	Infection	Allergy, multifactorial inflammation

	ANTROCHOANAL POLYP	ETHMOIDAL POLYPS
Number and side	Single, one-sided	Multiple, both sides
Origin	Maxillary sinus, near the accessory ostium	Ethmoid cells, uncinate, middle turbinate, middle meatus
Growth	Backwards to the choana; may hang behind the soft palate	Forwards; may appear at the nostril
Shape	Three parts: antral, choanal and nasal	Small, grape-like masses
Recurrence	Uncommon after complete removal	Common

3.2 Antrochoanal polyp

Also called *Killian's polyp*. It arises near the accessory ostium of the maxillary sinus and has three parts: a thin antral stalk, a round choanal part, and a flattened nasal part. It causes one-sided blockage that may become bilateral when the choanal part fills the nasopharynx, with a hyponasal voice. It can be missed on anterior rhinoscopy and is seen with an endoscope or hanging behind the soft palate. CT shows an opaque antrum and the mass passing through the ostium.

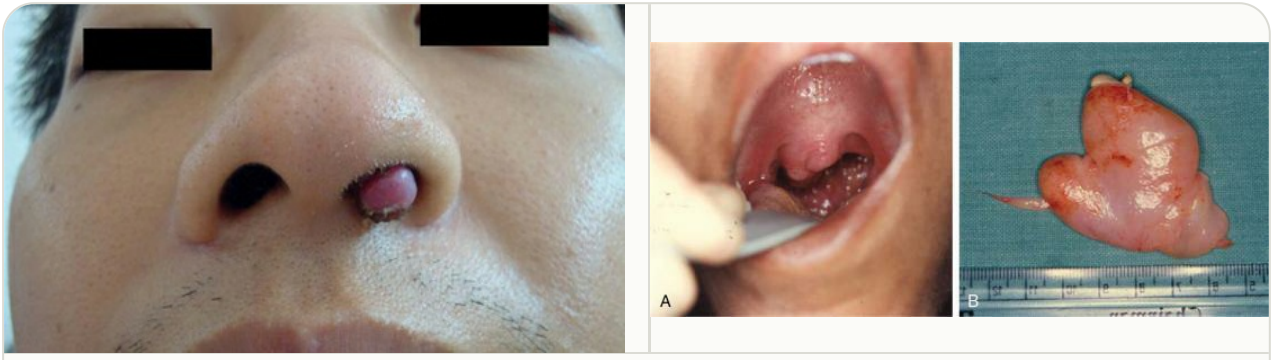


FIGURE 3 First, an antrochoanal polyp at the left nostril of a 14-year-old. Second, one hanging into the oropharynx behind the soft palate, and the polyp after removal.
Dhingra, figs 32.1 and 32.2

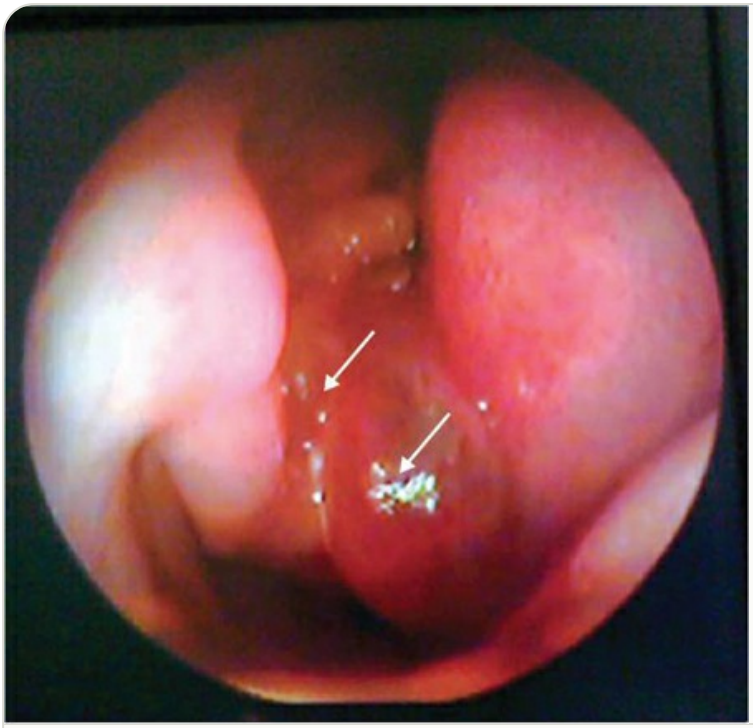


FIGURE 4 Endoscopic view of a choanal polyp.
Dhingra, fig. 32.3

3.2.1 Treatment

Endoscopic removal including its antral attachment, which prevents recurrence. Caldwell-Luc surgery is avoided, especially in children, because it damages the developing teeth.

3.3 Ethmoidal (bilateral) polyps

3.3.1 Associations

- Chronic rhinosinusitis, allergic and non-allergic, and NARES.
- Asthma (about 7% of asthmatics have polyps).
- Aspirin intolerance: about a third; *Samter's triad* (aspirin-exacerbated respiratory disease) is polyps, asthma and aspirin sensitivity.
- Cystic fibrosis (about 20%); polyps in a child should prompt a sweat test.
- Allergic fungal sinusitis (almost all).
- Kartagener and Young syndromes, Churg-Strauss syndrome (eosinophilic granulomatosis with polyangiitis), and nasal mastocytosis.

3.3.2 Pathology

Oedematous mucosa of the middle meatus and turbinate prolapses into the nose; sessile polyps become pedunculated with gravity and sneezing. Large fluid-filled spaces and eosinophils lie in the stroma; the surface may become squamous where exposed.

3.3.3 Site

Always from the lateral wall, usually the middle meatus (uncinate, bulla, sinus ostia, middle turbinate); allergic polyps almost never arise from the septum or floor.

3.3.4 Clinical picture

Progressive bilateral blockage, loss of smell, headache from sinusitis, sneezing and watery discharge. Polyps are smooth, glistening, pale, grape-like, insensitive to probing and do not bleed. Long-standing polyposis broadens the nose and widens the intercanthal distance (the *frog face*, or Woakes syndrome in its destructive form).

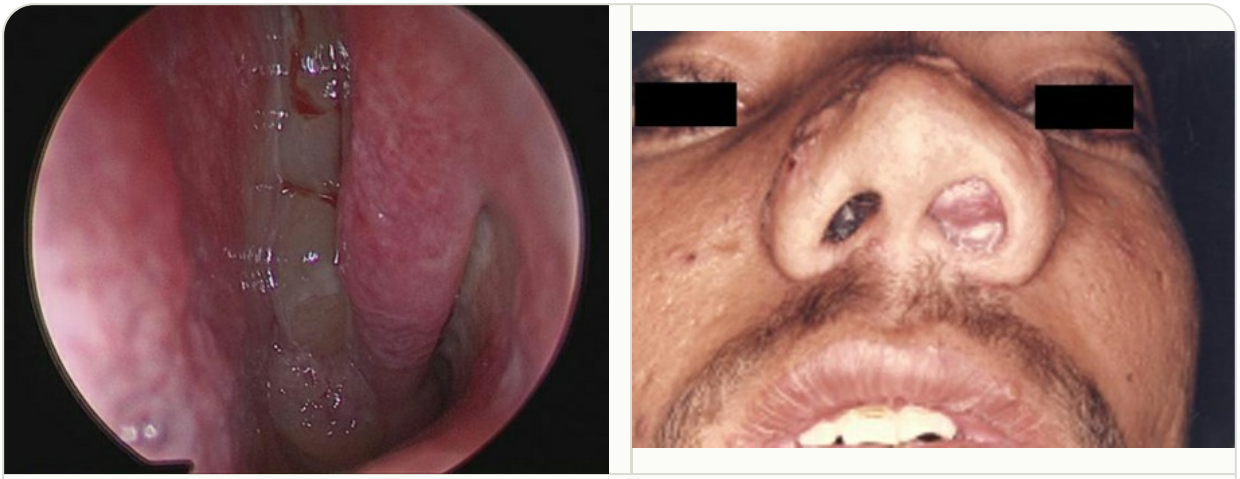


FIGURE 5 First, multiple polyps on endoscopy. Second, a polyp protruding from the left nostril. Dhingra, figs 32.4 and 32.5

3.3.5 Differential diagnosis

A blob of mucus (disappears on blowing), a hypertrophied middle turbinate (pink and bony on probing), angiofibroma (bleeds, firm), inverted papilloma and malignancy (one-sided, fleshy, friable).

ONE-SIDED POLYP IN AN ADULT

> A unilateral polyp is not simple polyposis until histology says so: inverted papilloma, carcinoma, esthesioneuroblastoma and encephalocele all present this way.

4 CHAPTER FOUR Assessment

Endoscopy and CT.

4.1 Nasal endoscopy

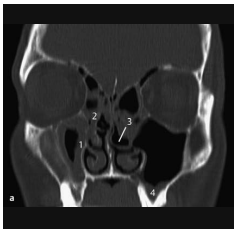
Before and after decongestion: oedema, polyps and pus in the middle meatus and sphenoethmoidal recess. Polyps are graded by extent (for example the Lund-Kennedy score).

4.2 CT of the sinuses

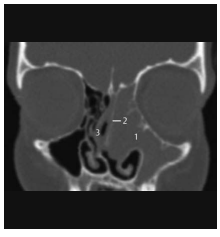
Non-contrast CT in three planes is the gold standard: it shows the extent of disease (scored by the Lund-Mackay system), anatomical variants and bony erosion, and maps the anatomy for surgery. Not for diagnosis alone, but before any operation.

IMAGING IN THE APP 5 cases

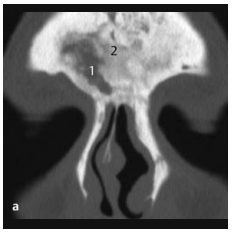
Nasal and unilateral polyps, cystic fibrosis and mucoceles.



Nasal Polyps
Infection & inflammation · 10 images



Unilateral Nasal Polyps
Infection & inflammation · 1 image



Cystic Fibrosis
Infection & inflammation · 3 images



Mucocele of the Frontal Recess
Infection & inflammation · 3 images



Mucocele
Infection & inflammation · 13 images

4.3 Other tests

Allergy testing, a sweat test or genetics for cystic fibrosis in children, ciliary studies, immunoglobulins, eosinophil count and ANCA when vasculitis is suspected, and culture from the middle meatus.

4.4 Symptom scores

Patient-reported scores such as SNOT-22 measure the burden and the response to treatment.

5 CHAPTER FIVE Medical treatment

Saline and steroids first.

5.1 CRS without polyps

- **Saline irrigation:** high-volume rinses clear mucus, bacteria and allergens and disrupt biofilms.
- **Intranasal corticosteroids:** sprays or drops; better after decongestion and with correct technique.
- **Antibiotics:** culture-guided for exacerbations; long-term low-dose macrolides (over 12 weeks) for their anti-inflammatory effect in selected patients.
- **Treat allergy:** antihistamines, leukotriene antagonists, immunotherapy.

5.2 CRS with polyps

- **Steroid sprays or drops,** long-term.
- **Short oral steroid courses** shrink polyps for a few weeks; limit their number because of side effects.
- **Doxycycline** for a few weeks reduces polyp size modestly.
- **Treat asthma** with the respiratory team.
- **Aspirin-exacerbated disease:** leukotriene modifiers, and aspirin desensitisation after surgery.

- **Biologics:** monoclonal antibodies against type 2 inflammation (anti-IL-4 receptor, anti-IgE, anti-IL-5) for severe, recurrent polyposis despite surgery and steroids, especially with asthma.

6 CHAPTER SIX Surgery

Opening the sinuses for the medicine.

6.1 Endoscopic sinus surgery

For patients whose symptoms persist despite adequate medical treatment. It removes polyps and diseased tissue, opens the natural ostia, corrects obstructing anatomy, and makes a cavity that saline and steroids can reach. Medical treatment continues afterwards. Revision is common in polyposis. In cystic fibrosis and ciliary disease, wide dependent drainage of the maxillary and sphenoid sinuses is made.

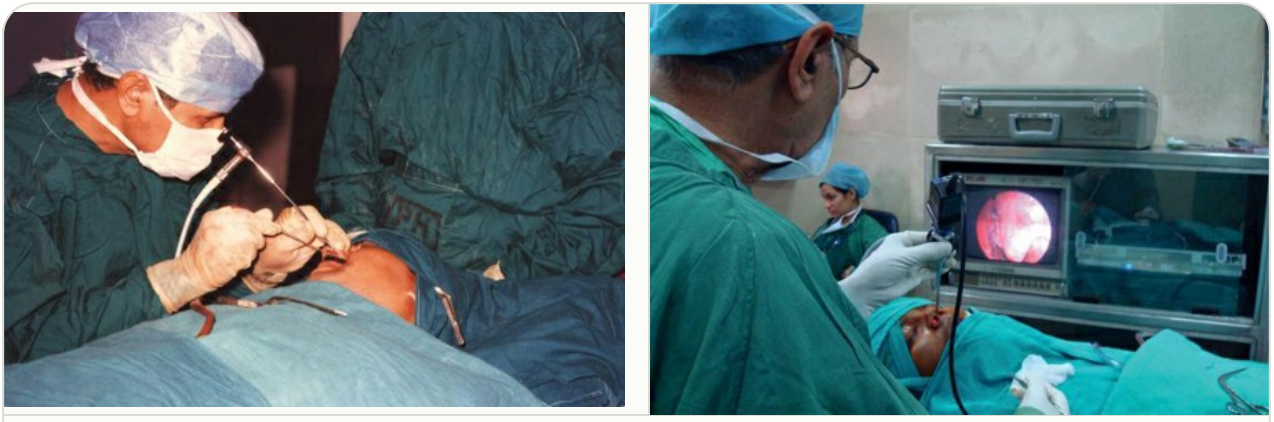


FIGURE 6 Functional endoscopic sinus surgery, without and with a camera and monitor.
Dhingra, figs 37.4 and 37.5

6.2 Historical operations

Intranasal polypectomy alone, Caldwell-Luc operation, intranasal and external ethmoidectomy, and external frontoethmoidectomy (Howarth or Lynch); now largely replaced by endoscopic surgery. See the nose and sinus operations topic.

OPERATIVE GUIDES IN THEATRE 5 operations

Antral Lavage; Endoscopic Middle Meatal Antrostomy/Antrochoanal Polyp
Antrostomy
Endoscopic Ethmoidectomy
Ethmoidectomy
Endoscopic Sphenoidotomy
Sphenoidotomy
Endoscopic Approach to the Frontal Sinus
Frontal approach
Draf III, Endoscopic Modified Lothrop
Draf III

CHAPTER SEVEN

7 Chronic sinusitis by sinus

Old names, still examined.

7.1 Chronic maxillary sinusitis

Purulent discharge, postnasal drip, cacosmia, often dental in origin; endoscopic middle meatal antrostomy has replaced antral washouts, inferior meatal antrostomy and Caldwell-Luc operations.

7.2 Chronic frontal sinusitis

Frontal headache and discharge; endoscopic frontal sinusotomy, extended to a Draf III (modified Lothrop) procedure when needed; external frontoethmoidectomy and osteoplastic obliteration are reserved for failures.

7.3 Chronic ethmoiditis and sphenoiditis

Usually part of pansinusitis; endoscopic ethmoidectomy and sphenoidotomy.

EPOS 2020: ENDOTYPES, IMAGING AND BIOLOGICS (OXFORD TEXTBOOK)

Classification. Whether polyps are present says little about the disease underneath, so EPOS 2020 classifies chronic rhinosinusitis by mechanism. *Primary* disease is split into **localised** (one side or one sinus system, such as isolated sphenoid disease or allergic fungal rhinosinusitis) and **diffuse** (both sides), and each into **type 2** (eosinophilic) and **non-type 2** endotypes. *Secondary* disease arises from something else: odontogenic sinusitis, a tumour, immunodeficiency, primary ciliary dyskinesia,

cystic fibrosis or eosinophilic granulomatosis with polyangiitis. Localised non-type 2 disease has the best prognosis; diffuse type 2 disease is part of a wider airway disorder and tends to recur.

Central compartment atopic disease is a pattern of allergic inflammation centred on the middle turbinate head, superior septum and posterosuperior nasal cavity, in contrast to the true polyps from the middle meatus seen in eosinophilic disease. Thick "chewing gum" eosinophilic mucin suggests eosinophilic chronic rhinosinusitis or allergic fungal disease. A blood eosinophil count above 0.25×10^9/L supports a type 2 endotype, and above 0.45×10^9/L predicts recurrence.

Imaging. Incidental changes are common: mucous retention cysts in 10 to 15% and mucosal thickening in a quarter of normal people, who score up to about 4 on Lund–Mackay; EPOS 2020 takes a score above 5 as strongly suggestive of disease. MRI is too sensitive to diagnose chronic rhinosinusitis, showing thickening in about half the general population.

Biologics. Monoclonal antibodies against type 2 inflammation (dupilumab against the IL-4 receptor, omalizumab against IgE, and mepolizumab against IL-5) are indicated by EPOS 2020 for bilateral polyps after sinus surgery (or in patients unfit for it) with at least three of: evidence of type 2 disease (tissue eosinophils ≥10 per high-power field, blood eosinophils ≥0.25×10^9/L or total IgE ≥100), need for systemic steroids (two or more courses a year or long-term use), SNOT-22 of 40 or more, anosmia on testing, and asthma needing inhaled steroids. Response is reviewed at 16 weeks and one year.

Surgery for diffuse type 2 disease aims for a complete, wide "neo-sinus" cavity that gives topical steroids access, rather than a limited opening of ostia, which suits localised disease. Long-term macrolides help selected non-type 2 patients with normal IgE.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Definition** and diagnostic criteria.
- **Predisposing factors,** especially anatomical variants and ciliary disorders.

- **Antrochoanal versus ethmoidal polyps.**
- **Samter's triad,** and polyps in a child (cystic fibrosis).
- **Medical treatment,** including biologics.
- **Indications for endoscopic sinus surgery.**

8.2 Common pitfalls

COMMON PITFALLS

- Repeated antibiotics without steroids or saline.
- Calling a one-sided polyp benign without histology.
- Operating before medical treatment has been tried properly.
- Forgetting the sweat test in a child with polyps.

Written for this app from Oxford Textbook of Otolaryngology ch. 37, 38, Oxford Specialist Handbook ch. 21, Oxford Handbook of ENT ch. 9, Dhingra ch. 32, 37 and K. J. Lee ch. 29. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Fungal sinusitis

Fungi in the sinuses cause two utterly different diseases. In a healthy host they sit in a sinus as a ball or provoke an allergic storm of polyps and thick mucin, and surgery cures or controls them. In a diabetic in ketoacidosis or a neutropenic patient they invade blood vessels, turn the turbinate black and can kill within days. The first question is always whether the fungus is invading tissue.

AT A GLANCE

- Invasive disease is the emergency.
- Biopsy a pale turbinate in the immunocompromised.
- Antifungals, debridement, reverse the cause.
- Non-invasive disease needs clearance, not antifungals.

1 CHAPTER ONE Overview

Invasive or not: everything follows.

1.1 Classification

FUNGAL RHINOSINUSITIS

- NON-INVASIVE **Fungal ball (mycetoma)** a mass in one sinus, healthy host
- NON-INVASIVE **Allergic fungal rhinosinusitis** hypersensitivity, polyps, eosinophilic mucin
- NON-INVASIVE **Saprophytic infestation** fungal growth on crusts after surgery
- INVASIVE **Acute fulminant** immunocompromised; Mucor, Aspergillus; days
- INVASIVE **Chronic invasive** slow, months; often diabetic
- INVASIVE **Granulomatous invasive** immunocompetent; Aspergillus flavus; Sudan, India, Pakistan

1.2 Organisms

Aspergillus (fumigatus, flavus, niger), the Mucorales (*Mucor*, *Rhizopus*), dematiaceous fungi (*Alternaria*, *Bipolaris*, *Curvularia*) and, rarely, *Candida*, *Histoplasma* and *Blastomyces*.

1.3 Mucor or Aspergillus?

MUCOR OR ASPERGILLUS?

	MUCOR	ASPERGILLUS
Hyphae	Broad	Narrow
Septa	Rarely septate	Always septate
Branching	At right angles	At acute angles, about 45 degrees
Typical disease	Rhino-orbito-cerebral invasion, black necrosis	Fungal ball, allergic disease, chronic invasive disease

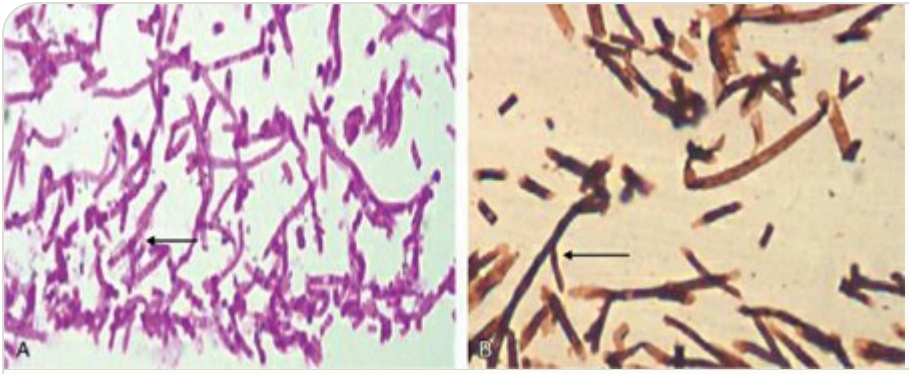


FIGURE 1 Aspergillus: septate hyphae branching at acute angles (arrow), on H&E and silver stain. Dhingra, fig. 37.3

2 CHAPTER TWO Fungal ball

A mass in a healthy sinus.

2.1 Picture

A dense, cheesy or greyish-black mass of fungal hyphae, usually in one maxillary sinus (then sphenoid, ethmoid, frontal), in an otherwise healthy, often older patient. Symptoms are those of one-sided chronic sinusitis, or none.

2.2 CT

A single opacified sinus with a hyperdense centre or calcific flecks, and no bone erosion; the wall may be thickened.

2.3 Treatment

Endoscopic removal of the ball and a wide opening of the sinus. No antifungal drugs are needed.

3 CHAPTER THREE Allergic fungal rhinosinusitis

An allergic storm.

3.1 What it is

A hypersensitivity (type I and III) reaction to fungi growing in the sinus mucin, without tissue invasion, in immunocompetent, often young, atopic patients; commoner in warm, humid climates.

3.2 Bent and Kuhn criteria

- Type I hypersensitivity to fungi.

- Nasal polyposis.
- Characteristic CT findings.
- Eosinophilic mucin without fungal invasion of tissue.
- Positive fungal stain of the mucin.

3.3 Clinical picture

Gradual, often one-sided polyposis, thick, dark, peanut-butter mucin, and slow expansion of the sinuses that can displace the eye (proptosis, telecanthus) or erode the skull base. The mucin contains eosinophils, *Charcot-Leyden crystals* and scattered hyphae.



FIGURE 2 Proptosis and lateral displacement of the eye from allergic fungal polyposis.
Dhingra, fig. 41.2

3.4 Imaging

CT shows expanded, opacified sinuses with heterogeneous high-density material (the *double density* sign), thinning and remodelling of bone. On MRI the mucin is dark on T2.

3.5 Treatment

Complete endoscopic clearance of mucin and polyps with wide drainage, combined with oral steroids before and after surgery, long-term topical steroids and saline irrigation, and close follow-up because recurrence is common. Immunotherapy and antifungals have a limited role.

CHAPTER FOUR

4 Acute invasive fungal sinusitis

A surgical and medical emergency.

4.1 Who gets it

Patients with uncontrolled diabetes, particularly with ketoacidosis (the acid environment favours Mucor), neutropenia, leukaemia, chemotherapy, organ transplants, HIV, and high-dose steroids. After the second wave of COVID-19, thousands of cases of rhino-orbito-cerebral mucormycosis, popularly called *black fungus*, followed steroid treatment in diabetics.

4.2 Mechanism

The hyphae invade arteries, causing thrombosis and ischaemic necrosis, and spread rapidly along vessels and nerves to the orbit, cribriform plate, cavernous sinus and brain.

4.3 Clinical picture

Fever, facial pain or numbness, nasal blockage and bloodstained discharge; then lid swelling, proptosis, ophthalmoplegia and loss of vision; palatal or facial skin discolouration; confusion. Endoscopy shows pale, insensate mucosa early, then *black necrotic eschar* on the turbinate, septum or palate.



FIGURE 3 Mucormycosis: (A) slight swelling of the right lower lid, (B) dark discolouration of the right palate and alveolus, (C) black necrotic tissue in the right nasal cavity.

Dhingra, fig. 28.4



FIGURE 4 Mucormycosis with cheek and orbital involvement, and destruction of the palate.

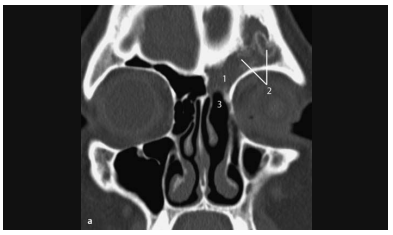
Dhingra, fig. 28.5

4.4 Diagnosis

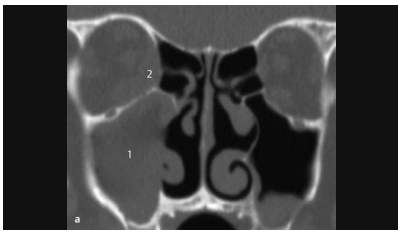
Urgent endoscopic biopsy of the middle turbinate or suspicious mucosa, with frozen section and fungal stains (silver, PAS), and culture. Contrast CT and MRI show the extent of soft tissue, orbital and intracranial spread; MRI may show non-enhancing, devitalised mucosa.

IMAGING IN THE APP 2 cases

Non-invasive and invasive fungal sinusitis.



Noninvasive Fungal Sinusitis
Infection & inflammation · 9 images



Invasive Fungal Sinusitis
Infection & inflammation · 10 images

4.5 Treatment

SUSPECTED INVASIVE FUNGAL SINUSITIS

1. Urgent endoscopy and biopsy with frozen section.
2. Start liposomal amphotericin B (about 5 mg/kg/day) immediately; monitor renal function.
3. Correct ketoacidosis and glucose; stop or reduce immunosuppression; support neutropenia.
4. Urgent surgical debridement back to bleeding tissue, endoscopic or open, repeated as needed.
5. Involve ophthalmology, infectious diseases and the medical team.

- **Antifungals:** liposomal amphotericin B for Mucor; posaconazole or isavuconazole for step-down or intolerance; voriconazole for invasive *Aspergillus* (ineffective against Mucor).
- **Surgery:** debridement may extend to medial or total maxillectomy and orbital exenteration in extensive disease, with later obturator rehabilitation of the palate.

4.6 Outcome

Mortality is high, especially with intracranial spread; survival depends on reversing the immune defect and early, complete debridement.

OPERATIVE GUIDES IN THEATRE 3 operations

Endoscopic Ethmoidectomy
Endoscopic debridement
Medial Maxillectomy
Medial maxillectomy
Complete Maxillectomy
Maxillectomy in extensive disease

5 CHAPTER FIVE Chronic invasive fungal sinusitis

Slow invasion.

5.1 Chronic invasive

Slowly progressive invasion over months, often in diabetics or patients on long-term steroids, presenting as chronic sinusitis with bone erosion and orbital apex syndrome or skull base spread. Biopsy shows fungal invasion of the submucosa. Surgical debridement and long-term antifungals.

5.2 Granulomatous invasive

In immunocompetent patients in Sudan, India and Pakistan, usually *Aspergillus flavus*: a slowly enlarging mass of the cheek, orbit or nose with non-caseating granulomas and giant cells. Surgery and long-term azoles.

6 CHAPTER SIX Differential diagnosis

Mimics.

- Bacterial chronic rhinosinusitis with polyps.
- Inverted papilloma and sinonasal malignancy (one-sided mass with bone erosion).
- Granulomatosis with polyangiitis and NK/T-cell lymphoma (destructive, crusted nose).
- Mucocele (expansion without the hyperdense mucin).

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Classification:** invasive versus non-invasive.
- **Mucor versus Aspergillus** hyphae.
- **Fungal ball:** CT and treatment without antifungals.
- **Allergic fungal sinusitis:** Bent and Kuhn criteria, mucin, double density.
- **Mucormycosis:** risk factors, black eschar, amphotericin B and debridement.

7.2 Common pitfalls

COMMON PITFALLS

- Treating a diabetic with facial numbness and a black turbinate as simple sinusitis.
- Waiting for culture before starting amphotericin.
- Giving antifungals for a fungal ball, or omitting steroids in allergic fungal disease.

Written for this app from Oxford Textbook of Otolaryngology ch. 38, Oxford Specialist Handbook ch. 21, Dhingra ch. 37 and K. J. Lee ch. 30. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Granulomatous disease of the nose

A crusted, destroyed nose that will not heal has a long differential: chronic infections such as rhinoscleroma, tuberculosis, leprosy, syphilis and rhinosporidiosis; inflammatory disease such as granulomatosis with polyangiitis and sarcoidosis; and NK/T-cell lymphoma, which looks like inflammation but is a cancer. Many are systemic diseases showing in the nose, and none can be named without a biopsy.

AT A GLANCE

- Crusting and destruction need a tissue diagnosis.
- Check kidneys, lungs and ANCA.
- Exclude lymphoma, infection and cocaine.
- Treat disease before repairing the nose.

1 CHAPTER ONE Overview

The destructive, crusting nose.

1.1 Classification

GRANULOMATOUS DISEASES OF THE NOSE

- BACTERIAL **Rhinoscleroma** tuberculosis, lupus vulgaris, leprosy, syphilis
- FUNGAL **Aspergillosis, mucormycosis** see the fungal sinusitis topic
- PARASITIC **Rhinosporidiosis** leishmaniasis
- INFLAMMATORY **Granulomatosis with polyangiitis** sarcoidosis, eosinophilic granulomatosis
- NEOPLASTIC MIMIC **NK/T-cell lymphoma** the old lethal midline granuloma

1.2 The rule

Always biopsy, and always look for the systemic disease: chest, kidneys, eyes, skin and lymph nodes. Many of these conditions mimic a neoplasm, and a neoplasm can

mimic them.

1.3 Common findings

Crusting, bloodstained discharge, nasal obstruction, granulations, septal perforation, saddle nose and, in some, palatal destruction.

2 CHAPTER TWO Bacterial infections

Chronic and destructive.

2.1 Rhinoscleroma

2.1.1 Cause

Klebsiella rhinoscleromatis (Frisch bacillus), a Gram-negative bacillus; endemic in parts of Africa, Central America, Eastern Europe, the Middle East (including Egypt) and North India.

2.1.2 Spread

Starts in the nose and spreads to the nasopharynx, oropharynx, larynx (especially the subglottis), trachea and bronchi.

2.1.3 Stages

- **Atrophic (catarrhal):** like atrophic rhinitis, with foul purulent discharge and crusts.
- **Granulomatous (proliferative):** painless, non-ulcerating nodules; infiltration of the lower external nose and upper lip gives a *woody* feel (the *Hebra nose*).
- **Cicatricial (sclerotic):** stenosis of the nostrils, distortion of the lip, adhesions in the nose and pharynx, and subglottic stenosis with stridor.

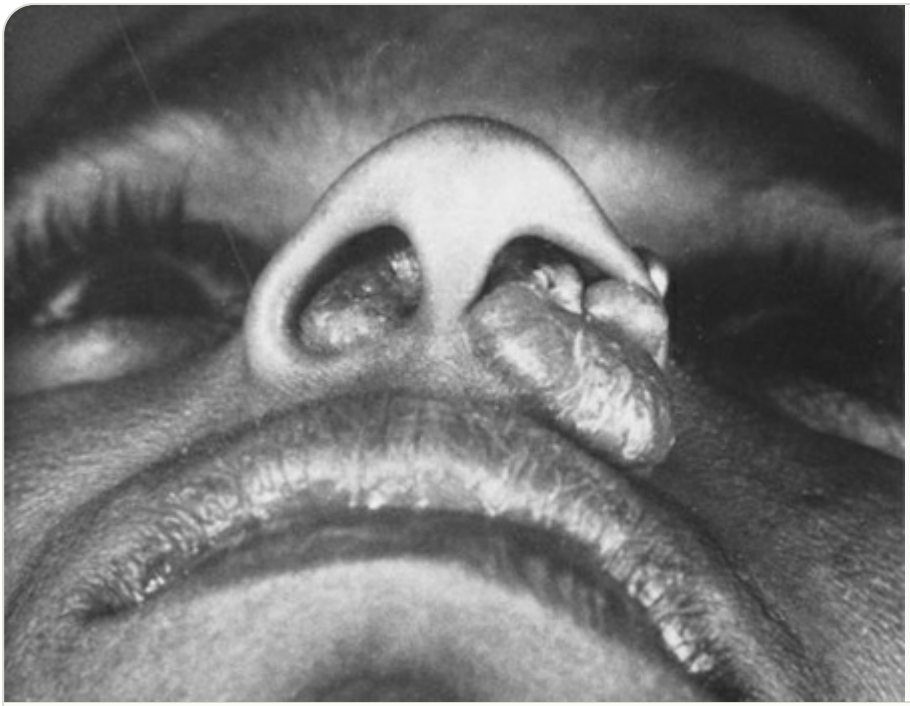


FIGURE 1 Rhinoscleroma of the nose.
Dhingra, fig. 28.1

2.1.4 Diagnosis

Biopsy: submucosal plasma cells, lymphocytes and eosinophils, with *Mikulicz cells* (large foamy macrophages containing the bacilli) and *Russell bodies* (eosinophilic inclusions of immunoglobulin in plasma cells), which are diagnostic. Culture of the biopsy.

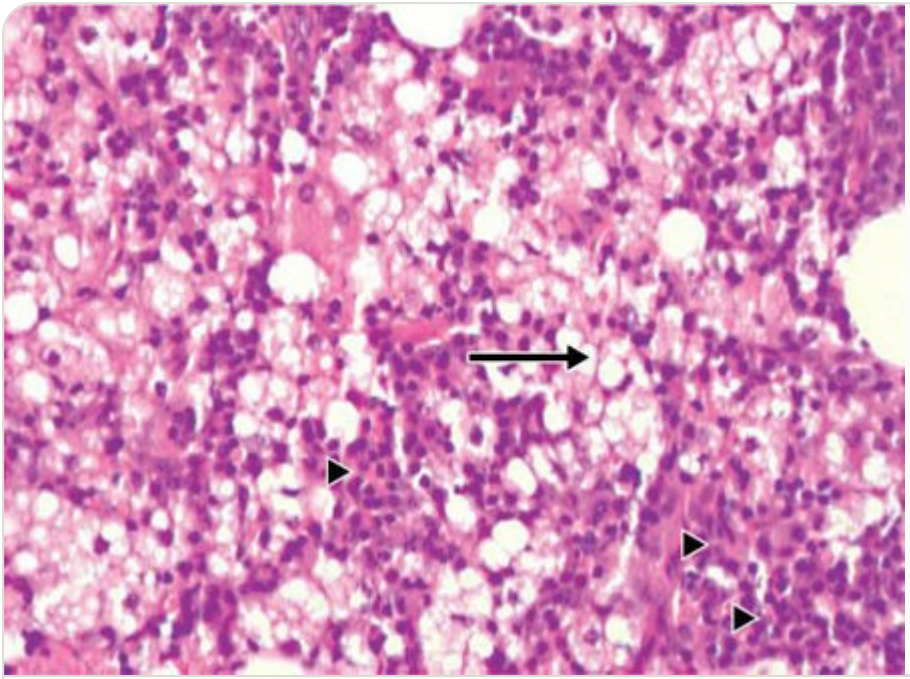


FIGURE 2 Rhinoscleroma: foamy Mikulicz cells (arrow) among lymphocytes (arrowheads).
Dhingra, fig. 28.2

2.1.5 Treatment

Long antibiotic courses until two consecutive biopsy cultures are negative.

WHERE THE BOOKS DIFFER · ANTIBIOTICS FOR RHINOSCLEROMA

Dhingra	Lee
Streptomycin 1 g/day with tetracycline 2 g/day for at least four to six weeks, repeated if needed; steroids to reduce fibrosis.	Fluoroquinolones such as ciprofloxacin, or tetracyclines, for months.

Surgery restores the airway (including tracheostomy for subglottic stenosis) and corrects deformity.

2.2 Tuberculosis

Usually secondary to pulmonary disease. Nodular infiltration, then ulceration of the anterior septum and anterior end of the inferior turbinate, with perforation of the cartilaginous septum. Biopsy with acid-fast stains and culture; antituberculous drugs.

2.3 Lupus vulgaris

A low-grade tuberculous infection of the vestibule and the skin of the nose and face, with brownish *apple-jelly nodules* and chronic vestibulitis; may perforate the cartilaginous septum. Biopsy; antituberculous treatment.

2.4 Leprosy

Mycobacterium leprae, mainly in the lepromatous form. The anterior septum and inferior turbinate become red and swollen with excess discharge, then crusting, bleeding and nodules that ulcerate and perforate. Late sequelae: atrophic rhinitis, a depressed bridge and loss of the anterior nasal spine with a retracted columella. Diagnosis from scrapings and biopsy (lepra bacilli in foamy *lepra cells*); multidrug therapy with dapsone and rifampicin; reconstruction when inactive.

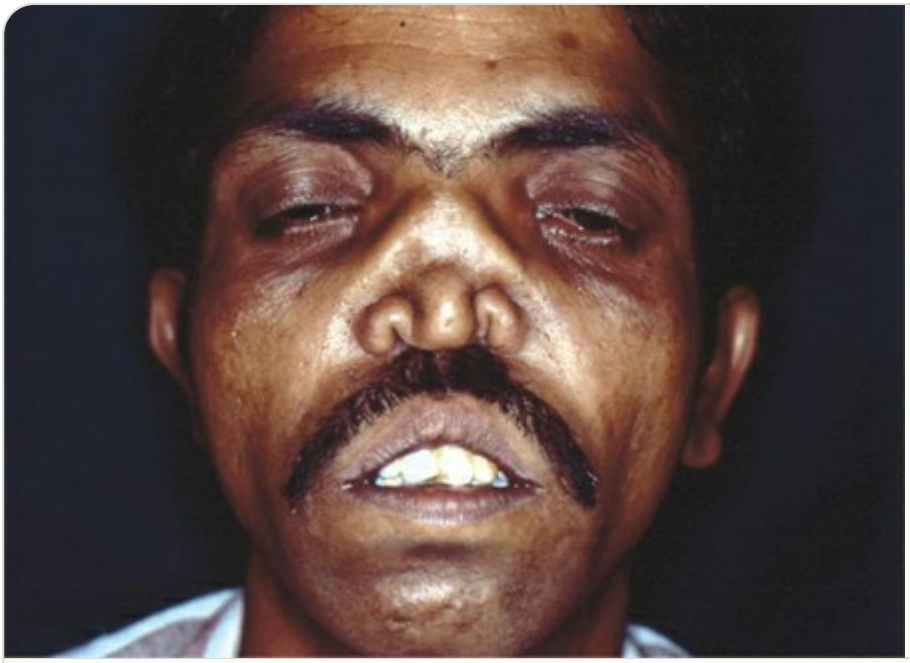


FIGURE 3 Leprosy of the nose.
Dhingra, fig. 28.3

2.5 Syphilis

2.5.1 Acquired

- **Primary:** a rare chancre of the vestibule.
- **Secondary:** rhinitis with crusting and fissuring, with mucous patches, rash and lymphadenopathy.
- **Tertiary:** the stage that involves the nose: a *gumma* destroys the septum, both bony and cartilaginous, with sequestra, foul crusts, palatal perforation and a saddle nose.

2.5.2 Congenital

- **Early:** *snuffles* in the first three months, becoming purulent, with excoriation of the vestibule and lip.
- **Late:** around puberty, with gummatous destruction like tertiary disease, plus corneal opacities, deafness and Hutchinson's teeth.

2.5.3 Diagnosis and treatment

Serology (VDRL with treponemal tests) and biopsy. Benzathine penicillin 2.4 million units weekly for three weeks; alkaline douches; removal of sequestra; reconstruction once inactive.

WHICH PART OF THE SEPTUM?

Tuberculosis, lupus and leprosy perforate the cartilaginous septum; syphilis destroys the bony septum.

3 CHAPTER THREE Parasitic infections

Rhinosporidiosis and leishmaniasis.

3.1 Rhinosporidiosis

3.1.1 Cause and spread

Rhinosporidium seeberi, once thought a fungus, now classed as an aquatic protistan parasite. Acquired from ponds shared with animals; common in South India and Sri Lanka.

3.1.2 Clinical picture

A leafy, polypoidal, pink to purple, strawberry-like mass on the septum or lateral wall, very vascular and bleeding easily, its surface studded with white dots (the sporangia). Bloodstained discharge or epistaxis may be the only complaint; it may hang into the nasopharynx.

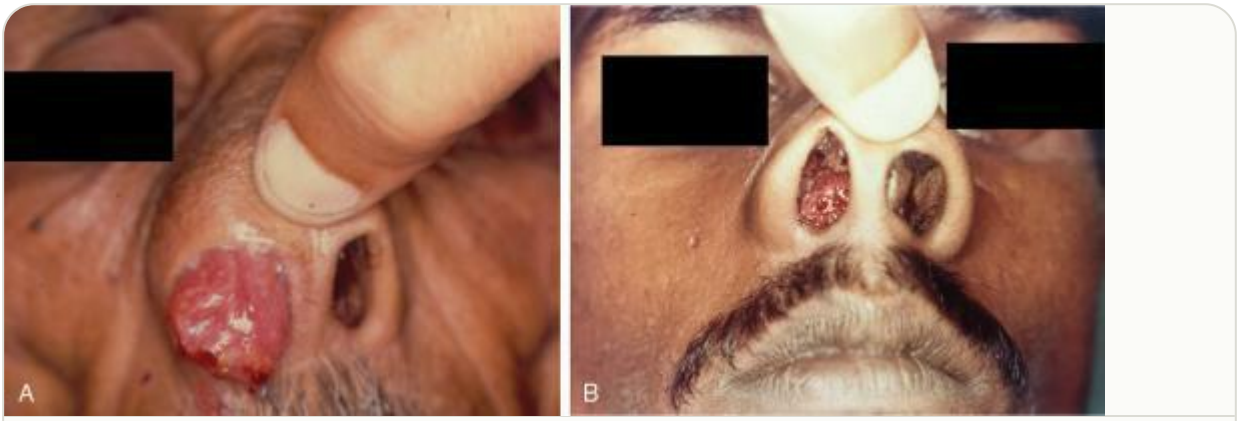


FIGURE 4 Rhinosporidiosis: a polypoidal mass protruding from the nostril, and its appearance inside the nose.
Dhingra, fig. 28.6

3.1.3 Life cycle

Endospores grow into trophocytes, which mature into *sporangia* 200 to 250 µm across containing thousands of endospores; the sporangium bursts through a pore and releases them.

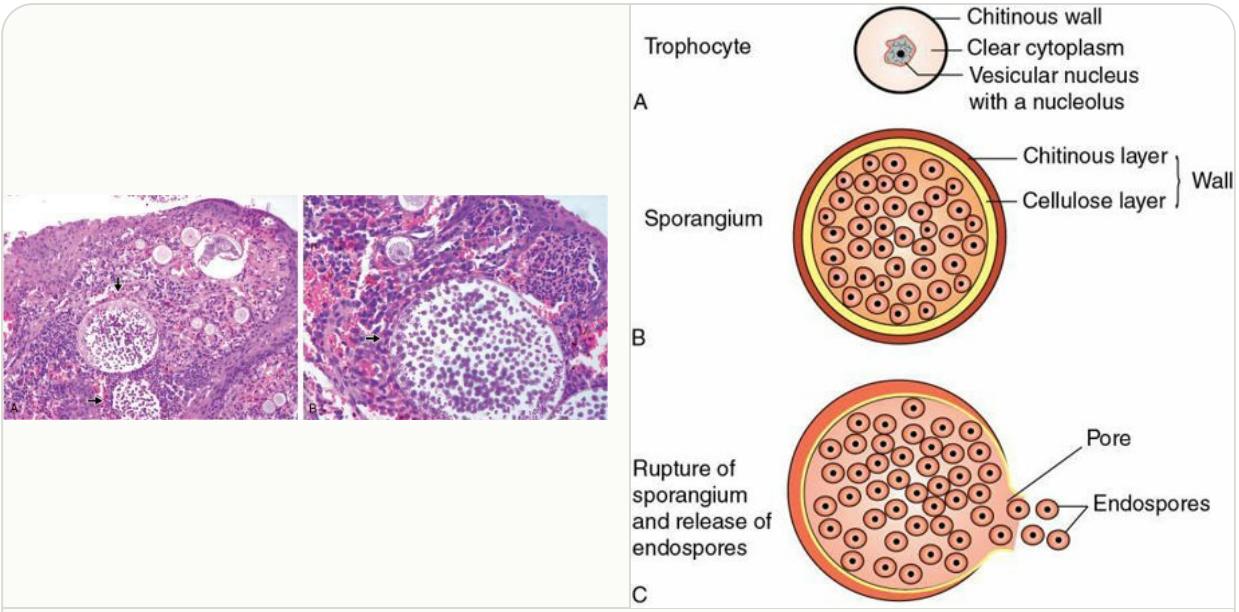


FIGURE 5 First, histology with sporangia packed with spores. Second, the life cycle of the organism.

Dhingra, figs 28.7 and 28.8

3.1.4 Diagnosis and treatment

Biopsy shows the sporangia; the organism cannot be cultured. Complete excision with a diathermy knife and cautery of the base; recurrence is common; dapsone helps some patients.

3.2 Leishmaniasis

Transmitted by the sandfly (*Phlebotomus*). Cutaneous lesions on exposed skin progress from papules to crusted ulcers; mucosal disease appears on the septum as a nodule or plaque that ulcerates and perforates, sometimes years after a skin lesion, and can spread to the palate, pharynx and larynx. Biopsy, culture and PCR; mucosal disease needs parenteral antimonials (sodium stibogluconate or meglumine antimoniate).

4 CHAPTER FOUR Inflammatory diseases

Vasculitis and sarcoid.

4.1 Granulomatosis with polyangiitis

4.1.1 What it is

Formerly Wegener's granulomatosis: a necrotising granulomatous vasculitis of small vessels affecting the upper airway, lungs and kidneys (the classic triad), with skin, eye and ear involvement.

4.1.2 Nose and sinuses

A persistent cold or sinusitis, bloodstained then purulent discharge, crusting, granulations, septal perforation and saddle nose; destruction may extend to the orbit, palate and middle ear. Subglottic stenosis develops in some.

4.1.3 Systemic features

Fatigue, night sweats, arthralgia, anaemia; cough and haemoptysis with cavitating lung nodules; glomerulonephritis with blood, casts and protein in the urine and a rising creatinine, the usual cause of death untreated.

4.1.4 Diagnosis

c-ANCA (anti-proteinase 3) is positive in most active systemic disease; raised ESR and CRP; urinalysis and creatinine; chest imaging. Nasal biopsy shows necrosis, granulomas and vasculitis, though often non-specific.

4.1.5 Treatment

Steroids with cyclophosphamide or rituximab to induce remission, then azathioprine or methotrexate to maintain it, with the rheumatology and renal teams. Nasal douching for crusts; reconstruction only when quiescent.

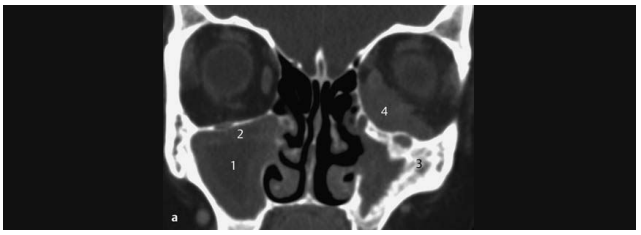
IMAGING IN THE APP 2 cases

Granulomatosis with polyangiitis in the nose, sinuses and orbit.



Granulomatosis with Polyangiitis (Wegener)

Infection & inflammation · 2 images



Granulomatosis with Polyangiitis, Orbital Involvement

Infection & inflammation · 2 images

4.2 Eosinophilic granulomatosis with polyangiitis

Churg-Strauss syndrome: asthma, nasal polyps, peripheral eosinophilia and vasculitis, often p-ANCA positive.

4.3 Sarcoidosis

A systemic non-caseating granulomatous disease. In the nose: submucosal nodules on the septum and inferior turbinate with obstruction, crusting, pain and bleeding; nodules on the vestibule and facial skin (*lupus pernio*). Chest imaging shows hilar adenopathy; serum calcium and *serum ACE* are raised in active disease. Topical and systemic steroids.

5 CHAPTER FIVE NK/T-cell lymphoma

The cancer that looks like inflammation.

5.1 What it is

Extranodal NK/T-cell lymphoma, nasal type, associated with Epstein-Barr virus; the condition once called *lethal midline granuloma* or polymorphic reticulosis. Commoner in East Asia and Latin America.

5.2 Clinical picture

Progressive, relentless destruction of the midline: nasal obstruction and discharge, ulceration and necrosis of the septum, palate and facial skin, with fever and weight loss.

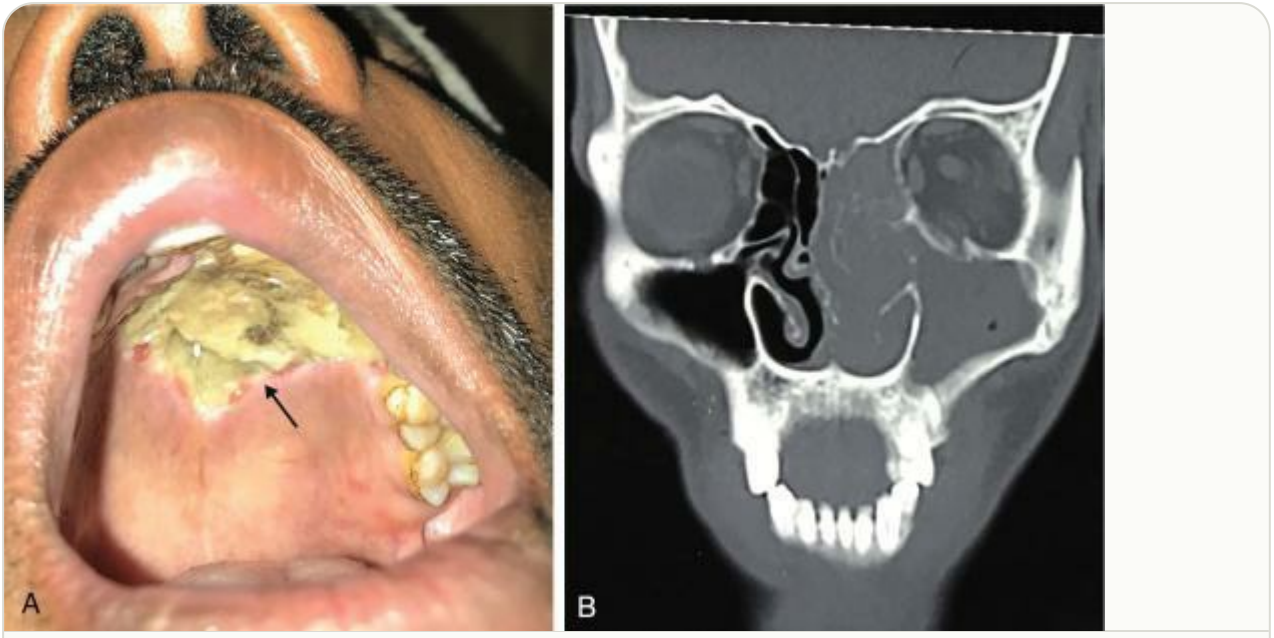


FIGURE 6 NK-cell lymphoma with midline destruction in a young man.
Dhingra, fig. 28.9

5.3 Diagnosis

Deep, repeated biopsies of viable tissue, with immunohistochemistry (CD56, cytotoxic markers) and EBV studies; superficial biopsies show only necrosis and inflammation.

5.4 Treatment

Radiotherapy with chemotherapy. It must be told apart from granulomatosis with polyangiitis, because the treatments are completely different.

6 CHAPTER SIX Assessment

The work-up of a destructive nose.

INVESTIGATIONS BY SUSPICION

SUSPECTED DISEASE	KEY TEST
Granulomatosis with polyangiitis	c-ANCA, urinalysis, creatinine, chest imaging, biopsy
Sarcoidosis	Serum ACE, calcium, chest imaging
Tuberculosis	Acid-fast stains, culture, PCR, chest imaging
Syphilis	VDRL with treponemal tests
Rhinoscleroma	Biopsy for Mikulicz cells and Russell bodies, culture
Lymphoma	Deep biopsy with immunohistochemistry and EBV studies
Cocaine damage	History and urine toxicology

COCAINE

Cocaine-induced midline destruction can mimic vasculitis, and some patients develop ANCA from levamisole in the drug. Ask, and test.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Classification** of granulomatous disease of the nose.
- **Rhinoscleroma:** organism, stages, Mikulicz cells and Russell bodies, treatment.
- **Rhinosporidiosis:** organism, appearance, sporangia, treatment.
- **Syphilis of the nose:** stages, snuffles, saddle nose, bony septal perforation.

- **Granulomatosis with polyangiitis:** triad, c-ANCA, treatment.
- **NK/T-cell lymphoma** versus vasculitis.

7.2 Common pitfalls

COMMON PITFALLS

- Treating a destructive nose with antibiotics for months without a biopsy.
- Superficial biopsies of necrotic tissue in suspected lymphoma.
- Forgetting the kidneys in granulomatosis with polyangiitis.

Written for this app from Oxford Textbook of Otolaryngology ch. 34, Oxford Specialist Handbook ch. 20, Oxford Handbook of ENT ch. 9, Dhingra ch. 28 and K. J. Lee ch. 12. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Juvenile nasopharyngeal angiofibroma

A teenage boy with a blocked nose and nosebleeds that keep coming back has a juvenile angiofibroma until proven otherwise. It is benign, but it bleeds torrentially, grows through the natural doorways of the skull base, and is diagnosed by imaging, never by a biopsy in clinic.

AT A GLANCE

- Adolescent boy, blockage and heavy bleeding.
- Image, don't biopsy.
- Embolise, then remove.
- Drill the pterygoid base.
- UPMC staging adds residual ICA vascularity after embolisation.

1 CHAPTER ONE Overview

The bleeding tumour of adolescent boys.

1.1 Definition

Juvenile nasopharyngeal angiofibroma (also called nasopharyngeal fibroma) is a benign, locally aggressive, highly vascular tumour arising at the back of the nasal cavity, near the sphenopalatine foramen.

1.2 Epidemiology

The commonest benign tumour of the nasopharynx, yet under 1% of head and neck tumours. Almost exclusively in boys and young men of 10 to 20 years.

1.3 Why it matters

Profuse epistaxis, anaemia, and spread into the orbit and skull base. Biopsy in clinic can cause catastrophic bleeding. It does not regress on its own, so waiting is not an

option.

2 CHAPTER TWO Aetiology and pathology

Hormones and malformed vessels.

2.1 Why adolescent boys

It is thought to be testosterone-dependent: a hamartomatous nidus of vascular tissue, perhaps a remnant of the first branchial arch artery, is activated at puberty. About three-quarters of tumours carry androgen receptors.

2.2 Molecular clues

Vascular endothelial growth factor and insulin-like growth factor II drive growth; stromal cells show beta-catenin activity, and C-MYC is found in large tumours. Angiofibroma is about 24 times commoner in familial adenomatous polyposis of the colon.

2.3 Site of origin

The posterior nasal cavity, at the superior margin of the sphenopalatine foramen, near the junction of the sphenoidal process of the palatine bone with the root of the pterygoid plate.

2.4 Gross and microscopic pathology

A reddish-purple, lobulated, sessile mass covered by mucosa, without a capsule. Two components: endothelium-lined vascular spaces lacking muscle and elastic tissue (so they cannot contract and bleed profusely), and a stroma of stellate and spindle cells laying down collagen, whose amount decides how firm the tumour is.

For personal study. Follow local protocols and senior advice for patients.

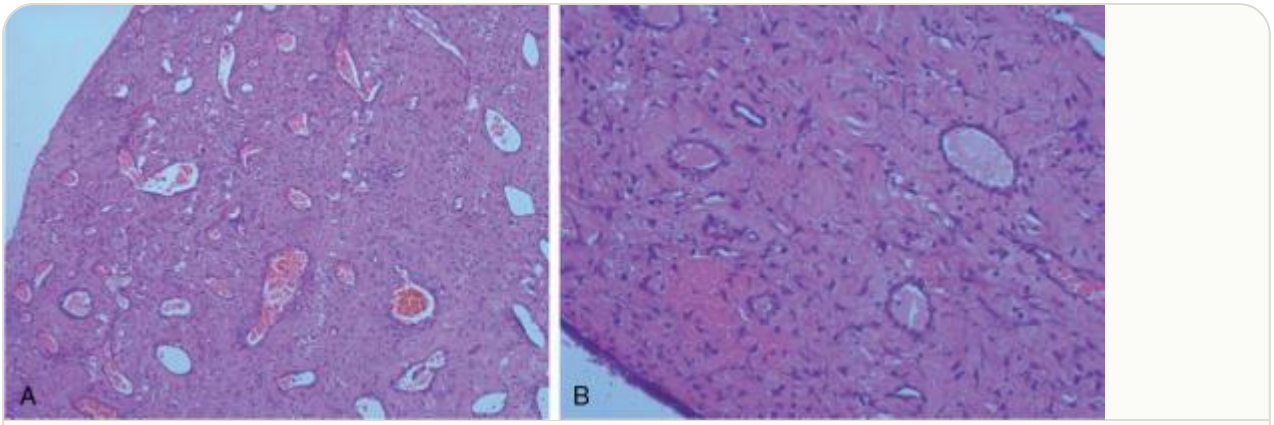


FIGURE 1 Angiofibroma: dilated vascular spaces in a fibrous stroma.
Dhingra, fig. 49.4

3 CHAPTER THREE Spread

Through the doorways of the skull base.

3.1 Three modes

- **Along paths of least resistance:** into the nose and nasopharynx, and through fissures and foramina: the foramen rotundum, vidian (pterygoid) canal, palatovaginal canal and inferior orbital fissure.
- **By expansion:** pushing the posterior wall of the maxillary sinus forwards from the pterygopalatine fossa (the antral sign).
- **By bone destruction:** at the root of the pterygoid and the greater wing of the sphenoid.

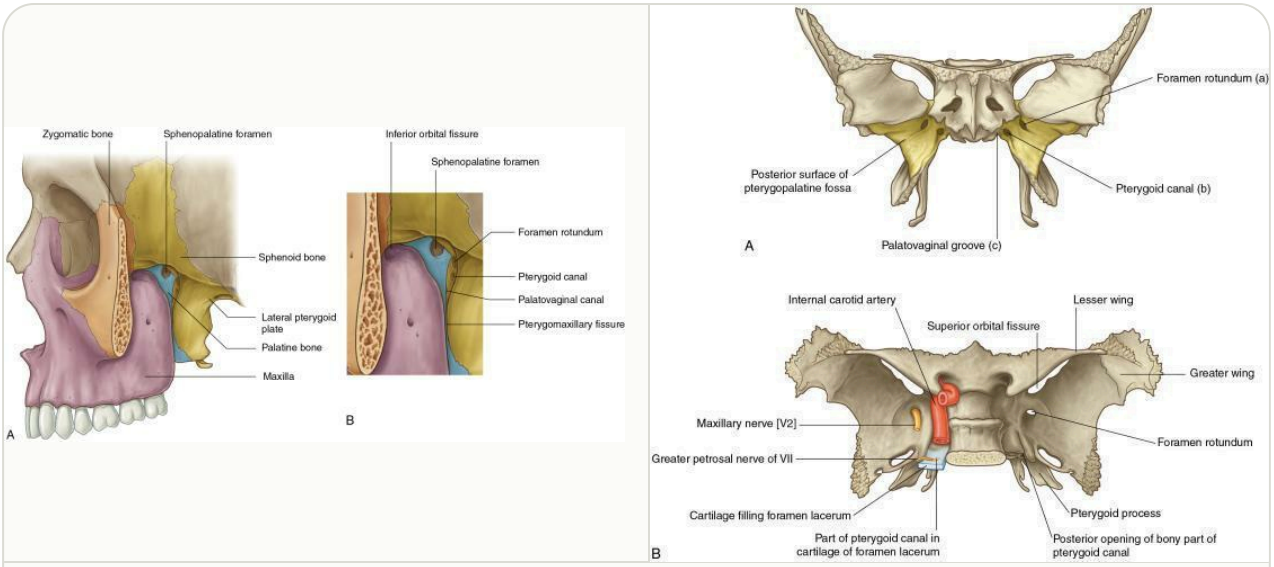


FIGURE 2 First, the pterygopalatine fossa from the side, with its routes to the orbit, infratemporal fossa and cranial cavity. Second, the openings in its posterior wall: foramen rotundum, vidian canal and palatovaginal canal.

Dhingra, figs 49.1 and 49.2

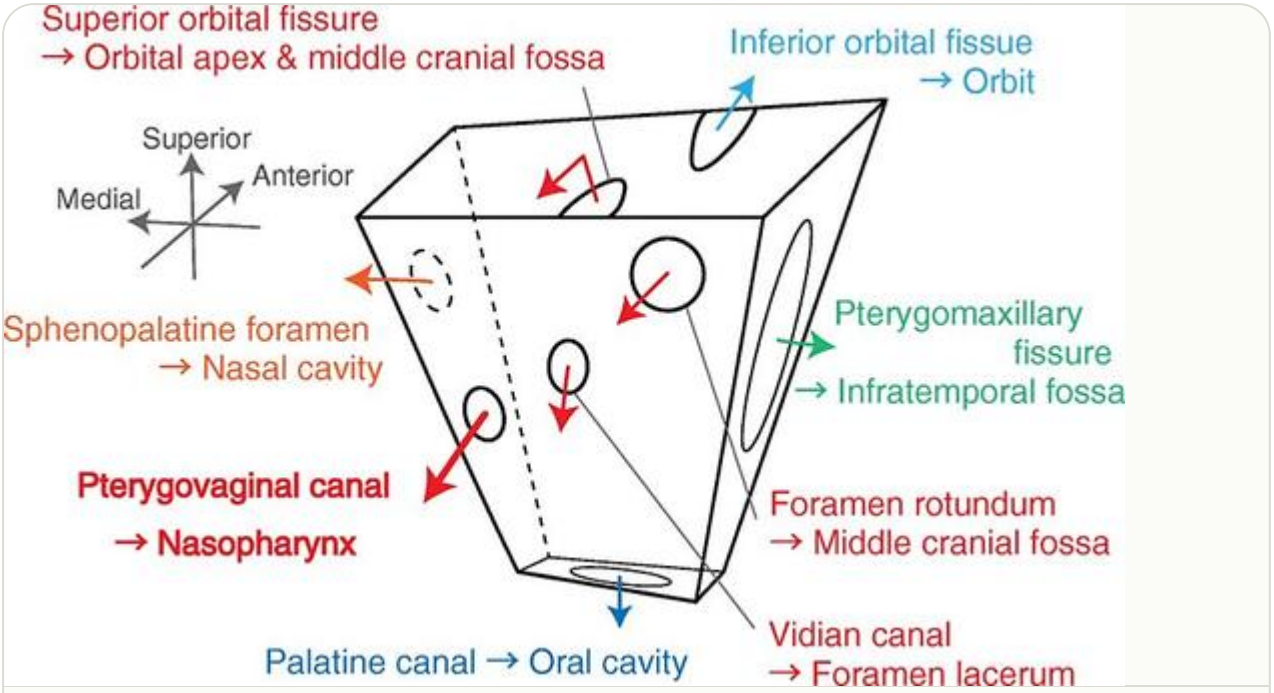


FIGURE 3 Communications of the pterygopalatine fossa, the crossroads through which the tumour spreads.

Anatomy atlas · ENT Steps p. 161

3.2 Extensions

1. Nasal cavity and nasopharynx, blocking the choanae and displacing the septum.
2. Paranasal sinuses: maxillary, sphenoid, ethmoid.

3. Pterygopalatine fossa, then through the pterygomaxillary fissure into the infratemporal and temporal fossae and the cheek.
4. Orbit, through the inferior orbital fissure: proptosis, diplopia, optic nerve compression.
5. Cranial cavity in 20 to 25%, mainly the middle fossa (lateral to the internal carotid if it erodes the floor, medial if it comes through the sphenoid and sella), less often the anterior fossa; usually extradural, sometimes into the cavernous sinus.

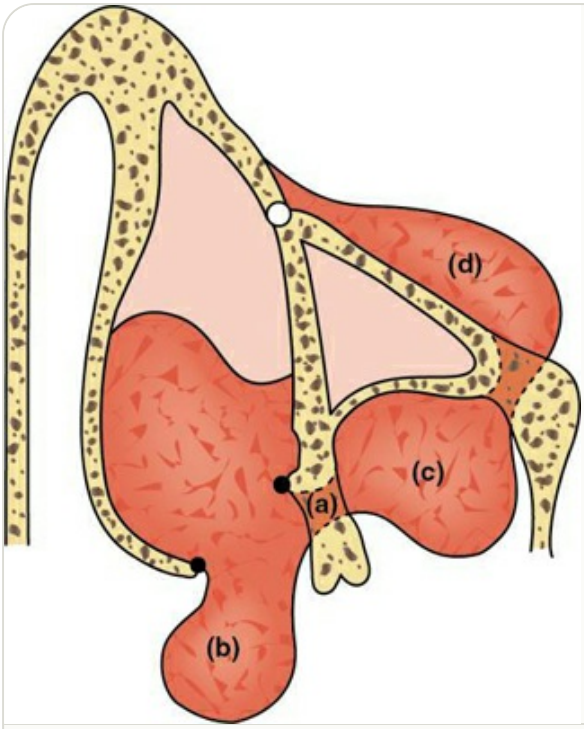


FIGURE 4 Axial view of the spread: into the nose, the nasopharynx, behind the antrum into the pterygopalatine fossa, and on into the cheek.

Dhingra, fig. 49.3

RECURRENCE HIDES IN CANALS

Tumour left in the vidian canal or the basisphenoid is the commonest source of recurrence: drill the canal at surgery.

4 CHAPTER FOUR Clinical features

Epistaxis and obstruction in a boy.

4.1 Nasal and nasopharyngeal

- **Profuse, recurrent, spontaneous epistaxis:** the commonest presentation, often with anaemia.
- **Progressive nasal obstruction** with a hyponasal voice.
- **Nasal discharge** from obstruction and sinusitis.

4.2 Ear

Conductive hearing loss and otitis media with effusion from tubal obstruction.

4.3 Eye and nerves

Proptosis and diplopia; involvement of the second, third, fourth and sixth nerves with orbital and intracranial spread.

4.4 Face

Broadening of the nasal bridge, swelling of the cheek, and a *frog face* with bilateral proptosis.

4.5 Examination

Endoscopy shows a smooth or lobulated, pink to purplish mass with prominent surface vessels in the back of the nose and nasopharynx, obstructing one or both choanae, sometimes pushing the soft palate down or hanging into the oropharynx. Never palpate it with a finger outside the operating theatre.

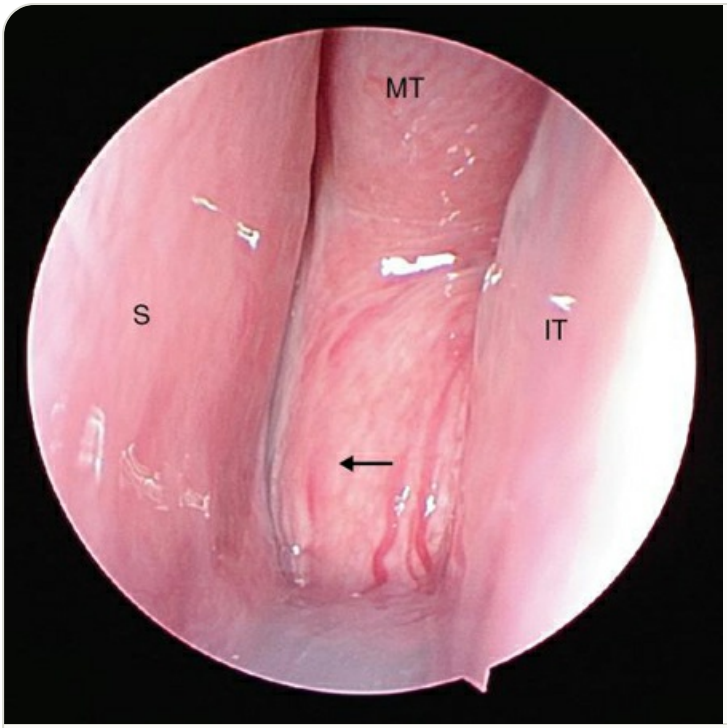


FIGURE 5 Endoscopic view of an angiofibroma (arrow) in the left nasal cavity, with prominent surface vessels; septum (S), inferior turbinate (IT), middle turbinate (MT).

Dhingra, fig. 49.5

NEVER BIOPSY IN CLINIC

Diagnosis is made by imaging. If tissue is truly needed to exclude another tumour, biopsy is done under general anaesthesia with blood cross-matched and everything ready to control bleeding.

5 CHAPTER FIVE Investigations

Imaging makes the diagnosis.

5.1 Contrast CT

The investigation of choice. It shows the enhancing mass, bone erosion and the extent of spread. *Anterior bowing of the posterior wall of the maxillary sinus* (the antral or Holman-Miller sign) is pathognomonic. Widening of the pterygopalatine fossa and erosion of the vidian canal are typical.

5.2 MRI

Complements CT for soft tissue extension: intracranial, cavernous sinus, orbit, infratemporal fossa. Flow voids give a *salt and pepper* appearance.

5.3 Angiography and embolisation

Digital subtraction angiography shows the feeders, mainly from the external carotid (the internal maxillary artery's sphenopalatine and posterior nasal branches, the ascending pharyngeal and ascending palatine arteries), and from the internal carotid in large or intracranial tumours. External carotid feeders can be embolised; surgery should follow within 24 to 48 hours, before collaterals develop. Embolisation carries risks of stroke, blindness and cranial nerve palsy, so some surgeons omit it for small tumours.

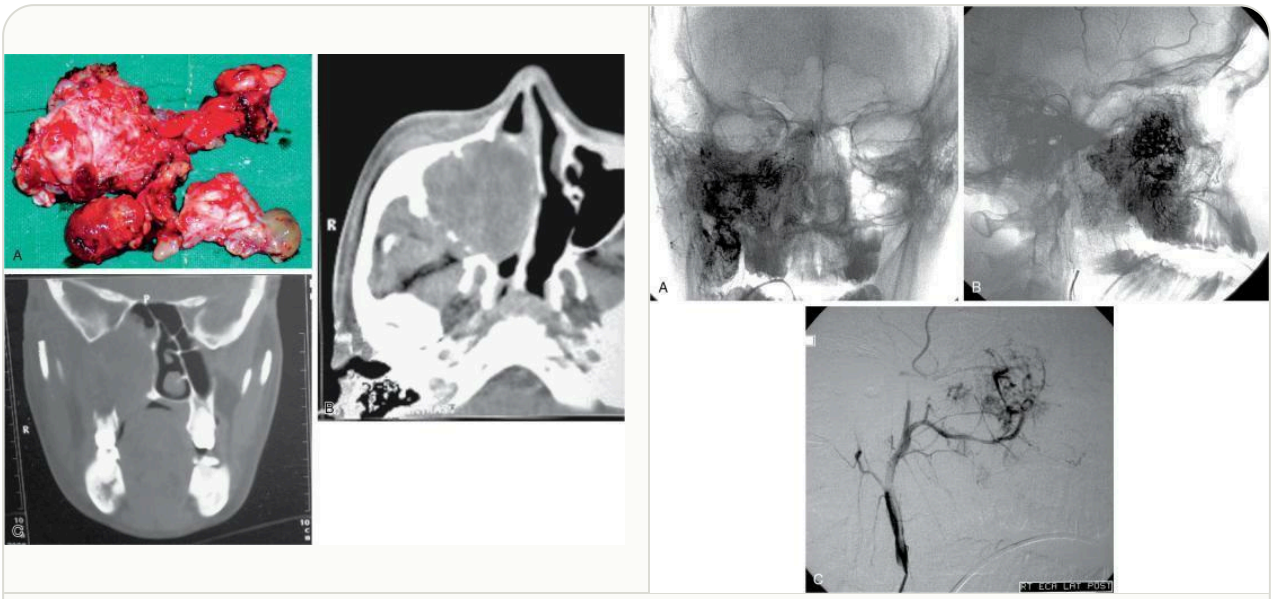
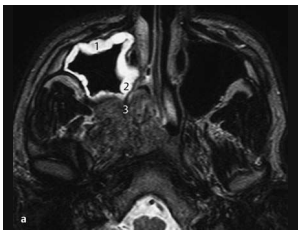


FIGURE 6 First, the specimen of an extensive angiofibroma and its CT, showing bone destruction and spread into the pterygopalatine fossa. Second, angiography before and after embolisation.
Dhingra, figs 49.6 and 49.7

IMAGING IN THE APP 1 case

Juvenile angiofibroma on CT, MRI and angiography.



Angiofibroma, Juvenile
Tumours · 12 images

5.4 Preparation

Group and cross-match two to three units of blood.

5.5 Differential diagnosis

Antrochoanal polyp, enlarged adenoids, olfactory neuroblastoma, rhabdomyosarcoma, lobular capillary haemangioma and haemangiopericytoma.

6 CHAPTER SIX Staging

Extent decides the approach.

Several systems exist (Sessions, Fisch, Chandler, Radkowski, Andrews, and newer endoscopic systems), all based on extension into the sinuses, pterygopalatine and infratemporal fossae, orbit and cranial cavity, and on residual vascularity after embolisation.

7 CHAPTER SEVEN Treatment

Surgery is the treatment.

7.1 Principles

Complete surgical removal is the treatment of choice; the tumour does not regress with age. Embolisation reduces blood loss. Hormonal treatment is no longer used.

7.2 Approaches by extent

EXTENT AND APPROACH

EXTENT	APPROACH
Nose and nasopharynx	Endoscopic, or transpalatal
Plus sinuses and pterygopalatine fossa	Endoscopic, lateral rhinotomy with medial maxillectomy, midfacial degloving, or Le Fort I
Plus infratemporal fossa	Extended endoscopic, infratemporal fossa approach, or maxillary swing
Plus cheek	Maxillary swing (facial translocation)
Plus intracranial extension	Combined endoscopic or open skull base surgery with neurosurgery; radiotherapy if inaccessible
Residual or recurrent	Observation, revision surgery, or radiotherapy

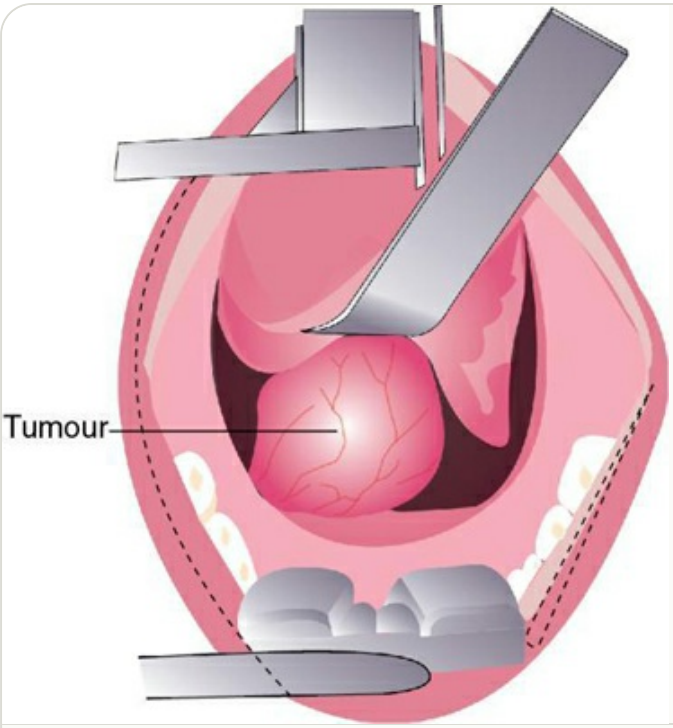


FIGURE 7 Angiofibroma exposed through a transpalatal approach.
Dhingra, fig. 49.8

7.2.1 Endoscopic surgery

Now used for most tumours, including many with infratemporal and limited intracranial extension, with less bleeding, no facial scars and fewer recurrences.

7.2.2 Open approaches

Transpalatal (with Sardana's sublabial extension), lateral rhinotomy and midfacial degloving with medial maxillectomy, the Le Fort I approach, the maxillary swing (Wei's facial translocation), the infratemporal fossa approach, and combined intracranial-extracranial resection.

OPERATIVE GUIDES IN THEATRE 2 operations

Juvenile Angiofibroma
Juvenile angiofibroma
Endoscopic Endonasal Transpterygoid Approaches to the Middle Cranial Fossa
Transpterygoid approach

7.3 Radiotherapy

For unresectable intracranial disease, residual tumour next to critical structures, and recurrence; stereotactic radiosurgery for small residuals. Long-term risks

include growth disturbance, cataract and radiation-induced malignancy in young patients.

7.4 Recurrence

Up to about a third, usually within two years, from residual tumour in the basisphenoid and vidian canal. Follow with contrast MRI.

UPMC STAGING AND CURRENT MANAGEMENT (OXFORD TEXTBOOK)

Older systems (Fisch, Radkowski, Andrews) stage angiofibroma by extent only. The **University of Pittsburgh Medical Center (UPMC)** system adds residual vascularity from the internal carotid artery after embolisation, and predicts blood loss and recurrence better. Surgery is the preferred treatment. Early tumours are removed endoscopically, often without embolisation; advanced tumours with internal carotid supply have the external carotid branches embolised first. Recurrence rates are similar after open and endoscopic surgery. Small residual tumour may be watched for growth; accessible recurrence is removed endoscopically, and disease in high-risk sites such as the cavernous sinus may be observed or treated with stereotactic radiosurgery. Follow-up with imaging continues for at least three years.

UPMC STAGING OF ANGIOFIBROMA

STAGE	EXTENT
I	Nasal cavity, medial pterygopalatine fossa
II	One or more sinuses, lateral pterygopalatine fossa; no residual vascularity
III	Skull base erosion, orbit or infratemporal fossa; no residual vascularity
IV	Skull base erosion, orbit or infratemporal fossa; residual vascularity
V	Intracranial extension with residual vascularity (medial or lateral)

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The patient:** adolescent boy, epistaxis and obstruction.
- **Site of origin** and routes of spread through the pterygopalatine fossa.
- **The antral (Holman-Miller) sign.**
- **Why no biopsy,** and the role of embolisation and its timing.
- **Surgical approaches** by extent.

8.2 Common pitfalls

COMMON PITFALLS

- Biopsying the mass in clinic.
- Treating recurrent epistaxis in a teenage boy without endoscopy.
- Leaving tumour in the vidian canal.

Written for this app from Oxford Textbook of Otolaryngology ch. 39, Oxford Specialist Handbook ch. 20, Dhingra ch. 49 and K. J. Lee ch. 32. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Sinonasal tumours

Tumours of the nose and sinuses hide behind the symptoms of ordinary sinusitis until they break out of their bony box into the cheek, palate, orbit or skull base. The warning is a one-sided story: blockage, bloodstained discharge, numbness or a loose tooth on one side in a middle-aged or older adult. Every one-sided nasal mass is sent for histology.

AT A GLANCE

- One-sided symptoms in an adult mean scope and biopsy.
- Staging with CT and MRI.
- Surgery then radiotherapy for most.
- Numbness and eye signs are alarming.
- SNUC responds to induction chemotherapy; endoscopic resection matches open results with clear margins.

1 CHAPTER ONE Overview

Silent until they escape.

1.1 The range

SINONASAL TUMOURS

- BENIGN EPITHELIAL **Papillomas** squamous, inverted, oncocytic, exophytic
- BENIGN OTHER **Haemangioma, schwannoma, meningioma** pleomorphic adenoma, chondroma, angiofibroma
- BENIGN BONY **Osteoma, fibrous dysplasia** ossifying fibroma, ameloblastoma
- MALIGNANT EPITHELIAL **Squamous cell carcinoma** adenocarcinoma, adenoid cystic carcinoma, undifferentiated carcinoma
- MALIGNANT OTHER **Melanoma, olfactory neuroblastoma** lymphoma, plasmacytoma, sarcomas, haemangiopericytoma

1.2 Epidemiology

Sinonasal cancer is rare (well under 1% of all cancers). The maxillary sinus is the commonest site, then the ethmoids; frontal and sphenoid primaries are rare.

1.3 Risk factors

- **Hardwood dust** (furniture workers): adenocarcinoma of the ethmoids and upper nose.
- **Nickel refining:** squamous and anaplastic carcinoma.
- **Leather work, chromium, mustard gas, formaldehyde.**
- **Snuff** rich in nickel and chromium (Bantu populations).
- **Smoking** and HPV for some squamous carcinomas.

1.4 Histology

More than 80% of malignant tumours are squamous cell carcinoma; the rest are adenocarcinoma, adenoid cystic carcinoma, melanoma, olfactory neuroblastoma, lymphoma and sarcomas.

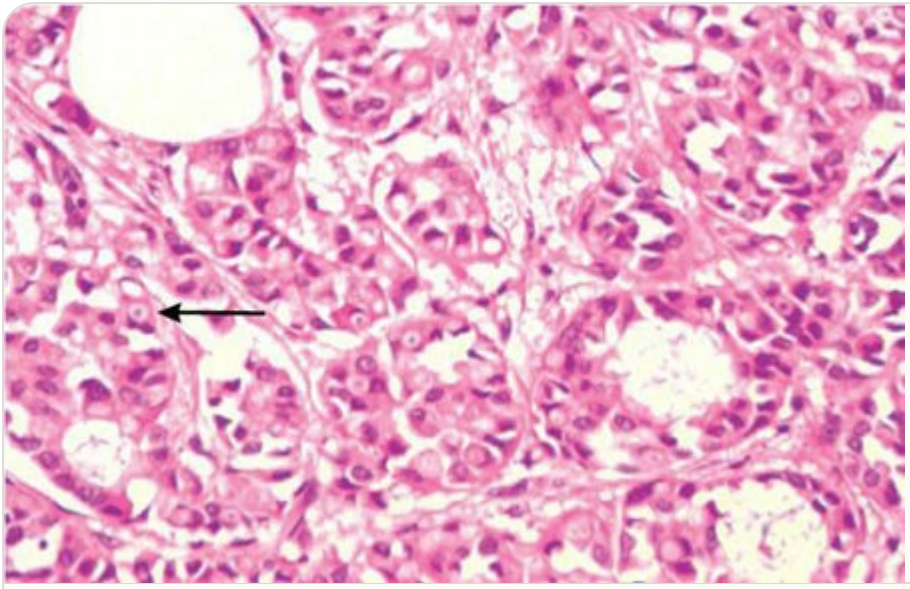


FIGURE 1 Adenocarcinoma with gland formation and intracytoplasmic mucin.
Dhingra, fig. 40.4

CHAPTER TWO

2 Benign tumours of the nasal cavity

And the one that isn't quite benign.

2.1 Squamous papilloma

A warty lesion of the vestibule or anterior septum; excised.



FIGURE 2 Squamous papilloma of the left nasal vestibule.
Dhingra, fig. 39.1

2.2 Inverted papilloma

2.2.1 What it is

A Schneiderian papilloma in which the epithelium grows inwards into the stroma. Typically in men of 40 to 70, almost always one-sided, arising from the lateral wall near the middle meatus and extending into the maxillary and ethmoid sinuses. HPV is found in some.

2.2.2 Three warnings

- It destroys bone by pressure.
- It recurs if incompletely removed.

- It is associated with squamous cell carcinoma in about 10%.

2.2.3 Clinical picture

One-sided obstruction, discharge and bleeding, with a firm, greyish, irregular mass that can masquerade as a simple polyp.

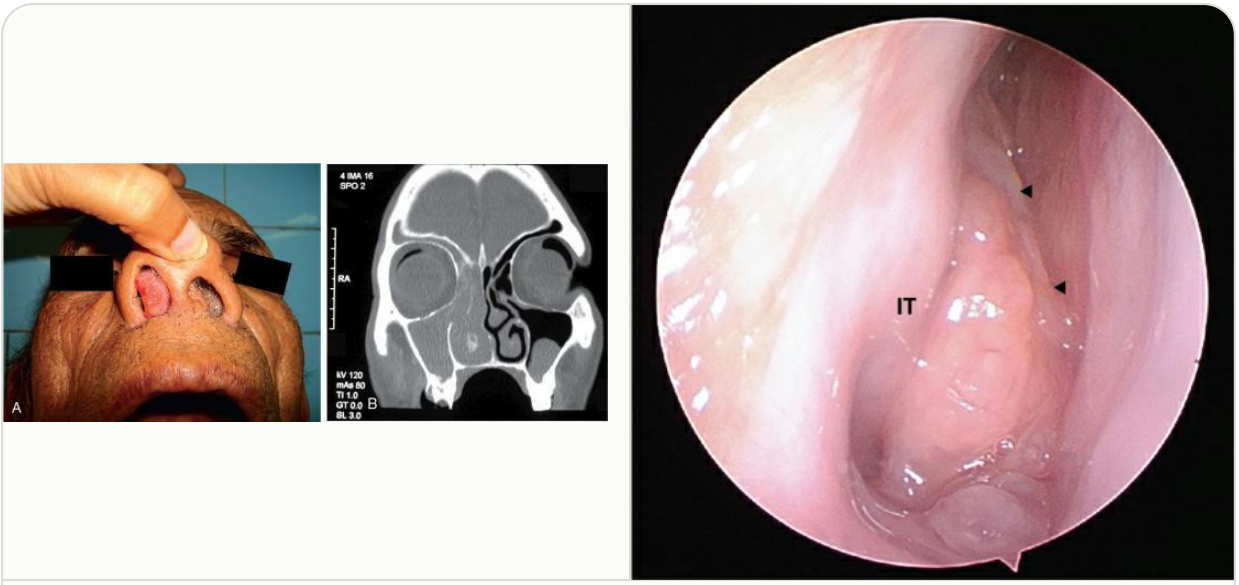


FIGURE 3 First, inverted papilloma in a 79-year-old man, with its CT. Second, an inverted papilloma in the right nose looking like a simple polyp.

Dhingra, figs 39.2 and 39.3

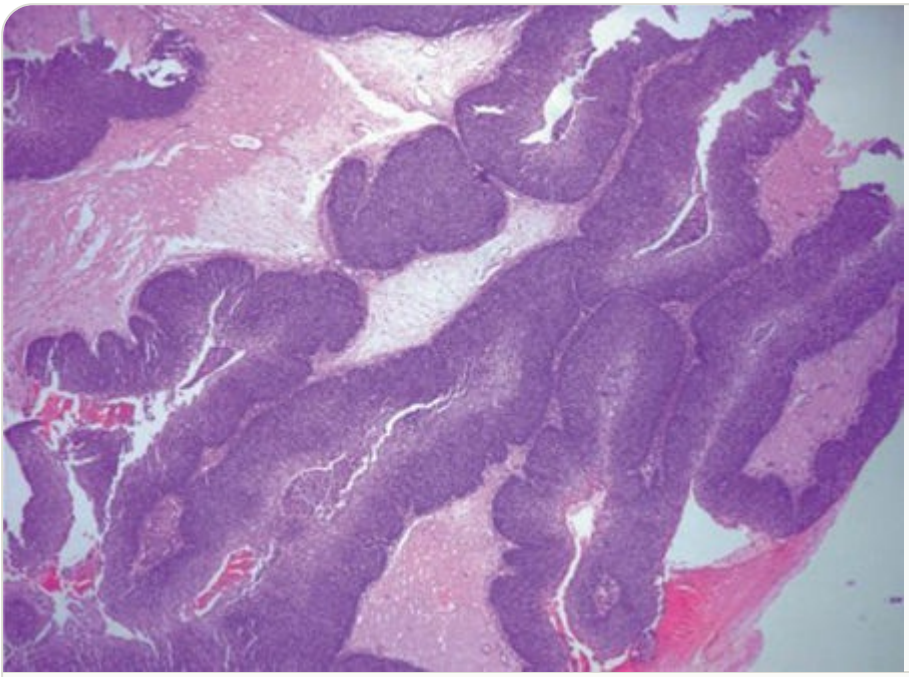


FIGURE 4 Histology of a Schneiderian papilloma: ribbons of thickened epithelium growing into the stroma.

Dhingra, fig. 39.4

2.2.4 Imaging

CT shows a one-sided mass and focal hyperostosis at the site of attachment; MRI shows a *convoluted cerebriform* pattern.

2.2.5 Treatment

Complete removal including the mucoperiosteum and drilling of the bone at the site of attachment, now usually endoscopic (endoscopic medial maxillectomy) and formerly by lateral rhinotomy or midfacial degloving. Long-term endoscopic follow-up.

2.3 Vascular and other benign tumours

- **Haemangioma** (lobular capillary haemangioma, *bleeding polyp of the septum*): a red, friable mass on the anterior septum, often in pregnancy; excised with its base.
- **Schwannoma, meningioma, chondroma, pleomorphic adenoma** of the septum.
- **Angiofibroma** (see its own topic).
- **Encephalocele, glioma and dermoid** (see the congenital nasal masses topic).



FIGURE 5 A bleeding polyp arising from the right side of the septum.
Dhingra, fig. 39.5

3 CHAPTER THREE
Benign bony tumours

Usually found by chance.

3.1 Osteoma

The commonest benign sinus tumour, usually in the frontal sinus, then ethmoid; often an incidental finding on CT. Symptoms arise when it blocks drainage (mucocele, sinusitis), bulges into the orbit or erodes the skull base. Multiple osteomas suggest *Gardner syndrome* (with colonic polyposis). Removed endoscopically or through an external approach when symptomatic.

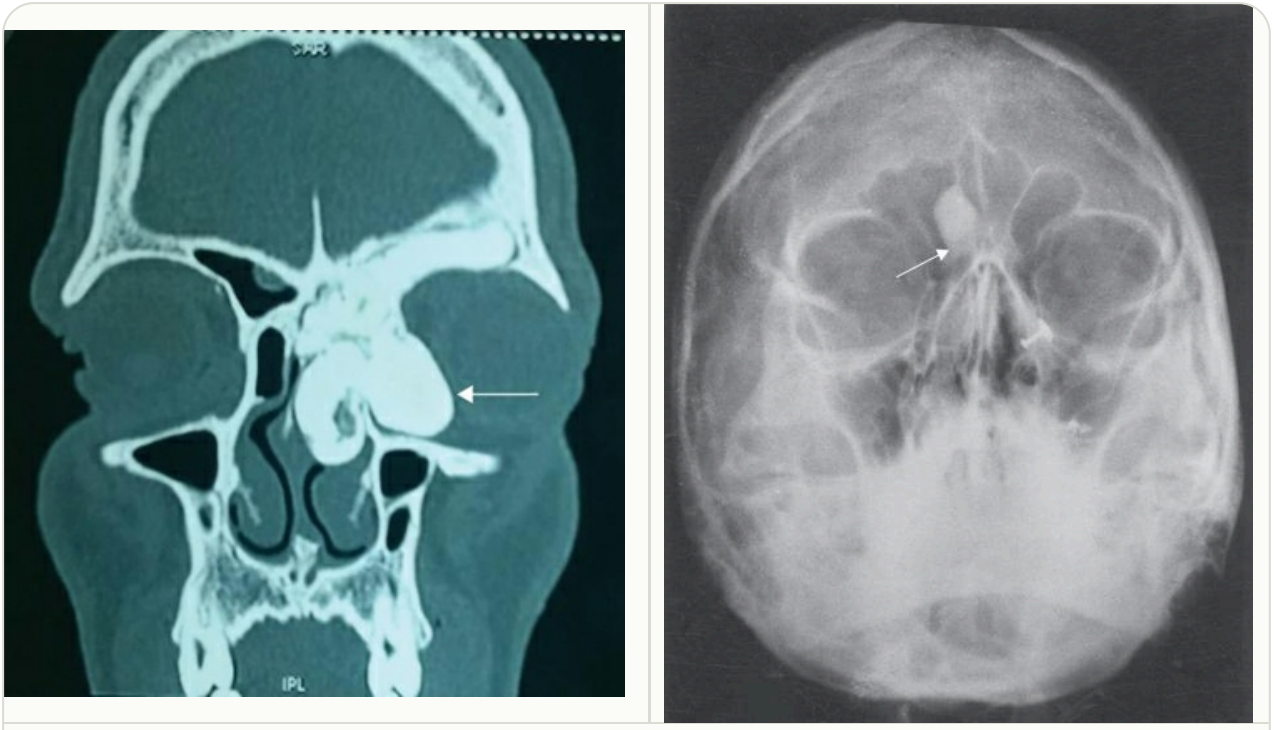


FIGURE 6 First, a frontoethmoidal osteoma invading the orbit. Second, an osteoma of the right frontal sinus.

Dhingra, figs 40.1 and 40.2

3.2 Fibrous dysplasia

Bone replaced by fibrous tissue and woven bone; *monostotic*, *polyostotic*, or with skin pigmentation and endocrine disorders (*McCune-Albright syndrome*). A painless swelling of the maxilla in a child or adolescent, with a *ground-glass* appearance on CT.

Growth usually slows after puberty; conservative recontouring for deformity or compression.



FIGURE 7 Fibrous dysplasia of the maxilla in a 13-year-old girl.
Dhingra, fig. 40.3

3.3 Ossifying fibroma

A well-defined, expansile lesion with a bony rim, more aggressive in its juvenile form; excised completely.

3.4 Ameloblastoma

An odontogenic tumour of the jaw, commonest in the mandible, that may invade the maxillary sinus; locally aggressive and recurrent unless widely excised.

IMAGING IN THE APP 4 cases

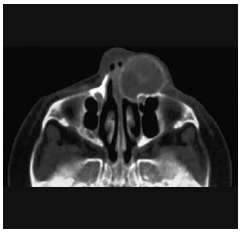
Inverted papilloma, osteoma, fibrous dysplasia and myxoma.



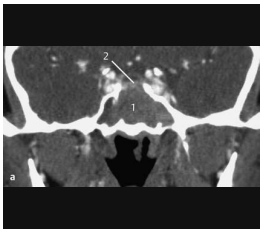
Osteoma
Bone disease · 3 images



Fibrous Dysplasia of the Sinuses (Monostotic and Polyostotic)
Bone disease · 5 images



Myxoma
Tumours · 1 image



Inverted Papilloma
Tumours · 12 images

4 CHAPTER FOUR Carcinoma of the maxillary sinus

The classic sinonasal cancer.

4.1 Clinical picture

Usually men of 40 to 60. Early symptoms are easily mistaken for sinusitis: nasal stuffiness, bloodstained discharge, facial pain or numbness, and epiphora. Late features depend on the direction of spread.

SPREAD OF MAXILLARY CARCINOMA

DIRECTION	STRUCTURES	FEATURES
Medial	Nasal cavity, ethmoids	Obstruction, discharge, epistaxis
Anterior	Cheek	Swelling, then skin invasion
Inferior	Alveolus, palate	Toothache, loose teeth, ill-fitting dentures, palatal swelling
Superior	Orbit	Proptosis, diplopia, pain, epiphora
Posterior	Pterygoid plates and muscles, infratemporal fossa	Trismus, spread to nasopharynx and skull base
Intracranial	Through the ethmoids, cribriform plate or foramen lacerum	Neurological signs



FIGURE 8 Carcinoma of the left maxilla in a 75-year-old man: (A) swelling and redness of the cheek, (B) expansion of the upper alveolus with an intraoral growth, (C) CT showing destruction of the maxilla and orbital floor.

Dhingra, fig. 40.6

4.2 Lymph node and distant spread

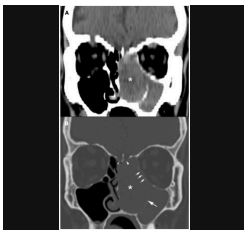
Nodes are uncommon and late (submandibular and upper jugular); the first-echelon retropharyngeal nodes cannot be felt. Distant metastases go to the lungs and bone.

4.3 Investigations

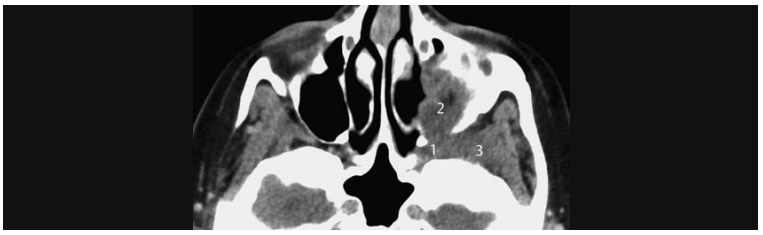
- **CT** in three planes: bone destruction and staging.
- **MRI:** separates tumour from retained secretions, and shows orbital, perineural and intracranial spread.
- **Biopsy:** endoscopic, or from the mouth or nose when the tumour presents there; the Caldwell-Luc route is no longer preferred.
- **Staging:** chest CT or PET-CT.

IMAGING IN THE APP 2 cases

Sinonasal malignancy and spread into the pterygopalatine and infratemporal fossae.



Sinonasal tumours
Tumours · 10 images



Malignancy of the Pterygopalatine and Infratemporal Fossa
Tumours · 2 images

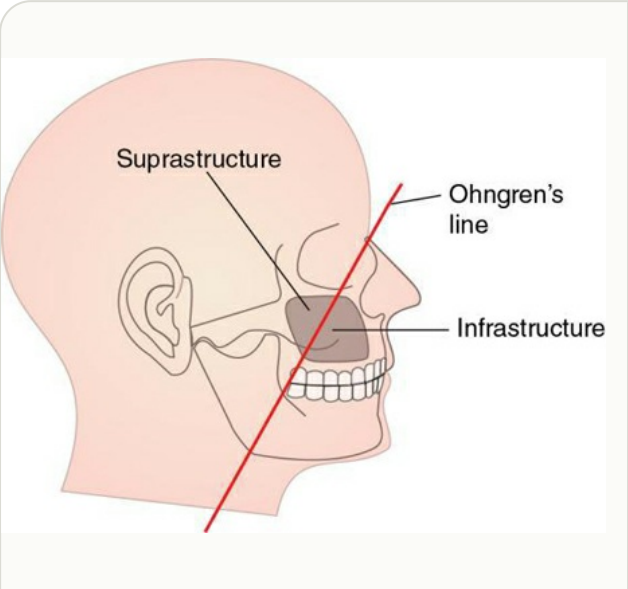
4.4 Classifications

4.4.1 Ohngren's line

An imaginary line from the medial canthus to the angle of the mandible. Tumours above and behind it (suprastructure) have a worse prognosis than those below and in front of it.

4.4.2 Lederman's classification

Two horizontal lines of Sebileau, through the floors of the orbits and the floors of the antra, divide the region into *suprastructure* (ethmoid, sphenoid, frontal sinuses and olfactory area), *mesostructure* (maxillary sinus and respiratory nose) and *infrastructure* (alveolus).



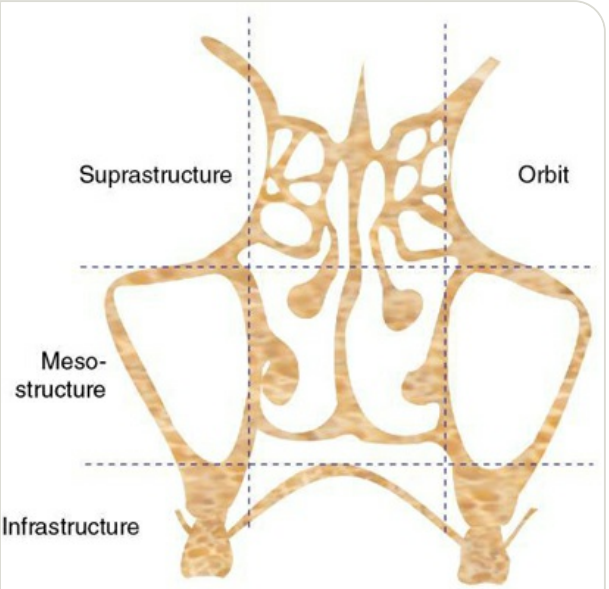


FIGURE 9 First, Ohngren's line. Second, Lederman's classification.
Dhingra, figs 40.7 and 40.8

4.4.3 AJCC TNM for the maxillary sinus

T STAGING OF MAXILLARY SINUS CARCINOMA

T	EXTENT
T1	Limited to the mucosa, no bone erosion
T2	Bone erosion including the hard palate or middle meatus, but not the posterior wall or pterygoid plates
T3	Posterior wall, subcutaneous tissue, floor or medial wall of the orbit, pterygoid fossa, ethmoids
T4a	Anterior orbit, cheek skin, pterygoid plates, infratemporal fossa, cribriform plate, sphenoid or frontal sinus
T4b	Orbital apex, dura, brain, middle cranial fossa, cranial nerves other than V2, nasopharynx, clivus

4.5 Treatment

4.5.1 Surgery

Resection with clear margins, followed by radiotherapy for most stages:

- **Medial maxillectomy** (endoscopic or open) for medial tumours.
- **Total maxillectomy** through a *Weber-Fergusson* incision (lateral rhinotomy with lip split, extended below the eye), with **orbital exenteration** when the orbit is invaded beyond the periosteum.
- **Craniofacial resection** when the cribriform plate or anterior skull base is involved.
- Endoscopic resection is increasingly used for selected tumours.



FIGURE 10 First, the Weber-Fergusson incision. Second, maxillectomy with orbital exenteration, before and after rehabilitation.
Dhingra, figs 40.10 and 40.9

4.5.2 Reconstruction and rehabilitation

A dental obturator closes the palatal defect, or a free flap reconstructs it; orbital prostheses after exenteration.

4.5.3 Radiotherapy and chemotherapy

Postoperative radiotherapy for most; primary chemoradiotherapy for unresectable tumours; neoadjuvant chemotherapy in selected cases.

4.5.4 Neck

Neck dissection or irradiation for nodal disease, and elective treatment in some advanced tumours.

OPERATIVE GUIDES IN THEATRE 4 operations

Medial Maxillectomy
Medial maxillectomy

Complete Maxillectomy
Complete maxillectomy

Transcranial and Endonasal Approaches to the Anterior Cranial Base
Anterior skull base approaches

Orbital Exenteration
Exenteration

4.6 Prognosis

Five-year survival is around a third to a half overall, worse with orbital apex, skull base or intracranial spread.

5 CHAPTER FIVE
Tumours of the ethmoid, frontal and sphenoid

Rarer sites.

5.1 Ethmoid

Often adenocarcinoma in woodworkers. Nasal obstruction, bloodstained discharge, a mass in the medial canthus, proptosis with lateral displacement of the eye, and early skull base spread. Craniofacial or endoscopic resection with radiotherapy.

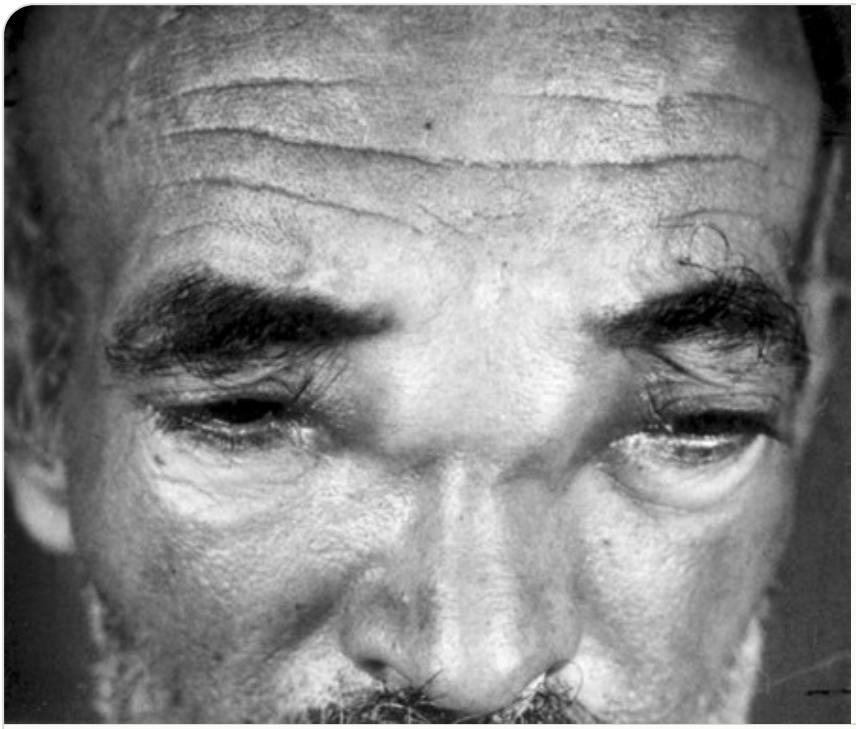


FIGURE 11 Carcinoma of the ethmoid.
Dhingra, fig. 40.11

5.2 Frontal

Very rare: forehead swelling, frontal pain, displacement of the eye downwards, and early intracranial spread.

5.3 Sphenoid

Usually secondary or from the nasopharynx; presents with cranial nerve palsies (second to sixth) and headache.

6 CHAPTER SIX
Other malignant tumours

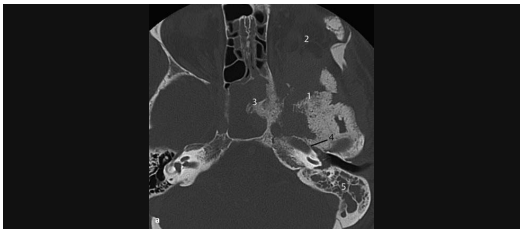
Each with its clue.

6.1 Olfactory neuroblastoma

Esthesioneuroblastoma, from the olfactory epithelium at the roof of the nose, at any age with peaks in the second and sixth decades. One-sided obstruction, epistaxis and loss of smell, with a mass high in the nose that may extend through the cribriform plate (a *dumb-bell* shape). Staged by the Kadish system (A nasal cavity, B sinuses, C beyond); treated by craniofacial or endoscopic resection with radiotherapy. Late neck metastases occur.

IMAGING IN THE APP 1 case

A meningioma destroying the skull base, a differential of olfactory neuroblastoma.



Meningioma with Skull Base Destruction

Tumours · 2 images

6.2 Mucosal melanoma

Older adults, on the septum or lateral wall; pigmented or amelanotic, bleeding; spreads by blood and lymph and has a poor prognosis; wide excision and radiotherapy.

6.3 Lymphoma, plasmacytoma and sarcomas

NK/T-cell lymphoma (see the granulomatous disease topic), extramedullary plasmacytoma, and rhabdomyosarcoma in children.



FIGURE 12 Rhabdomyosarcoma of the nose in a two-and-a-half-year-old boy.
Dhingra, fig. 39.7

THE NEW CLASSIFICATION AND TREATMENT TRENDS (OXFORD TEXTBOOK)

The fourth WHO classification reorganised sinonasal tumours and added new entities, so histology now carries more weight than ever. **Sinonasal undifferentiated carcinoma (SNUC)** is aggressive, often presents with intracranial and nodal spread, and responds well to **induction chemotherapy** followed by chemoradiotherapy, with surgery when the response is poor. **NUT carcinoma**, defined by a *NUTM1* rearrangement, affects young adults and has a very poor prognosis. **Neuroendocrine carcinoma** (small and large cell types) is treated with combined chemotherapy and radiotherapy. **Intestinal-type adenocarcinoma**, linked to hardwood dust, looks like bowel cancer under the microscope but is not a metastasis; it is often multifocal, so wide removal of the sinus mucosa is advised. Low-grade non-intestinal adenocarcinoma may need surgery alone; high-grade disease needs radiotherapy too.

SCC arising in an inverted papilloma behaves like de novo SCC; about 5% of inverted papillomas harbour dysplasia or carcinoma, so the whole specimen must be examined.

Endoscopic resection now achieves oncological outcomes equivalent to open surgery for most tumours, with less morbidity, provided clear margins can be obtained; advanced tumours involving the orbit or skull base may first have induction chemotherapy, with the response guiding radiotherapy or surgery. Nodal metastases at presentation suggest a high-grade tumour such as SCC or SNUC, and up to 20% of olfactory neuroblastomas spread to the neck, so long-term neck surveillance is needed. Somatostatin-receptor PET (DOTATATE) helps stage and follow olfactory neuroblastoma and neuroendocrine tumours. The Kadish system for olfactory neuroblastoma discriminates poorly, and the Dulguerov TNM-based system is often preferred.

CHAPTER SEVEN

7 For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Occupational risks:** wood dust and adenocarcinoma, nickel and squamous carcinoma.
- **Inverted papilloma:** site, three warnings, imaging and treatment.
- **Maxillary carcinoma:** symptoms by direction of spread.
- **Ohngren's line and Lederman's classification.**
- **T staging,** and maxillectomy with or without orbital exenteration.
- **Olfactory neuroblastoma** and Kadish staging.

7.2 Common pitfalls

COMMON PITFALLS

- Treating one-sided sinus symptoms with antibiotics for months.

- Removing a one-sided polyp without sending it for histology.
- Calling numbness of the cheek sinusitis.

Written for this app from Oxford Textbook of Otolaryngology ch. 39, Oxford Specialist Handbook ch. 21, Oxford Handbook of ENT ch. 9, Dhingra ch. 39, 40 and K. J. Lee ch. 32. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Acute, chronic and atrophic rhinitis

Beyond allergy lies a family of infective and chronic rhinitis: the common cold and its complications, the bacterial and diphtheritic forms, and the chronic inflammations that thicken the turbinates or, at the other extreme, waste them away. Atrophic rhinitis, with its wide, crusted, foul-smelling nose and a patient who cannot smell it, is the one examiners never forget.

AT A GLANCE

- Colds are viral: treat symptoms, not with antibiotics.
- Simple rhinitis pits and shrinks; hypertrophic rhinitis does not.
- Atrophic rhinitis: wide crusted nose, foul smell, merciful anosmia.
- Douches and glucose-glycerine first; Young's operation for resistant cases.

1 CHAPTER ONE Overview

Infective and chronic rhinitis.

1.1 The family

INFECTIVE AND CHRONIC RHINITIS

- ACUTE **Viral** common cold, influenza, exanthemas, COVID-19
- ACUTE **Bacterial and irritative** non-specific bacterial, diphtheritic, chemical
- CHRONIC **Simple and hypertrophic** swollen or thickened turbinates
- CHRONIC **Atrophic and crusting** atrophic rhinitis (ozaena), rhinitis sicca, rhinitis caseosa

Allergic and vasomotor rhinitis, and the specific granulomatous infections, have their own topics.

2 CHAPTER TWO Acute rhinitis

The common cold and its relatives.

2.1 Common cold (coryza)

2.1.1 Cause

Droplet spread of rhinovirus, adenovirus, coxsackie and other enteroviruses and coronaviruses. Incubation is one to four days; the illness lasts two to three weeks.

2.1.2 Clinical picture

Burning at the back of the nose, then stuffiness, sneezing and profuse watery rhinorrhoea, chills and low fever. The discharge becomes mucopurulent if bacteria (streptococci, pneumococci, staphylococci, *H. influenzae*, *Moraxella*) invade secondarily.

2.1.3 Treatment

Rest, fluids, analgesics (non-aspirin preferred, as aspirin increases viral shedding), antihistamines and short-term decongestants. Antibiotics only for bacterial complications.

2.1.4 Complications

Sinusitis, pharyngitis, tonsillitis, otitis media, bronchitis and pneumonia.

2.2 Other viral rhinitis

Influenza A, B and C cause a similar but more severe illness, with more frequent bacterial complications. Measles, rubella and chickenpox cause rhinitis two to three days before the rash. COVID-19 begins like a cold and may cause sudden loss of smell and taste; during its second wave it was followed by a surge of rhino-orbito-cerebral mucormycosis in diabetics and steroid-treated patients (see the fungal sinusitis topic).

2.3 Bacterial rhinitis

2.3.1 Non-specific

Primary infection in children with pneumococcus, streptococcus or staphylococcus, sometimes with a greyish-white membrane that bleeds when removed; or secondary infection after a cold.

2.3.2 Diphtheritic

Rare with immunisation. A greyish, tenacious membrane over the inferior turbinate and floor that bleeds when removed, with excoriation of the nostrils and upper lip. Isolate; give penicillin or erythromycin and diphtheria antitoxin.

2.4 Irritative rhinitis

From dust, smoke, ammonia, formalin or acid fumes, or trauma during intranasal procedures: immediate sneezing, rhinorrhoea and congestion that settle once the irritant is removed.

3 CHAPTER THREE Chronic simple and hypertrophic rhinitis

Swollen, then thickened.

3.1 Chronic simple rhinitis

3.1.1 Causes

Repeated colds with persistent infection in the sinuses, tonsils or adenoids; smoke and dust; nasal obstruction from a deviated septum or synechiae; vasomotor rhinitis; hypothyroidism.

3.1.2 Pathology

Hyperaemic, oedematous mucosa with hypertrophied glands, more goblet cells and distended sinusoids over the turbinates: an early stage of hypertrophic rhinitis.

3.1.3 Clinical picture

Blockage worse on lying, affecting the dependent side; thick mucoid or mucopurulent discharge and postnasal drip; headache. The turbinates are dull red and swollen, but they *pit on pressure and shrink with a vasoconstrictor*.

3.1.4 Treatment

Treat the cause (sinuses, tonsils, adenoids, allergy, smoking, workplace), saline or alkaline irrigation, short-term decongestants, and antibiotics for infection.

3.2 Hypertrophic rhinitis

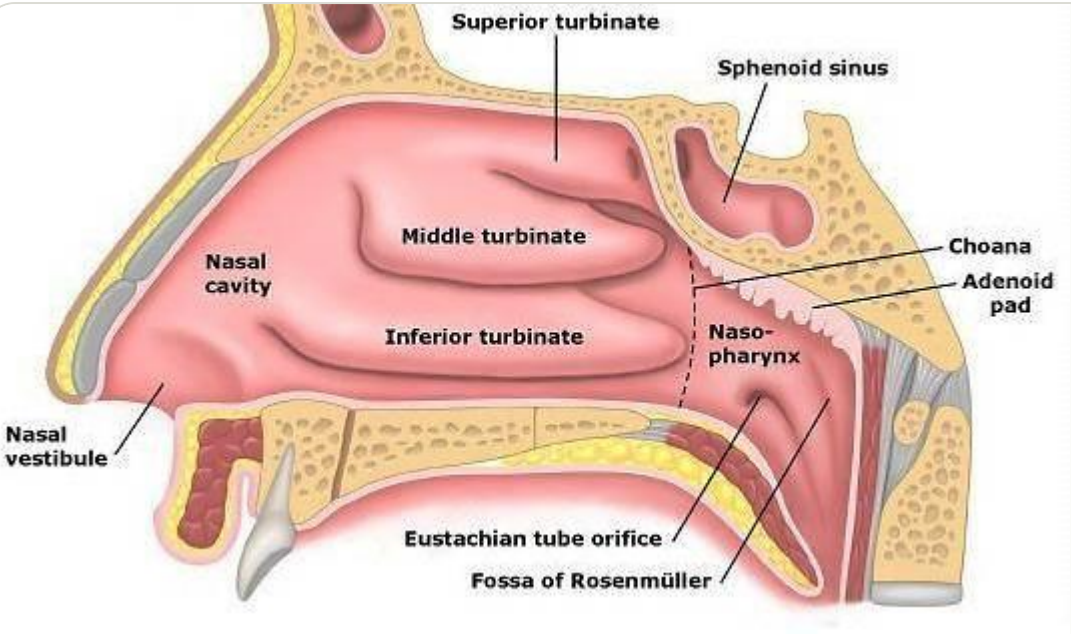


FIGURE 1 The lateral nasal wall with the turbinates.

Anatomy atlas · ENT Steps p. 98

3.2.1 Pathology

Thickening of the mucosa, submucosa, glands, periosteum and bone, most marked on the inferior turbinate.

3.2.2 Causes

Recurrent infections, chronic sinusitis, irritants, long-term nasal drops, and vasomotor or allergic rhinitis.

3.2.3 Clinical picture

Obstruction is the main symptom, with thick sticky discharge and headache. The turbinate mucosa is thick, does *not pit on pressure* and shrinks little with vasoconstrictors, because of fibrosis. The inferior turbinate may be enlarged throughout, or at its anterior end, posterior end or lower border, giving a *mulberry* appearance.

3.2.4 Treatment

> Remove the cause, then reduce the turbinates: linear cautery, submucosal diathermy, cryosurgery, laser, coblation or radiofrequency, submucous resection of the turbinate bone, or partial turbinectomy. Avoid excessive removal, which leaves crusting and an empty nose.

OPERATIVE GUIDES IN THEATRE 1 operation

Surgery of the Turbinate
Turbinate reduction for hypertrophic rhinitis

3.3 Compensatory hypertrophy

With a severely deviated septum, the turbinates on the wider side enlarge to narrow the airway and protect it from drying. The change is not reversible after septoplasty, so the turbinate is often reduced at the same operation.

SIMPLE OR HYPERTROPHIC?

	CHRONIC SIMPLE RHINITIS	HYPERTROPHIC RHINITIS
Mucosa	Dull red, oedematous	Thick, fibrosed
Probe	Pits on pressure	Does not pit
Vasoconstrictor	Shrinks well	Shrinks little
Treatment	Medical	Turbinate reduction

4 CHAPTER FOUR Atrophic rhinitis

Ozaena.

4.1 Definition

A chronic inflammation with atrophy of the nasal mucosa and turbinate bones. The nasal cavities become roomy and fill with foul-smelling crusts.

4.2 Primary atrophic rhinitis

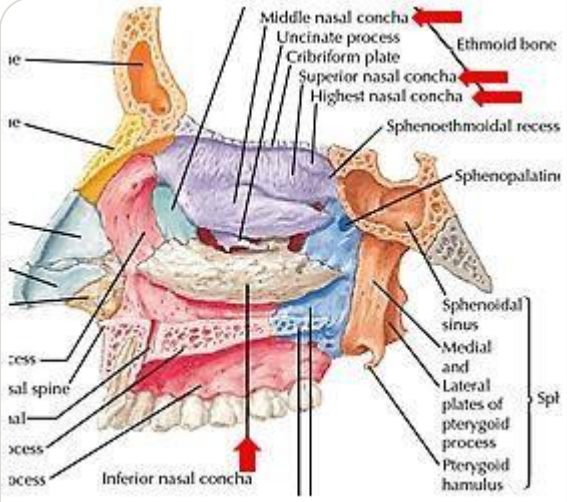


FIGURE 2 The lateral nasal wall: in atrophic rhinitis the turbinates shrink and the cavity widens.
Anatomy atlas · ENT Steps p. 99

4.2.1 Causes

Unknown; several theories (Dhingra's mnemonic *HERNIA*):

- **Hereditary:** several family members affected.
- **Endocrine:** begins at puberty, commoner in women, tends to settle after the menopause.
- **Racial:** commoner in white and East Asian populations than in equatorial Africa.
- **Nutritional:** deficiency of vitamins A and D or iron; rarer in well-off families.
- **Infective:** *Klebsiella ozaenae* (Perez bacillus) is the commonest organism cultured, with diphtheroids, *Proteus*, *E. coli* and cocci, but these are secondary invaders responsible for the smell.
- **Autoimmune:** a destructive reaction to antigens from the nasal mucosa.

4.2.2 Pathology

Ciliated columnar epithelium becomes squamous; seromucinous glands, venous sinusoids and nerves atrophy; arteries show obliterative endarteritis; the turbinate bones are resorbed, widening the nose. The sinuses stay small and thick-walled because their development was arrested.

4.2.3 Clinical picture

- Usually a young woman, starting at puberty.

- A foul smell that makes the patient a social outcast, though the patient cannot smell it (*merciful anosmia*).
- *Paradoxical nasal obstruction:* the nose is wide, but damaged sensory nerves cannot feel the airflow.
- Epistaxis when crusts are removed.

4.2.4 Signs

Greenish or greyish-black crusts cover the turbinates and septum; when removed, the cavities are roomy, the turbinates reduced to ridges, the mucosa pale, and the posterior nasopharyngeal wall easily seen. There may be a septal perforation, vestibular dermatitis or saddle nose, with similar changes in the pharynx (atrophic pharyngitis) and larynx (atrophic laryngitis), and ear effusion from tubal blockage.

4.2.5 Prognosis

Persists for years, with a tendency to improve in middle age.

4.3 Treatment

4.3.1 Medical: nasal hygiene

- **Irrigation:** warm saline, or an alkaline douche (sodium bicarbonate, sodium biborate and sodium chloride in water), two or three times a day at first, then every few days; loosen and remove hard crusts.
- **25% glucose in glycerine:** painted on after the crusts are removed; it inhibits the proteolytic organisms that cause the smell.
- **Antibiotics:** topical, or a 7 to 10 day systemic course (for example a fluoroquinolone with metronidazole) for active infection; not long-term.
- **Mucolytics** such as potassium iodide.
- **Diet** rich in vitamins A and D, folic acid, iron and protein.

4.3.2 Surgical

- **Young's operation:** both nostrils are closed with flaps just inside the vestibule and reopened after six months or more; the mucosa may recover. A *modified Young's operation* closes them partially.

- **Narrowing the nasal cavities:** submucosal implants of fat, cartilage, bone or acellular dermis under the floor, lateral wall and septum, or medial displacement of the lateral wall (*Lautenschlaeger operation*).

4.4 Secondary atrophic rhinitis

Destruction from syphilis, lupus, leprosy or rhinoscleroma; long-standing purulent sinusitis; radiotherapy; and excessive turbinate surgery (the *empty nose syndrome*). A severely deviated septum can cause one-sided atrophic rhinitis on the wide side.

PEARLS

- Merciful anosmia: the patient is the only one who cannot smell it.
- Paradoxical obstruction: the widest nose in clinic, yet it feels blocked.
- Klebsiella ozaenae is a secondary invader, not the cause.

5 CHAPTER FIVE Rhinitis sicca and caseosa

Other crusting conditions.

5.1 Rhinitis sicca

A crusting disease of people who work in hot, dry, dusty places (bakers, ironsmiths, goldsmiths), confined to the anterior septum. Squamous metaplasia leads to crusts whose removal causes ulceration, bleeding and perforation. Change the working environment, apply a bland or antibiotic-steroid ointment, use douches, and stop picking.

5.2 Rhinitis caseosa

Rare, usually one-sided and in men: the nose fills with offensive pus and cheesy material, probably from chronic sinusitis; the sinus walls may be eroded, mimicking malignancy. Remove the debris and granulation and drain the sinus; the outlook is good.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Chronic simple versus hypertrophic rhinitis:** the probe and vasoconstrictor tests.
- **Atrophic rhinitis:** theories, pathology, merciful anosmia, paradoxical obstruction, treatment including glucose-glycerine and Young's operation.
- **Diphtheritic rhinitis:** membrane and treatment.
- **Rhinitis sicca** and caseosa.

6.2 Common pitfalls

COMMON PITFALLS

- Removing too much turbinate and creating an empty nose.
- Forgetting that a crusted, destroyed nose may be secondary to a granulomatous disease or lymphoma.

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CLINICAL · NOSE & SINUSES

Nasal physiology and disorders of smell

The nose is the lungs' air conditioner: in a quarter of a second it warms, humidifies and filters every breath, and it carries trapped dust and microbes away on a moving sheet of mucus. It is also the organ of smell, whose loss takes much of the pleasure from food and can be the first sign of disease elsewhere.

AT A GLANCE

- The nose conditions air and clears it on a mucus conveyor belt.
- Conductive smell loss is blockage; sensorineural is nerve or brain.
- Commonest causes: viral infection, sinonasal disease, head injury.
- Test smell, scope the olfactory cleft, and image when the cause is unclear.

1 CHAPTER ONE Overview

What the nose does.

1.1 Functions

- Respiration.
- Air conditioning: filtering, warming and humidifying.
- Protection of the lower airway.
- Vocal resonance.
- Nasal reflexes.
- Olfaction.

1.2 Why it matters

Understanding normal function explains the symptoms of nasal disease: dryness and crusting after excessive turbinate surgery, hypoxia with nasal packs, sinusitis when cilia fail, and loss of smell with polyps.

2 CHAPTER TWO Respiration

The natural airway.

2.1 Nose breathing

Breathing through the nose is the natural route; mouth breathing is learned. A newborn with bilateral choanal atresia can asphyxiate. The nose also lets us breathe and eat at the same time.

2.2 The airflow

In quiet breathing, inspired air passes mainly through the middle of the nose, between the turbinates and septum; little goes through the inferior meatus or the olfactory cleft, which is why weak odours must be sniffed. On expiration, friction at the limen nasi creates eddies beneath the turbinates that ventilate the sinuses.

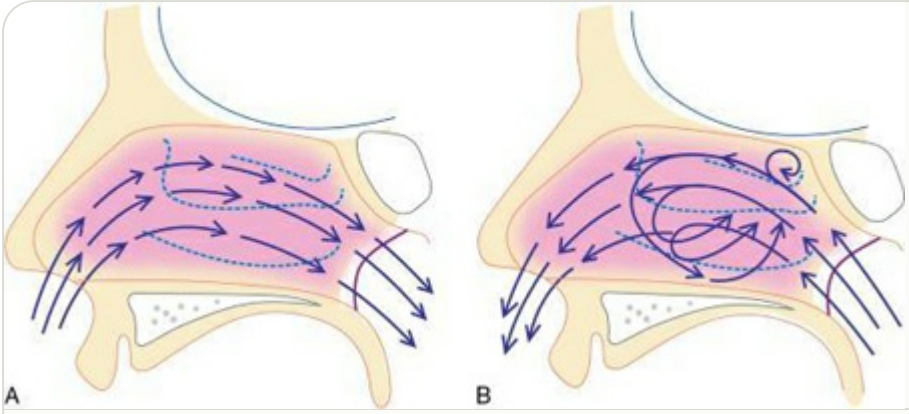


FIGURE 1 Nasal airflow during (A) inspiration and (B) expiration.
Dhingra, fig. 24.1

2.3 The nasal valve and turbinates

The swelling and shrinking of the anterior end of the inferior turbinate regulate inflow; the nasal valve is the narrowest point and the main resistance.

2.4 The nasal cycle

Rhythmic alternating congestion and decongestion of the two sides every 2.5 to 4 hours; total airflow stays constant because the open side compensates.

2.5 Measuring the airway

Rhinomanometry measures pressure and flow to give nasal resistance; *acoustic rhinometry* uses reflected sound to map the cross-sectional area along the nose; peak nasal inspiratory flow is a simple bedside measure.

3 CHAPTER THREE Air conditioning

The lungs' radiator.

3.1 Filtration

Vibrissae at the entrance catch large particles; the mucus traps particles down to about 0.5 µm; smaller particles pass on to the lungs.

3.2 Temperature

The vascular turbinate and septal mucosa, with its cavernous sinusoids, acts as a radiator: air at 0°C or below reaches nearly body temperature by the time it reaches the nasopharynx, about a quarter of a second later; hot air is cooled.

3.3 Humidification

Relative humidity is raised to 75% or more, using about a litre of water a day from the mucosal glands. At 50% relative humidity cilia stop working within minutes, so dry air predisposes to infection.

3.4 The lungs depend on it

Nasal obstruction alters gas exchange: packing both sides lowers oxygen tension (the *nasopulmonary reflex*), and long-standing obstruction from adenotonsillar hypertrophy can cause pulmonary hypertension in children.

4 CHAPTER FOUR Protection

The mucociliary conveyor belt.

4.1 The mucous blanket

Goblet cells and glands secrete a two-layer blanket: thick mucus on top, a thin serous layer beneath. Cilia beat 10 to 20 times a second in the serous layer, with a fast effective stroke that reaches the mucus and a slow recovery stroke within the serous layer, moving the blanket in one direction towards the nasopharynx at 5 to 10 mm a minute. The whole sheet is cleared every 10 to 20 minutes and swallowed. About 600 to 700 mL of secretion is made each day.

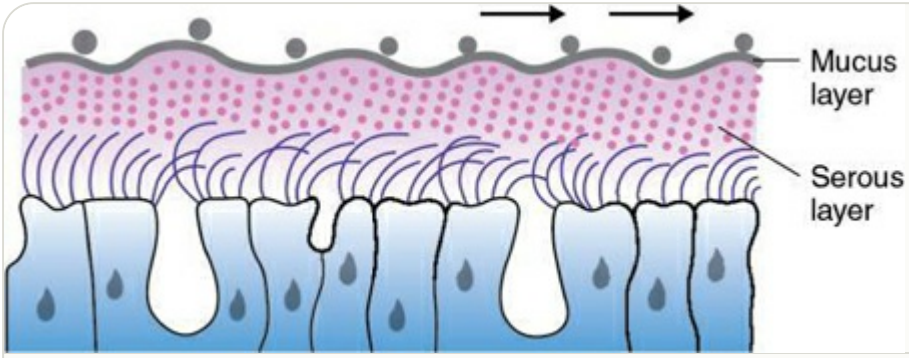


FIGURE 2 The conveyor belt of the mucous blanket, carrying trapped organisms and dust to the nasopharynx.
Dhingra, fig. 24.2

4.2 Sinus clearance

In each sinus, cilia sweep mucus along fixed paths towards the natural ostium, even if a second opening has been made; this is why endoscopic surgery enlarges natural ostia.

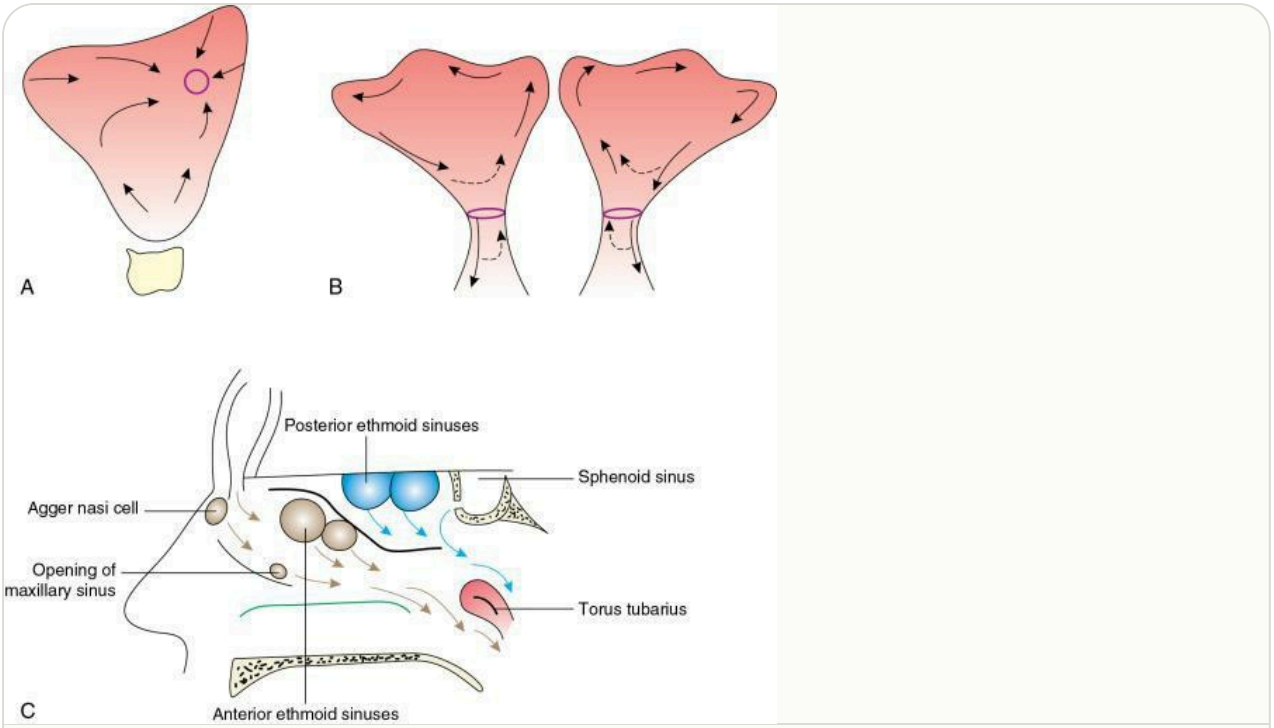


FIGURE 3 Mucociliary clearance of the sinuses: (A) maxillary, (B) frontal, (C) ethmoid and sphenoid.

Dhingra, fig. 35.4

4.3 What harms the cilia

Drying, heat or cold, smoking, infection, noxious gases, some drugs (including topical adrenaline and preservatives), and a change in pH from the normal of about 7. In *primary ciliary dyskinesia* the cilia are immotile, causing chronic rhinosinusitis, otitis media and bronchiectasis.

4.4 Chemical defences

Lysozyme (muramidase), secretory IgA, IgE and interferon in the secretions.

4.5 Sneezing

A protective reflex that expels irritants, followed by a flood of secretions to wash them out. The *saccharin test* measures mucociliary transit: a particle placed on the inferior turbinate is normally tasted within about 20 minutes.

5 CHAPTER FIVE Resonance and reflexes

Voice and neural links.

5.1 Vocal resonance

The nasal consonants M, N and NG pass through the nose. A blocked nose or nasopharynx causes *hyponasal* (denasal) speech, so M, N and NG sound like B, D and G; velopharyngeal insufficiency causes the opposite, *hypernasal* speech.

5.2 Reflexes

Sneezing; salivary and gastric secretion at the smell of food; and nasobronchial and nasopulmonary reflexes linking nasal obstruction with bronchial resistance and lower oxygen levels.

6 CHAPTER SIX Olfaction

The sense of smell.

6.1 The olfactory pathway

The olfactory epithelium lies high in the nose, over the cribriform plate, the upper septum and the superior turbinate. Bipolar receptor neurons carry cilia into the mucus; their axons gather into olfactory nerve bundles that pass through the cribriform plate to the olfactory bulb, then along the olfactory tract to the piriform cortex, amygdala and entorhinal cortex. Olfactory receptor neurons regenerate throughout life.

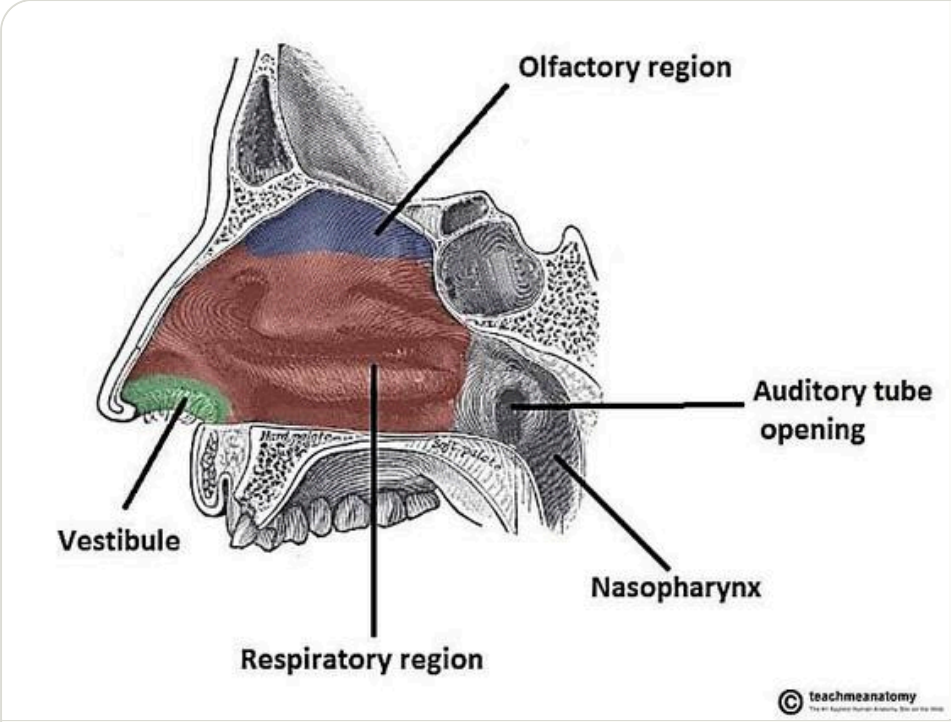


FIGURE 4 The vestibule, respiratory and olfactory regions of the nasal cavity.
Anatomy atlas · ENT Steps p. 97

6.2 Smell and flavour

Taste (sweet, salty, sour, bitter, umami) comes from taste buds supplied by the seventh, ninth and tenth nerves; *flavour* is taste plus smell, so patients who lose smell often complain that food has no taste. Ammonia is not used to test smell, because it stimulates the trigeminal nerve.

6.3 Terms

- **Anosmia:** total loss of smell; **hyposmia:** partial loss.
- **Parosmia:** a smell present but perceived wrongly, often as unpleasant; common during recovery from post-viral loss, from misdirected regeneration.
- **Phantosmia:** smelling an odour that is not there; intracranial causes (tumour, temporal lobe epilepsy) must be excluded.
- **Ageusia, hypogeusia, dysgeusia:** absent, reduced and distorted taste.

6.4 Epidemiology

Smell declines with age: in Lee's figures, most people over 80 have impaired smell and about half are anosmic. Smoking and poor nutrition add to the risk.

CHAPTER SEVEN

7 Disorders of smell

Conductive or sensorineural.

7.1 Conductive loss

Odorants cannot reach the olfactory cleft:

- Chronic rhinosinusitis with polyps (the commonest treatable cause), allergic and viral rhinitis.
- Septal deviation, turbinate hypertrophy, tumours.
- Atrophic rhinitis.

7.2 Sensorineural loss

Damage to the receptors or the pathway:

- **Post-viral:** after a cold or influenza (parainfluenza 3 especially), and COVID-19, often sudden.
- **Head injury:** up to 15% of significant head injuries, from shearing of olfactory fibres at the cribriform plate or frontal contusion; usually anosmia, and only about 10% recover.
- **Tumours:** olfactory groove meningioma, olfactory neuroblastoma, frontal lobe tumours.
- **Neurodegenerative disease:** Parkinson's and Alzheimer's disease, where smell loss may be the first sign.
- **Congenital:** *Kallmann syndrome* (anosmia with hypogonadotropic hypogonadism, from failed migration of olfactory and gonadotropin-releasing neurons).
- **Toxins, drugs, radiation, endocrine and metabolic disease:** smoking, hypothyroidism, diabetes, zinc or B12 deficiency.

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8 CHAPTER EIGHT Assessment

History, endoscopy, testing, imaging.

8.1 History

Timing points to the cause: sudden after a cold or injury, gradual or fluctuating with sinus disease and allergy, lifelong in congenital loss. Separate true taste loss from flavour loss.

8.2 Examination

Head and neck and neurological examination, including cognition and mood. Endoscopy of the olfactory cleft and middle meatus for polyps and inflammation; avoid topical anaesthetic before testing, as it blunts smell temporarily.

8.3 Smell testing

- **Bedside:** familiar odours (coffee, clove, peppermint, lemon) to each nostril with the eyes closed; the alcohol sniff test.
- **Sniffin' Sticks:** odour pens testing threshold, discrimination and identification.
- **University of Pennsylvania Smell Identification Test:** a 40-item scratch-and-sniff test scored out of 40.

8.4 Imaging

CT of the sinuses for inflammatory disease; MRI of the anterior cranial fossa and olfactory bulbs when the cause is unclear, after trauma, or with phantosmia or neurological signs.

9 CHAPTER NINE Management

Treat what can be treated.

9.1 Conductive loss

Treat the inflammation: topical steroids (drops in the head-down position reach the cleft better), a short course of oral steroids to test reversibility, and endoscopic sinus surgery for polyps.

9.2 Sensorineural loss

Olfactory training, sniffing a set of strong odours twice a day for months, improves recovery after post-viral and post-traumatic loss. Counsel about safety: smoke and gas alarms, food dates, and personal hygiene.

PEARLS

- A patient who says food has lost its taste usually has lost smell.
- Parosmia during recovery is often a good sign.
- Unexplained phantosmia or smell loss with neurological signs needs MRI.

10 CHAPTER TEN For the examination

What the examiner wants to hear.

10.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Functions of the nose,** with numbers: humidity, temperature, mucus speed.
- **The mucociliary blanket** and what damages it.
- **The nasal cycle.**
- **The olfactory pathway** and why ammonia is not used.

- **Causes of anosmia,** conductive and sensorineural.
- **Kallmann syndrome.**

10.2 Common pitfalls

COMMON PITFALLS

- Calling smell loss a taste disorder.
- Missing polyps as a treatable cause of anosmia.
- Forgetting imaging in unexplained phantosmia.

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CLINICAL · NOSE & SINUSES

External nose and nasal vestibule

The external nose sits in the middle of the face, so its problems are seen by everyone: infections of the skin and vestibule, deformities after injury, the bulbous rhinophyma of rosacea, and skin cancers that love the sun-exposed tip and ala. Two rules stand out: never squeeze a nasal furuncle, and biopsy any lesion that does not heal.

AT A GLANCE

- A furuncle can spread to the cavernous sinus: antibiotics, never squeeze.
- Saddle, hump and crooked nose have different causes.
- Rhinophyma is acne rosacea of the nose.
- Basal cell carcinoma is the commonest cancer of the nasal skin.

1 CHAPTER ONE Overview

Skin, cartilage and the vestibule.

1.1 Applied anatomy

The upper third of the nose is bony (paired nasal bones and the frontal processes of the maxillae); the lower two-thirds is cartilaginous (upper lateral cartilages, lower lateral or alar cartilages, and the septal cartilage). The skin is thin and mobile over the dorsum and thick and sebaceous over the tip and alae. The *vestibule* is the skin-lined entrance, with hairs (vibrissae), up to the limen nasi.

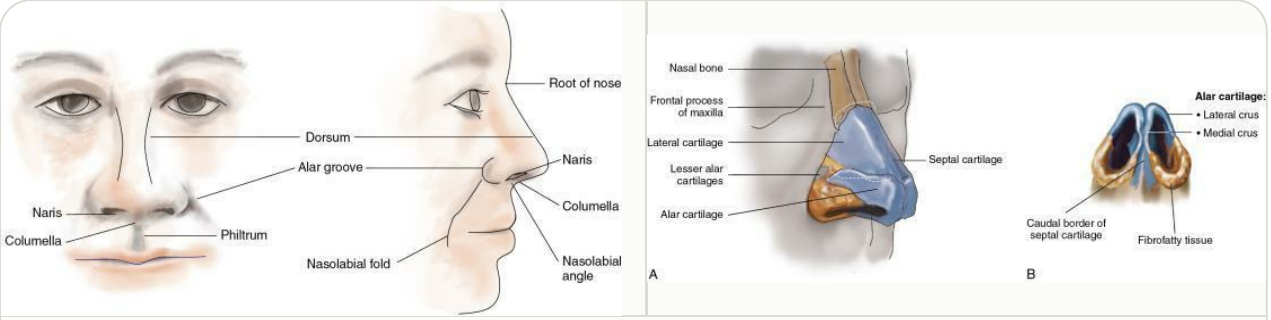


FIGURE 1 First, the parts of the nose and related facial structures. Second, the osteocartilaginous framework.
Dhingra, figs 23.1 and 23.2

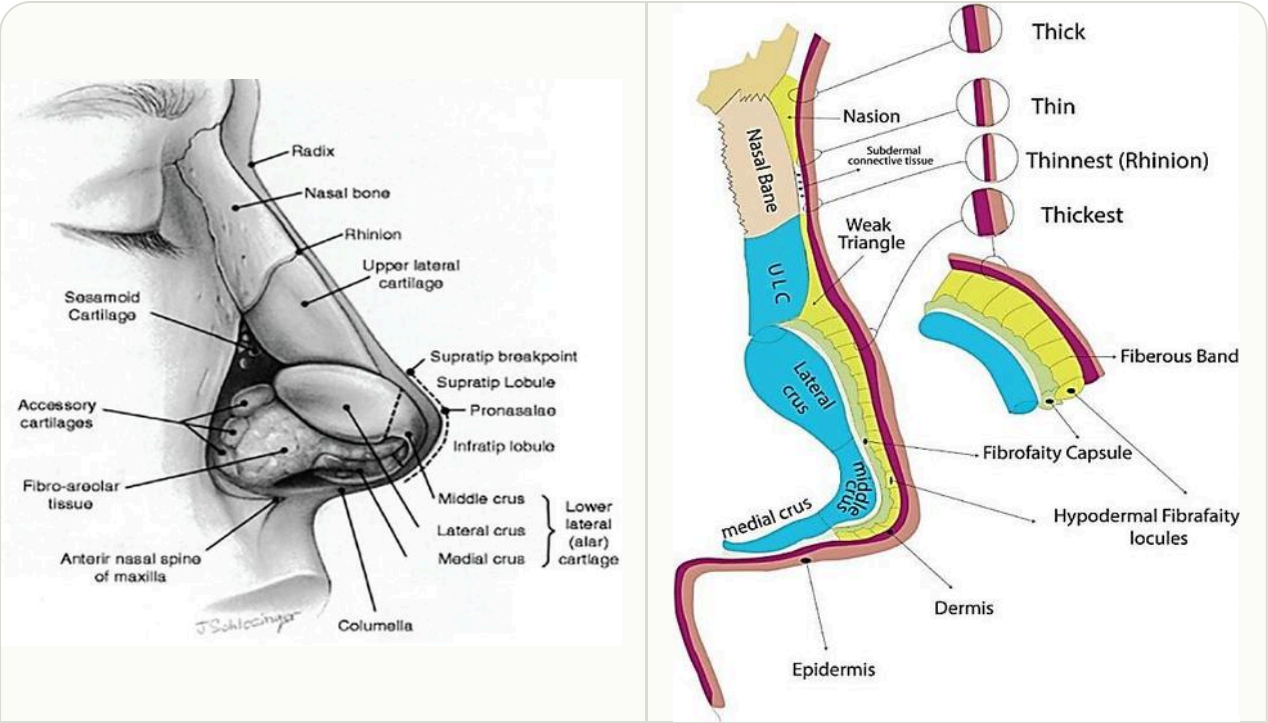


FIGURE 2 The osteocartilaginous framework of the nose, and the change in skin thickness from nasion to tip.

Anatomy atlas · ENT Steps p. 93, ENT Steps p. 96

1.2 The danger area

Veins of the upper lip and nose drain through the facial and angular veins and the ophthalmic veins to the cavernous sinus, without valves. Infection here can spread to the cavernous sinus.

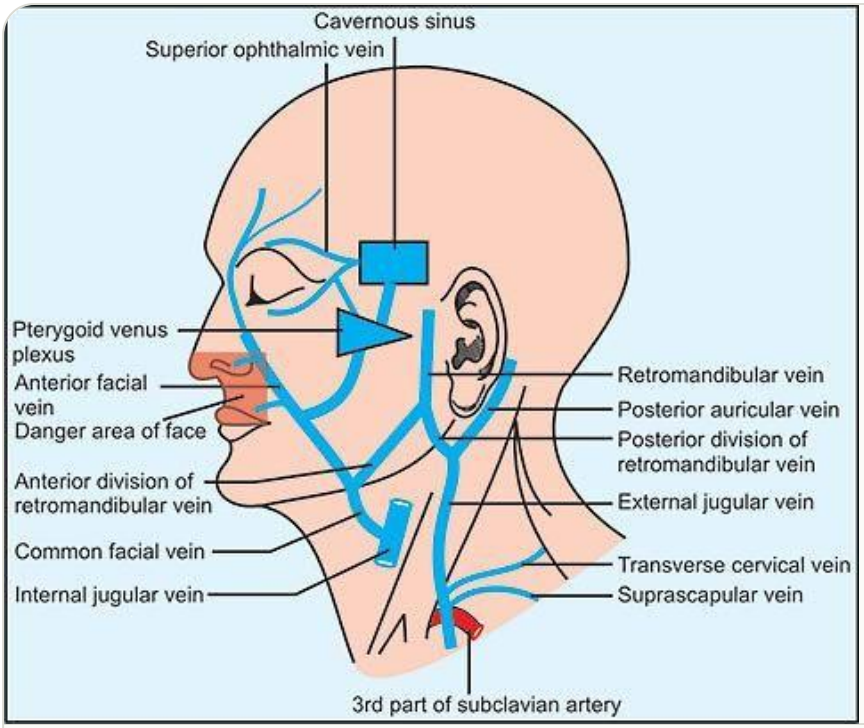


FIGURE 3 The danger area of the face and its venous drainage to the cavernous sinus.

Anatomy atlas · ENT Steps p. 96

2 CHAPTER TWO Infections

Skin and vestibule.

2.1 Furuncle of the vestibule

2.1.1 Cause

Acute staphylococcal infection of a hair follicle, usually after nose picking or plucking hairs.

2.1.2 Clinical picture

A small, exquisitely tender red swelling in the vestibule, with redness and swelling spreading to the tip and dorsum; it may point and rupture.

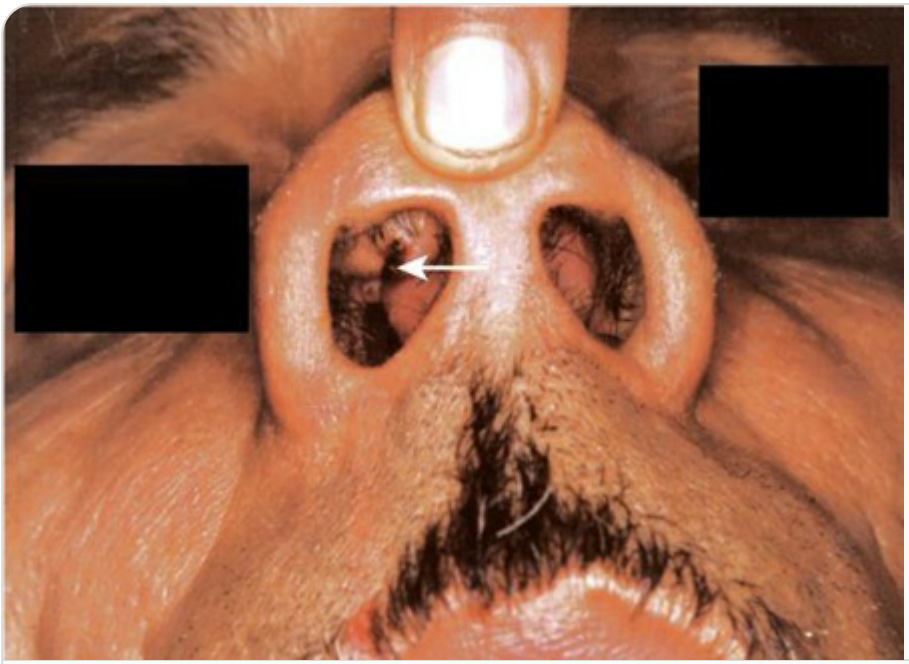


FIGURE 4 Furuncle of the right nasal vestibule.
Dhingra, fig. 25.9

2.1.3 Treatment

Warm compresses, analgesia, and topical and systemic antistaphylococcal antibiotics; incise only when fluctuant.

NEVER SQUEEZE A NASAL FURUNCLE

Squeezing or early incision can spread infection through the angular and ophthalmic veins to the cavernous sinus. Other complications: cellulitis of the upper lip and septal abscess. Recurrent furuncles mean checking for diabetes and nasal staphylococcal carriage.

2.2 Vestibulitis

2.2.1 Cause

Diffuse dermatitis of the vestibular skin, usually *Staphylococcus aureus*, encouraged by nasal discharge from rhinitis or sinusitis and the trauma of wiping.

2.2.2 Clinical picture

Acute: red, swollen, tender skin with crusts over erosions, sometimes involving the upper lip. Chronic: thickened skin with painful fissures and crusts.



FIGURE 5 Acute vestibulitis on the left side.
Dhingra, fig. 25.10

2.2.3 Treatment

Clean the crusts with hydrogen peroxide, apply an antibiotic-steroid ointment and continue it for several days after apparent cure; cauterise a chronic fissure with silver nitrate; treat the source of discharge.

2.3 Cellulitis

Streptococcal or staphylococcal infection making the nose red, swollen and tender, sometimes spreading from the vestibule; systemic antibiotics, warmth and analgesia.

2.4 Specific infections

Lupus vulgaris (apple-jelly nodules), leprosy and syphilis also affect the external nose (see the granulomatous disease topic).

3 CHAPTER THREE Deformities

Shape and function.

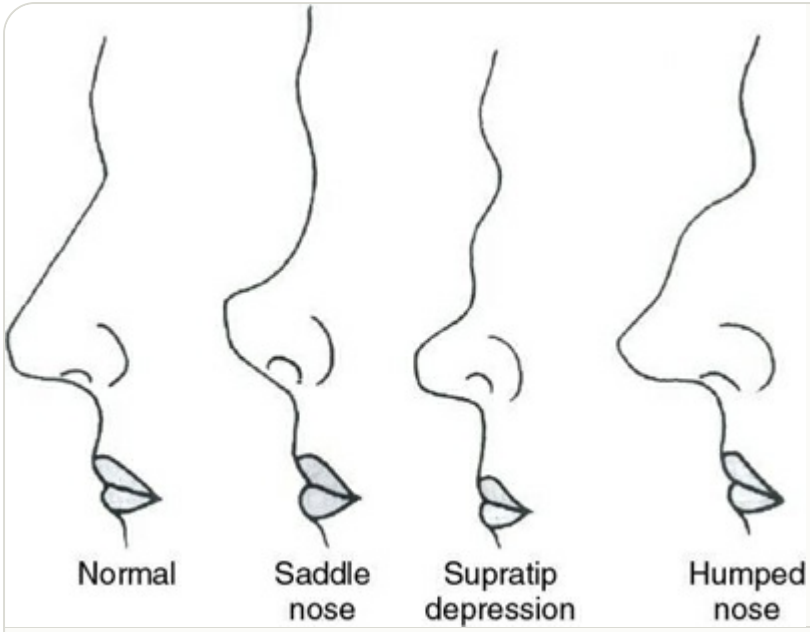


FIGURE 6 Deformities of the nose.
Dhingra, fig. 25.1

3.1 Saddle nose

A depressed dorsum, bony, cartilaginous or both.

3.1.1 Causes

Depressed fractures (the commonest), excessive removal of septal cartilage at submucous resection, septal haematoma or abscess, granulomatosis with polyangiitis, cocaine, leprosy, tuberculosis, syphilis and relapsing polychondritis.

3.1.2 Treatment

Augmentation rhinoplasty: septal, auricular or rib cartilage for a cartilaginous depression; bone (iliac crest or calvarial) for bony and cartilaginous loss. Autografts are preferred; synthetic implants tend to extrude.

3.2 Hump nose

A dorsal hump of bone, cartilage or both, corrected by *reduction rhinoplasty*: removal of the hump and lateral osteotomies to close the open roof.

3.3 Crooked and deviated nose

In a *crooked* nose, the dorsum from the root to the tip is C- or S-shaped; in a *deviated* nose, the dorsum is straight but lies to one side. Both usually follow injury, often in childhood. Correction by *septorhinoplasty* restores both appearance and airway.

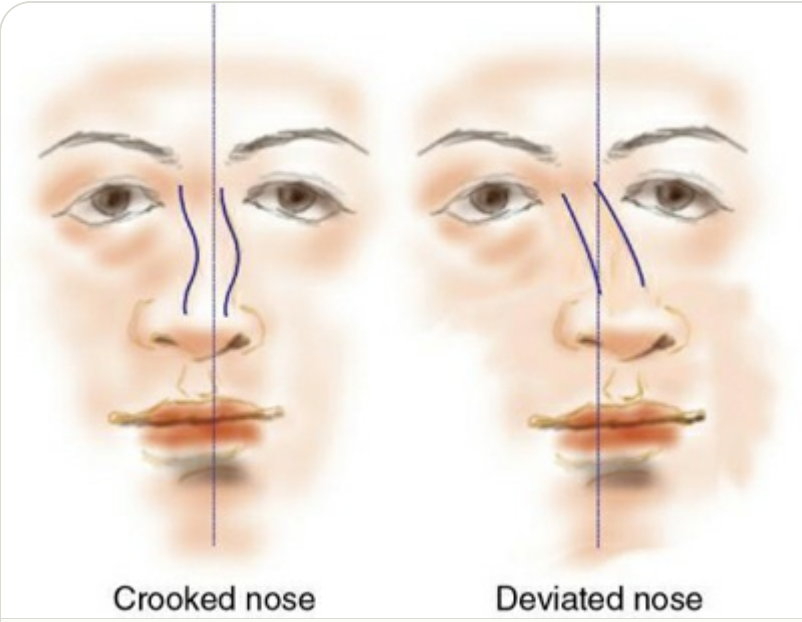


FIGURE 7 The nasal bridge is S-shaped in a crooked nose, and straight but displaced in a deviated nose.
Dhingra, fig. 25.2

3.4 Nasal valve collapse

Weak or scarred lateral walls collapse on inspiration; Cottle's test is positive. Spreader grafts, alar batten grafts or functional rhinoplasty.

OPERATIVE GUIDES IN THEATRE 3 operations

Rhinoplasty and Surgery of the Nasal Valve
Rhinoplasty and the nasal valve

Rhinectomy
Rhinectomy for cancer

Nasal Reconstruction
Nasal reconstruction

CHAPTER FOUR

4 Tumours of the external nose

Benign and malignant.

4.1 Benign

Papilloma (wart), haemangioma, pigmented naevus, seborrhoeic keratosis, neurofibroma and sweat gland tumours. Congenital masses (dermoid, glioma, encephalocele) are covered in their own topic.

4.2 Rhinophyma

The *potato nose*: a slow, lobulated, pink overgrowth of sebaceous glands and connective tissue over the tip and alae, with dilated vessels, in men past middle age with long-standing acne rosacea. It is cosmetic, but may block breathing or vision. Treatment is shaving it down with a scalpel, dermabrasion or carbon dioxide laser, letting the surface re-epithelialise from the remaining glands.



FIGURE 8 Rhinophyma.
Dhingra, fig. 25.4

4.3 Basal cell carcinoma

The commonest malignant tumour of the nasal skin (about 87%), at 40 to 60 years, on the tip and ala: a pearly nodule or an ulcer with rolled edges (*rodent ulcer*). Slow, locally destructive, almost never metastasises. Excision with 3 to 5 mm margins, Mohs surgery for high-risk sites and recurrences, or radiotherapy or cryotherapy for early lesions; reconstruction with local flaps.



FIGURE 9 Basal cell carcinoma of the nose.
Dhingra, fig. 25.6

4.4 Squamous cell carcinoma

The second commonest (about 11%), an infiltrating nodule or ulcer with everted edges on the side of the nose or columella; nodal metastases in about 20%. Radiotherapy for early lesions; wide excision with reconstruction, and neck dissection for nodes.



FIGURE 10 Squamous cell carcinoma of the nose, and of the nasal tip.
Dhingra, fig. 25.7

4.5 Melanoma

The least common: superficial spreading or nodular; wide excision.

4.6 Total rhinectomy

For advanced cancers, followed by a prosthesis or a forehead flap reconstruction.

5 CHAPTER FIVE The vestibule

Narrowing and swellings.

5.1 Stenosis and atresia of the nares

Trauma, burns, surgery, destructive inflammation (formerly smallpox) and Young's operation can narrow the nostrils; congenital atresia from failure of canalisation is rare. Corrected by excision of scar with flaps or grafts and stenting.



FIGURE 11 Stenosis of the left naris after smallpox (A) and after trauma (B).
Dhingra, fig. 25.11

5.2 Nasoalveolar (nasolabial) cyst

A smooth swelling in the floor and lateral wall of the vestibule, raising the ala and filling the nasolabial fold; excised through a sublabial incision.



FIGURE 12 A nasoalveolar cyst at operation (arrow).
Dhingra, fig. 25.12

5.3 Vestibular papilloma and carcinoma

Warts are excised. Squamous carcinoma of the vestibule may spread to the floor,

columella and upper lip, and to the parotid and submandibular nodes.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Furuncle:** the danger area, and why not to squeeze.
- **Vestibulitis:** cause and treatment.
- **Causes of saddle nose,** and its correction.
- **Crooked versus deviated nose.**
- **Rhinophyma.**
- **Basal versus squamous cell carcinoma** of the nose.

6.2 Common pitfalls

COMMON PITFALLS

- Squeezing or incising an early furuncle.
- Treating a non-healing ulcer with creams without biopsy.
- Using synthetic implants that later extrude.

Written for this app from Oxford Textbook of Otolaryngology ch. 35, Oxford Specialist Handbook ch. 22, Dhingra ch. 25 and K. J. Lee ch. 31, 54. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Choanal atresia and congenital nasal masses

Newborns breathe through the nose. A baby who turns blue when quiet or feeding and pink when crying has bilateral choanal atresia until proven otherwise, and needs an oral airway now. A midline mass on a child's nose may connect to the brain: it is imaged, never biopsied or probed, before a plan is made.

AT A GLANCE

- Bilateral choanal atresia is an airway emergency: oral airway, then repair.
- Unilateral atresia presents later with one-sided discharge.
- Dermoid, glioma and encephalocele: image before biopsy.
- Furstenberg sign suggests encephalocele.
- Stents after choanal atresia repair do not reduce revision; piriform aperture stenosis needs pituitary assessment.

1 CHAPTER ONE Overview

Developmental anomalies of the nose.

1.1 The conditions

CONGENITAL NASAL ANOMALIES

- AIRWAY **Choanal atresia** pyriform aperture stenosis, atresia of the nares
- MIDLINE MASSES **Nasal dermoid** glioma, encephalocele
- CYSTS **Nasolacrimal duct cyst** nasoalveolar cyst
- OTHER **Haemangioma** proboscis lateralis, arhinia (rare)

1.2 Why it matters

Obligate nasal breathing in the first weeks of life makes bilateral obstruction an emergency; midline masses carry a risk of meningitis and CSF leak.

2 CHAPTER TWO Choanal atresia

The closed back door.

2.1 Embryology

Persistence of the *bucconasal membrane* (or abnormal mesoderm) between the nasal cavity and the nasopharynx.

2.2 Types

- One-sided or both; one-sided is commoner and may go unnoticed until adulthood.
- Complete or incomplete.
- Bony (about 90% in Dhingra; modern series show most are mixed bony and membranous) or purely membranous.

2.3 Associations

Associated anomalies are common, especially with bilateral atresia: *CHARGE* syndrome (coloboma, heart defects, choanal atresia, retarded growth and development, genital and ear anomalies), Treacher Collins and Crouzon syndromes.

2.4 Clinical picture

2.4.1 Bilateral

Respiratory distress at birth: *cyclical cyanosis*, blue at rest or when feeding and pink when crying (the baby breathes through the mouth while crying), with difficulty feeding.

2.4.2 Unilateral

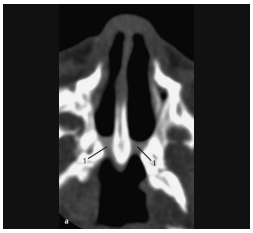
One-sided nasal obstruction and persistent thick discharge from that side, often discovered in later childhood.

2.5 Diagnosis

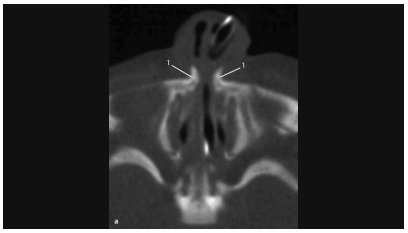
- Mucoid discharge without air bubbles.
- Failure to pass a soft catheter (a size 6 to 8 French) more than about 3 cm into the nose.
- Methylene blue in the nose fails to reach the pharynx; a wisp of cotton under the nostril does not move.
- Flexible nasal endoscopy.
- **CT** in the axial plane is the definitive test: it shows the atretic plate, narrowing of the posterior nasal cavity and thickening of the vomer, and separates bony from membranous atresia.

IMAGING IN THE APP 2 cases

Choanal atresia and pyriform aperture stenosis on CT.



Choanal Atresia
Congenital · 10 images



Piriform Aperture Malformation
Congenital · 2 images

2.6 Management

2.6.1 Emergency airway in bilateral atresia

An oral airway, or a feeding nipple with the end cut off taped in the mouth (*McGovern's technique*), keeps the mouth open and avoids tracheostomy; orogastric feeding. Intubation if needed.

NEWBORN WITH CYCLICAL CYANOSIS

1. Insert an oral airway or McGovern nipple.
2. Try to pass a catheter through each side of the nose.
3. Arrange CT of the nasal cavity after suction and decongestion.

4 Look for CHARGE anomalies: heart, eyes, ears, genitalia.

2.6.2 Definitive repair

- **Endoscopic transnasal repair** with a drill: removal of the atretic plate and the posterior part of the vomer, with or without mucosal flaps and stents. The standard approach now.
- **Transpalatal repair:** better exposure, historically at about 18 months, but risks palatal growth disturbance.
- Unilateral atresia can be repaired electively in childhood.

2.6.3 After repair

Restenosis is common, so regular review, dilatation and revision may be needed.

OPERATIVE GUIDES IN THEATRE 2 operations

Choanal Atresia/Pyriform Aperture Stenosis
Choanal atresia and pyriform aperture stenosis

Midline Nasal Masses
Midline nasal masses

2.7 Pyriform aperture stenosis

Narrowing of the bony nasal entrance from overgrowth of the nasal processes of the maxilla, presenting like bilateral choanal atresia but at the front of the nose. A single central maxillary incisor suggests holoprosencephaly and pituitary problems. Mild cases are managed medically; severe cases by drilling through a sublabial approach.

3 CHAPTER THREE Congenital midline nasal masses

Dermoid, glioma and encephalocele.

3.1 Embryology

In early development a projection of dura passes through the *foramen caecum* (between the frontal and ethmoid bones) into the prenasal space, or between the frontal and nasal bones through the *fonticulus frontalis*. Normally it regresses. If it

does not, ectodermal elements (a dermoid), isolated glial tissue (a glioma) or a herniation of brain and meninges (an encephalocele) remain, sometimes with an intracranial connection. Incidence is about 1 in 20,000 to 40,000 births.

3.2 Nasal dermoid

3.2.1 Picture

The commonest midline nasal mass: a firm, non-tender, non-pulsatile, non-compressible swelling anywhere from the glabella to the columella, often with a *midline pit* that may sprout a hair. It may become infected repeatedly.

3.2.2 Types

- Simple dermoid beneath the skin, in front of the nasal bones, without an opening.
- Dermoid with a sinus, whose tract runs under the nasal bones to the upper septum.
- Dermoid with a tract through the foramen caecum or cribriform plate to the dura, or intradural extension, which risks meningitis.

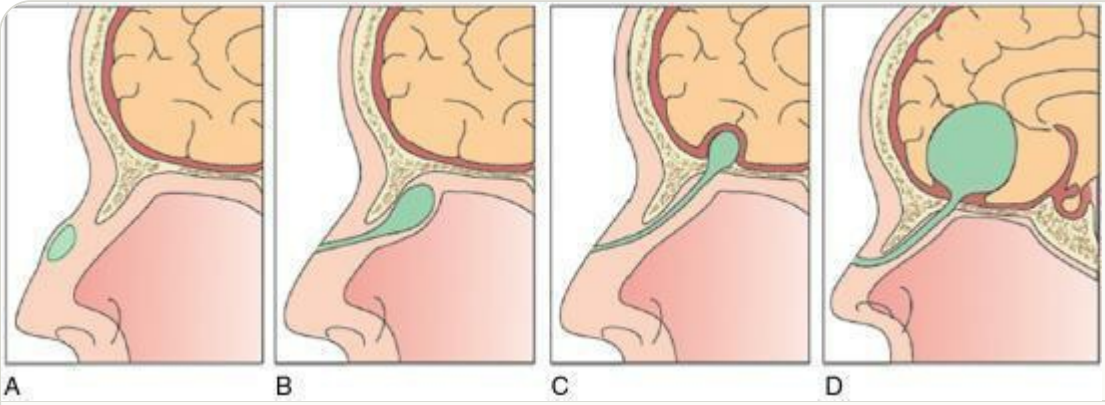


FIGURE 1 Types of nasal dermoid: (A) simple, (B) with an external pit, lying deep to the nasal bones, (C) with a connection to the dura, (D) intradural.
Dhingra, fig. 25.3

3.2.3 Treatment

Complete excision of the cyst, the tract and any involved bone, cartilage and septum, often by an open rhinoplasty approach; combined neurosurgical repair when the tract reaches the dura.

3.3 Nasal glioma

3.3.1 Picture

A mass of ectopic glial tissue, not a true tumour, sometimes called a nasal glial heterotopia. About 60% are extranasal (firm, reddish swelling on the bridge or side of the nose with telangiectasias), 30% intranasal (a pale, firm polypoid mass, mistaken for a polyp) and 10% both. About 15 to 20% have a fibrous stalk to the dura. Non-pulsatile and non-compressible.



FIGURE 2 Extranasal glioma on the right side of a child's nose.
Dhingra, fig. 25.4

3.3.2 Treatment

Complete excision by an external approach; craniotomy if there is an intracranial connection.

3.4 Encephalocele

3.4.1 Picture

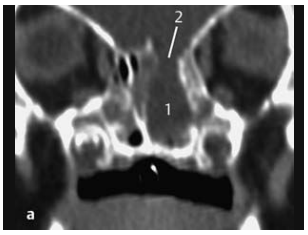
A herniation of meninges (meningocele), or meninges with brain (meningoencephalocele), through a skull base defect, communicating with the subarachnoid space. It is soft, bluish, *pulsatile* and *compressible*, enlarges on crying, straining or jugular compression (*Furstenberg sign*), and may transilluminate.

3.4.2 Types by site

- **Sincipital (frontoethmoid), presenting outside the nose:** nasofrontal (glabella), nasoethmoidal (side of the nose) and naso-orbital (medial orbit).
- **Basal, presenting inside the nose or nasopharynx:** transethmoidal, sphenoethmoidal, transsphenoidal and sphenomaxillary, often mistaken for a polyp.

IMAGING IN THE APP 1 case

Meningoencephalocele through the skull base.



Meningoencephalocele

Congenital · 7 images

3.4.3 Treatment

Neurosurgical or endoscopic: dividing the stalk, excising the extracranial part and repairing the dural and bony defect.

IMAGE BEFORE YOU TOUCH

A midline nasal mass in a child may connect with the brain. Biopsy or simple removal can cause CSF leak and meningitis. Obtain MRI (with CT for the bone) first.

TELLING THE MIDLINE MASSES APART

	DERMOID	GLIOMA	ENCEPHALOCELE
Consistency	Firm	Firm	Soft
Pulsatile or compressible	No	No	Yes
Furstenberg sign	Negative	Negative	Positive
Pit or hair	Often	No	No
Intracranial connection	Sometimes	About 15 to 20%, fibrous stalk	Always, with CSF

4 CHAPTER FOUR Other congenital anomalies

Less common.

4.1 Nasolacrimal duct cyst

A cystic dilatation of the lower nasolacrimal duct when Hasner's valve fails to open, seen as a bluish swelling under the inferior turbinate; bilateral cysts can obstruct a newborn's breathing. Endoscopic marsupialisation.

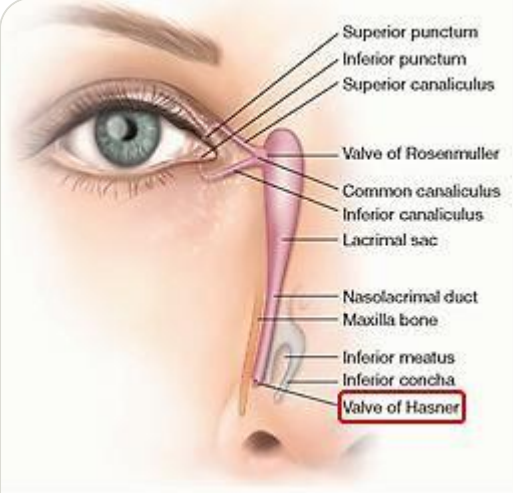


FIGURE 3 The lacrimal drainage system and Hasner's valve in the inferior meatus.
Anatomy atlas · ENT Steps p. 100

4.2 Haemangioma

A strawberry swelling of the nasal tip that grows in the first year and regresses over several years; propranolol for problem lesions.

4.3 Rare anomalies

Arhinia (absent nose), proboscis lateralis, bifid nose and congenital atresia of the nares.

CURRENT PAEDIATRIC PRACTICE (OXFORD TEXTBOOK)

Neonates breathe preferentially, not obligatorily, through the nose for the first weeks to months, which is why bilateral nasal obstruction presents with distress at birth. About half of children with CHARGE syndrome have bilateral choanal atresia, and craniosynostoses such as Crouzon and Apert are rarer associations. Unilateral atresia is commoner, usually isolated and presents later with a one-sided discharge; CT is best kept for planning once surgery is decided, to avoid repeated radiation and sedation.

Stents. After endoscopic repair of bilateral atresia some surgeons stent for two to six weeks with cut endotracheal tubes, but recent series show no difference in revision rates with or without stents, and because stents need intensive care by parents and sometimes removal under anaesthesia, their use is declining; unilateral stenting is not worthwhile. Postoperative suction, saline and topical steroids reduce granulation and early scarring.

Piriform aperture stenosis, an overgrowth of the nasal process of the maxilla, mimics bilateral choanal atresia (a catheter cannot be passed) but is anterior; it may accompany a single central maxillary incisor and holoprosencephaly, so pituitary and brain assessment is needed. Endoscopy and CT distinguish the two. Surgery, a sublabial drilling of the aperture, is reserved for persistent respiratory distress or problems with sleep, growth or feeding. **Mid-nasal stenosis** can coexist. **Septal surgery** in children was long avoided for fear of affecting facial growth, but conservative septoplasty for significant obstruction appears safe. Adenoids are largest in absolute terms at 6 to 10 years, but largest relative to the nasopharynx, and so most obstructive, at 2 to 6 years.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Choanal atresia:** embryology, cyclical cyanosis, tests, McGovern nipple, repair.
- **CHARGE syndrome.**
- **Dermoid, glioma and encephalocele:** embryology and differences.
- **Furstenberg sign.**

- **Why not to biopsy** a midline nasal mass.

5.2 Common pitfalls

COMMON PITFALLS

- Missing unilateral atresia in a child with one-sided discharge.
- Taking a nasal mass in a child for a polyp.
- Forgetting the associated anomalies of CHARGE.

Written for this app from Oxford Textbook of Otolaryngology ch. 78, Oxford Specialist Handbook ch. 23, Oxford Handbook of ENT ch. 17, Dhingra ch. 25, 29 and K. J. Lee ch. 31, 53. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Nasal foreign bodies, rhinoliths and myiasis

A child with a one-sided, foul-smelling, sometimes bloodstained nasal discharge has a foreign body in that nostril until proven otherwise. Most are harmless beads and seeds; a button battery is an emergency. In adults the same story may be a rhinolith built around a forgotten object, and in a neglected, foul nose flies may lay their eggs.

AT A GLANCE

- One-sided foul discharge in a child: foreign body.
- Button batteries and magnets: remove urgently.
- Rhinolith: a stone around an old foreign body.
- Maggots: kill them first, then remove; look for the cause.

1 CHAPTER ONE Overview

Things that should not be in the nose.

1.1 The conditions

- Foreign bodies, mostly in children.
- Rhinoliths: stones formed around a nucleus in the nose.
- Nasal myiasis: maggots in the nose.
- Nasal synechiae: adhesions, often the aftermath of injury or surgery.

1.2 The dictum

If a child presents with unilateral, foul-smelling nasal discharge, a foreign body must be excluded. Other causes of one-sided bloodstained discharge in a child are rhinolith, nasal diphtheria, myiasis, sinusitis and, rarely, a tumour.

2 CHAPTER TWO Foreign bodies

Mostly children.

2.1 Types

- **Inorganic:** beads, buttons, pebbles, chalk, pieces of plastic and toys; non-irritant, may stay silent for a long time.
- **Organic:** seeds, peas, beans, nuts, paper and sponge; they absorb water and swell, and cause early inflammation.
- **Dangerous:** button (disc) batteries leak alkali and produce electrical currents that destroy mucosa within hours, causing septal perforation; paired magnets on either side of the septum cause pressure necrosis.
- **Iatrogenic:** cotton pledgets or swabs left behind.

2.2 Clinical picture

A history of insertion if it was seen; otherwise one-sided obstruction and foul, sometimes bloodstained discharge with excoriation of the nostril. The object often lies on the floor or between the septum and inferior turbinate, buried in granulation and discharge.

2.3 Investigations

Usually none. A radiograph helps with metallic or radio-opaque objects and button batteries (a double ring on the radiograph).

2.4 Removal

- **Soft objects** (paper, cotton): forceps under good light.
- **Round, smooth objects:** a blunt hook or a Eustachian catheter passed beyond the object and drawn forwards along the floor; never forceps, which push it deeper.
- **Positive pressure:** in a cooperative child, a parent blowing into the mouth while the clear nostril is blocked (*mother's kiss*), or a bag-valve device.

- **Under general anaesthesia:** for uncooperative children and posterior or impacted objects, with a cuffed tube, the head extended (Rose's position) and a pack in the nasopharynx to prevent inhalation; an endoscope helps to find and remove it.

BUTTON BATTERIES

Remove immediately. Do not instil drops or water, which speed the electrical burn. Check afterwards for septal necrosis and perforation.

2.5 Complications

Rhinitis and sinusitis, epistaxis, septal perforation, rhinolith formation, and inhalation into the airway during removal.

3 CHAPTER THREE Rhinolith

A stone in the nose.

3.1 Formation

Slow deposition of calcium and magnesium salts around a nucleus: a small foreign body (*exogenous*), or a blood clot or dried secretions (*endogenous*). Over years it grows into a large, irregular mass that can erode the septum or lateral wall.

3.2 Clinical picture

More common in adults: one-sided obstruction with a foul, often bloodstained discharge, sometimes epistaxis or neuralgic pain. A grey-brown or greenish-black, irregular, stony-hard and brittle mass lies between the septum and turbinates, often surrounded by granulation. A radiograph or CT shows the calcified mass.

3.3 Treatment

Removal under general anaesthesia, usually through the nostril, breaking large

stones into pieces; rarely a lateral rhinotomy for very large, hard stones.

4 CHAPTER FOUR Nasal myiasis

Maggots in the nose.

4.1 How it happens

Flies, especially *Chrysomya* species, are attracted by the foul discharge of atrophic rhinitis, syphilis, leprosy, rhinosporidiosis or infected wounds. They lay about 200 eggs at a time, which hatch into larvae within a day. In India it is seen mostly from August to October.

4.2 Clinical picture

For the first three or four days: intense irritation, sneezing, watering eyes, headache and a thin bloodstained discharge, often presenting as epistaxis. The lids and lips swell; then maggots crawl out of the nose, and the patient smells foul. They burrow and destroy the nose, sinuses, face, palate and eye, form fistulae, and can cause meningitis and death.

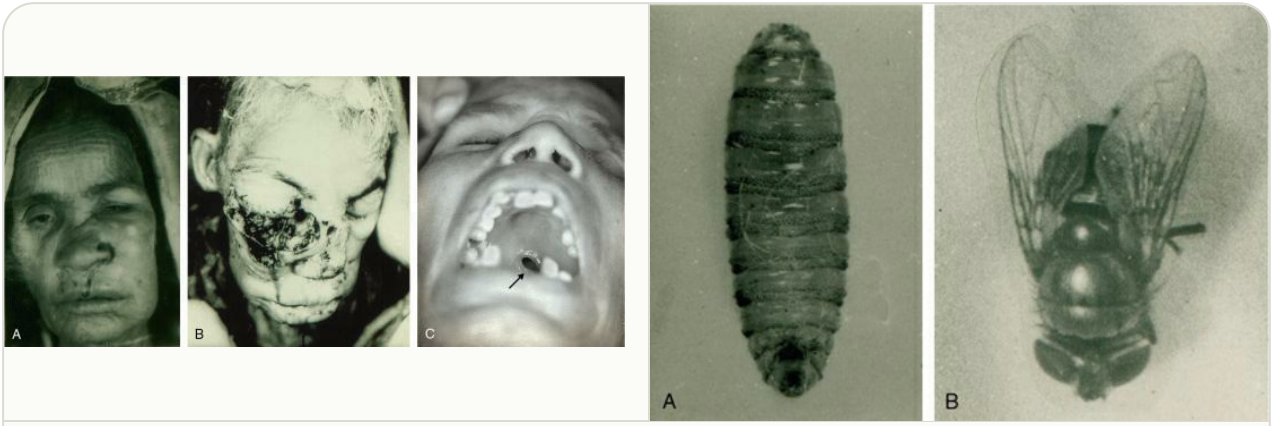


FIGURE 1 First, nasal myiasis: swelling of the nose and lids with bloodstained discharge, destruction of the cheek and eye in a neglected case, and a palatal perforation (arrow). Second, the maggot and the fly that lays it.

Dhingra, figs 29.1 and 29.2

4.3 Treatment

- Pick out every visible maggot with forceps; they retreat from light into dark recesses.
- Instil chloroform water (or turpentine, or ivermectin) to kill or suffocate them, and remove them over several sessions.
- Warm saline douches to clear slough and dead maggots; systemic ivermectin and antibiotics.
- Isolate the patient under a net, away from flies.
- Endoscopy and imaging to find hidden maggots, and treat the underlying disease.

5 CHAPTER FIVE Nasal synechiae

Adhesions.

5.1 Cause

Scar bands between the septum and lateral wall where two raw mucosal surfaces touch: after septal or sinus surgery, polypectomy, removal of foreign bodies, reduction of fractures, cautery, packing, or ulcerating infections.

5.2 Effects

Nasal obstruction, and blocked sinus drainage with sinusitis, headache and discharge.

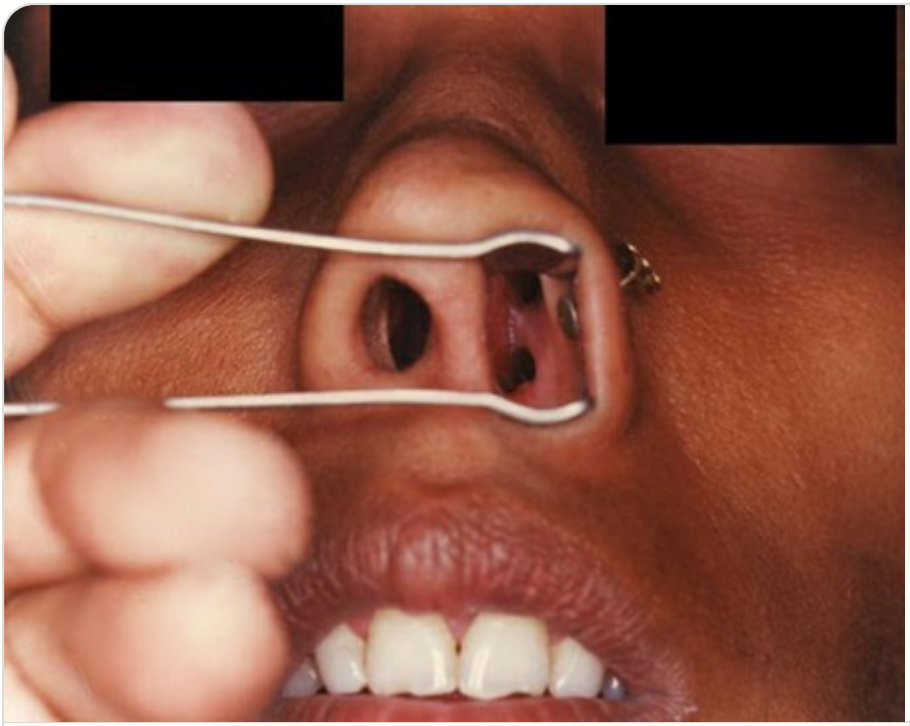


FIGURE 2 A synechia in the left nasal cavity.
Dhingra, fig. 29.3

5.3 Treatment

Divide the adhesion and keep the raw surfaces apart with a thin silastic sheet (splint) until healed, about one to three weeks. Prevention: avoid opposing raw areas and use splints after septal and turbinate surgery.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The dictum:** one-sided foul discharge in a child.
- **Removal technique** for round objects, and why not forceps.
- **Button battery** as an emergency.
- **Rhinolith:** formation and appearance.

- **Myiasis:** predisposing diseases and treatment.

6.2 Common pitfalls

COMMON PITFALLS

- Treating one-sided discharge in a child with antibiotics without looking.
- Grasping a bead with forceps and pushing it backwards.
- Putting drops into a nose with a button battery.

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CLINICAL · NOSE & SINUSES

CSF rhinorrhoea

A clear, watery drip from one nostril that gushes when the patient bends forward, after a head injury, after sinus or pituitary surgery, or for no obvious reason, is cerebrospinal fluid until proven otherwise. It is an open door to meningitis. Prove it with beta-2 transferrin, find the hole, and close it.

AT A GLANCE

- Beta-2 transferrin confirms CSF.
- High-resolution CT finds the bony defect; cisternography or MRI if needed.
- Most traumatic leaks stop with conservative care.
- Endoscopic repair succeeds in over 90%.

1 CHAPTER ONE Overview

The skull base that leaks.

1.1 Definition

Leakage of cerebrospinal fluid from the subarachnoid space into the nose through a defect in the dura and the bone of the skull base.

1.2 CSF physiology

CSF is secreted by the choroid plexuses at about 20 mL an hour (350 to 500 mL a day) and absorbed by the arachnoid villi into the venous sinuses; the total volume is 90 to 150 mL, replaced three to five times a day. Normal lumbar pressure is 60 to 250 mm of water, rising with coughing, sneezing, nose blowing, straining and lifting, all of which a patient with a leak should avoid.

1.3 Why it matters

Ascending infection causes meningitis, often pneumococcal and recurrent; air can

enter the cranium (pneumocephalus).

2 CHAPTER TWO Causes

Trauma first.

CAUSES OF CSF RHINORRHOEA

- TRAUMA **Accidental** anterior skull base and temporal bone fractures
- TRAUMA **Surgical** endoscopic sinus surgery, pituitary and skull base surgery
- INFLAMMATION AND EROSION **Mucoceles, polyposis** fungal disease, osteomyelitis
- TUMOURS **Benign and malignant** invading the skull base; pituitary tumours
- CONGENITAL **Meningocele, encephalocele** skull base defects
- SPONTANEOUS **Raised intracranial pressure** often obese middle-aged women; empty sella

2.1 Traumatic

The commonest cause: about 80% of leaks follow accidental injury, mostly fractures of the anterior skull base. Surgical injury, especially at the lateral lamella of the cribriform plate during ethmoid surgery, may leak at once or later.

2.2 Spontaneous

Many patients have *idiopathic intracranial hypertension*: typically obese middle-aged women, with an *empty sella* on MRI, and often small encephaloceles in the cribriform plate or lateral sphenoid recess. Their leaks recur unless the pressure is also treated.

2.3 Sites

From the anterior cranial fossa: the cribriform plate (commonest), the ethmoid roof or the frontal sinus. From the middle cranial fossa: through the sphenoid sinus. After

a temporal bone fracture, CSF passes into the middle ear and down the Eustachian tube (*CSF otorhinorrhoea*).

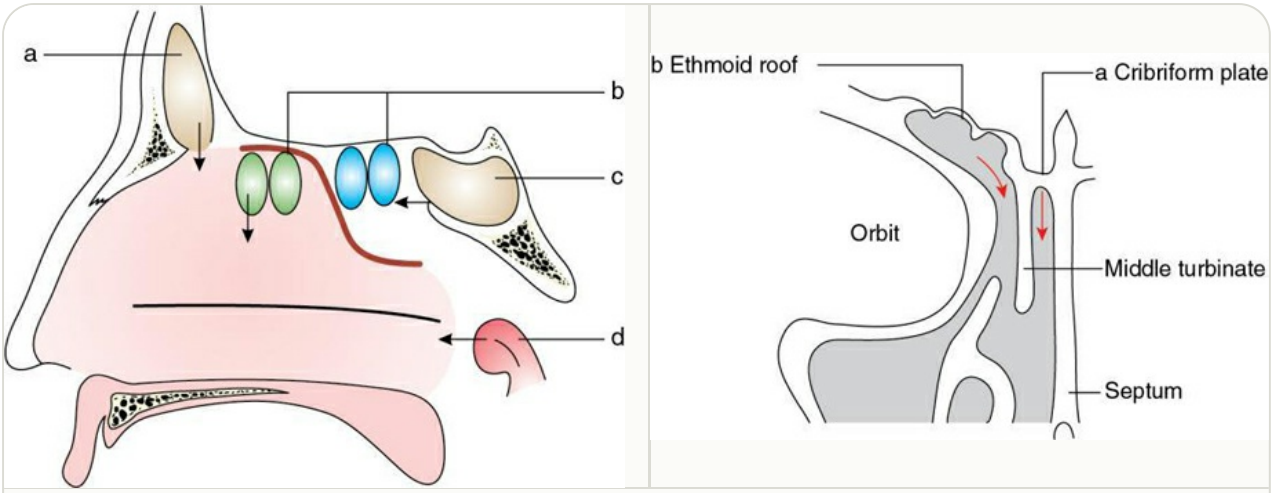


FIGURE 1 First, sites of leakage: (a) frontal sinus, (b) ethmoid, © sphenoid, (d) Eustachian tube after a temporal bone fracture. Second, bony defects in (a) the cribriform plate and (b) the ethmoid roof.

Dhingra, figs 29.4 and 29.5

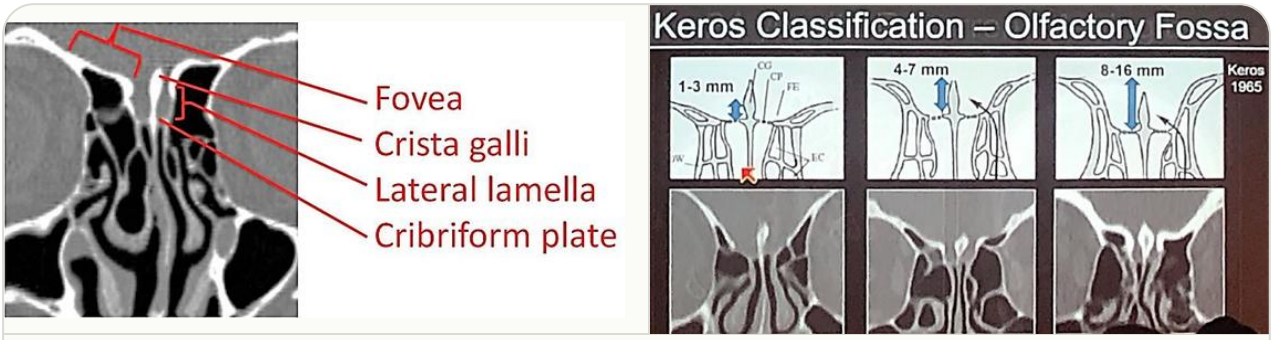


FIGURE 2 The fovea ethmoidalis, lateral lamella and cribriform plate on coronal CT, and the Keros classification of the olfactory fossa: a deep fossa (Keros 3) has a long, thin lateral lamella, the commonest site of surgical injury.

Anatomy atlas · ENT Steps p. 139, ENT Steps p. 143

3 CHAPTER THREE Clinical picture

The drip that cannot be sniffed back.

3.1 History

A clear, watery discharge from one side, increasing on bending forward, straining or lifting, sometimes in the morning when the patient first bends down and a sinus reservoir empties (the *reservoir sign*, typical of sphenoid leaks). It may taste salty or sweet, and cannot be sniffed back. There may be headaches (low pressure from the leak, or high pressure in idiopathic intracranial hypertension), loss of smell after trauma, or a history of meningitis.

3.2 Halo sign

After trauma, bloodstained fluid on filter paper gives a central red spot with a clear outer ring (the *double ring* or *target sign*); it suggests, but does not prove, CSF.

CSF OR NASAL SECRETION?

	CSF	NASAL SECRETION
History	Head injury, sinus or skull base surgery, tumour	Sneezing, itching, blockage
Flow	Drops or a stream on bending or straining; cannot be sniffed back	Continuous; can be sniffed back
Character	Thin, watery, clear	Mucoid, or watery like tears
Handkerchief	Stays soft	Stiffens (mucus)
Glucose	Higher (over about 30 mg/dL)	Lower
Beta-2 transferrin	Present	Absent

3.3 Examination

Endoscopy may show fluid from the olfactory cleft, the middle meatus, the sphenoethmoidal recess or the Eustachian tube, or an encephalocele; look in the ear for fluid behind the drum.

CHAPTER FOUR

4 Investigations

Confirm, then localise.

4.1 Confirming CSF

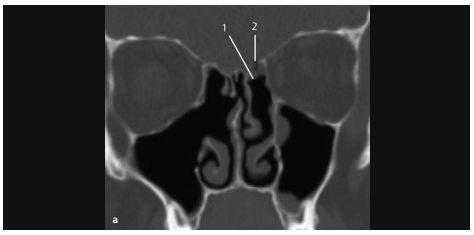
- **Beta-2 transferrin:** a protein found only in CSF, perilymph and aqueous humour; the specific and sensitive test, needing only a few drops.
- **Beta-trace protein:** an alternative, also specific, used widely in Europe.
- Glucose testing is unreliable and no longer recommended.

4.2 Localising the leak

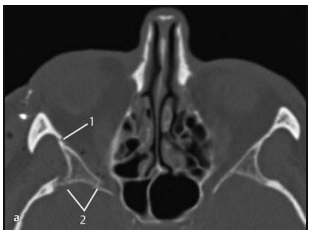
- **High-resolution CT** of the skull base in fine coronal and axial slices: the first test, showing bony defects in the cribriform plate, ethmoid roof, frontal and sphenoid sinuses.
- **MRI (heavily T2-weighted):** shows fluid tracking through the defect and any encephalocele or intracranial pathology, without injections; also shows an empty sella.
- **CT cisternography:** intrathecal contrast with CT, when the leak is active but not localised.
- **Intrathecal fluorescein:** a small dose of diluted fluorescein by lumbar puncture, seen endoscopically as yellow or, with a blue filter, fluorescent green; used before or during surgery to find the site.

IMAGING IN THE APP 3 cases

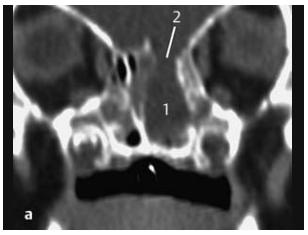
A dural defect after sinus surgery, anterior skull base fractures, and a meningoencephalocele.



Dural Defect after Sinonasal Surgery
After surgery · 3 images



Anterior Skull Fractures
Trauma · 3 images



Meningoencephalocele
Congenital · 7 images

5 CHAPTER FIVE Management

Most traumatic leaks stop.

5.1 Conservative treatment

For early traumatic leaks: bed rest with the head raised, stool softeners, no nose blowing, sneezing through an open mouth, no straining. Acetazolamide reduces CSF production; a lumbar drain lowers the pressure. Most traumatic leaks close within one to two weeks. Routine prophylactic antibiotics are debated; pneumococcal vaccination is advised.

5.2 Surgical repair

5.2.1 Indications

A leak that persists beyond about one to two weeks, recurs, follows surgery and is recognised at the time, or is spontaneous; meningitis; a large defect or an encephalocele.

5.2.2 Endoscopic repair

The standard approach, with success over 90% at the first attempt. The defect is exposed, mucosa removed from its edges, any encephalocele reduced or cauterised, and the defect closed in layers:

- **Inlay (underlay) graft** of fascia, fat or a dural substitute between the dura and the bone.

- **Onlay graft** of mucosa or fascia over the defect.
- **Vascularised flap** for large defects, most commonly the *nasoseptal (Hadad-Bassagasteguy) flap* on the posterior septal artery.
- Fibrin glue and absorbable packing to support the repair.

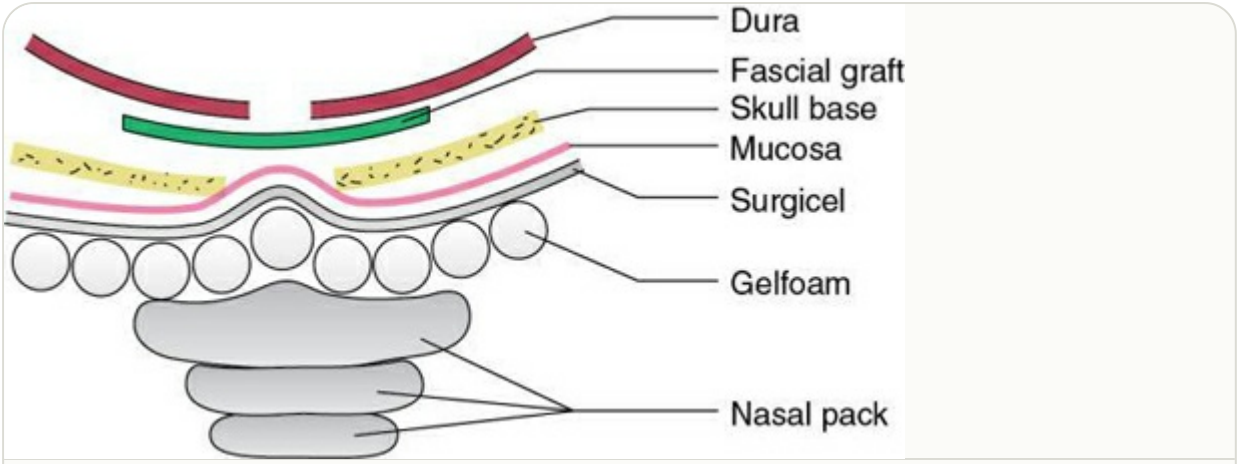


FIGURE 3 Repair of a CSF fistula with fascia and a mucosal graft; frontal sinus leaks often need an external approach.

Dhingra, fig. 29.6

5.2.3 Open approaches

Intracranial repair by craniotomy for very large defects, multiple fractures or associated brain injury; external approaches for leaks high in the frontal sinus.

OPERATIVE GUIDES IN THEATRE 2 operations

Endoscopic Management of Cerebrospinal Fluid Leaks
Endoscopic repair of CSF leaks

Reconstruction After Skull Base Surgery
Reconstruction after skull base surgery

5.2.4 Afterwards

Bed rest with the head up, avoidance of straining, stool softeners and no nose blowing for several weeks. In spontaneous leaks, treat raised pressure (weight loss, acetazolamide, sometimes a shunt) to prevent recurrence.

PEARLS

- Beta-2 transferrin is the test; the halo sign is only a hint.
- Spontaneous leak: think of idiopathic intracranial hypertension.

- The lateral lamella of the cribriform plate is the weak spot in ethmoid surgery.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Causes** and sites of CSF rhinorrhoea.
- **CSF versus nasal secretions.**
- **Beta-2 transferrin,** and localisation by CT, MRI, cisternography and fluorescein.
- **Conservative treatment,** and indications for repair.
- **Endoscopic multilayer repair** and the nasoseptal flap.

6.2 Common pitfalls

COMMON PITFALLS

- Treating a one-sided clear drip as rhinitis.
- Relying on glucose testing.
- Repairing a spontaneous leak without treating raised intracranial pressure.

Written for this app from Oxford Textbook of Otolaryngology ch. 29, 45, Oxford Specialist Handbook ch. 21, Dhingra ch. 29 and K. J. Lee ch. 34. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Nasal and facial fractures

Facial fractures arrive with the drama of trauma: bleeding, swelling and a frightened patient. The order of work never changes. Airway, breathing, circulation and the cervical spine come first; then the eye and any CSF leak; then the septum, looking for a haematoma; and only then the shape of the nose and face, much of which can wait for the swelling to settle.

AT A GLANCE

- Look for a septal haematoma in every nasal injury.
- Reduce a displaced nasal fracture within about two weeks.
- Le Fort I, II and III separate the maxilla at different levels.
- Blow-out fracture: diplopia on upgaze, entrapment, enophthalmos.

1 CHAPTER ONE Overview

Thirds of the face.

1.1 Regions

- **Upper third:** frontal bone, frontal sinus and supraorbital ridges.
- **Middle third:** nasal bones and septum, naso-orbito-ethmoid complex, zygoma, orbital floor and maxilla.
- **Lower third:** mandible.

1.2 General management

FACIAL INJURY IN THE EMERGENCY DEPARTMENT

1. Airway (with cervical spine control), breathing and circulation; a midface fracture can drop back and block the airway.
2. Control bleeding: packs, and reduction of a mobile maxilla.
3. Examine the eyes now: acuity, pupils, movements, proptosis; retrobulbar haemorrhage needs a lateral canthotomy within minutes.

4. Look for CSF rhinorrhoea and otorrhoea, and other injuries.
5. Examine the septum for a haematoma in every nasal injury.
6. CT of the face (with 3D reconstruction) for anything beyond a simple nasal fracture.

1.3 Soft tissue injuries

Clean and close lacerations early, preserving tissue; look for injury to the parotid duct and the facial nerve in cheek wounds and repair them primarily.

2 CHAPTER TWO Nasal fractures

The commonest facial fracture.

2.1 Mechanism and types

The projecting nose is the most frequently fractured bone of the face.

- **Frontal blow:** depressed fracture of the thinner lower nasal bones; with more force, an *open-book* fracture (septum collapsed, nasal bones splayed) and comminution with a flat, wide dorsum.
- **Lateral blow:** depression of one nasal bone, or fracture of both with the septum, deviating the bridge.

The septum is often buckled, dislocated or fractured, and a septal haematoma may form.

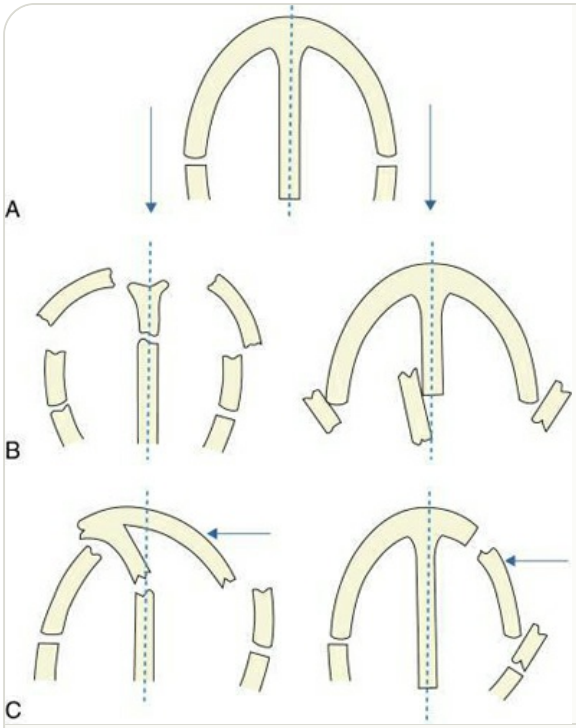


FIGURE 1 Types of nasal fracture: (A) normal, (B) a frontal blow with depressed or open-book fracture, (C) a lateral blow deviating the bridge or depressing one nasal bone.

Dhingra, fig. 34.1

2.2 Clinical features

Swelling within hours, periorbital bruising, tenderness, deformity, crepitus and mobility, epistaxis, obstruction from septal injury or haematoma, and lacerations exposing bone in compound fractures.



FIGURE 2 A nasal fracture with the bridge displaced to the right (A), and after manual correction (B).

Dhingra, fig. 34.2

2.3 Diagnosis

Clinical. Radiographs are not needed for an isolated nasal fracture and may miss it; CT is used when other facial fractures are suspected.

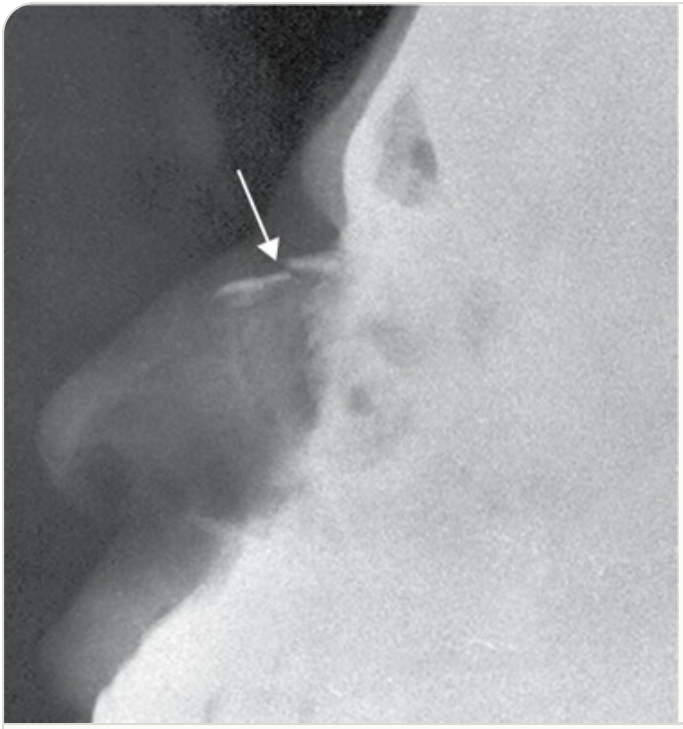
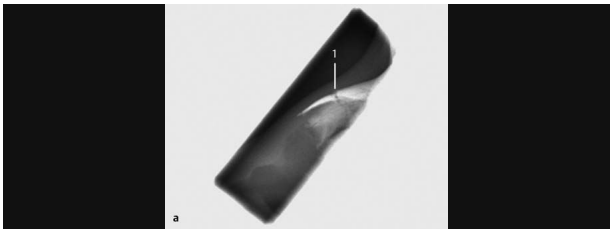


FIGURE 3 A fractured nasal bone (arrow) on a radiograph.
Dhingra, fig. 34.3

IMAGING IN THE APP 1 case

A nasal bone fracture on imaging.



Fracture of the Nasal Bone and Retention Cysts
Trauma · 3 images

2.4 Treatment

2.4.1 Timing

Reduce before swelling develops, or after it settles at about 5 to 7 days. After about two weeks the bones have set and reduction becomes difficult; healing is faster in children, so reduce them sooner.

2.4.2 Closed reduction

Under local or general anaesthesia: a blunt elevator lifts depressed bones, guided by fingers outside; digital pressure realigns a deviated bridge; *Walsham's forceps* disimpact the nasal bones and *Asch's forceps* straighten the septum. Drain any septal haematoma. External splint, and packing for unstable fractures.

2.4.3 Open reduction

When closed reduction fails, and for severe septal fractures; late deformity is corrected by septorhinoplasty after six months or more.

PEARL

Always look inside the nose. A septal haematoma missed today becomes a septal abscess and a saddle nose (see the septum topic).

3 CHAPTER THREE Naso-orbito-ethmoid fractures

The flattened midline.

3.1 Mechanism

A blow over the nasion drives the nasal bones, perpendicular plate of the ethmoid, ethmoid cells and medial orbital walls backwards. The cribriform plate, frontal sinus outflow, lacrimal system, medial canthal tendon and eye may all be injured.

3.2 Clinical features

Telecanthus (widened distance between the medial canthi), a flattened bridge with an upturned *pug* tip, periorbital bruising, orbital haematoma from the ethmoidal arteries, CSF rhinorrhoea, epiphora and displacement of the eye.

3.3 Treatment

CT to map the injury; open reduction and fixation with miniplates through existing lacerations, a coronal or local incision, with repair of the medial canthal tendon and the lacrimal system as a priority over the nasal contour. Treat CSF leaks (see that topic).

4 CHAPTER FOUR Zygoma and orbital floor

Cheek and eye.

4.1 Zygomatic (tripod) fracture

4.1.1 Mechanism

The second commonest facial fracture. The zygoma separates at its frontal, temporal and maxillary attachments, usually rotated inwards and backwards.

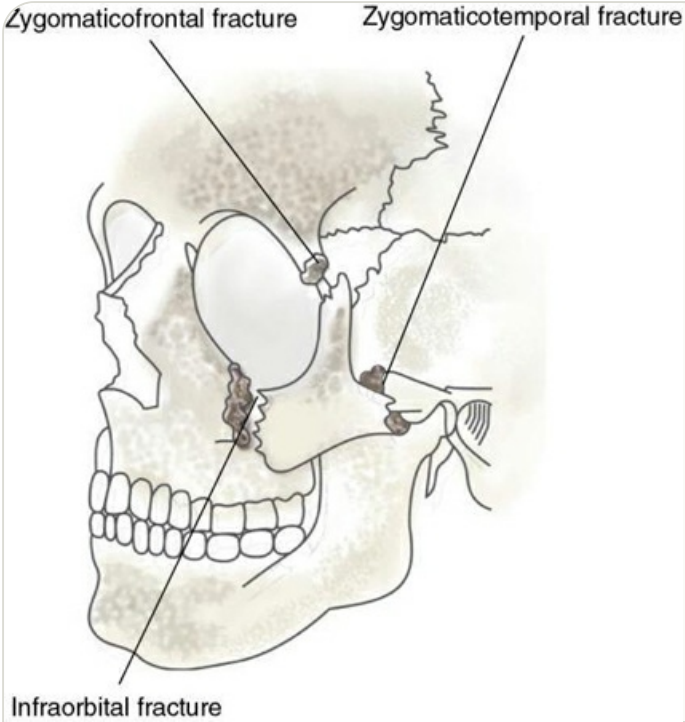


FIGURE 4 A fractured left zygoma.
Dhingra, fig. 34.4

4.1.2 Clinical features

- Flattening of the cheek.

- Step at the infraorbital margin.
- Numbness of the cheek, side of the nose and upper lip (infraorbital nerve).
- Trismus, as the depressed zygoma presses on the coronoid process.
- A slanted palpebral fissure from displacement of the lateral canthus.
- Diplopia from entrapment of the inferior rectus.
- Surgical emphysema of the lid after nose blowing.

4.1.3 Diagnosis

CT in axial and coronal planes, or 3D reconstruction; Waters' view in the past.

4.1.4 Treatment

Only displaced fractures need treatment: elevation through a temporal (Gillies) or intraoral approach, and open reduction with plate fixation at the frontozygomatic suture, infraorbital rim and zygomaticomaxillary buttress, repairing the orbital floor if needed.

4.2 Zygomatic arch fracture

The arch breaks in three places, depressing two fragments: a dimple over the arch, pain on chewing and trismus as fragments touch the coronoid. Elevated through a temporal (*Gillies*) incision with an elevator passed deep to the temporalis fascia; usually stable without fixation.

4.3 Orbital floor blow-out fracture

4.3.1 Mechanism

A blunt object larger than the orbit (a fist or ball) raises orbital pressure, and the thin floor gives way; orbital fat and the inferior rectus herniate into the maxillary sinus while the rim stays intact.

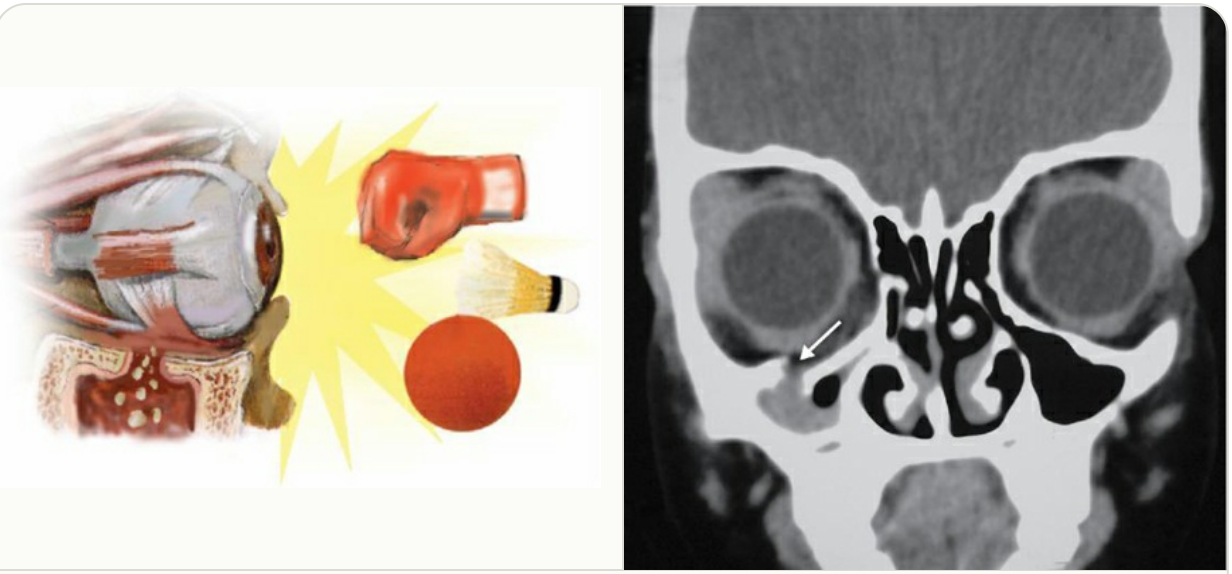


FIGURE 5 First, a blow-out fracture with orbital contents herniating into the maxillary sinus. Second, CT of a blow-out fracture of the right orbital floor.

Dhingra, figs 34.5 and 34.6

4.3.2 Clinical features

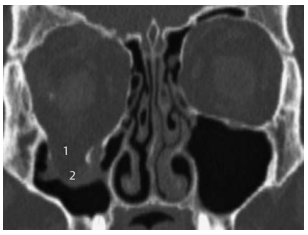
- Bruising of the lid, conjunctiva and sclera.
- *Diplopia*, especially on looking up, from entrapment or swelling of the inferior rectus and oblique.
- *Enophthalmos* and downward displacement of the eye, apparent once the swelling settles.
- Numbness of the cheek and upper lip.

4.3.3 Diagnosis

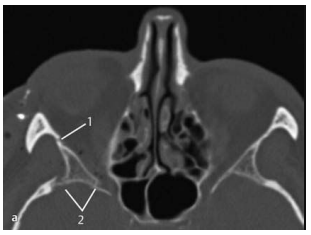
CT in coronal planes. A *forced duction (traction) test* distinguishes mechanical entrapment from nerve or muscle injury. Ophthalmology review for globe injury. In children, a *white-eyed* blow-out with a trapped muscle and vomiting (a trapdoor fracture) needs urgent release.

IMAGING IN THE APP 2 cases

Blow-out fracture, and anterior skull base fractures.



Blow-out Fracture
Trauma · 5 images



Anterior Skull Fractures
Trauma · 3 images

4.3.4 Treatment

Surgery for persistent diplopia with entrapment, enophthalmos over about 2 mm, or a large floor defect: the floor is explored through a lower lid or transconjunctival incision (or endoscopically through the maxillary sinus), herniated tissue is freed, and the floor is reconstructed with titanium mesh, bone or a synthetic plate.

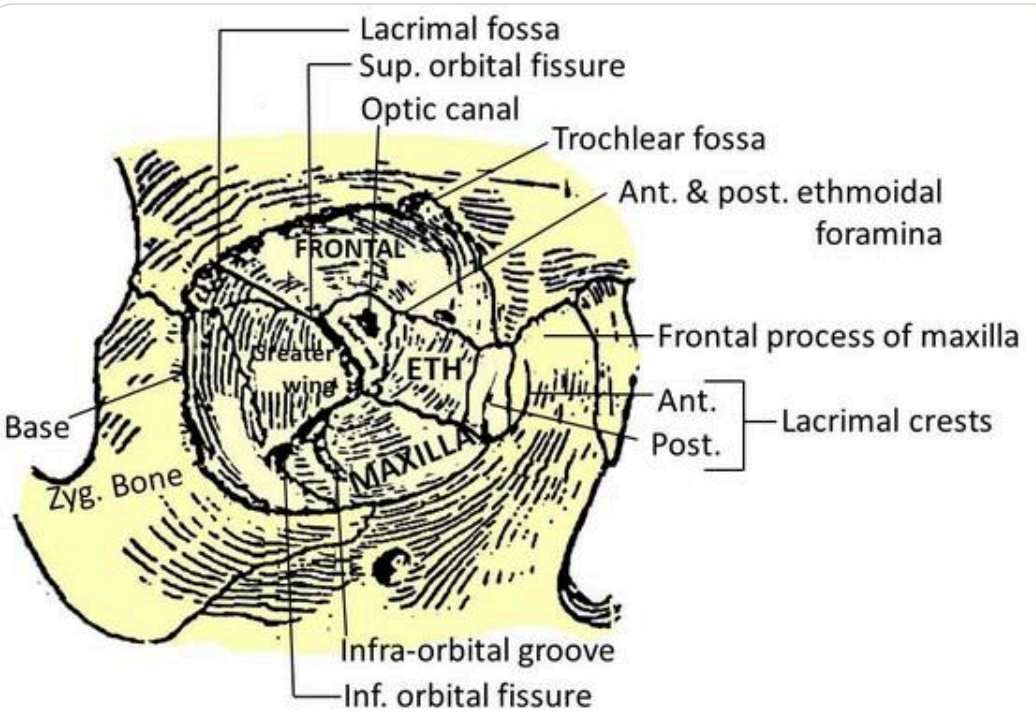


FIGURE 6 The bony orbit, its walls, fissures and foramina.
Anatomy atlas · Doss p. 90

5 CHAPTER FIVE Maxillary fractures

Le Fort's lines.

5.1 Classification

LE FORT FRACTURES

TYPE	LINE	MOVES ON TESTING
Le Fort I (transverse)	Above and parallel to the palate, across the lower septum, maxillary sinuses and pterygoid plates	Tooth-bearing segment only
Le Fort II (pyramidal)	Root of nose, lacrimal bone, orbital floor, upper maxilla, pterygoid plates	The nose and maxilla together
Le Fort III (craniofacial dysjunction)	Root of nose, ethmofrontal junction, orbit, frontozygomatic and zygomatic arch sutures, pterygoid plates	The whole midface

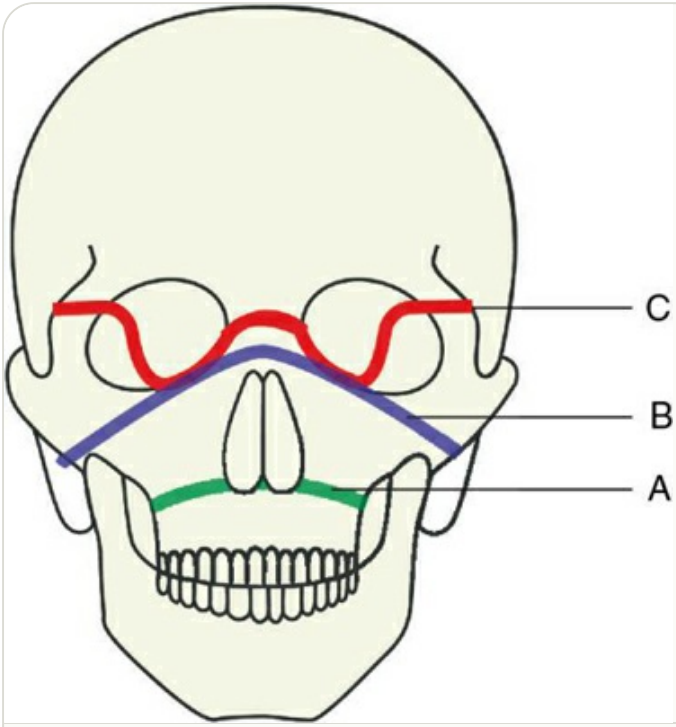


FIGURE 7 Fractures of the maxilla: (A) Le Fort I, (B) Le Fort II, (C) Le Fort III.
Dhingra, fig. 34.7

5.2 Clinical features

Malocclusion with an anterior open bite, elongation and flattening of the midface (*dish face*), mobility of the maxilla, CSF rhinorrhoea in Le Fort II and III (the cribriform plate is involved), bilateral periorbital bruising, numbness, and airway compromise as the maxilla slides back.

5.3 Treatment

Secure the airway (tracheostomy may be needed); restore occlusion with maxillomandibular fixation; open reduction and internal fixation with plates along the facial buttresses.

OPERATIVE GUIDES IN THEATRE 5 operations

Fractures of the Cranio-Orbital and Midfacial Skeleton
Cranio-orbital and midfacial fractures

Fractures of the Orbit
Fractures of the orbit

Fractures of the Mandible
Fractures of the mandible

Septoplasty—Classic and Endoscopic
Septoplasty for septal fractures

Rhinoplasty and Surgery of the Nasal Valve
Rhinoplasty for late deformity

6 CHAPTER SIX Mandible and frontal sinus

Lower and upper thirds.

6.1 Mandibular fractures

The condyle is the commonest site, then the angle, body and symphysis, often fractured in two places. Malocclusion, pain, trismus, step deformity, numbness of the lower lip (inferior alveolar nerve) and sublingual haematoma. Orthopantomogram and CT; treated by open reduction and plating or maxillomandibular fixation.

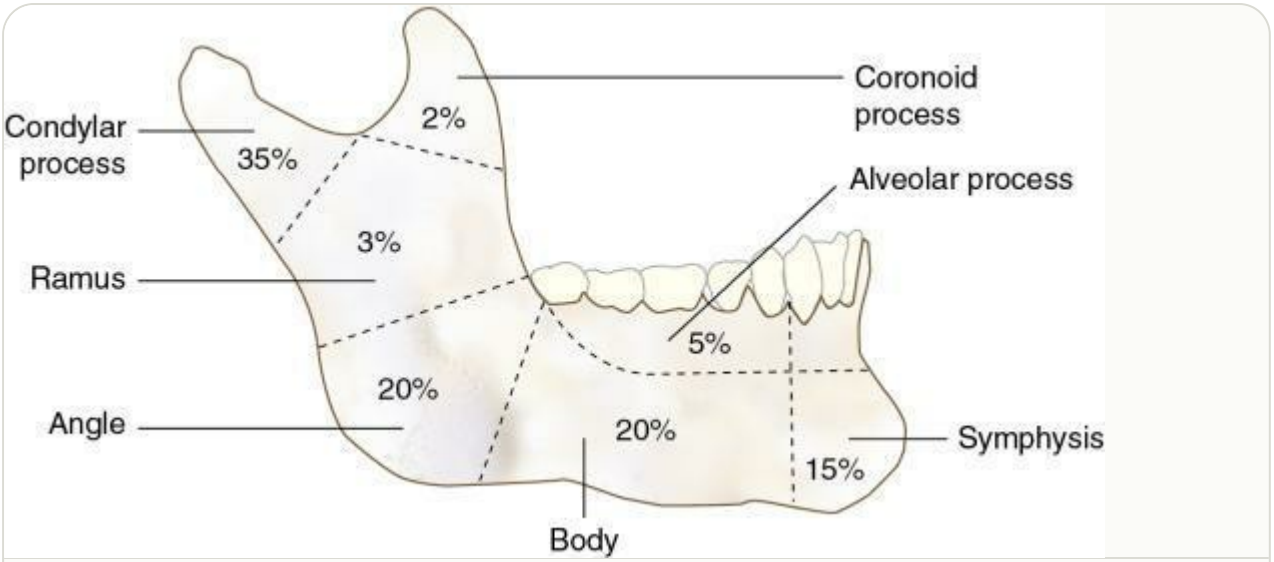


FIGURE 8 Fractures of the mandible by site (Dingman classification); condylar fractures are the commonest.

Dhingra, fig. 34.8

6.2 Frontal sinus fractures

- **Anterior table:** cosmetic depression; reduce and plate if displaced.
- **Frontal sinus outflow injury:** risk of mucocele; obliteration or endoscopic drainage.
- **Posterior table:** dura and brain at risk, with CSF leak; cranialisation of the sinus with neurosurgery for displaced fractures.

Supraorbital ridge and frontal bone fractures are managed with the neurosurgeons.

7 CHAPTER SEVEN Complications

Early and late.

- Airway obstruction and haemorrhage.
- Retrobulbar haemorrhage and blindness.
- CSF leak, meningitis and loss of smell.
- Diplopia, enophthalmos and epiphora.
- Malocclusion and trismus.
- Facial asymmetry, saddle or crooked nose, and nasal obstruction.

8 CHAPTER EIGHT
For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Nasal fracture:** timing of reduction, Walsham's and Asch's forceps, septal haematoma.
- **Naso-orbito-ethmoid fracture:** telecanthus.
- **Tripod fracture:** features and treatment.
- **Blow-out fracture:** mechanism, diplopia on upgaze, enophthalmos, forced duction test, indications for repair.
- **Le Fort I, II and III.**

8.2 Common pitfalls

COMMON PITFALLS

- Missing a septal haematoma.
- Waiting more than two weeks to reduce a displaced nasal fracture.
- Missing a trapdoor fracture in a child with vomiting and a quiet eye.
- Forgetting the cervical spine and the airway in a midface fracture.

Written for this app from Oxford Textbook of Otolaryngology ch. 53, Oxford Specialist Handbook ch. 22, Oxford Handbook of ENT ch. 25, Dhingra ch. 34 and K. J. Lee ch. 56, 57. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Proptosis and the orbit

The orbit shares thin walls with three sinuses, so the rhinologist meets its diseases often: the eye pushed out by a mucocele, a sinus infection or a tumour, the protruding eyes of thyroid disease that the nose can decompress, and the watering eye that a dacryocystorhinostomy can drain. Proptosis is a sign; its direction, speed and sidedness point to the cause.

AT A GLANCE

- Thyroid eye disease is the commonest cause in adults.
- Orbital cellulitis is the commonest cause in children.
- Direction of displacement points to the source.
- Retrobulbar haemorrhage: lateral canthotomy within minutes.
- Drain a subperiosteal abscess for age ≥9, large or non-medial abscess, frontal sinusitis, visual change or no improvement in 48 h.

1 CHAPTER ONE Overview

The eye pushed forward.

1.1 Definitions

Proptosis is forward displacement of the eyeball; *exophthalmos* is often used for proptosis from thyroid eye disease. Measured with an exophthalmometer (Hertel): a difference of more than 2 mm between the eyes, or a reading above about 20 mm, is abnormal. *Pseudoproptosis* (a large eye in myopia, lid retraction, or a small opposite eye) must be excluded.

1.2 Applied anatomy

The orbit is a four-sided pyramid. Its medial wall (the lamina papyracea) separates it from the ethmoids, its floor from the maxillary sinus, its roof from the frontal sinus and anterior cranial fossa; the sphenoid lies at the apex beside the optic canal. The

orbital periosteum and septum limit the spread of infection; the valveless ophthalmic veins drain to the cavernous sinus.

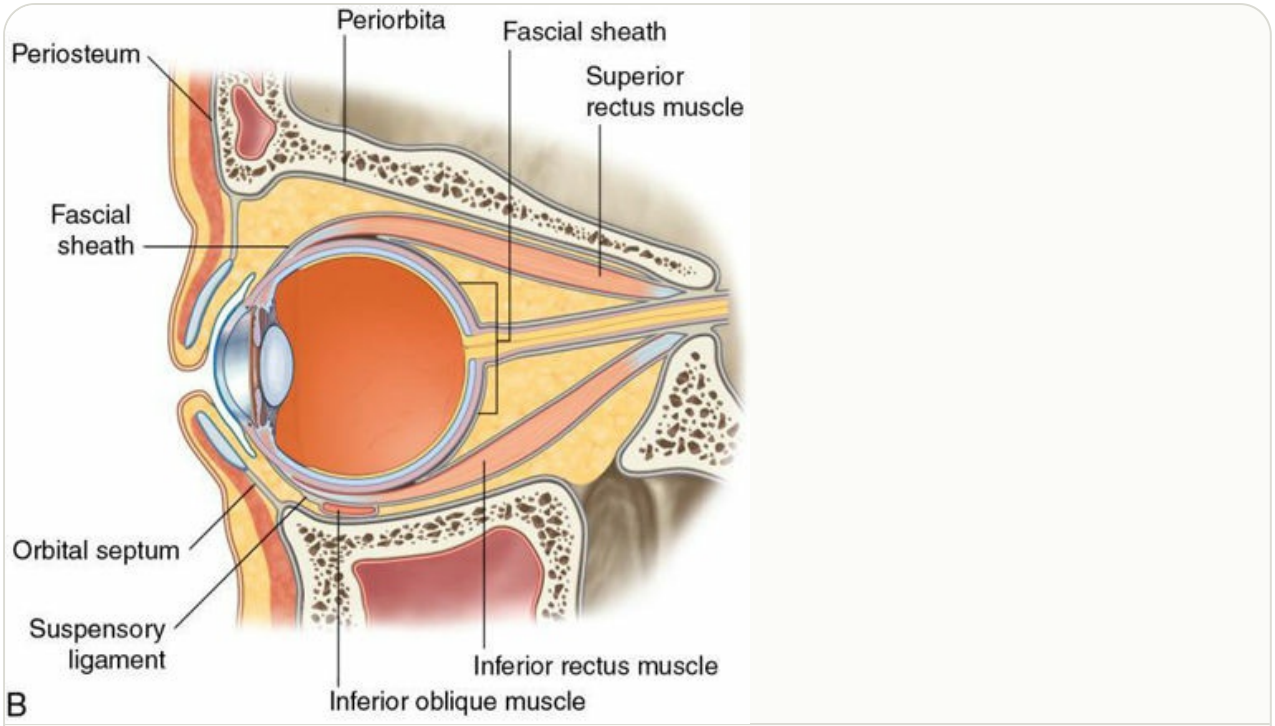


FIGURE 1 Anatomy of the orbit: its bones and relations.

Dhingra, fig. 41.1

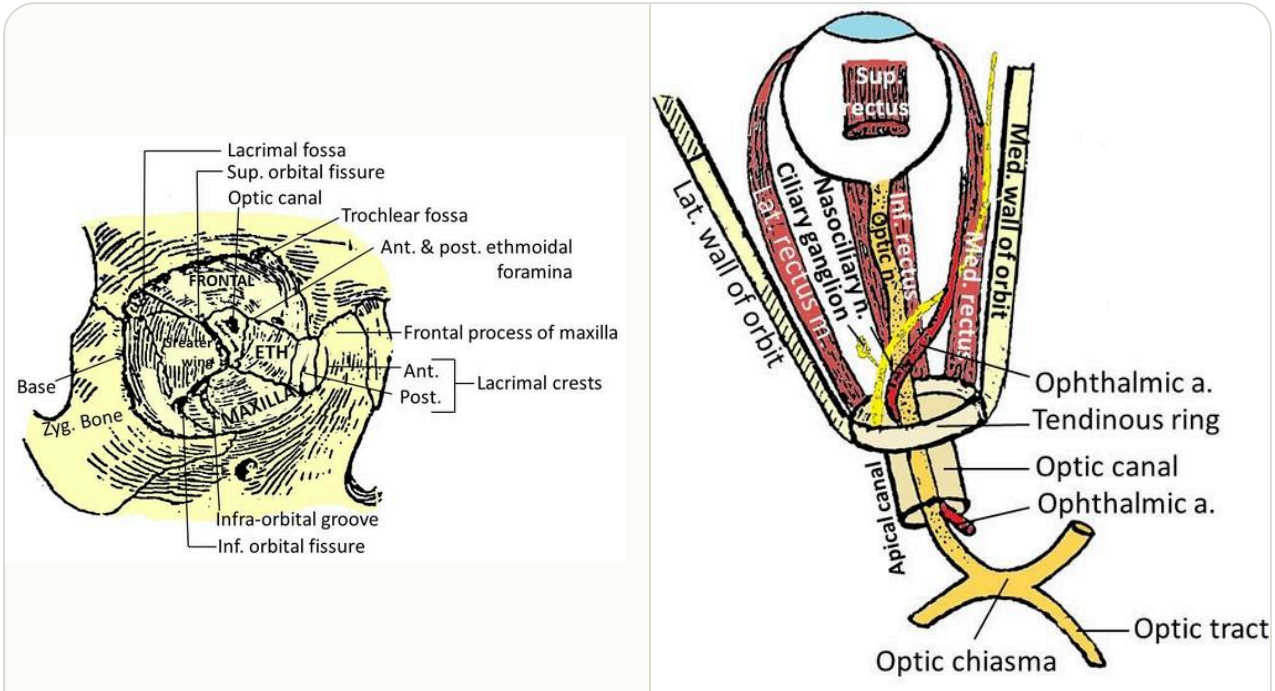


FIGURE 2 The bony orbit with its walls, fissures and foramina, and the optic canal and tendinous ring at the apex.

Anatomy atlas · Doss p. 90, Doss p. 95

1.3 Why it matters for ENT

Sinus infections, mucoceles, fungal disease, polyps, angiofibroma, inverted papilloma and sinonasal cancers all invade the orbit; endoscopic surgery can injure it; and the orbit can be decompressed and drained through the nose.

2 CHAPTER TWO Causes

VEIN.

CAUSES OF PROPTOSIS (VEIN)

GROUP	EXAMPLES
Vascular	Venous varix, cavernous haemangioma, carotid-cavernous fistula, orbital haematoma
Endocrine	Thyroid eye disease (Graves ophthalmopathy)
Inflammatory and infective	Orbital cellulitis and abscess, idiopathic orbital inflammation, fungal sinusitis, granulomatosis with polyangiitis, dacryoadenitis
Neoplastic	Primary orbital tumours, spread from the sinuses or cranium, metastases

2.1 From the sinuses

Mucoceles of the frontal and ethmoid sinuses, allergic fungal polyposis, inverted papilloma, angiofibroma, olfactory neuroblastoma and sinus carcinoma, and orbital complications of sinusitis (see those topics).



FIGURE 3 First, proptosis with lateral displacement of the eye from allergic fungal polyposis. Second, a left ethmoid mucocele displacing the eye.

Dhingra, fig. 41.2

2.2 Clues from the history and examination

- **Unilateral or bilateral:** thyroid eye disease is the commonest cause of both bilateral and unilateral proptosis in adults; orbital cellulitis is the commonest cause in children.
- **Speed:** sudden (haemorrhage, cellulitis), weeks (inflammation, malignancy), years (benign tumours, mucoceles, fibrous dysplasia).
- **Direction:** *axial* proptosis suggests a lesion inside the muscle cone (haemangioma, optic nerve tumour, thyroid eye disease); *non-axial* displacement points away from the lesion: down and out with a frontoethmoid mucocele, laterally with an ethmoid lesion, upwards with a maxillary tumour.
- **Variable proptosis** with straining or bending: venous varix.
- **Pulsatile proptosis** with a bruit and arterialised conjunctival vessels: carotid-cavernous fistula.
- **Pain:** inflammation, infection or malignancy.

3 CHAPTER THREE Important conditions

Inside and around the orbit.

3.1 Thyroid eye disease

The commonest cause of proptosis in adults, bilateral or one-sided, in hyperthyroid, euthyroid or hypothyroid patients. Lid retraction, lid lag, chemosis, restricted movement and exposure keratopathy; CT shows enlargement of the muscle bellies with sparing of their tendons. Optic nerve compression at the apex threatens vision.

3.2 Idiopathic orbital inflammation

Orbital pseudotumour: painful proptosis, worse on eye movement; may involve the muscles (myositis, including the tendons, unlike thyroid disease), lacrimal gland, sclera or optic nerve. Biopsy shows non-specific inflammation; treated with steroids, immunosuppression or radiotherapy.

3.3 Vascular lesions

- **Cavernous haemangioma:** the commonest benign orbital tumour of adults; painless, slowly progressive axial proptosis from a well-defined, enhancing intraconal mass; excised by lateral orbitotomy or endoscopically.
- **Capillary haemangioma:** the commonest orbital tumour of infants; usually involutes by about seven; propranolol or steroids.
- **Venous varix:** proptosis that comes and goes with straining.
- **Carotid-cavernous fistula:** pulsatile proptosis, bruit, chemosis and arterialised vessels; endovascular closure.

3.4 Tumours

- **Lymphoma:** the commonest malignant orbital tumour of adults, often anterior and palpable; biopsy, radiotherapy and chemotherapy.
- **Rhabdomyosarcoma:** the commonest primary orbital malignancy of children (around

six or seven years), with rapid proptosis; chemoradiation.

- **Dermoid cyst:** the commonest benign orbital tumour of children, at the frontozygomatic suture or deeper; excised.
- **Optic nerve glioma** (children, neurofibromatosis) and **optic nerve sheath meningioma** (adults).
- **Lacrimal gland tumours:** pleomorphic adenoma and adenoid cystic carcinoma in the upper outer orbit.
- **Metastases:** breast most often, then lung, prostate and kidney; neuroblastoma in children.

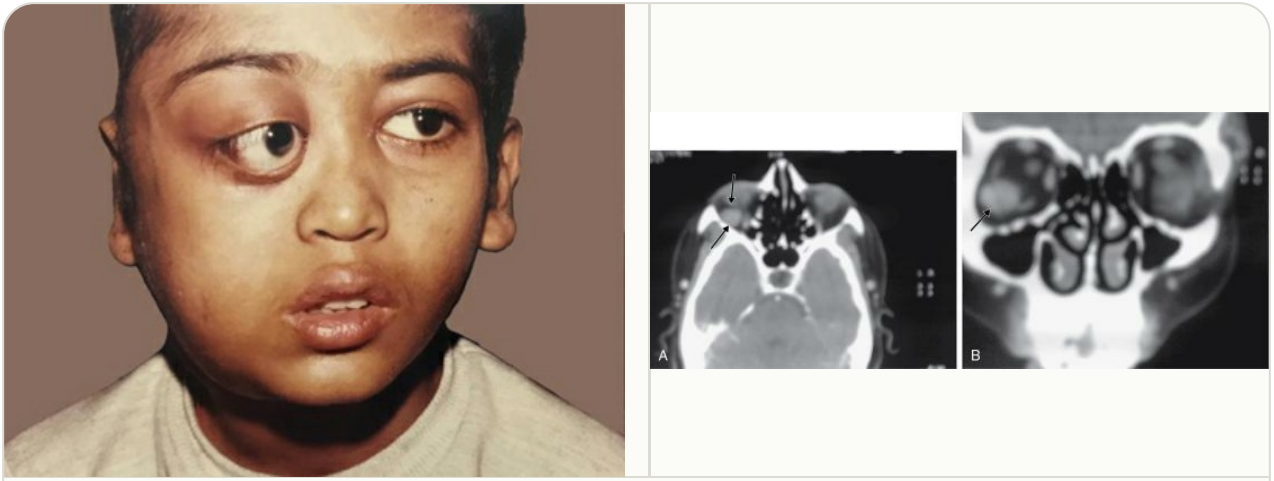
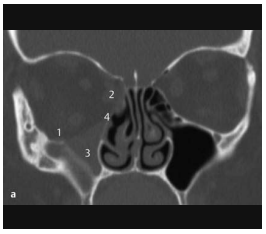


FIGURE 4 First, proptosis with medial displacement of the eye from lymphoma in a child. Second, a dermoid cyst of the right orbit on axial and coronal CT.

Dhingra, figs 41.4 and 41.5

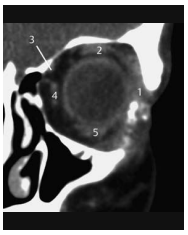
IMAGING IN THE APP 5 cases

Orbital decompression for Graves disease, orbital amyloidosis, mucocele, osteoma and orbital cellulitis.



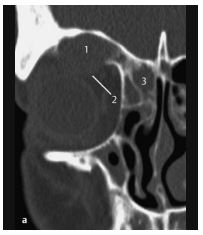
Orbital Decompression in Graves Disease

After surgery · 2 images



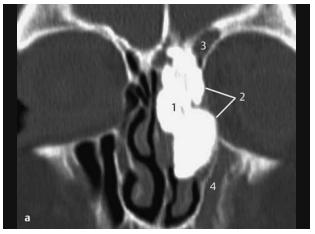
Amyloidosis in the Orbit

Infection & inflammation · 1 image



Mucocele

Infection & inflammation · 13 images



Osteoma, Gardner Syndrome, and Complications

Bone disease · 3 images



Orbital Cellulitis

Infection & inflammation · 6 images

4 CHAPTER FOUR Evaluation

A joint ENT and eye assessment.

4.1 Examination

Visual acuity, colour vision and pupils (a relative afferent pupillary defect means optic nerve compromise), eye movements and diplopia, the degree and direction of proptosis, lids, conjunctiva and cornea, fundus, and nasal endoscopy.

4.2 Investigations

- **CT** of the orbits and sinuses: bone, sinus disease, fractures, muscle size.
- **MRI** for soft tissue, the optic nerve, apex and intracranial extension.
- Thyroid function and antibodies; blood tests for vasculitis and lymphoma as indicated.
- Biopsy for undiagnosed masses, endoscopically or by orbitotomy.

5 CHAPTER FIVE Procedures

What the nose can do for the eye.

5.1 Endoscopic orbital decompression

Removal of the medial wall (lamina papyracea) and the medial part of the floor, opening the periorbita so orbital fat can prolapse into the ethmoid and maxillary spaces. For thyroid eye disease with optic neuropathy, severe exposure or disfiguring proptosis. Diplopia may follow.

5.2 Optic nerve decompression

Removal of the bone of the optic canal through the sphenoid in traumatic or compressive optic neuropathy; its value in trauma is debated.

5.3 Dacryocystorhinostomy

For epiphora from obstruction of the nasolacrimal duct: a window is made from the lacrimal sac into the middle meatus, through the lacrimal bone and frontal process of the maxilla, endoscopically or through an external skin incision, sometimes with silicone tubes.

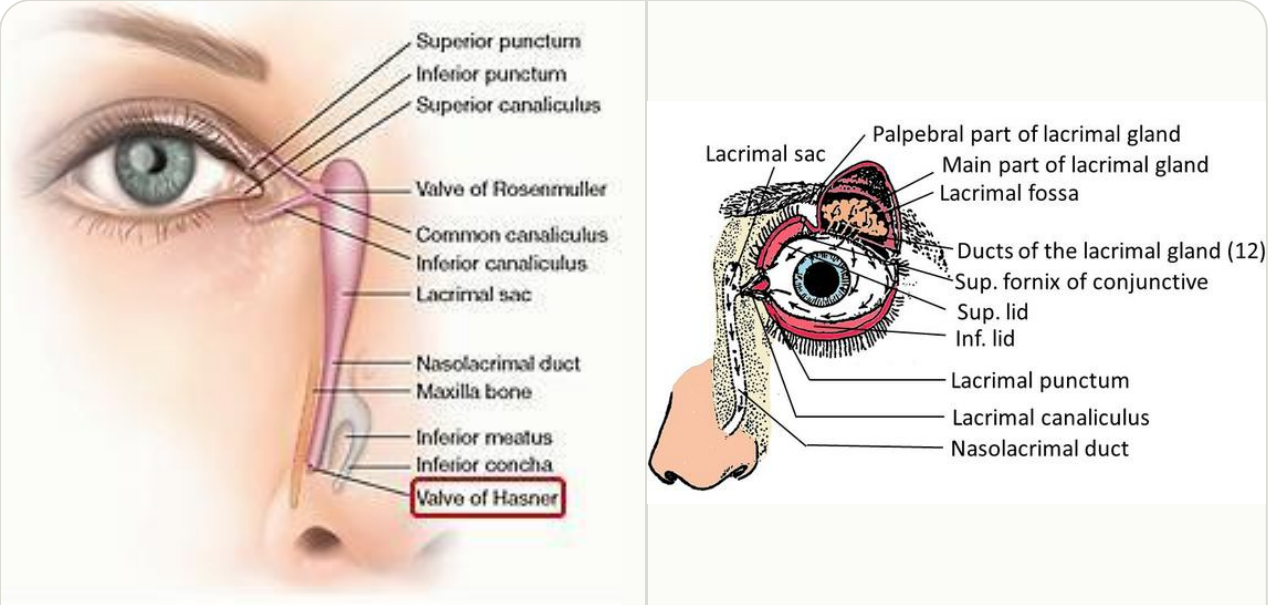


FIGURE 5 The lacrimal drainage system, and the lacrimal apparatus.
Anatomy atlas · ENT Steps p. 100, Doss p. 92

5.4 Drainage of orbital abscess

Endoscopic or external drainage of a subperiosteal or orbital abscess complicating sinusitis.

5.5 Lateral canthotomy and cantholysis

An emergency release for retrobulbar haemorrhage after trauma or sinus surgery: a tense, proptosed eye with rising pressure and falling vision. Cut the lateral canthal tendon and its inferior crus within minutes.

RETROBULBAR HAEMORRHAGE

Pain, tense proptosis, a hard globe, ophthalmoplegia and loss of vision after injury or sinus surgery. Do not wait for imaging: perform lateral canthotomy and cantholysis at once, then call ophthalmology.

5.6 Orbital exenteration

Removal of the orbital contents for malignancy invading beyond the periorbita.

OPERATIVE GUIDES IN THEATRE 6 operations

Orbital Decompression
Orbital decompression
Optic Nerve Decompression
Optic nerve decompression
Dacryocystorhinostomy
Dacryocystorhinostomy
Orbital Abscess
Orbital abscess drainage
Lateral Canthotomy
Lateral canthotomy
Orbital Exenteration
Orbital exenteration

WHO NEEDS DRAINAGE IN CHILDREN

Most children with orbital complications are treated medically at first, with intravenous antibiotics covering *Streptococcus anginosus* (milleri), the commonest organism in complicated sinusitis in young people, plus nasal decongestant, and with vision, pupils, colour vision and eye movements checked at least every few hours. A small **medial subperiosteal abscess** in a child with normal vision often resolves without surgery. Features that favour early drainage (based on the criteria of Garcia and Harris) are age nine years or older, frontal sinusitis, a non-medial or large abscess, gas in the abscess, a dental source, any reduction in visual acuity or colour vision, an afferent pupillary defect, intracranial complications, or failure to improve within 24 to 48 hours. Adolescents with frontal sinusitis and orbital involvement need imaging of the brain as well, because *S. anginosus* readily forms intracranial abscesses and Pott's puffy tumour.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Causes of proptosis** (VEIN) and the commonest in adults and children.
- **Direction of displacement** as a clue.
- **Thyroid eye disease** and endoscopic orbital decompression.
- **Commonest orbital tumours** in adults and children.

- **Retrobulbar haemorrhage** and lateral canthotomy.
- **Dacryocystorhinostomy.**

6.2 Common pitfalls

COMMON PITFALLS

- Waiting for a scan in retrobulbar haemorrhage.
- Missing a sinus cause in one-sided proptosis.
- Forgetting colour vision and the pupils when assessing the optic nerve.

Written for this app from Oxford Textbook of Otolaryngology ch. 36, Oxford Specialist Handbook ch. 21, Oxford Handbook of ENT ch. 9, Dhingra ch. 41 and K. J. Lee ch. 58. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Nose and sinus operations: principles

Nose and sinus operations are best learned by what each one opens, removes and risks. The old operations (antral puncture, inferior meatal antrostomy, Caldwell-Luc, submucous resection) are still examined and occasionally used; endoscopic sinus surgery has replaced most of them by restoring drainage through the natural openings. The step-by-step guides are in Theatre.

AT A GLANCE

- Septoplasty has replaced submucous resection.
- Endoscopic sinus surgery restores drainage through the natural ostia.
- Major complications: orbital injury, CSF leak, carotid injury.
- Know the danger areas: lamina papyracea, cribriform plate, anterior ethmoidal artery.
- Appropriateness: Lund–Mackay ≥1, 8 weeks of maximal medical therapy, SNOT-22 ≥20.

1 CHAPTER ONE Overview

From puncture to endoscope.

1.1 The operations

NOSE AND SINUS OPERATIONS

- DIAGNOSTIC **Nasal endoscopy** antral puncture (proof puncture)
- SEPTUM AND TURBINATES **Septoplasty** submucous resection, turbinate reduction
- MAXILLARY SINUS **Middle meatal antrostomy** inferior meatal antrostomy, Caldwell-Luc
- ETHMOID, FRONTAL, SPHENOID **Endoscopic sinus surgery** external ethmoidectomy, frontal trephine, Lynch-Howarth, osteoplastic flap, Draf procedures
- SKULL BASE **Endoscopic skull base surgery** CSF leak repair, pituitary surgery
- OTHER **Oroantral fistula repair** balloon sinuplasty

1.2 The danger areas

The lamina papyracea (orbit), the cribriform plate and its thin lateral lamella (CSF leak), the anterior ethmoidal artery in the skull base (orbital haematoma), the optic nerve and carotid artery in the sphenoid and Onodi cells, and the nasolacrimal duct in front of the maxillary ostium.

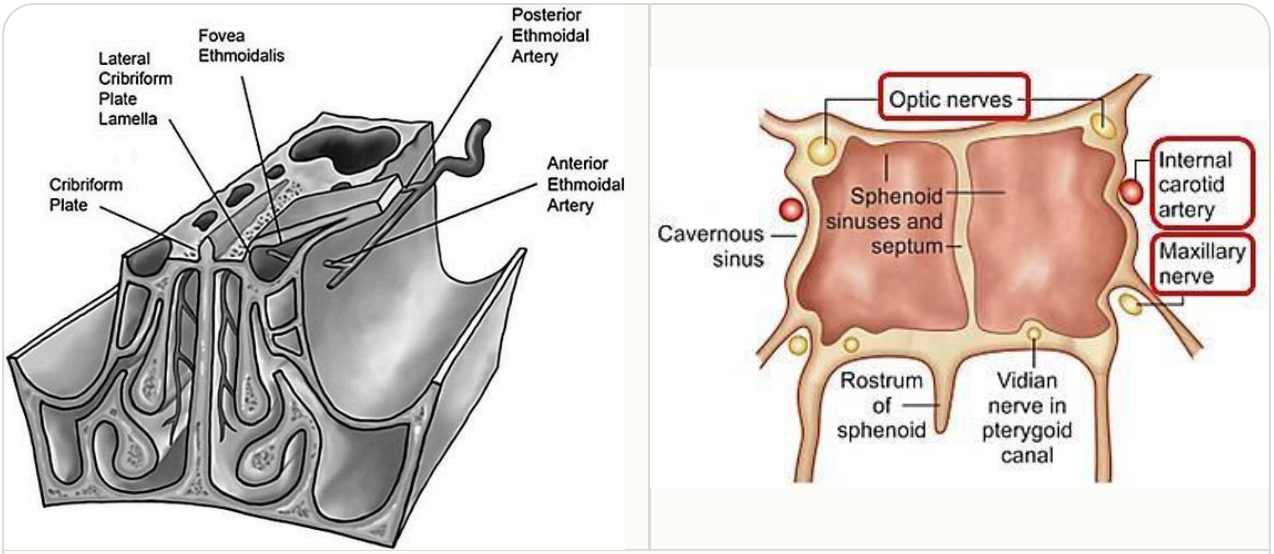


FIGURE 1 The ethmoid roof and anterior ethmoidal artery, and the relations of the sphenoid sinus: optic nerve, internal carotid artery and vidian nerve.

Anatomy atlas · ENT Steps p. 139, ENT Steps p. 154

2 CHAPTER TWO Diagnostic nasal endoscopy

The modern examination.

2.1 Indications

Every chronic nasal complaint: obstruction, discharge, epistaxis, smell loss, headache, suspected tumour or CSF leak, and follow-up after surgery; biopsies and swabs under vision.

2.2 Technique

After topical decongestant and anaesthetic (4% lignocaine with a vasoconstrictor), with the patient sitting or lying, a 4 mm 0° or 30° rigid endoscope makes three

passes:

- **First pass:** along the floor to the nasopharynx: inferior turbinate and meatus, Eustachian tube, fossa of Rosenmüller.
- **Second pass:** between the middle and inferior turbinates to the sphenoethmoidal recess and superior meatus.
- **Third pass:** under the middle turbinate into the middle meatus: uncinate process, bulla, hiatus semilunaris and accessory ostia.

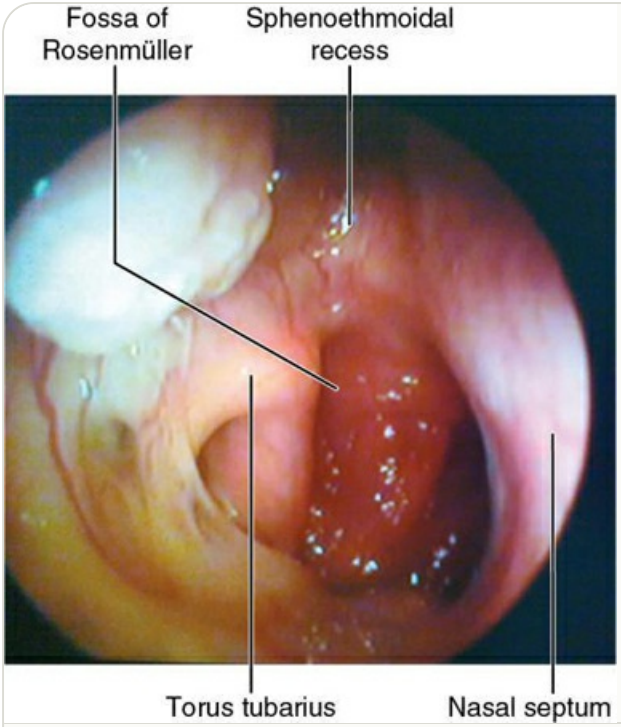


FIGURE 2 The region of the Eustachian tube, sphenoethmoidal recess, choana and fossa of Rosenmüller.

Dhingra, fig. 89.1

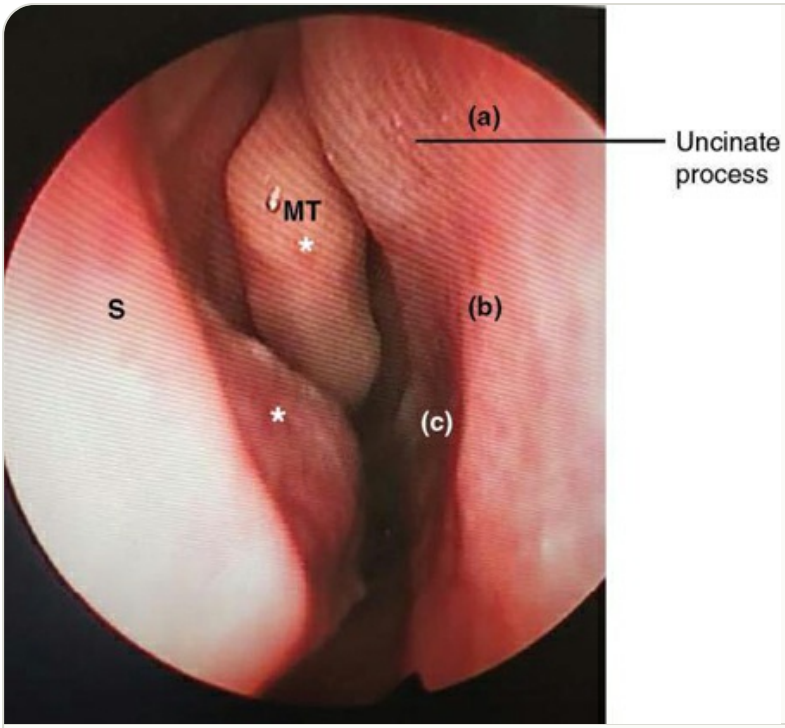


FIGURE 3 Endoscopic view of the left nasal cavity showing the uncinate process, with sites for injection on the lateral wall and middle turbinate.

Dhingra, fig. 90.2

CHAPTER THREE

3 Old maxillary operations

Still examined.

3.1 Antral puncture (proof puncture)

3.1.1 Indications

Chronic or subacute maxillary sinusitis, to confirm the diagnosis and wash out pus; to obtain a specimen for culture or cytology. Contraindicated in acute sinusitis for fear of osteomyelitis.

3.1.2 Technique

Under local anaesthesia, a Lichtwitz trocar and cannula pierce the thin lateral wall of the inferior meatus 1.5 to 2 cm behind the front of the inferior turbinate, directed towards the ear on the same side; the sinus is irrigated with warm saline until the return is clear.

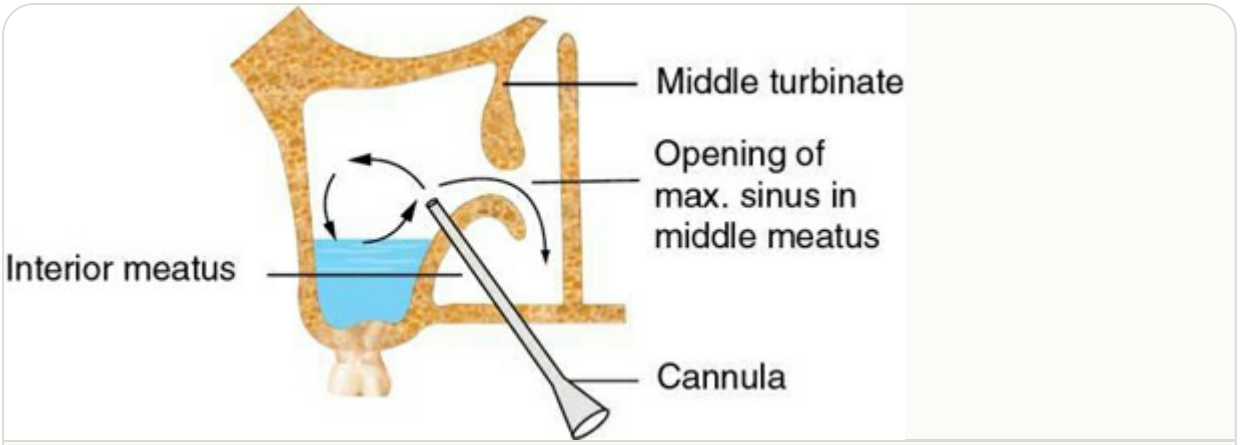


FIGURE 4 Antral puncture through the inferior meatus.
Dhingra, fig. 84.1

3.1.3 What comes back

Thin amber fluid with cholesterol crystals means a cyst; blobs of mucopus mean hyperplastic sinusitis; foul pus mixing with the fluid means suppuration.

3.1.4 Complications

Cheek swelling (the cannula in the soft tissues), orbital injury through the roof, puncture of the posterior wall, bleeding and, rarely, fatal *air embolism* (never blow air into the antrum).

3.2 Inferior meatal antrostomy

A window under the inferior turbinate for drainage. It fails because mucociliary clearance carries mucus to the natural ostium, not to the new opening; risks bleeding and nasolacrimal duct injury. Now obsolete.

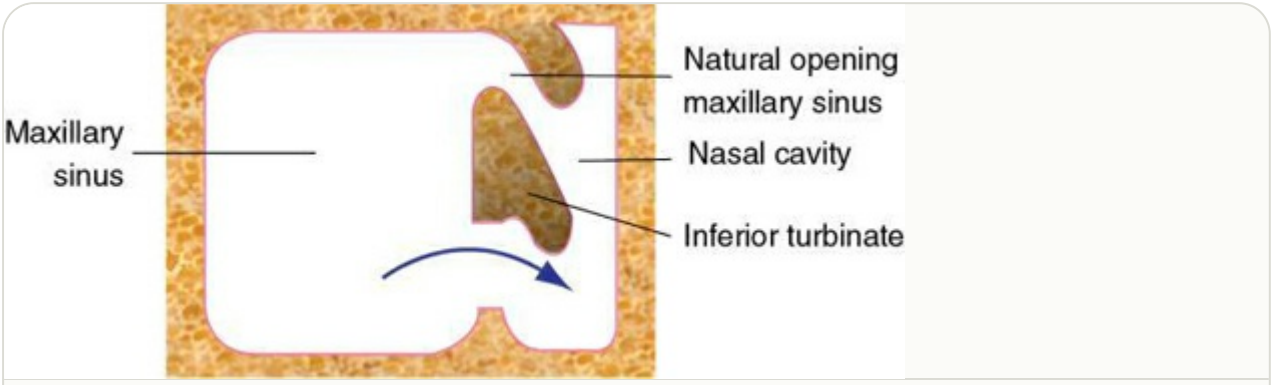


FIGURE 5 Intranasal antrostomy in the inferior meatus.
Dhingra, fig. 85.1

3.3 Caldwell-Luc operation

3.3.1 What it is

Opening the maxillary sinus through its anterior wall in the canine fossa via a sublabial incision, with a nasoantral window.

3.3.2 Indications (historical and current)

Irreversible sinus disease, foreign bodies and tooth roots, dental cysts, oroantral fistula, biopsy of antral tumours, recurrent antrochoanal polyp, maxillary and orbital floor fractures, access to the ethmoids (Horgan) or pterygopalatine fossa (maxillary artery ligation, vidian neurectomy). Avoided in children under about 17, whose developing teeth lie in the anterior wall.

3.3.3 Complications

Bleeding, numbness of the cheek and teeth (infraorbital and superior alveolar nerves), nasolacrimal duct injury, sublabial (oroantral) fistula and, rarely, osteomyelitis.

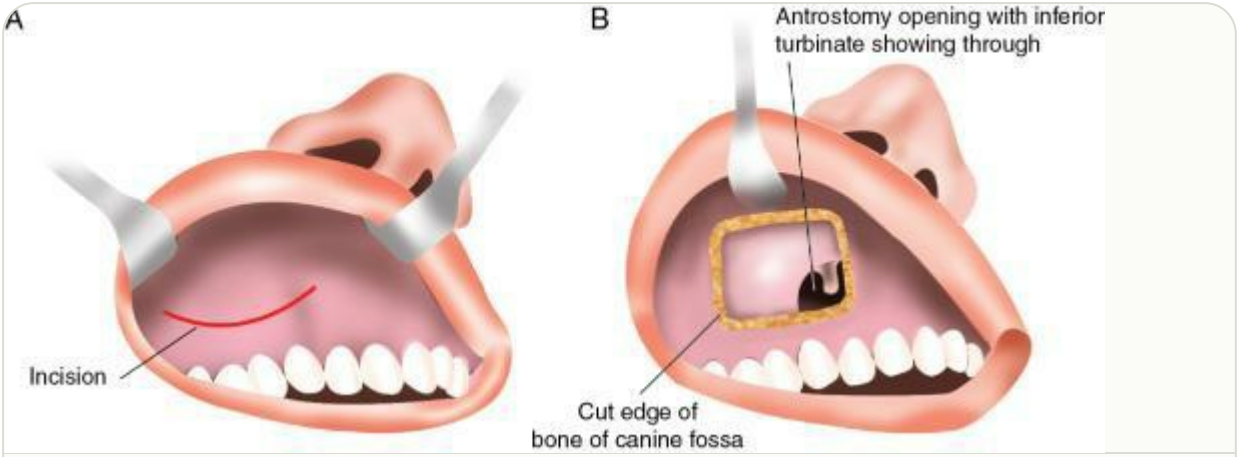


FIGURE 6 The Caldwell-Luc operation: (A) incision, (B) inside view and position of the antrostomy.

Dhingra, fig. 86.1

3.3.4 Present status

Most indications are now handled endoscopically; it remains useful for disease in the anterior and lateral recesses of the sinus and combined approaches.

4 CHAPTER FOUR Septal and turbinate surgery

The airway.

4.1 Submucous resection

Flaps raised on both sides of the septum and all deformed cartilage and bone removed, leaving only dorsal and caudal struts. Not done before about 17.

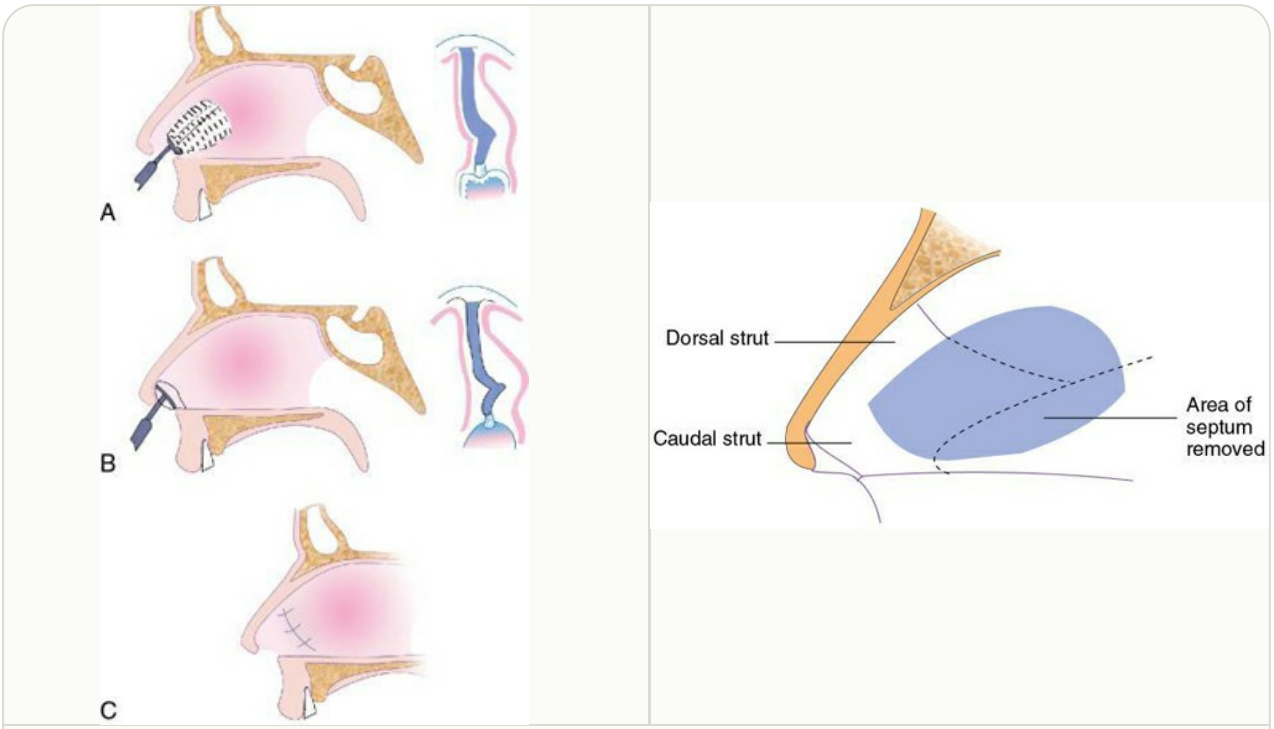


FIGURE 7 First, submucous resection: incision and elevation of the flaps. Second, the dorsal and caudal struts that must be preserved to avoid a supratip depression.

Dhingra, figs 87.1 and 87.2

4.2 Septoplasty

The conservative operation that has replaced submucous resection: a Killian (or hemitransfixion for caudal deviation) incision, a flap on one side, limited removal of bone and cartilage, and correction of the rest by scoring, cross-hatching, shaving or wedge excision and repositioning on the maxillary crest; can be done in children.

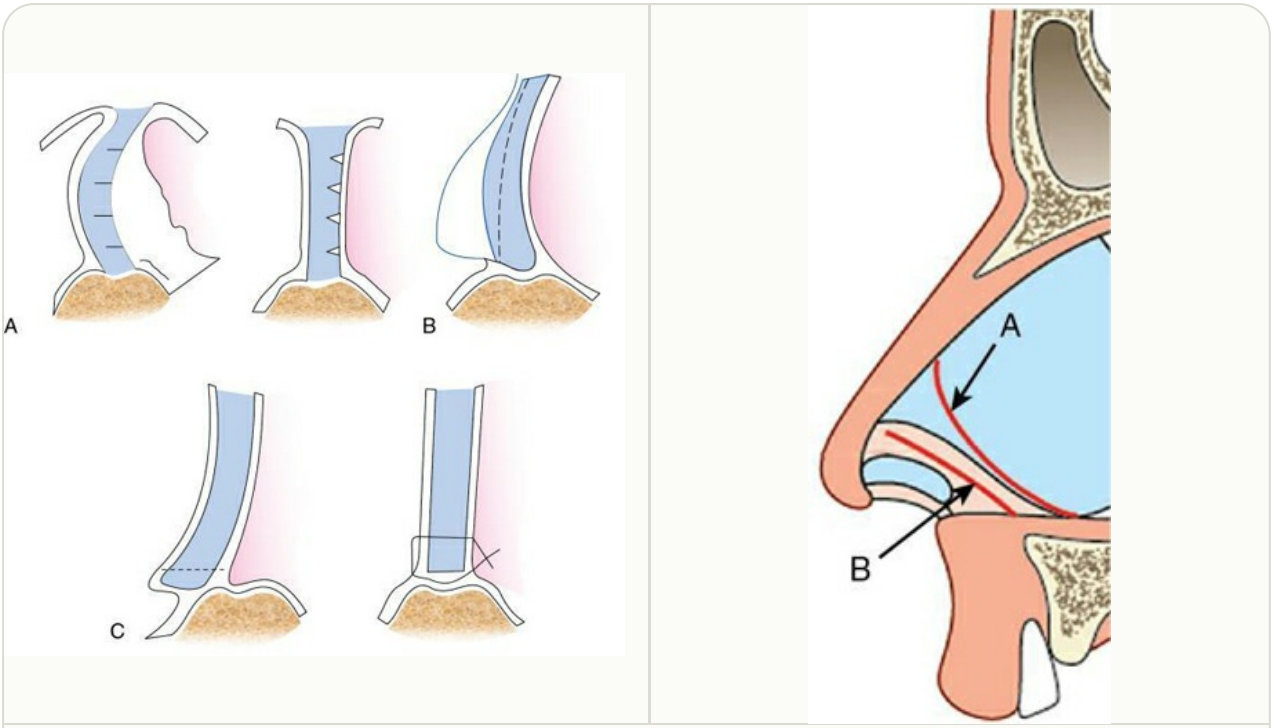


FIGURE 8 First, straightening septal cartilage by scoring the concave side, shaving the convex side, or repositioning a dislocated cartilage. Second, septal incisions: (A) Killian's, (B) hemitransfixion.

Dhingra, figs 88.1 and 88.2

SUBMUCOUS RESECTION OR SEPTOPLASTY?

	SUBMUCOUS RESECTION	SEPTOPLASTY
Dissection	Extensive, flaps on both sides	Limited, one flap
Cartilage	Removed, leaving dorsal and caudal struts	Mostly preserved and repositioned
Age	Not before about 17	Children too
Complications	Supratip depression, columellar retraction, flapping septum, perforation	Fewer
Revision	Difficult	Easier

4.3 Indications for septoplasty

Obstruction from a deviated septum, as part of septorhinoplasty, epistaxis from a spur, sinusitis or contact headache from deviation, access for endoscopic sinus, lacrimal or pituitary surgery, and sleep-disordered breathing.

4.4 Complications of septal surgery

Bleeding, septal haematoma and abscess, perforation, saddle nose and supratip depression, columellar retraction, synechiae, anosmia, and rarely CSF leak.

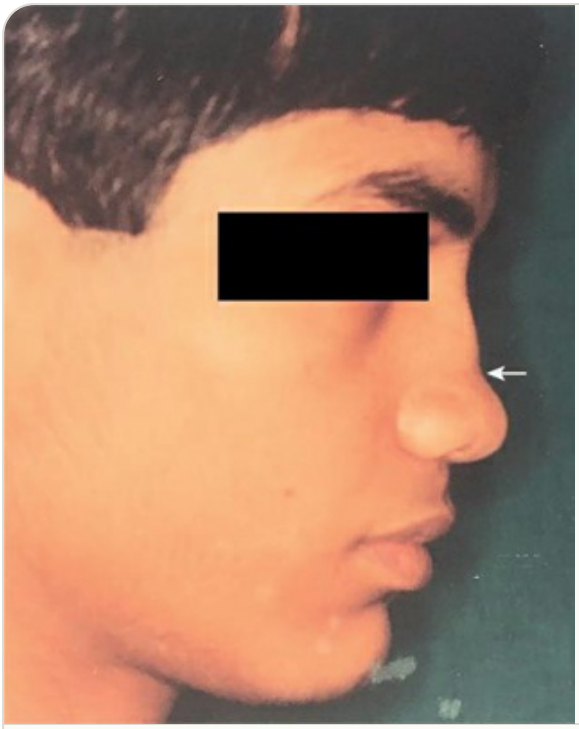


FIGURE 9 A supratip depression after extensive septal surgery.
Dhingra, fig. 87.3

4.5 Turbinate surgery

Submucosal reduction (microdebrider, radiofrequency, coblation), outfracture, partial turbinectomy, laser; preserve mucosa to avoid an empty nose.

5 CHAPTER FIVE Endoscopic sinus surgery

Restoring drainage.

5.1 Principle

Functional endoscopic sinus surgery removes disease in the ostiomeatal complex and enlarges the natural ostia, so that ventilation and mucociliary clearance

recover; the mucosa is preserved.

5.2 Indications

Chronic rhinosinusitis with or without polyps after failed medical treatment, recurrent acute sinusitis, complications of sinusitis (orbital abscess, mucocele), fungal sinusitis, antrochoanal polyp, and access for extended procedures: CSF leak repair, orbital and optic nerve decompression, dacryocystorhinostomy, sphenopalatine artery ligation, choanal atresia, tumours and the skull base.

5.3 Preparation

CT of the sinuses reviewed systematically before every operation (the *CLOSE* checklist: cribriform plate, lamina papyracea, Onodi cell, sphenoid and carotid, ethmoidal artery); topical decongestion and infiltration.

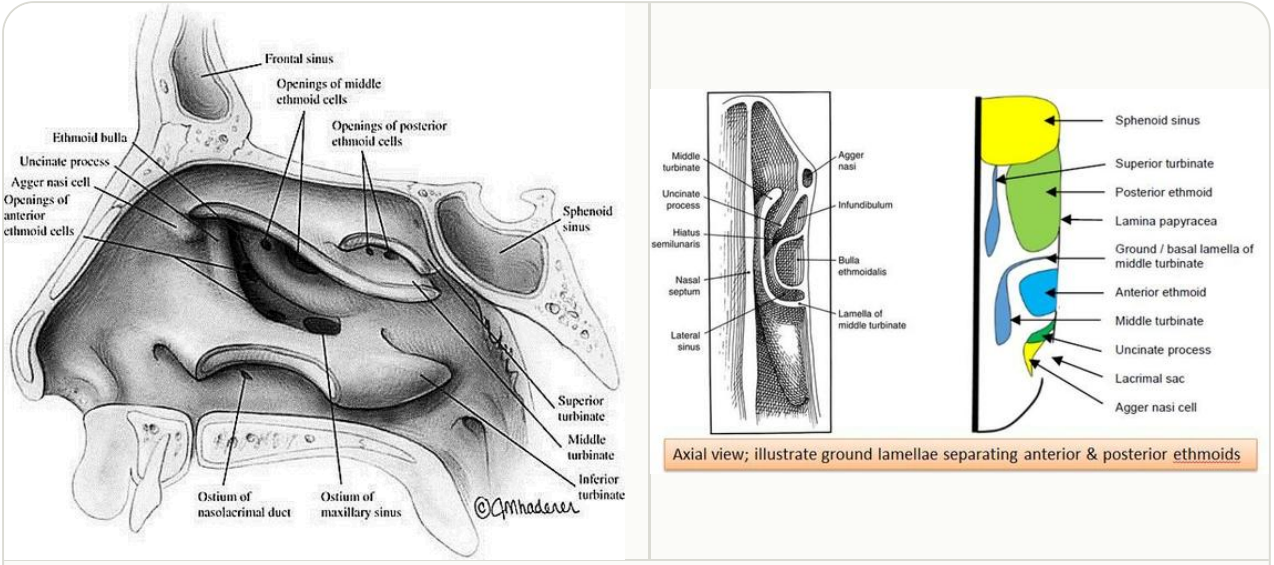


FIGURE 10 The lateral nasal wall with the middle turbinate removed (uncinate, bulla and hiatus), and the ground lamella separating anterior from posterior ethmoid cells.

Anatomy atlas · ENT Steps p. 108, ENT Steps p. 142

5.4 Steps (anterior to posterior, Messerklinger-Stammberger)

1. Medialise the middle turbinate; trim a concha bullosa by removing its lateral lamella.
2. **Uncinectomy.**

3. **Middle meatal antrostomy:** find and enlarge the natural maxillary ostium, which lies behind the lower uncinate, above the inferior turbinate.
4. **Bullectomy** (anterior ethmoidectomy), staying away from the orbit, skull base and anterior ethmoidal artery.
5. **Posterior ethmoidectomy** through the lower, medial part of the basal (ground) lamella; beware an Onodi cell around the optic nerve.
6. **Frontal recess clearance and frontal sinusotomy** if diseased: the opening lies lateral to the middle turbinate attachment, medial to the orbit, in front of the anterior ethmoidal artery and behind the agger nasi; respect the mucosa to prevent stenosis.
7. **Sphenoidotomy** through the natural ostium, about 1 cm above the choana, medial to the superior turbinate.

The posterior-to-anterior (*Wigand*) technique starts at the sphenoid and follows the skull base forwards, for extensive polyposis and revisions.



FIGURE 11 Endoscopic surgery in progress, with instruments passed through the nose and the view on a monitor.
Dhingra, fig. 90.1

5.5 Frontal sinus procedures

Draf I (frontal recess clearance), Draf IIa and IIb (enlarging the frontal ostium), and Draf III (the endoscopic modified Lothrop, joining both frontal sinuses across the midline). External options: trephination, Lynch-Howarth frontoethmoidectomy, the osteoplastic flap with obliteration, and cranialisation after posterior wall fractures.

5.6 Balloon sinuplasty

Dilatation of the natural ostia with a balloon catheter, for selected patients with limited disease.

5.7 After surgery

Saline irrigation, steroid sprays for polyposis or allergy, antibiotics as needed, and endoscopic cleaning of crusts and division of adhesions at weekly visits until the cavity heals.

5.8 Complications

COMPLICATIONS OF ENDOSCOPIC SINUS SURGERY

TYPE	EXAMPLES
Minor	Synechiae, crusting, bleeding, middle turbinate lateralisation, periorbital bruising, surgical emphysema of the lid, dental pain
Orbital	Breach of the lamina papyracea, orbital haematoma (anterior ethmoidal artery), injury to the medial rectus, nasolacrimal duct injury and epiphora, optic nerve injury and blindness
Intracranial	CSF leak, meningitis, pneumocephalus, brain injury
Vascular	Internal carotid artery injury, sphenopalatine and ethmoidal haemorrhage
Other	Anosmia, frontal sinus stenosis and mucocele

ORBITAL HAEMATOMA DURING SURGERY

Proptosis, a tense eye and falling vision: stop, remove packs, and perform a lateral canthotomy and cantholysis; consider medical treatment and orbital decompression.

OPERATIVE GUIDES IN THEATRE 16 operations

Office-Based Diagnosis of Sinonasal Disorders
Office endoscopy

Septoplasty—Classic and Endoscopic
Septoplasty

Surgery of the Turbinate
Turbinates

Balloon Sinus Dilation
Balloon dilation
Antral Lavage; Endoscopic Middle Meatal Antrostomy/Antrochoanal Polyp
Antrostomy
Anterior Antrostomy: The Caldwell-Luc Operation
Caldwell-Luc
Oroantral Fistulas
Oroantral fistula
Endoscopic Ethmoidectomy
Ethmoidectomy
External Ethmoidectomy
External ethmoidectomy
Endoscopic Sphenoidotomy
Sphenoidotomy
Trephination of the Frontal Sinus
Frontal trephination
Endoscopic Approach to the Frontal Sinus
Endoscopic frontal sinus
Draf III, Endoscopic Modified Lothrop
Draf III
Osteoplastic Obliteration of the Frontal Sinus
Osteoplastic flap
Frontal Sinus Cranialization
Cranialisation
Endonasal Approach to the Sella and Parasellar Areas
Endonasal pituitary surgery

CHAPTER SIX

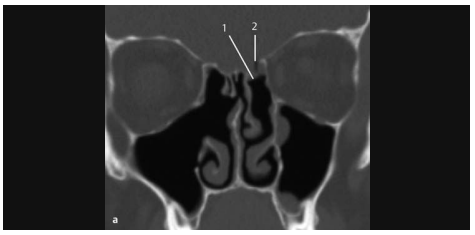
6 Endoscopic skull base surgery

Through the nose to the brain.

Extended endonasal approaches reach the sella (pituitary tumours), the anterior skull base and cribriform plate (olfactory neuroblastoma, meningioma), the clivus and the pterygopalatine and infratemporal fossae. The defect is closed in layers, and the *nasoseptal flap* on the posterior septal artery has greatly reduced postoperative CSF leaks (see the CSF rhinorrhoea topic).

IMAGING IN THE APP 3 cases

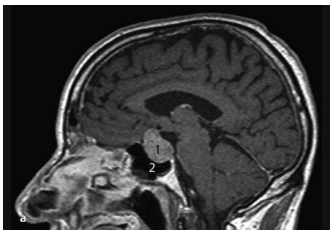
A dural defect after sinus surgery, a frontal recess mucocele, and pituitary pathology.



Dural Defect after Sinonasal Surgery
After surgery · 3 images



Mucocele of the Frontal Recess
Infection & inflammation · 3 images



Pituitary Gland Pathology
Tumours · 3 images

7 CHAPTER SEVEN Oroantral fistula

A hole between mouth and sinus.

7.1 Causes

Extraction of the second premolar or upper molars, whose roots lie close to the sinus floor (especially with an apical abscess); a Caldwell-Luc incision that fails to heal; carcinoma; fractures; osteitis, syphilis and granulomatous disease.

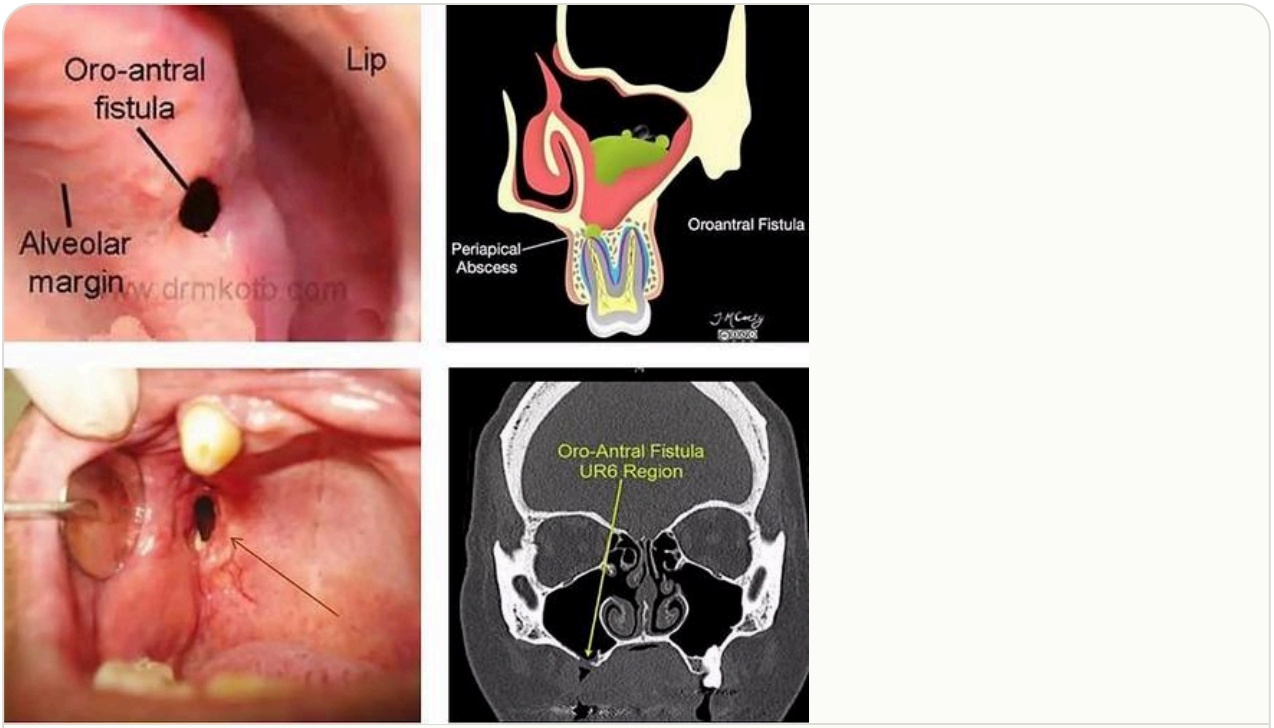


FIGURE 12 An oro-antral fistula: the thin floor of the maxillary sinus lies over the roots of the upper teeth.

Anatomy atlas · ENT Steps p. 132

7.2 Clinical picture

Fluids and food regurgitate into the nose; a foul discharge from the infected sinus into the nose or mouth; the patient cannot drink through a straw or blow a wind instrument. A probe passes from the socket into the antrum.

7.3 Treatment

- **Fresh, small fistula** without infection or a retained root: suture the gum margins and give antibiotics.
- **Chronic or large fistula:** treat the sinusitis (endoscopic antrostomy, removal of any root), excise the epithelialised track and close it with a buccal advancement or palatal rotation flap; or an obturator after cancer surgery.

SELECTION, OUTCOMES AND REVISIONS (OXFORD TEXTBOOK)

Who should have surgery. An international appropriateness consensus (RAND method) defined a minimum below which endoscopic sinus surgery is unlikely to help: in adults with uncomplicated chronic rhinosinusitis, a CT Lund–Mackay score of at least 1, a trial of at least eight weeks of topical nasal steroid plus a short course of

systemic steroid (with polyps) or antibiotics (without polyps), and a SNOT-22 score of 20 or more after that treatment. Up to one in three patients do not achieve a clinically meaningful improvement after surgery, and the **preoperative SNOT-22** is the best predictor of benefit, so selection matters. Equally, repeated rounds of medical treatment that delay surgery in patients who meet the criteria may worsen the final result.

Technique. Revision cases most often show a residual uncinate process, a missed natural ostium with a separate antrostomy (causing mucus recirculation) and residual anterior ethmoid cells, so the basic steps done well prevent many revisions. Polyp disease does better with more complete surgery (wide antrostomies, clearing all ethmoid partitions, extended frontal and sphenoid surgery) to give topical steroids access. Balloon dilatation cannot treat the ethmoids and suits only limited disease without polyps. Total intravenous anaesthesia gives a drier field; routine preoperative steroids and antibiotics are not needed. Image guidance is useful in revisions and abnormal anatomy, but has not been shown to reduce complications, and over-reliance can mislead. Steroid-eluting stents may help selected patients.

Revision rates are about 11% at three years, 19% at five and 17% at ten. A patient who develops proptosis during or after surgery needs urgent assessment of vision, pupils and intraocular pressure, and immediate decompression of an orbital haematoma (lateral canthotomy and cantholysis) if pressure is raised.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Antral puncture:** site, direction, findings and complications, including air embolism.
- **Caldwell-Luc operation:** incision, indications, complications, why not in children.
- **Submucous resection versus septoplasty.**
- **Steps of endoscopic sinus surgery** and the danger areas.
- **Complications** of endoscopic sinus surgery and their management.

- **Oroantral fistula.**

8.2 Common pitfalls

COMMON PITFALLS

- Operating without reviewing the CT for the dangerous variants.
- Making a second maxillary opening that does not join the natural ostium (recirculation).
- Stripping mucosa in the frontal recess.

Written for this app from Oxford Textbook of Otolaryngology ch. 37, Oxford Specialist Handbook ch. 21, Oxford Handbook of ENT ch. 20, Dhingra ch. 84 to 90 and K. J. Lee ch. 33, 34. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Skull base and the parapharyngeal space

The skull base is the floor of the brain and the roof of the nose, sinuses, nasopharynx, ear and neck. Tumours here present to the ENT surgeon as a blocked nose, a unilateral middle ear effusion, facial numbness, diplopia, hoarseness or a lump behind the tonsil. Modern skull base surgery is a team effort with neurosurgeons. Endoscopic endonasal routes now reach the sella, clivus and much of the anterior skull base without a facial incision. The parapharyngeal space, just below, has a logic of its own: decide whether the mass lies in front of or behind the styloid.

AT A GLANCE

- Abducens runs free in the cavernous sinus: often the first nerve to fail.
- Prestyloid mass is usually pleomorphic adenoma; poststyloid is neural or paraganglioma.
- Never biopsy a parapharyngeal mass through the mouth.
- Nasoseptal flap is pedicled on the posterior septal artery.
- Multilayer closure; vascularised nasoseptal flap for high-flow leaks; lumbar drains used selectively.

1 CHAPTER ONE Applied anatomy

Three fossae and the spaces below.

1.1 Anterior skull base

The anterior skull base separates the frontal lobes from the nose, ethmoid sinuses and orbits. In the midline lies the *cribriform plate*, its thinnest part, perforated by the olfactory fila; beside it the thin *lateral lamella* is a common site of CSF leak. The *crista galli*, where the falx attaches, rises in front, and the *planum sphenoidale* lies behind. Laterally are the orbital roofs and the *fovea ethmoidalis* of the frontal bone, with the anterior and posterior ethmoidal arteries crossing the ethmoid roof.

1.2 Middle cranial fossa and the central skull base

The middle fossa holds the temporal lobes. Centrally, the body of the sphenoid contains the sphenoid sinus, with the *sella turcica* above it holding the pituitary and a *cavernous sinus* on each side. In the lateral wall of the cavernous sinus run III, IV, V1 and V2, while VI runs free within the sinus beside the internal carotid artery, which is why VI is so often the first nerve to fail.

The foramina of the region carry predictable contents: the superior orbital fissure (III, IV, V1 and VI), the optic canal, the foramen rotundum (V2), the foramen ovale (V3), the foramen spinosum (the middle meningeal artery), and the foramen lacerum with the carotid canal. The **clivus** slopes from the dorsum sellae to the foramen magnum; its upper part is the basisphenoid and its lower part the basiocciput.

Seen from the sphenoid sinus, the carotid and optic nerve make bulges in the lateral wall, separated by the *optico-carotid recess*, and the bone over the carotid is dehiscent in some people. An **Onodi cell** is a posterior ethmoid cell that extends above the sphenoid and may contain the optic nerve.

1.3 Posterior fossa and lateral skull base

The petrous temporal bone contains the cochlea, labyrinth, facial nerve and internal carotid artery. The anterior surface of the petrous apex forms the floor of the middle fossa, with the arcuate eminence, the greater superficial petrosal nerve and *Kawase's triangle*, and its posterior surface carries the internal auditory canal. The jugular foramen transmits IX, X and XI with the jugular bulb, and the hypoglossal canal carries XII below and medial to it.

1.4 Infratemporal and pterygopalatine fossae

The **infratemporal fossa** lies below the middle cranial fossa, behind the maxilla and deep to the ramus, and contains the pterygoid muscles, V3, the maxillary artery and the pterygoid venous plexus.

The **pterygopalatine fossa** is a small inverted cone behind the maxillary sinus that opens in every direction: to the orbit through the inferior orbital fissure, to the

infratemporal fossa through the pterygomaxillary fissure, to the nose through the sphenopalatine foramen, to the middle cranial fossa through the foramen rotundum and vidian canal, and to the mouth through the greater palatine canal. It is therefore the crossroads by which juvenile angiofibroma and perineural cancer spread.

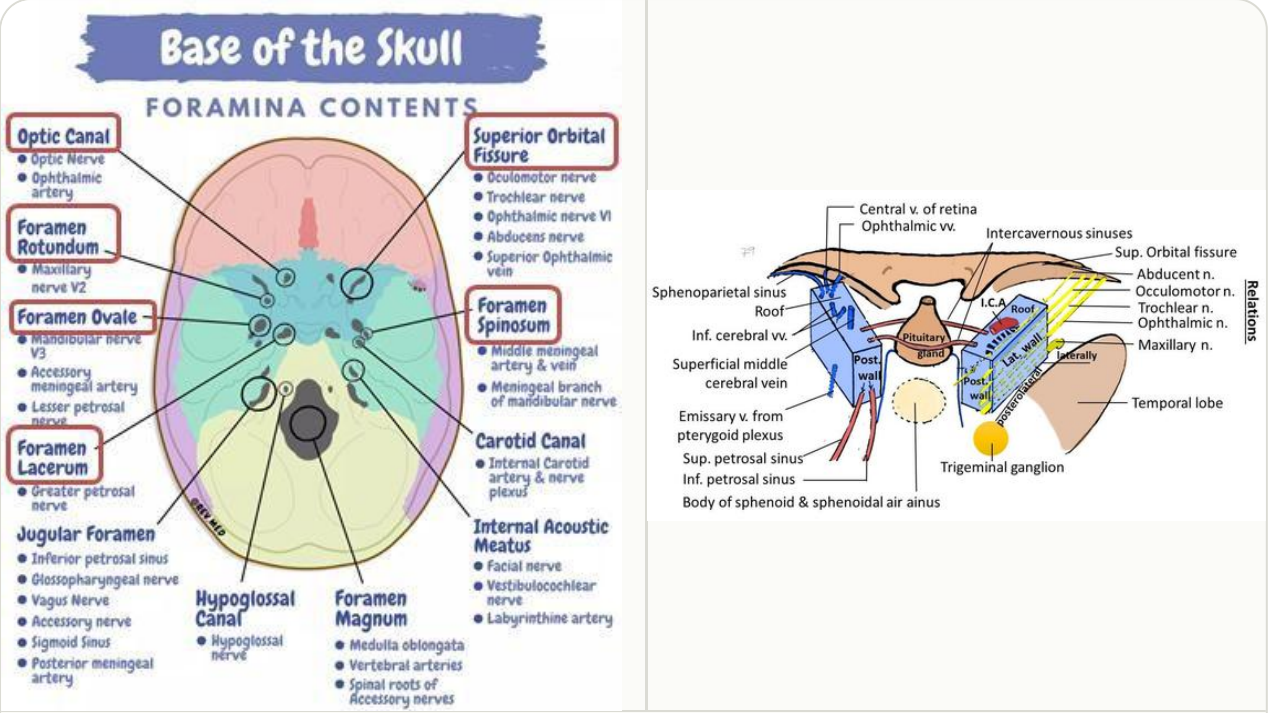


FIGURE 1 Foramina of the skull base and their contents, and the cavernous sinus with its contents.

Anatomy atlas · ENT Steps p. 152, Doss p. 85

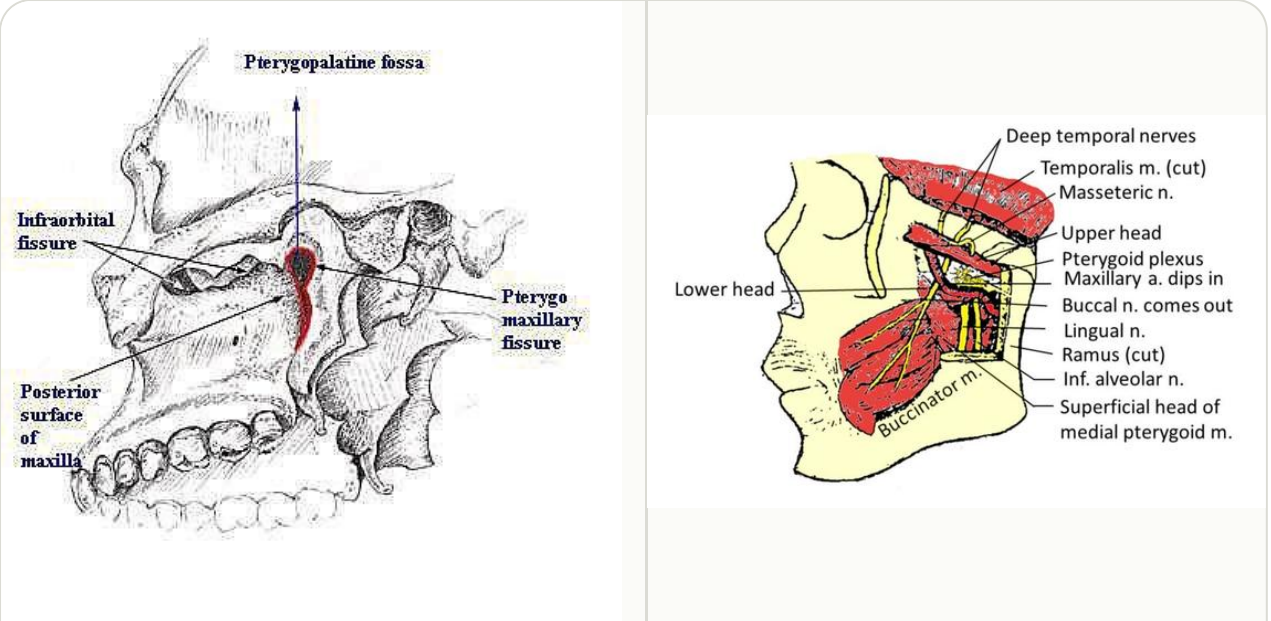


FIGURE 2 The pterygopalatine fossa, and the infratemporal fossa after removal of the ramus.
Anatomy atlas · ENT Steps p. 160, Doss p. 54

CHAPTER TWO

2 The parapharyngeal space

An inverted pyramid.

2.1 Boundaries and compartments

The parapharyngeal space is an inverted pyramid with its base on the skull base and its apex at the greater cornu of the hyoid. Medially lie the buccopharyngeal fascia and superior constrictor, and so the tonsil; laterally the medial pterygoid, the ramus and the deep lobe of the parotid; posteriorly the prevertebral fascia; and anteriorly the pterygomandibular raphe.

The *tensor–styloid fascia*, formed by the styloid process and its muscles, divides the space into two compartments. The **prestyloid compartment** contains fat, the deep lobe of the parotid, minor salivary rests, the maxillary artery and branches of V3. The **poststyloid (carotid) compartment** contains the internal carotid artery, the internal jugular vein, nerves IX to XII, the sympathetic chain and nodes.

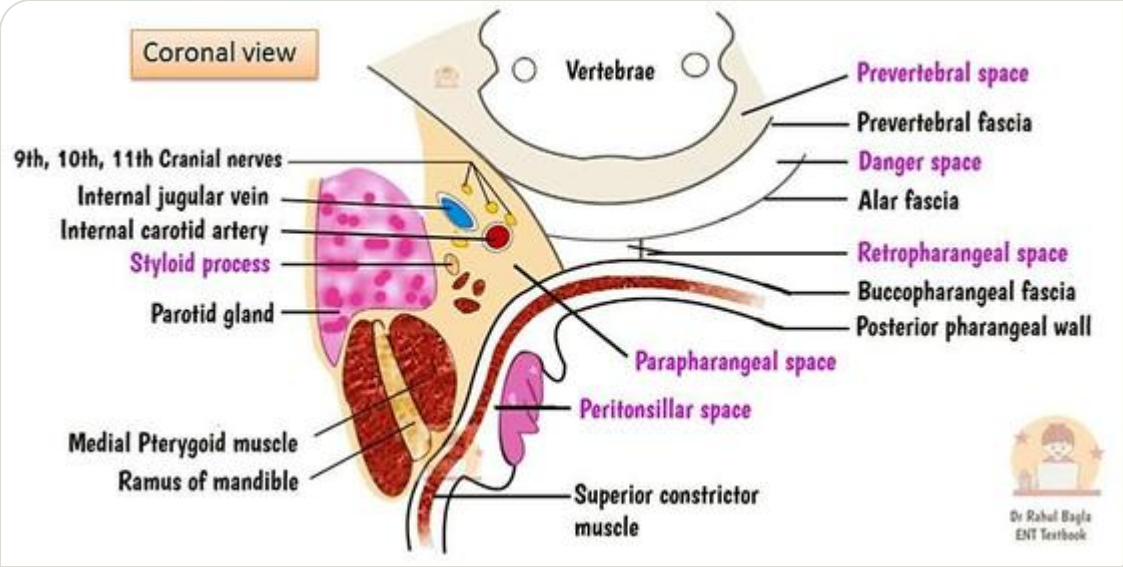


FIGURE 3 The parapharyngeal and retropharyngeal spaces.
Anatomy atlas · ENT Steps p. 198

2.2 Tumours

PARAPHARYNGEAL SPACE TUMOURS BY COMPARTMENT

COMPARTMENT	COMMON TUMOURS	IMAGING HINT
Prestyloid	*Pleomorphic adenoma* (deep lobe or minor salivary), other salivary tumours, lipoma	Fat plane between tumour and parotid lost (deep lobe origin) or preserved (minor salivary); carotid pushed *posteriorly*
Poststyloid	*Schwannoma* (vagus, sympathetic chain), *paraganglioma* (vagal, carotid body), neurofibroma, nodes (metastases, lymphoma), meningioma	Carotid pushed *anteriorly* or splayed; flow voids in paragangliomas

About half of parapharyngeal tumours are salivary, mostly pleomorphic adenoma, a fifth to a third are neurogenic, and the rest are paragangliomas and others; about 80% are benign. They are often silent until large, then appear as a smooth bulge pushing the *tonsil and soft palate medially* under normal mucosa, or as fullness behind the angle of the jaw, with a muffled voice, sleep apnoea, Eustachian tube obstruction or dysphagia. Cranial nerve palsies, Horner syndrome or pain suggest malignancy or a neural tumour.

MRI, with CT, places the tumour in its compartment and shows the vessels; metanephrines are measured whenever a paraganglioma is possible, and prestyloid

and nodal masses can be sampled by image-guided FNA. A *transoral biopsy is never done*, because it seeds a pleomorphic adenoma and makes a paraganglioma bleed.

The **transcervical** approach through the submandibular triangle, dividing the stylomandibular ligament, is the workhorse. Deep lobe tumours need a **transparotid–transcervical** approach, large tumours reaching the skull base a **transcervical approach with mandibulotomy**, and selected prestyloid tumours can be removed by **transoral robotic** surgery. The old blind transoral enucleation risks rupture, bleeding and incomplete excision.

OPERATIVE GUIDES IN THEATRE 3 operations

Tumours of the prestyloid parapharyngeal space, the transoral approach, and excision of neural tumours.

Management of Tumors in the Prestyloid Parapharyngeal Space
Prestyloid parapharyngeal
Transoral Approach to the Parapharyngeal Space
Excision of Neural Tumors of the Parapharyngeal Space
Neural tumours

3 CHAPTER THREE Skull base tumours

What arises here.

3.1 Anterior skull base

The malignant tumours of the anterior skull base are mostly sinonasal: squamous carcinoma, adenocarcinoma in woodworkers, adenoid cystic carcinoma, *olfactory neuroblastoma*, *sinonasal undifferentiated carcinoma*, neuroendocrine carcinoma and mucosal melanoma, together with sarcomas and lymphoma. The benign lesions are meningioma, inverted papilloma, juvenile angiofibroma, fibro-osseous lesions, mucoceles and encephaloceles.

3.2 Sella and parasellar region

The **pituitary adenoma** is the commonest sellar tumour; microadenomas are under 10 mm and macroadenomas larger. Mass effect produces *bitemporal hemianopia*,

headache and hypopituitarism, and hormone excess produces the prolactinoma, with amenorrhoea, galactorrhoea and impotence, acromegaly or Cushing disease. A prolactinoma is treated first with *cabergoline or bromocriptine*, not surgery. The endocrine work-up measures prolactin, IGF-1, cortisol and ACTH, thyroid function, gonadotropins and sodium, with formal visual fields.

A **Rathke cleft cyst** is usually incidental and observed. The **craniopharyngioma** has two peaks, in children and in late middle age, is calcified and cystic, and is hard to remove completely. **Meningiomas** arise from the tuberculum, planum and olfactory groove and show a "dural tail". An **empty sella** is associated with idiopathic intracranial hypertension and spontaneous CSF leaks.

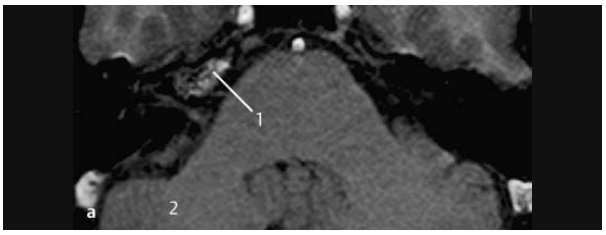
3.3 Clivus and petrous apex

The **chordoma** arises from notochordal remnants at the spheno-occipital synchondrosis in the midline. It is bright on T2 with bone destruction, causes diplopia from a VI palsy and headache, and is treated by maximal resection followed by high-dose, often proton, radiotherapy; recurrence is common. The **chondrosarcoma** arises at the petroclival synchondrosis, off the midline, and has a better prognosis.

At the petrous apex, the cholesterol granuloma is bright on both T1 and T2; cholesteatoma, mucocele and petrous apicitis also occur, the last producing *Gradenigo syndrome* of otorrhoea, VI palsy and retro-orbital pain. Asymmetric pneumatisation or retained fluid can only *look* like a lesion, and MRI settles the question.

IMAGING IN THE APP 1 case

Petrous apex lesions and pseudotumours on imaging.



Pseudotumors in the Apex of the Petrous Bone

Tumours · 12 images

3.4 Lateral skull base and posterior fossa

Here lie the paragangliomas (glomus jugulare, tympanicum and vagale), vestibular schwannoma and other cerebellopontine angle lesions, temporal bone carcinoma, schwannomas of the facial and lower cranial nerves, meningiomas, and the endolymphatic sac tumours of von Hippel–Lindau syndrome.

MRI SIGNAL OF CEREBELLOPONTINE ANGLE AND PETROUS APEX LESIONS

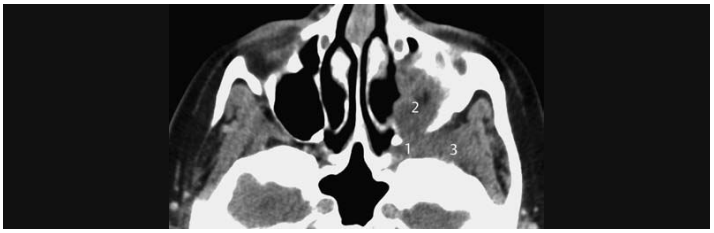
LESION	T1	T1 WITH GADOLINIUM	T2	DIFFUSION
Vestibular schwannoma	Iso or low	Marked enhancement	Moderately high	—
Meningioma	Iso or low	Marked, with a dural tail	Variable	—
Epidermoid (cholesteatoma)	Low	None	High	*Restricted* (bright)
Arachnoid cyst	Low (like CSF)	None	High (like CSF)	Not restricted
Cholesterol granuloma	*High*	None	High	—
Lipoma	High (suppressed by fat saturation)	None	Intermediate	—

3.5 Infratemporal fossa

Primary infratemporal tumours are sarcomas, neurogenic tumours and salivary tumours, but more often the fossa is invaded from its neighbours by nasopharyngeal carcinoma, maxillary sinus cancer, juvenile angiofibroma or parotid cancer. The signs are *trismus* from invasion of the pterygoids, *numbness of V3* over the lower lip and chin, serous otitis media from Eustachian tube invasion, and a mass deep to the ramus.

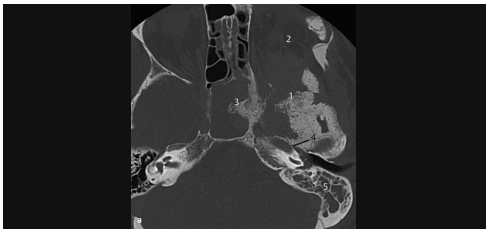
IMAGING IN THE APP 2 cases

Malignancy of the pterygopalatine and infratemporal fossae, and a meningioma destroying the skull base.



Malignancy of the Pterygopalatine and Infratemporal Fossa

Tumours · 2 images



Meningioma with Skull Base Destruction

Tumours · 2 images

4 CHAPTER FOUR Assessment

Nerves, images, tissue.

4.1 Clinical picture

Skull base disease announces itself through whichever structure it reaches first. In the nose it causes unilateral obstruction, bleeding and anosmia; in the eye proptosis, diplopia and visual loss; in the face numbness of V2 or V3 and pain, which suggest *perineural spread*; and in the jaw trismus. In the ear, a unilateral conductive loss from a *middle ear effusion* in an adult means the nasopharynx must be examined and biopsied, and pulsatile tinnitus suggests a vascular tumour. In the throat,

hoarseness, dysphagia and aspiration mean IX to XI are involved, and tongue weakness means XII. CSF rhinorrhoea and headache complete the picture.

The examination is a complete cranial nerve examination, nasal and nasopharyngeal endoscopy, otomicroscopy, and examination of the larynx and neck.

JUGULAR FORAMEN SYNDROMES

- **Vernet syndrome:** IX, X and XI together, from jugular foramen lesions (paragangliomas, schwannomas, metastases).
- **Collet–Sicard syndrome:** adds XII.
- **Villaret syndrome:** adds Horner syndrome, from the retroparotid space.

4.2 Imaging and tissue

CT and MRI are complementary and both are needed. **CT** shows bone detail: erosion, remodelling and widening of foramina. **MRI** with gadolinium and fat suppression shows soft tissue, dura, brain, orbit and perineural spread. Dural invasion is suggested by dural thickening of more than 5 mm or nodular thickening, and periorbital invasion by enhancement or enlargement of the extraocular muscles, stranding of the orbital fat and a nodular interface. **Angiography** is used for vascular tumours and embolisation, and *balloon test occlusion* of the internal carotid when it may have to be sacrificed; **PET-CT** stages disease and detects recurrence.

Most lesions need tissue, often by *endoscopic biopsy*, but not all: paragangliomas, schwannomas, typical meningiomas and juvenile angiofibromas can be diagnosed on imaging alone.

4.3 Unresectability

A tumour is generally unresectable when it invades the brainstem or spinal cord, both internal carotids, both cavernous sinuses or extensive brain parenchyma. Involvement of the optic chiasm, distant metastases and poor general health are relative contraindications. Lymphoma and other chemo- or radiosensitive tumours are not resected at all.

5 CHAPTER FIVE Approaches

Endoscopic and open.

5.1 Endoscopic endonasal surgery

Endoscopic skull base surgery works through both nostrils with the "*binostril, four-handed*" technique, using a wide sphenoidotomy and removal of the posterior septum, with ethmoidectomy, medial maxillectomy, transpterygoid access or a Draf III as needed. The **transsphenoidal** route reaches the sella and is the standard for pituitary surgery; the **transplanum and transtuberculum** routes reach the suprasellar region; the **transcribriform** route exposes the anterior skull base from the frontal sinus to the planum between the orbits; the **transclival** route reaches the clivus down to the foramen magnum and odontoid; and the **transpterygoid** route opens the pterygopalatine and infratemporal fossae, Meckel's cave and the petrous apex.

The approach avoids facial and scalp incisions and brain retraction, gives magnified and angled views and speeds recovery, and for selected sinonasal cancers survival is comparable to open craniofacial resection. Its limits are lateral extension beyond the mid-orbit, involvement of the orbit, the skin or the anterior or lateral wall of the frontal or maxillary sinus, dural or brain invasion lateral to the orbit, and encasement of the carotid.

OPERATIVE GUIDES IN THEATRE 3 operations

Anterior cranial base, transpterygoid and petrosal endoscopic approaches.

Transcranial and Endonasal Approaches to the Anterior Cranial Base
Anterior cranial base
Endoscopic Endonasal Transpterygoid Approaches to the Middle Cranial Fossa
Transpterygoid
Endoscopic Petrosal Approaches

5.2 Reconstruction

Small defects are closed with free grafts of fascia, fat or mucosa in several layers, an *inlay* graft beneath the dura and an *onlay* graft over the bone. Large defects need

vascularised tissue. The *nasoseptal (Hadad–Bassagasteguy) flap*, pedicled on the posterior septal artery from the sphenopalatine artery, has cut postoperative leak rates from over 20% to about 5%, and a "rescue flap" incision at the start of surgery preserves its pedicle in case it is needed. Inferior and middle turbinate flaps, the *pericranial flap* (open or endoscopically harvested), the temporoparietal fascia flap and free flaps for open defects are the alternatives. Lumbar drains are used for selected high-flow leaks, and the patient avoids nose blowing, straining and mask ventilation.

OPERATIVE GUIDES IN THEATRE 2 operations

Reconstruction after skull base surgery, and endoscopic repair of CSF leaks.

Reconstruction After Skull Base Surgery
Skull base reconstruction
Endoscopic Management of Cerebrospinal Fluid Leaks

5.3 Open approaches

Craniofacial resection combines a transfacial route, by lateral rhinotomy or midface degloving, with a bifrontal craniotomy to remove the cribriform plate, ethmoids and involved dura en bloc, and reconstructs with a *pericranial flap* on the supraorbital and supratrochlear vessels. The orbit is exenterated if the periorbital fat, extraocular muscles or orbital apex are invaded.

Fisch's **infratemporal fossa approaches** close the ear canal and remove the middle ear. **Type A** reaches the jugular foramen and infralabyrinthine region with anterior rerouting of the facial nerve, and is used for glomus jugulare; **type B** reaches the clivus and petrous apex; and **type C** the parasellar region, nasopharynx and pterygopalatine fossa.

Other routes include the lateral (preauricular) subtemporal–infratemporal approach, which removes the zygoma and condyle; the far lateral approach to the foramen magnum and lower clivus; the translabyrinthine, retrosigmoid, middle fossa, transotic and transcochlear approaches to the temporal bone and cerebellopontine angle, described with vestibular schwannoma and temporal bone tumours; and the maxillary swing, transpalatal and midface degloving routes to the nasopharynx.

OPERATIVE GUIDES IN THEATRE 3 operations

Approaches to the infratemporal fossa, the maxillary swing and transpalatal routes, and endoscopic nasopharyngectomy.

Surgical Approaches to the Infratemporal Fossa
Infratemporal fossa
Maxillary Swing/Transpalatal/Midface Degloving With Le Fort I Osteotomy Approach for Nasopharyngectomy
Endoscopic Nasopharyngectomy

5.4 Complications

CSF leak is the commonest complication, and meningitis may follow. **Tension pneumocephalus** presents with sudden confusion and a falling conscious level, and CT shows the "*Mount Fuji sign*". **Carotid injury** is managed by packing with pressure, then a crushed muscle patch, then urgent angiography to look for a pseudoaneurysm. Stroke, cranial nerve deficits and visual loss can occur, sellar surgery can cause hypopituitarism and diabetes insipidus (so sodium and urine output are monitored), and the nose is left with crusting, synechiae and anosmia. Trismus and wound or flap failure complete the list.

Overall five-year survival for skull base malignancy is about 50%. Clear margins are the strongest predictor of survival, and postoperative radiotherapy is given for high-risk features.

RECONSTRUCTION OF THE ANTERIOR SKULL BASE (OXFORD TEXTBOOK)

The aim is a durable barrier between the nose and the intradural space to prevent CSF leak, pneumocephalus and meningitis. **Multilayer closure** is standard: an *inlay* graft (usually fascia lata, sometimes with fat) tucked 5 to 10 mm under the dural edges for a watertight seal, then an *onlay* of free mucosa or, preferably, a **vascularised flap**, secured with sealant and supported by dressings or a balloon. Mucosa is stripped from the bone under the onlay so that it adheres and no mucocoele forms.

The choice depends on defect size and site, whether the intraoperative leak is **low-flow** (dura opened without entering a cistern or ventricle) or **high-flow** (a cistern or ventricle opened), and patient factors such as previous radiotherapy, long-term steroids and malnutrition. Small, low-flow leaks close reliably with free grafts, for example the *bath-plug* technique with a fat plug and a free mucosal onlay. Large or high-flow defects, high-risk sites and poor healers need a vascularised flap. The **Hadad–Bassagasteguy nasoseptal flap**, on the posterior septal branch of the

sphenopalatine artery, transformed endoscopic skull base surgery by cutting leak rates; it should be planned (or its pedicle preserved as a "rescue" flap) before the sphenoidotomy. Alternatives on sphenopalatine branches are inferior and middle turbinate flaps and lateral nasal wall flaps; regional options are the **pericranial flap**, on the supraorbital and supratrochlear vessels and passed through a window above the frontal sinus, and the temporoparietal fascia flap.

Lumbar drains reduce pressure on the repair, but with reliable multilayer closure and flaps many surgeons now use them only to treat a postoperative leak, for high-flow defects or in patients with raised intracranial pressure, or to give intrathecal fluorescein.

CHAPTER SIX

6 For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Contents of the cavernous sinus;** why VI fails first; the skull base foramina.
- **The pterygopalatine fossa:** its communications and what spreads through it.
- **The parapharyngeal space:** prestyloid versus poststyloid, medial displacement of the tonsil, never biopsy through the mouth, the transcervical approach.
- **Adult unilateral middle ear effusion:** examine the nasopharynx.
- **Chordoma versus chondrosarcoma;** the MRI table for petrous apex and cerebellopontine angle lesions.
- **Endoscopic endonasal approaches,** their limits, and the nasoseptal flap and its pedicle.
- **Craniofacial resection;** the Fisch infratemporal fossa approaches.
- **Complications:** CSF leak, pneumocephalus, carotid injury.

6.2 Common pitfalls

COMMON PITFALLS

- Taking a transoral biopsy of a parapharyngeal mass.
- Treating a prolactinoma surgically before trying a dopamine agonist.
- Missing a VI palsy or facial numbness as signs of skull base spread.
- Calling an asymmetric petrous apex "a lesion" without MRI.

Written for this app from Oxford Textbook of Otolaryngology ch. 44, 45, Oxford Specialist Handbook ch. 19, K. J. Lee ch. 34, 49. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NOSE & SINUSES

Facial plastic surgery

Facial plastic surgery begins with analysis: proportions, angles, the skin and the structures beneath it. It then asks what the patient wants and whether that is achievable and wise. For the ENT surgeon the core is the nose, where form and airway must be fixed together. The other procedures are the ear, the ageing face and the injectables, each with its own anatomy and its own complications to avoid.

AT A GLANCE

- Major tip supports: lower lateral cartilages, footplate to caudal septum, scroll.
- Spreader grafts for the internal valve; batten grafts for the external valve.
- Retrobulbar haematoma after blepharoplasty: lateral canthotomy now.
- Mustarde sutures make the antihelical fold; Furnas sets back the concha.
- Preservation rhinoplasty lowers the dorsum without resecting it; screen for body dysmorphic disorder.

1 CHAPTER ONE Facial analysis

Measure before you cut.

1.1 Landmarks

Analysis uses a small vocabulary of points. In the midline from above down lie the **trichion** (the hairline), the **glabella** (the most prominent point of the forehead) and the **nasion**, the deepest point of the nasofrontal angle at the nasofrontal suture. On the dorsum, the **rhinion** marks the junction of bone and cartilage, where the skin is thinnest, and the light reflexes of the domes are the **tip-defining points**. The **subnasale** is where the columella meets the lip and the **stomion** where the lips meet. On the chin, the **pogonion** is the most anterior point and the **menton** the lowest, and the **cervical point** is the deepest point between the submental region and the neck.

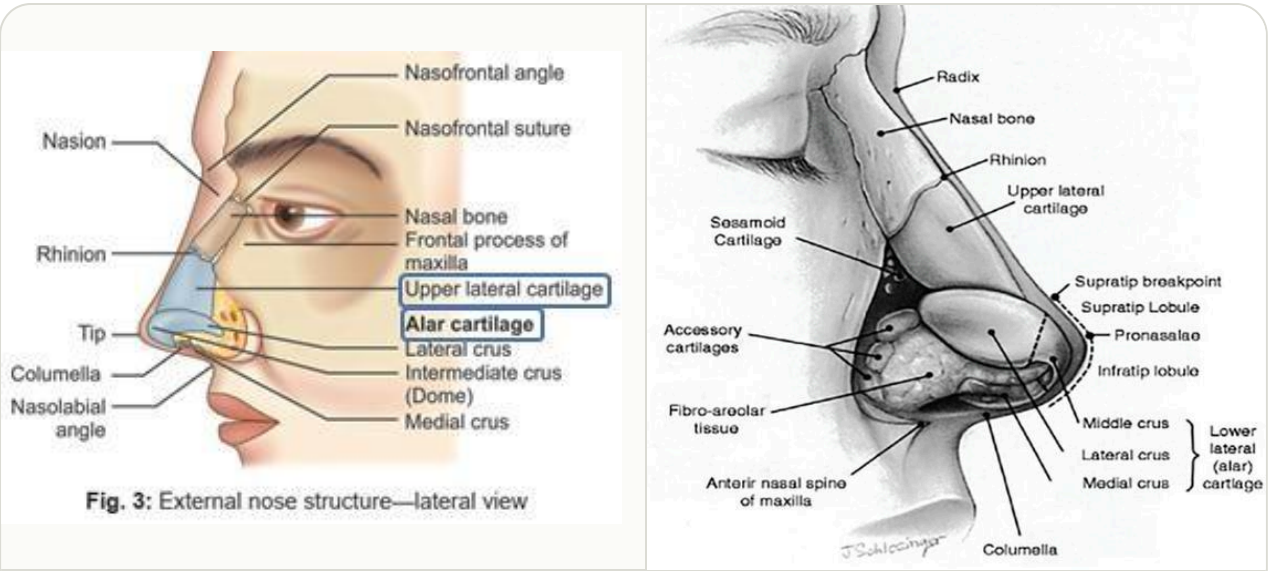


FIGURE 1 Surface landmarks and the osteocartilaginous framework of the external nose.
Anatomy atlas · ENT Steps p. 92, ENT Steps p. 93

1.2 Proportions

In height the face divides into **thirds**: trichion to glabella, glabella to subnasale, and subnasale to menton are about equal, and within the lower third the upper lip takes a third and the lower lip and chin two-thirds. In width it divides into **fifths**, each about one eye's width; the intercanthal distance equals an eye width and also the width of the nasal base. Width to height is about 3:4.

FACIAL ANGLES

ANGLE	DEFINED BY	IDEAL
Nasofrontal	Glabella–nasion and nasion–tip	115 to 130°
Nasofacial	Facial plane (glabella–pogonion) and the dorsum	30 to 40°
Nasolabial	Columella and upper lip at subnasale	90 to 95° in men, 95 to 110° (up to 120°) in women
Nasomental	Dorsum and tip–pogonion	120 to 132°
Mentocervical	Facial plane and the submental line	80 to 95°

1.3 The chin and occlusion

In profile the chin should sit on a vertical line dropped from the lower lip, or just behind it in women. A weak chin is either **microgenia**, a small chin with normal occlusion that is corrected with an implant or a *sliding genioplasty*, or **retrognathia**, a retruded mandible with a class II bite that needs orthognathic surgery such as a sagittal split osteotomy. The distinction is made by looking at the bite.

Angle's classification describes occlusion by the upper first molar. In **class I**, the normal state, its mesiobuccal cusp sits in the buccal groove of the lower first molar; in **class II** the upper molar sits forward of the lower, as in retrognathia; and in **class III** it sits behind, as in prognathism.

Chin implants of silastic, porous polyethylene or ePTFE are placed in a subperiosteal pocket through a submental or intraoral incision, and can displace, become infected, cause bone resorption or injure the mental nerve. A sliding genioplasty adds both projection and height, at the risk of injury to the mental nerve and the tooth roots.

1.4 Skin

The **Fitzpatrick types** run from I, which always burns and never tans, to VI, which never burns; types III to VI are at more risk of pigment change after peels and lasers. **Glogau's photoageing classes** progress from I (no wrinkles) through II (wrinkles in motion) and III (wrinkles at rest) to IV (only wrinkles). Nasal skin is thick and sebaceous at the tip and supratip and thinnest at the rhinion: thick skin hides tip work and swells for months, while thin skin shows every irregularity.

2 CHAPTER TWO Rhinoplasty

Form and function together.

2.1 Surgical anatomy

The nose is best thought of in thirds. The **upper, bony third** is the nasal bones. The **middle third** is the upper lateral cartilages fused to the dorsal septum, which together form the *internal nasal valve*, normally an angle of 10 to 15°. The **lower third** is the lower lateral (alar) cartilages, with their medial, intermediate and lateral crura and the domes, and the external valve. The junction of the upper and lower laterals is the **scroll**, and the area where the nasal bones meet the upper lateral cartilages and septum is the **keystone**, whose disruption lets the middle vault collapse.

Tip support comes from three major mechanisms: the size, shape and strength of the lower lateral cartilages; the attachment of the medial crural footplates to the caudal septum; and the attachment of the upper to the lower laterals at the scroll. The minor mechanisms are the interdomal ligament, the dorsal cartilaginous septum, the sesamoid complex, the skin and soft-tissue envelope, the nasal spine and the membranous septum. Every incision or resection that disrupts one of these must be compensated for.

The **tripod concept** makes tip surgery predictable. The two lateral crura and the conjoined medial crura form the three legs of a tripod; shortening or lengthening the legs changes *projection* and *rotation* in ways that can be planned.

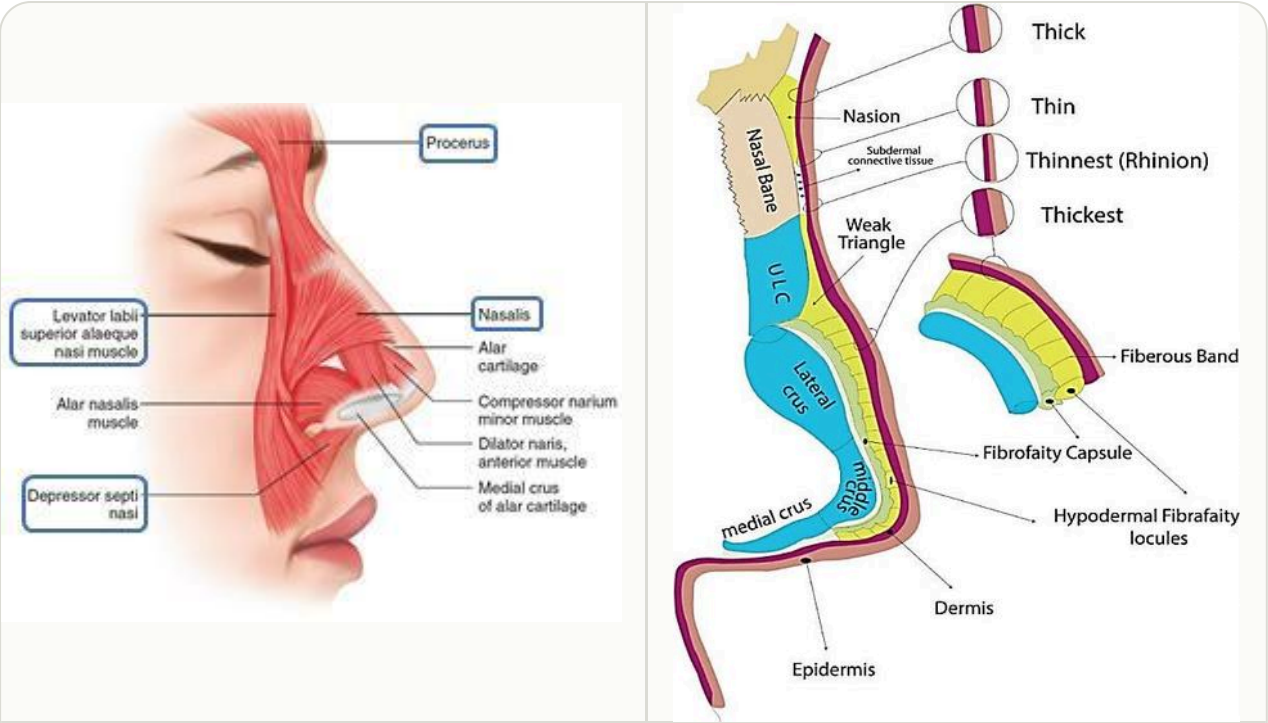


FIGURE 2 Muscles of the nose, and nasal skin thickness from nasion to tip.
Anatomy atlas · ENT Steps p. 95, ENT Steps p. 96

2.2 Analysis of the nose

On the **frontal view**, assess deviation by thirds, the width of the bony base compared with the alar base, smooth *brow–tip aesthetic lines* running from the medial brow to the tip-defining point, tip definition, and the "gull in flight" outline of the alae and columella.

On **profile**, note the radix position, which should lie between the lash line and the upper lid crease, any dorsal hump, and a supratip break, which is desirable in women. Tip *projection* is judged by the *Goode ratio*, in which projection is 0.55 to 0.6 of nasal length, or by the 3:4:5 triangle; *rotation* is read from the nasolabial angle; and 2 to 4 mm of *columellar show* is normal.

On the **base view**, the nose should form an equilateral triangle with a columella-to-lobule ratio of 2:1, symmetrical nostrils and no caudal septal deviation. **Inside the nose**, examine the septum, turbinates and valves; the *Cottle manoeuvre*, in which lateral traction on the cheek improves the airway, suggests valve collapse, and the modified Cottle test with a cotton bud localises it.

Take standard photographs, and always weigh the patient as well as the nose: body dysmorphic disorder and unrealistic expectations are the commonest causes of an unhappy outcome after technically sound surgery.

2.3 Approaches and incisions

The incisions are the **marginal**, along the caudal edge of the lateral crus; the **intercartilaginous**, between the upper and lower laterals; the **transcartilaginous**, through the lateral crus; the **transcolumellar**, an inverted V or stair step; and the **transfixion or hemitransfixion**, between the caudal septum and the medial crura.

Endonasal (closed) rhinoplasty uses either a non-delivery approach, cartilage-splitting or retrograde, or a delivery approach in which the alar cartilages are brought out as bipedicled flaps. It leaves no external scar and causes less swelling, but exposure is limited.

External (open) rhinoplasty joins transcolumellar and marginal incisions and lifts the whole skin envelope. It gives the best exposure for precise sutures and grafts and is preferred for revision, cleft and crooked noses, major tip work and teaching. Its costs are a columellar scar, more oedema and the disruption of a minor tip support.

COMMON MANOEUVRES

PROBLEM	MANOEUVRE
Dorsal hump	Component reduction: rasp the bone, cut the cartilage; then osteotomies to close the open roof; spreader grafts to keep the middle vault and valve
Wide or crooked bony vault	Medial, lateral (low-to-high along the nasomaxillary groove, starting above the inferior turbinate attachment) and intermediate osteotomies
Crooked middle third	Septoplasty, asymmetric spreader grafts or spreader flaps, camouflage onlay grafts
Bulbous tip	Cephalic trim, keeping at least 6 to 10 mm of lateral crus; dome-binding (transdomal, interdomal) sutures; lateral crural strut grafts
Under-projected tip	Columellar strut, caudal septal extension graft, dome sutures, tip (shield, cap) grafts, lateral crural steal
Over-projected tip	Full transfixion, medial or lateral crural overlay, reducing the footplates

PROBLEM	MANOEUVRE
Under-rotated (drooping) tip	Trim the caudal septum, tongue-in-groove, lateral crural steal or overlay, cephalic trim
Over-rotated tip	Caudal extension graft, medial crural overlay
Internal valve collapse	Spreader grafts, flaring sutures, butterfly graft
External valve collapse	Alar batten grafts, lateral crural strut grafts, alar rim grafts
Saddle nose	Dorsal onlay or reconstruction with septal, ear or *rib cartilage*; rebuild the L-strut

Septal cartilage is the first-choice graft, provided a dorsal and caudal *L-strut* of 10 to 15 mm is left to hold the nose up. **Conchal cartilage** from the ear suits alar grafts, and **costal cartilage** is plentiful for revisions and saddle noses but can warp. Fascia, and diced cartilage wrapped in fascia, smooth the dorsum. The guiding principle is to "preserve and conserve, not resect and regret": reshape with sutures and grafts rather than by removing support.

OPERATIVE GUIDES IN THEATRE 2 operations

Rhinoplasty and surgery of the nasal valve, and septoplasty.

Rhinoplasty and Surgery of the Nasal Valve
Rhinoplasty
Septoplasty—Classic and Endoscopic

2.4 Complications and revision

Between 5 and 15% of rhinoplasties are revised, and revision should wait about a year for the tissues to settle. The named deformities each have a mechanism and a remedy.

A **pollybeak** is fullness above the tip, from an under-resected cartilaginous dorsum, supratip dead space and scar, or loss of tip support; it is treated with steroid injection, trimming or restoring support. A **saddle nose** follows over-resection of the dorsum, septal collapse, a septal abscess or haematoma, or granulomatous disease such as granulomatosis with polyangiitis or cocaine use, and needs grafts.

An **inverted-V deformity** appears when the upper lateral cartilages collapse away from the nasal bones and is corrected with spreader grafts. An **open roof** results from omitting osteotomies after hump removal, and a **rocker deformity** from carrying an osteotomy too high into the thick nasal bone. **Alar retraction and notching** follow over-resection of the lateral crus and need alar rim or composite grafts. Tip bossae, a pinched tip and tip ptosis are the other tip problems.

Functionally, the nose may end up more obstructed through valve collapse, synechiae or residual septal deviation. Septal haematoma or perforation, epistaxis, a rare CSF leak from a high septal or osteotomy fracture, and dissatisfaction complete the list.

3 CHAPTER THREE The ageing face

Upper, middle and lower thirds.

3.1 What ages

Every layer ages. The **skin** develops a thinner dermis with less collagen and elastin, photodamage and wrinkles. The **soft tissues** descend and the fat pads lose volume. The **bone** resorbs, especially at the orbital rims, piriform aperture and maxilla and in the pre-jowl mandible.

The changes show differently in each third. The upper third gains forehead lines and brow ptosis; the middle third shows eyelid laxity and fat bulges, a tear trough and deeper nasolabial folds; and the lower third develops jowls, marionette lines, neck laxity and platysmal bands.

3.2 Brow and forehead lift

The ideal brow begins over the medial canthus and ends on a line from the alar base through the lateral canthus. In women it arches above the orbital rim, peaking between the lateral limbus and the lateral canthus; in men it lies flatter, at the rim.

A **coronal** lift hides the scar well but raises the hairline and leaves the scalp numb, so a **pretrichial** incision is chosen for a high hairline. The **endoscopic** lift uses small

incisions and subperiosteal release with fixation. **Direct** and **midforehead** lifts suit older men with deep forehead lines and patients with facial palsy.

The danger is the **frontal branch of the facial nerve**, which runs in the temporoparietal fascia along *Pitanguy's line*, from 0.5 cm below the tragus to 1.5 cm above the lateral brow. Lateral dissection is therefore kept *deep*, on the deep temporal fascia.

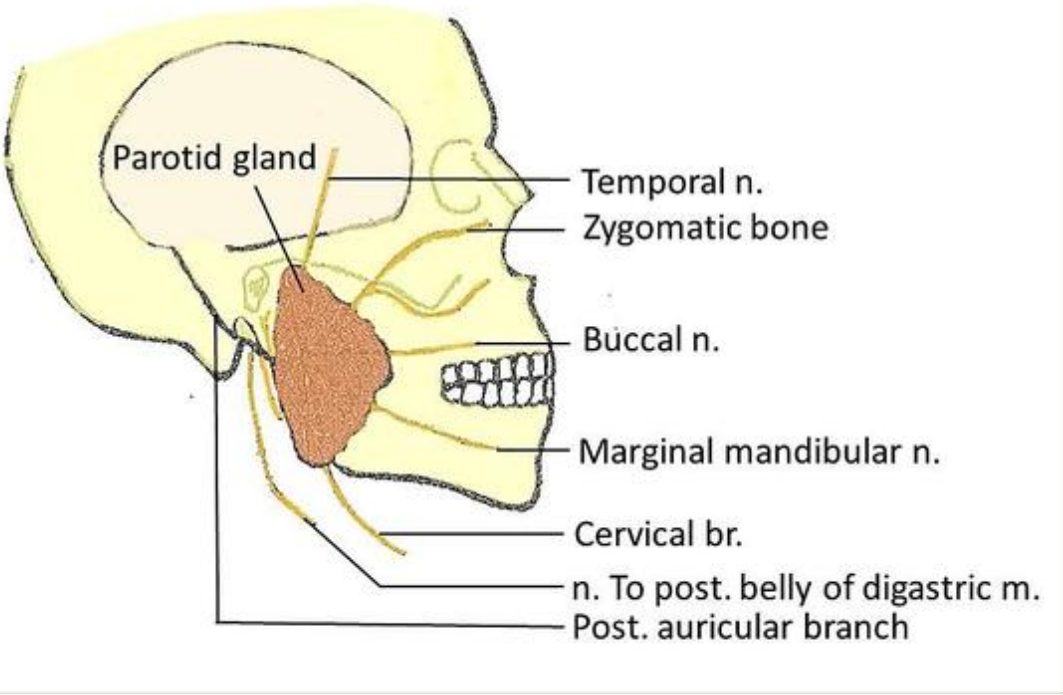


FIGURE 3 Extracranial branches of the facial nerve.
Anatomy atlas · Doss p. 149

3.3 Blepharoplasty

Upper lid blepharoplasty removes excess skin (*dermatochalasis*) with a strip of orbicularis and a conservative amount of fat from the medial and central compartments, through a crease incision about 8 to 10 mm above the lashes. At least *20 mm* of skin must remain between the lash line and the brow, and a ptotic brow, rather than the lid, may be the real problem.

Before surgery, record vision and fields, test for dry eye with the *Schirmer test*, and look for thyroid eye disease, previous surgery and true *blepharoptosis*, which needs levator surgery rather than skin removal.

In the **lower lid**, fat alone is removed through a *transconjunctival* approach, and excess skin through a *subciliary* skin or skin–muscle flap. Lid laxity is tested with the *snap test* and the distraction test, in which more than 6 to 10 mm is abnormal; a lax lid needs a canthopexy or canthoplasty to avoid ectropion. A *negative vector*, with the globe projecting ahead of the cheek, is a warning against removing fat.

The feared complication is a **retrobulbar haematoma**, which can blind; it presents with pain, proptosis and falling vision, and is treated by emergency *lateral canthotomy and cantholysis*, opening the wound, and mannitol, acetazolamide and steroids. The others are lagophthalmos and dry eye, ectropion and scleral show, ptosis, diplopia from inferior oblique injury, and milia.

PAIN AND PROPTOSIS AFTER EYELID SURGERY

Severe pain, a tense proptosed eye and falling vision after blepharoplasty mean a retrobulbar haematoma. Release the sutures and perform a lateral canthotomy and cantholysis at the bedside within minutes, then call ophthalmology. Vision can be lost within about 90 minutes.

3.4 Rhytidectomy (facelift)

The facelift depends on the **SMAS**, the superficial musculoaponeurotic system, continuous with the platysma below and the temporoparietal fascia above. The facial nerve branches run *deep* to it, which is why dissection above the SMAS is safe and dissection beneath it demands care.

Techniques are graded by depth: a **skin-only** lift; **SMAS plication**; **SMAS imbrication** or lateral SMASectomy; the **deep-plane** lift, which goes beneath the SMAS and carries the malar fat to give the strongest midface effect at more risk to the nerves; and the **composite** lift. The neck is treated with liposuction, a midline platysmaplasty and, in selected patients, contouring of the submandibular gland or digastric. Dedo's classification guides the neck: class I minimal deformity, II skin laxity, III fat, IV platysmal banding, V micrognathia and VI a low hyoid.

Haematoma is the commonest complication, at about 3 to 15%, more often in men and hypertensive patients, and must be evacuated promptly. Skin necrosis occurs behind the ear, especially in smokers. The **great auricular nerve** is the nerve most often injured, leaving the ear lobe numb; facial nerve injury, usually of the temporal

or marginal mandibular branch, is usually temporary. Alopecia, a shifted hairline, the pulled-down "pixie ear" and poor scars are the cosmetic failures.

OPERATIVE GUIDES IN THEATRE 3 operations

Brow lift, facelift and blepharoplasty.

Brow Lift Benjamin W. Light, Paul Leong
Brow lift
Face lift
Face lift
Blepharoplasty
Blepharoplasty

4 CHAPTER FOUR The ear

Otoplasty.

4.1 The prominent ear

The prominent ear is the commonest auricular deformity. It results from a deficient *antihelical fold*, a deep *conchal bowl*, or both, sometimes with a protruding lobule. Normally the auriculocephalic angle is about 20 to 30°, the helix sits 15 to 20 mm from the scalp at the top and about 20 mm at mid-ear, and the ear is about 6 cm long and 55% as wide as it is long.

Surgery is usually done at five to six years, when the ear has reached 85 to 90% of adult size and before teasing at school begins. In newborns, *ear moulding* splints in the first weeks correct many deformities without surgery.

4.2 Techniques and complications

Mustardé horizontal mattress sutures create the antihelical fold, and **Furnas** conchomastoid sutures set back the concha; they are placed posteriorly so as not to narrow the canal. Stiff cartilage is weakened by scoring or excision (the Chongchet, Stenström and Converse techniques), and the conchal bowl or lobule may be reduced directly.

Haematoma is the urgent complication, because it leads to perichondritis and a cauliflower ear. Infection and chondritis, suture extrusion and asymmetry also occur. Under-correction or recurrence is the commonest problem overall, and over-correction of the middle third with prominent poles produces the "*telephone ear*".

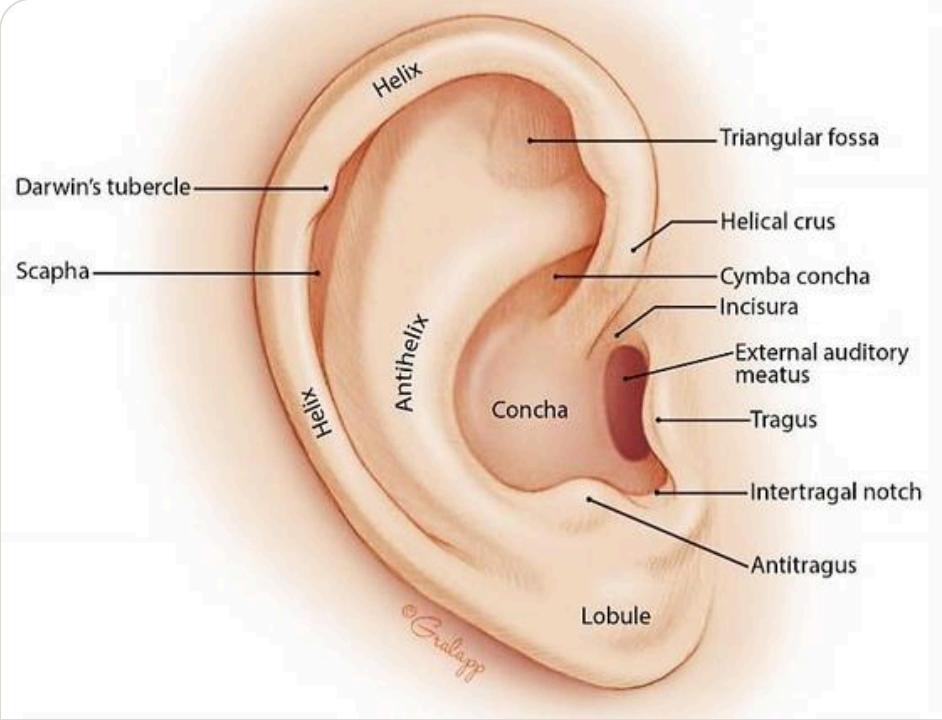


FIGURE 4 Surface landmarks of the auricle.

Anatomy atlas · ENT Steps p. 29

OPERATIVE GUIDES IN THEATRE 1 operation

Otoplasty.

Otoplasty
Otoplasty

5 CHAPTER FIVE Skin and injectables

Resurfacing, toxin, fillers.

5.1 Resurfacing

Every resurfacing method works the same way: it removes the epidermis and injures the papillary dermis, so that new collagen and epidermis form. Injury that reaches

the reticular dermis scars.

Chemical peels are superficial (glycolic acid, Jessner solution, 10 to 25% TCA), medium (Jessner solution followed by 35% TCA) or deep (Baker–Gordon phenol). Phenol is cardiotoxic, causing *arrhythmias*, as well as nephrotoxic and hepatotoxic, so the ECG is monitored, the patient hydrated and one facial unit treated at a time. **Dermabrasion** works mechanically down to the papillary dermis, signalled by pinpoint bleeding, and suits acne scars, rhinophyma and scar revision. Among **lasers**, the ablative CO2 laser penetrates deeper and tightens, while Er:YAG is shallower and heals faster; fractional lasers make microscopic columns of injury that heal rapidly, and non-ablative lasers treat beneath an intact epidermis.

Resurfacing is avoided within 6 to 12 months of isotretinoin (some say two years), and in patients with keloids, previous irradiation or scleroderma. Active herpes needs antiviral prophylaxis, and darker skin types risk pigment change.

5.2 Botulinum toxin

Botulinum toxin blocks the presynaptic release of acetylcholine by cleaving the SNARE proteins, SNAP-25 in the case of type A. It acts within 2 to 3 days, peaks at 1 to 2 weeks and lasts 3 to 4 months.

Cosmetically it treats glabellar lines (corrugator and procerus), forehead lines (frontalis), crow's feet (orbicularis oculi), platysmal bands and the gummy smile. In ENT it treats spasmodic dysphonia, hemifacial spasm and blepharospasm, synkinesis after facial palsy and the opposite side for symmetry, Frey syndrome, sialorrhoea and sialoceles, cricopharyngeal dysfunction, palatal myoclonus and migraine.

Its complications are bruising and headache, *eyelid ptosis* from spread to the levator (treated with apraclonidine drops), brow ptosis, an asymmetric smile and dysphagia. It is avoided in myasthenia gravis, Lambert–Eaton syndrome and other neuromuscular diseases, in patients taking aminoglycosides, and in pregnancy and breastfeeding.

5.3 Fillers and fat

Hyaluronic acid is the commonest filler. It lasts 6 to 18 months and, importantly, can be dissolved with *hyaluronidase*. Calcium hydroxylapatite and poly-L-lactic acid, which stimulates collagen over months, last longer, and permanent fillers such as PMMA and silicone carry a risk of late granulomas. **Autologous fat** is harvested, centrifuged and injected in small aliquots in several layers, with unpredictable survival.

The danger zones are the glabella, nasal dorsum, nasolabial fold and temple, where filler can enter an artery.

VASCULAR OCCLUSION FROM FILLER

- **Skin occlusion:** blanching, pain, then a livedo or dusky pattern after injection. Stop injecting, flood the area with high-dose hyaluronidase (for hyaluronic acid fillers), massage, and apply warm compresses.
- **Visual loss:** retrograde embolism into the ophthalmic artery, from glabellar, nasal or nasolabial injection. This is an ophthalmic emergency.

OPERATIVE GUIDES IN THEATRE 2 operations

Facial resurfacing, and cosmetic botulinum toxin and fillers.

Facial Resurfacing: Peels and Lasers
Cosmetic Botox and Fillers
Botox and fillers

5.4 Scar revision

Scars hide best in *relaxed skin tension lines*, facial creases and the borders of aesthetic units, so revision aims to move them there.

The **Z-plasty** lengthens a scar and changes its direction. With limbs at 30°, 45° and 60° it lengthens the scar by about 25%, 50% and 75%, and turns it by up to 90°, which makes it the tool for contracted scars and for scars lying 30 to 60° off the tension lines. Long scars that run against the lines are broken up with a **W-plasty** or a geometric broken-line closure. Dermabrasion or laser at 6 to 8 weeks refines the result, and steroid injection or silicone controls hypertrophic scars and keloids.

5.5 Hair restoration

Male pattern baldness is androgenetic alopecia, graded by the *Norwood classification*. Minoxidil, finasteride and platelet-rich plasma slow it. Transplantation moves follicular unit grafts from the androgen-resistant occipital scalp, harvested as a strip or by *follicular unit extraction*, with single-hair units placed at the hairline. Transplanted hairs shed after a few weeks and regrow from about three months.

PRESERVATION RHINOPLASTY AND POWERED TOOLS (OXFORD TEXTBOOK)

Removing a dorsal hump in one piece (the Joseph technique) destroys the junction between the nasal bones, upper lateral cartilages and septum, which can lead to middle vault collapse, an *inverted V* deformity and internal valve narrowing; hence the routine use of **spreader grafts** or **spreader flaps** (folding the upper lateral cartilages inwards) after hump reduction, and preserving the mucoperichondrium under the dorsum as a soft-tissue spreader. **Preservation rhinoplasty**, revived in recent years, aims to avoid these problems. As described by Daniel, it has three parts: subperichondrial elevation of the soft tissue envelope, preservation of the nasal ligaments, and **dorsal preservation**, in which the hump is lowered by removing a strip of septum and bone beneath it ("push-down" or "let-down") rather than by resection; it suits a well-shaped dorsum with a hump. **Piezoelectric** instruments and powered burs allow precise osteotomies under direct vision after wide subperiosteal exposure. Every rhinoplasty assessment should include a patient-reported outcome measure and a screening questionnaire for body dysmorphic disorder, at least six standard photographs, and a functional examination of the valves.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Facial proportions** (thirds and fifths) and the main angles.
- **Major and minor tip support mechanisms;** the tripod concept.

- **The internal and external nasal valves:** the Cottle test, spreader and batten grafts.
- **Open versus closed rhinoplasty;** osteotomies; the L-strut.
- **Complications:** pollybeak, saddle nose, inverted V.
- **The frontal branch** and its safe plane; the SMAS.
- **Facelift haematoma;** great auricular nerve injury.
- **Blepharoplasty:** tests, retrobulbar haematoma.
- **Otoplasty:** Mustardé and Furnas sutures; the telephone ear.
- **Botulinum toxin:** mechanism, uses, complications.
- **Filler vascular occlusion and hyaluronidase.**
- **Z-plasty geometry.**

6.2 Common pitfalls

COMMON PITFALLS

- Removing a dorsal hump without osteotomies or spreader grafts (open roof, inverted V).
- An aggressive cephalic trim leading to alar collapse and notching.
- Blepharoplasty without recognising true ptosis, dry eye or a lax lower lid.
- Injecting filler in danger zones without aspiration, slow low-pressure technique, or hyaluronidase available.
- Operating on a patient with body dysmorphic disorder.

Written for this app from Oxford Textbook of Otolaryngology ch. 35, 40, 41, Oxford Specialist Handbook ch. 22, K. J. Lee ch. 54. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

Imaging for this region

The core imaging conditions of this region, with their key facts and a representative scan. For the normal anatomy and a search pattern, read Lesson 8, Sinus CT before surgery; Lesson 9, The orbit in the imaging volume.

NOSE & NASAL CAVITY · CONGENITAL

Meningoencephalocele

IMAGING FINDINGS

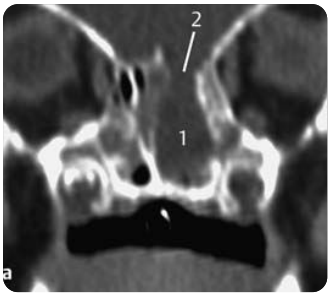
- Bony defect in the cribriform plate, lateral lamella, ethmoid roof or sphenoid
- Soft tissue or fluid through the defect
- MRI: CSF, meninges or brain in continuity with the intracranial contents

TOP DIFFERENTIALS

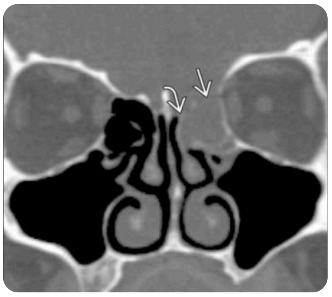
- Skull base defect without leak
- Sinus opacification from mucus

CHECKLIST

- Lateral sphenoid recess and multiple defects
- Signs of raised intracranial pressure: empty sella



CT, coronal. The left nasal cavity is filled with a soft-tissue mass (1) that seems to be connected to brain through a defect in the roof of the…



Coronal bone CT shows large bony defect in left ethmoid roof, lateral to insertion of middle turbinate. Because there is complete opacification of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 920.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Maxillary Sinusitis

IMAGING FINDINGS

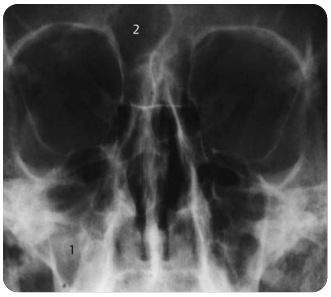
- Air–fluid level, frothy or bubbly secretions
- Mucosal thickening
- Chronic: sclerotic thickened walls (osteitis)

TOP DIFFERENTIALS

- Retention cyst or polyp
- Post-traumatic blood
- Odontogenic source (periapical lucency, oroantral fistula)

CHECKLIST

- Look at the teeth and the ostiomeatal unit
- Contrast CT only for complications



Caldwell view. This 26-year-old patient had pain in the right maxillary region. The right maxillary sinus seems to be opacified in the lower part (1). The lef…



Axial NECT in patient with purulent nasal discharge and headache shows patchy bilateral ethmoid sinus disease, an air-fluid level in the right sphenoid…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 688.

NOSE & NASAL CAVITY · INFECTION & INFLAMMATION

Nasal Polyps

IMAGING FINDINGS

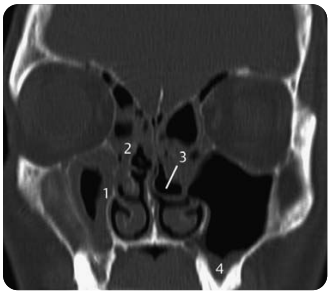
- Polypoid soft tissue filling the nose and all sinuses, both sides
- Widening of the ethmoid labyrinth and thinned septa
- Hyperdense contents suggest fungus or inspissated mucus

TOP DIFFERENTIALS

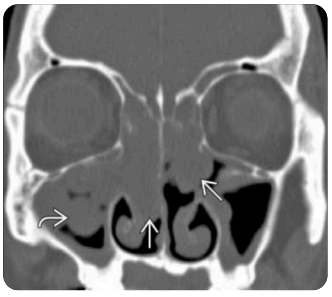
- Allergic fungal sinusitis
- Cystic fibrosis
- Inverted papilloma (unilateral)

CHECKLIST

- Bone erosion or asymmetry: think tumour or fungus
- Pre-operative CLOSE checklist



CT, coronal. The scan shows mucosal thickening in the right maxillary sinus (1), partial opacification of the ethmoid cells (2) on both sides, pneumatizati…



Coronal bone CT shows the classic appearance of sinonasal polyposis
Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 708

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 708.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Noninvasive Fungal Sinusitis

IMAGING FINDINGS

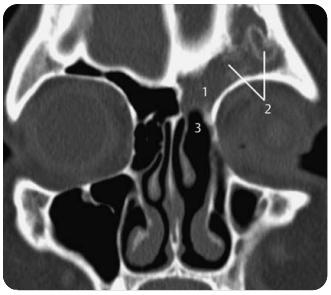
- Fungal ball: single sinus with dense contents and calcification, sclerotic walls
- Allergic fungal: expanded sinuses with central hyperdensity and low-density rim
- Very dark on T2

TOP DIFFERENTIALS

- Polyposis
- Mucocele
- Inspissated secretions

CHECKLIST

- Lamina papyracea and skull base remodelling
- Usually unilateral or asymmetric



Patient with constant left frontal sinus pressure. CT, coronal (a), sagittal (b), and axial ©. There is complete opacification of the left frontal sinus…



Axial NECT shows unilateral involvement of allergic fungal sinusitis. The involved sinuses are expanded, with centrally dense inspissated…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 700.

NOSE & NASAL CAVITY · CONGENITAL

Choanal Atresia

IMAGING FINDINGS

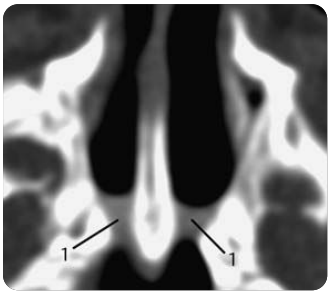
- Narrowed posterior choanae
- Thickened vomer; medially bowed posterior maxilla
- Bony, membranous or mixed closure; fluid in the nose

TOP DIFFERENTIALS

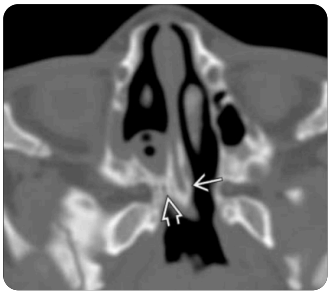
- Choanal stenosis
- Pyriform aperture stenosis
- Nasolacrimal duct mucocele

CHECKLIST

- CT after suction and decongestion
- CHARGE: look at the inner ears



CT, axial. Both choanal areas demonstrate narrowing of the bony passage, but congenital fibrous closure (1) is the cause of the distres…



Axial bone CT through the upper choana in a child with mixed bony/osseous choanal atresia shows osseous choanal obstruction…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 670.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Orbital Cellulitis

IMAGING FINDINGS

- Stranding of the postseptal fat
- Swollen extraocular muscles; proptosis
- No discrete collection

TOP DIFFERENTIALS

- Subperiosteal abscess
- Invasive fungal infection
- Idiopathic orbital inflammation

CHECKLIST

- Separate preseptal from postseptal
- Repeat imaging if not improving



CT, axial with intravenous contrast. Bilateral opacification of the ethmoid (1). The lamina papyracea is intact without signs of periosteal…



Axial CECT in a patient with facial impetigo shows edema and enhancement of the preseptal soft tissues with normal appearance of th…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 814.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Subperiosteal Orbital Abscess

IMAGING FINDINGS

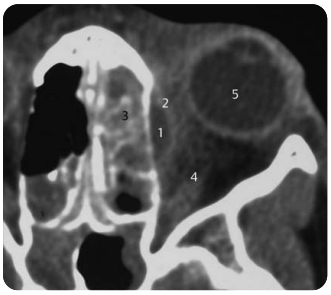
- Lentiform rim-enhancing collection along the lamina papyracea
- Medial rectus and globe displaced laterally
- Adjacent ethmoid opacification; diffusion restriction

TOP DIFFERENTIALS

- Orbital cellulitis (phlegmon)
- Idiopathic orbital inflammation
- Mucocele

CHECKLIST

- Size, posterior extent and vision
- Superior ophthalmic vein and cavernous sinus



CT, axial with intravenous contrast enhancement. The CT scan shows a subperiosteal orbital abscess (1) with typical rim enhancement (2). The…



Axial CECT shows asymmetric opacification of the left ethmoid sinuses with a large subperiosteal abscess extending into the medial…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 810.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Invasive Fungal Sinusitis

IMAGING FINDINGS

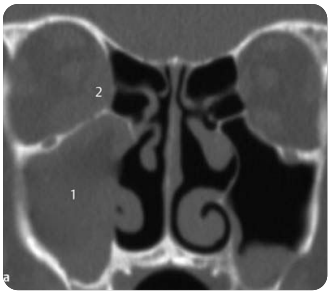
- Unilateral mucosal thickening with fat stranding outside the sinus
- Subtle bone erosion; loss of fat in the pterygopalatine fossa or orbit
- Non-enhancing mucosa ("black turbinate") on MRI

TOP DIFFERENTIALS

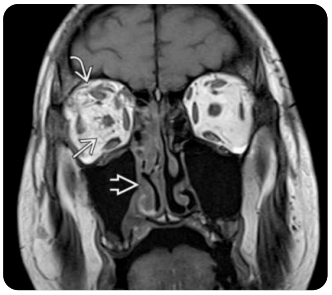
- Acute bacterial sinusitis with complication
- Lymphoma or carcinoma

CHECKLIST

- Immunocompromised patient: endoscopy and biopsy without delay
- Orbit, cavernous sinus and brain on MRI



CT, coronal, bone-tissue setting. The bone-tissue setting reveals complete opacification of the right maxillary sinus (1), as well as very doubtful…



Coronal T1WI MR in patient with chronic myelogenous leukemia complicated by Rhizopus acute invasive fungal rhinosinusitis…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 704.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Mucocele

IMAGING FINDINGS

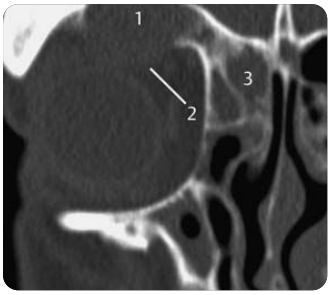
- Expanded, airless sinus with smooth remodelled walls
- Variable density and signal with protein content
- Rim enhancement only

TOP DIFFERENTIALS

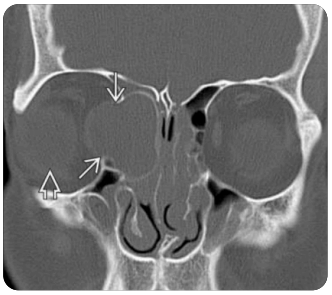
- Allergic fungal sinusitis
- Polyposis
- Tumour (solid enhancement, destruction)

CHECKLIST

- Orbit and skull base erosion
- Frontal recess obstruction as the cause



CT, coronal. The left frontal sinus is opacified with expansion of its inferior bony outline through the roof of the orbit (1). Although this is a bone-…



Coronal bone CT shows the typical features of a right ethmoid mucocele. The sinus is opacified, and there is remodeling of the surrounding bony…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 716.

ORBIT & SINUS RELATIONS · TRAUMA

Blow-out Fracture

IMAGING FINDINGS

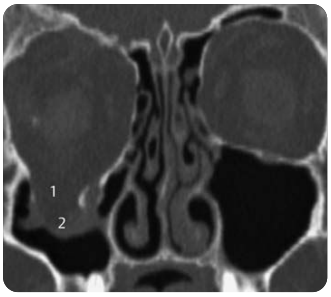
- Fracture of the orbital floor or medial wall with the rim intact
- Herniated fat and muscle; rounded inferior rectus
- Blood in the maxillary sinus; orbital emphysema

TOP DIFFERENTIALS

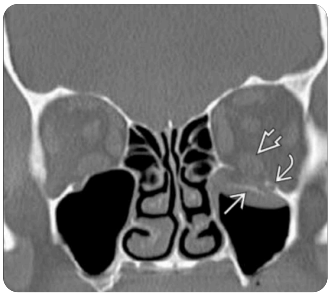
- Dehiscent lamina papyracea
- Old fracture

CHECKLIST

- Muscle entrapment, especially the trapdoor fracture in children
- Defect size (enophthalmos) and retrobulbar haemorrhage



CT, coronal. Traumatic disruption of the bony floor of the orbit (1), and prolapse of orbital fat into the maxillary sinus (2). The eyeball was at a lower…



Coronal bone CT demonstrates a mildly depressed left orbital floor orbital blowout fracture
Koch et al., Diagnostic Imaging: Head…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 980.

NOSE & NASAL CAVITY · TUMOURS

Inverted Papilloma

IMAGING FINDINGS

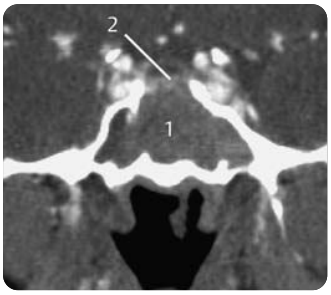
- Unilateral mass on the lateral nasal wall near the middle meatus
- Focal hyperostosis at the attachment
- Convoluted cerebriform pattern on T2 and post-contrast MRI

TOP DIFFERENTIALS

- Antrochoanal or solitary polyp
- Squamous cell carcinoma (in up to about 10%)
- Juvenile angiofibroma

CHECKLIST

- MRI separates tumour from secretions
- Bone destruction raises concern for carcinoma



Patient operated before because of nasal polyps. Histologic evaluation revealed inverted papilloma. Follow-up scans are shown.…



Axial bone CT shows a mass in the nasal cavity along the lateral wall near the middle meatus. The maxillary sinus is opacified, but it is difficult to…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 744.

NOSE & NASAL CAVITY · TUMOURS

Angiofibroma, Juvenile

IMAGING FINDINGS

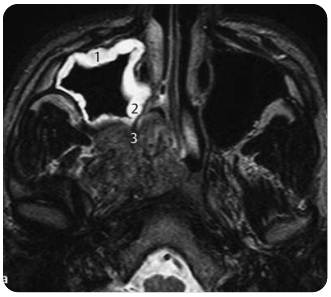
- Enhancing mass at the sphenopalatine foramen in an adolescent male
- Widened pterygopalatine fossa; bowing of the posterior maxillary wall
- Flow voids; extension to the sphenoid, infratemporal fossa and skull base

TOP DIFFERENTIALS

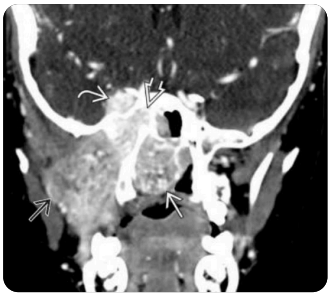
- Hypervascular polyp
- Rhabdomyosarcoma
- Antrochoanal polyp

CHECKLIST

- No biopsy: angiography and embolisation before surgery
- Vidian canal and cavernous sinus spread



A young man with obstruction and bloodtinged nasal discharge from the right side of the nose.
Dunnebier, Imaging for…



Coronal CECT in a 13-year-old shows typical growth pattern of a large, hypervascular angiofibroma occluding the nasopharynx, extending into righ…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 740.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Sinusitis and its orbital and intracranial complications

IMAGING FINDINGS

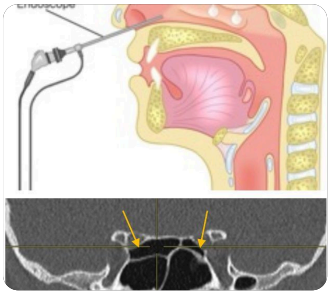
- Orbit: preseptal or postseptal stranding, subperiosteal or orbital abscess
- Intracranial: epidural or subdural empyema, abscess, meningitis
- Venous: superior ophthalmic vein and cavernous sinus thrombosis

TOP DIFFERENTIALS

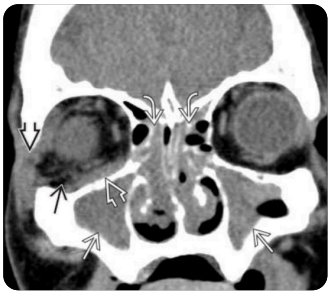
- Idiopathic orbital inflammation
- Tolosa-Hunt syndrome

CHECKLIST

- Contrast CT for the orbit, MRI for intracranial disease
- Frontal sinusitis: look for Pott puffy tumour



Coronal reconstructed image from a high-resolution CT. Bilateral Onodi cells (arrows). This puts the optic nerves at risk of damage during FESS…



Coronal NECT in a man with acute sinusitis complicated by orbital cellulitis and subperiosteal abscess shows extensive opacification of…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 696.

PARANASAL SINUSES · TUMOURS

Sinonasal tumours

IMAGING FINDINGS

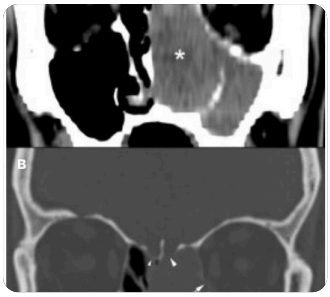
- Unilateral solid mass with bone destruction
- Intermediate T2, solid enhancement, distinct from bright secretions
- Spread to orbit, skull base, pterygopalatine fossa, cheek

TOP DIFFERENTIALS

- Inverted papilloma
- Lymphoma
- Olfactory neuroblastoma, adenocarcinoma, melanoma

CHECKLIST

- Periorbita and dura
- Perineural spread along V2 and the vidian nerve; nodes



Coronal reformatted images from a noncontrast CT in soft tissue window (a) and bone window settings (b). The mass (asterisk) causes resorption of…



Axial CECT shows the typical location and appearance of an antral SCCa. There is extension into the premaxillary soft tissues anteriorly an…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 754.